1. On the Formation of the form System of the Ternary Biquadratic form

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(Extract from the author's dissertation.)

Work by Gordan, Maisano, and Pascal[1] deals with the form system of the ternary biquadratic form. Gordan sets up the complete form system, consisting of 54 formations, for the special automorphic form

$$f = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1,$$

on the basis of principles similar to those which he had given for form systems in the binary domain.

In Maisano's work, for the general biquadratic form, the forms up to and including the fifth order[2] are set up, together with some invariants, covariants, and contravariants of higher order, according to the method used by Gordan in volume I of the *Mathematische Annalen* for the ternary cubic form. Pascal, using Maisano's results, is chiefly concerned with the question of the decomposition of the biquadratic form into factors[3].

The aim of my investigations is to set up the form system for the general ternary biquadratic form; to begin with, only the main foundations are given, and a so-called "relatively complete system"[4] is established.

The basic idea for forming systems of ternary forms is the same as in the binary domain. Starting from a first relatively complete system – the system of forms taken over from the binary case – one passes, according to a definite law, to systems with ever higher modulus, until either the system of a modulus – taking the modulus as ground form – becomes finite and known, or until a modulus can be reduced to forms having invariants as factors. By transvection of the relatively complete system with the system of the modulus, in the first case one obtains the absolutely complete system, while in the second case the relatively complete and absolutely complete systems coincide. By Hilbert's general proof the finiteness of form systems, this procedure must necessarily come to an end.

In our case, the modulus (abc) of the first relatively complete system can be led back to the moduli

$$\Delta = (abc)^2 a_x^2 b_x^2 c_x^2 \quad \text{and} \quad \nu = (abu)^4;$$

from this the relatively complete system modulo ν is readily obtained. As the sequence of moduli we now choose the forms

$$\nu; \qquad \nu^{(\nu)} = (\nu\nu_1 x)^4 = s_x^4,$$
$$\nu^{(s)} = (ss'u)^4 \ldots,$$

and adjoin, as further moduli, two quadratic forms occurring in the formation of the relatively complete system modulo s, namely u_σ^2 and t_x^2. One can then show that the modulus following s, namely $(ss'u)^4$, is reducible to the modulus (ϱ, t). Since, however, the simultaneous system of two quadratic forms is finite and known, the termination described above has thereby been reached. As the relatively complete system modulo (ϱ, t) one obtains 331 formations. —

[1] P. Gordan, "Über the volle Formensystem of the ternären biquadratischen form" $f = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1$ (Math. Annalen vol. XVII, pp. 217–233, 1880); G. Maisano, 1. "Sistemi completi dei primi cinque gradi della forma ternaria biquadratica e degl' invarianti, covarianti e contravarianti di sesto grado" (Giorn. di Battaglini XIX). 2. "Sui covarianti indipendenti di 6° grado nei coefficienti della forma biquadratica ternaria" (Rend. Circ. Mat. di Palermo I, 1887); E. Pascal, "Contributo alla teoria della forma ternaria biquadratica e delle sue varie decomposizioni in fattori" (Memoria premiata dalla R. Accademia delle scienze fisiche e matematiche di Napoli, 1905).

[2] By "order" the dimension in the coefficients is to be understood, and by "degree" the dimension in the variables.

[3] In his second short note Maisano attempts to prove a linear dependence among the three covariants of order 6 and degree 6. In contrast to this not quite complete proof, Pascal believes he has proved the linear independence of the three covariants of order 6, as well as that of the three invariants of order 9. That, however, a linear relation does in fact exist between the three covariants on the one hand and the three invariants on the other, emerged in the course of my computations; in Pascal's notation the explicit formulas are

$$0 = 20\Omega_1 + 6\Omega_2 - 3\Omega_3 + 4fC_1 - 12fC_2 - 4A \cdot \Delta,$$

$$0 = 4A^3 - 15AB + 30C - 90D + 9E.$$

[4] For the terminology, see Gordan–Kerschensteiner, *Vorlesungen über Invariantentheorie*, vol. II, p. 227.

I add a few details concerning the method used.

The fundamental process for producing forms, the folding process, can be defined in the ternary domain as follows:

If in a symbolic product

$$s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$$

one replaces the pairs of factors

$$\begin{array}{r|c|c|c|c} & s_x t_x & u_\sigma u_\tau & s_x u_\sigma \text{ or } s_x u_\tau & t_x u_\tau \text{ or } t_x u_\sigma \\ \text{respectively by} & (stu) & (\sigma\tau x) & s_\sigma \text{ or } s_\tau & t_\tau \text{ or } t_\sigma \\ \text{folding} & \text{I} & \text{II} & \text{III} & \text{IV,} \end{array}$$

then the forms so arising have been obtained from the original form by folding[5].

Here one may still always put $s_\sigma = 0$; $t_\tau = 0$. For the relation among the individual foldings, the following theorem holds:

The foldings I and II are fundamental foldings, from which, independently of the order of composition, the foldings III and IV can be composed. In other words: to form all forms arising from a given expression by folding, one need only apply foldings I and II[6].

By a "form sequence" we mean an initial form together with all forms arising from it by folding into itself. By the theorem, a form sequence $s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$ can be arranged in a rectangular scheme. In proceeding, one obtains:

1. by moving one column to the right, all forms arising by folding I from the adjacent forms;

2. by moving one row downward, all forms arising by folding II from the forms standing above, and hence all forms arising by folding.

Under these assumptions, the theorems on reducers can be stated in their most general form, under the definition

"A reducer is a reducible form sequence,"

as follows:

If the initial form of a form sequence is reducible because one of its terms has arisen by folding with a reducer, and if the final form of the form sequence has arisen from this same term by folding, then the whole form sequence is reducible.

Further reduction methods to be mentioned are:

1. the so-called "double reduction," that is, the reduction of a form in two ways in order to obtain a relation between the higher forms;

2. "folding with decomposed forms," which partly has the character of reduction by reducers and partly that of "double reduction."

[5] Gordan, Math. Annalen, vol. XVII, p. 219.

[6] The theorem has an exception in the case where n and μ, respectively m and ν, vanish simultaneously; the "form sequence" then reduces to the diagonal term.

2. On the Formation of the form System of the Ternary Biquadratic form

Journal for the reine and angewandte Mathematik 134 (1908), pp. 23–90 and two tables

Introduction.

Work by Gordan, Maisano, and Pascal[7][8] deals with the form system of the ternary biquadratic form. Gordan sets up the complete form system, consisting of 54 formations, of the special automorphic form

$$f = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1$$

on the basis of principles similar to those that he had given for form systems in the binary domain.

In Maisano's work, for the general biquadratic form, the forms up to and including the fifth order[9] are set up, together with some invariants, covariants, and contravariants of higher order, according to the method used by Gordan in volume I of the *Mathematische Annalen* for the ternary cubic form. Pascal, using Maisano's results, is chiefly concerned with the question of the decomposition of the biquadratic form into factors[10].

The aim of the following investigations is to set up the form system for the general ternary biquadratic form; in this work, namely, only the principal foundations are given, and a so-called "relatively complete system"[11] *is established.*

The work is closely connected with Gordan's work; however, the principles that were stated there only quite generally first had to be worked out in detail. With the aid of a theorem on the connection among the foldings and by introducing the "form sequence" (§ 1), the reduction theorems are sharply formulated and completed (§ 3), while the recursive establishment of special series expansions (§ 2) gives the computational means for actually carrying out the reductions.

The basic idea for forming systems of ternary forms is the same as in the binary domain. Starting from a first relatively complete system – the system of forms taken over from the binary case – one passes, according to a definite law, to systems with ever higher modulus, until either the system of a modulus – the modulus taken as ground form – becomes finite and known, or until a modulus can be reduced to forms having invariants as factors. By transvection of the relatively complete system with the system of the modulus, in the first case the absolutely complete system arises, while in the second case the relatively complete and absolutely complete systems coincide. By Hilbert's general proof the finiteness of form systems, this procedure must necessarily come to an end.

In our case the modulus (abc) of the first relatively complete system (§ 4) is reduced to the moduli

$$\Delta = (abc)^2 a_x^2 b_x^2 c_x^2 \quad \text{and} \quad \nu = (abu)^4$$

(§ 5), and then the relatively complete system modulo ν is found (§ 6). These results are already given in Gordan's work for the general biquadratic form; our manner of deriving them, however, is more easily transferred to forms of higher degree.

As the sequence of moduli we now choose the forms (§ 7)

$$\nu, \qquad \nu(\nu) = (\nu\nu_1 x)^4 = s_x^4, \qquad \nu(s) = (ss'u)^4, \ldots.$$

[7] A short extract from the Introduction and Chapter I appeared in the *Sitzungsberichte of the physikalisch-medizinischen Sozietät Erlangen 1907*, pp. 176–179.

[8] P. Gordan, on the complete form system of the ternary biquadratic form

$$f = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1.$$

(*Math. Annalen*, vol. XVII (1880), pp. 217–233.) G. Maisano, 1) *Sistemi completi dei primi cinque gradi della forma ternaria biquadratica e degl' invarianti, covarianti e contravarianti di sesto grado.* (Giorn. di Battaglini XIX.) 2) *Sui covarianti indipendenti di* 6° *grado nei coefficienti della forma biquadratica ternaria.* (Rend. Circ. Mat. di Palermo I, 1887.) E. Pascal, *Contributo alla teoria della forma ternaria biquadratica e delle sue varie decomposizioni in fattori.* (Memoria premiata dalla R. Accademia delle scienze fisiche e matematiche di Napoli. 1905.)

[9] By "order" the dimension in the coefficients is to be understood, and by "degree" the dimension in the variables.

[10] In his second short note, Maisano attempts to prove a linear dependence of the three covariants of order 6 and degree 6. In contrast to this not quite complete proof, Pascal believes he has proved the linear independence of the three covariants of order 6, and likewise that of the three invariants of order 9. That, however, a linear relation does in fact exist between the three covariants on the one hand and the three invariants on the other is shown by formulas (13), § 11, and (22), § 17 note, of the present work, where these relations are given explicitly. According to a communication from Pascal, the contradiction is explained by a numerical error in the expression for his contravariant p (called ϱ here), p. 46 of his paper.

[11] For the terminology see § 3a.

In forming the relatively complete system modulo s, two quadratic forms occurring in the system, u_ϱ^2 and t_x^2, are adjoined as moduli (§§ 9 and 10), and accordingly the relatively complete system modulo (s, ϱ, t) is formed. It is then shown (§ 17) that the modulus $(ss'u)^4$ is reducible to the moduli ϱ and t. The next higher system thereby passes over into a relatively complete system modulo (ϱ, t). Since, however, the simultaneous system of two quadratic forms is finite and known, and in the general case consists of 20 formations[12], the termination described above is reached with the establishment of the relatively complete system modulo (ϱ, t). The transvection of this system with the system of (ϱ, t) in order to form the absolutely complete system is reserved.

As the relatively complete system modulo (ϱ, t) one finds 331 formations, which in the appended table are ordered according to their degree in the variables x and u.

Finally we give an overview of the symbols introduced:

$$\begin{aligned}
f &= a_x^4,\\
\theta &= \theta_x^4 u_\vartheta^2 = (abu)^2 a_x^2 b_x^2,\\
K &= K_x^6 u_k^3 = (a\theta u) u_\vartheta^2 a_x^3 \theta_x^3,\\
j &= u_j^6 = (a\theta u)^4 u_\vartheta^2,\\
\Delta &= \Delta_x^6 = a_\vartheta^2 a_x^2 \theta_x^4 = (abc)^2 a_x^2 b_x^2 c_x^2,\\
N &= N_x^8 u_\eta = (a\Delta u) a_x^3 \Delta_x^5,\\
\nu &= u_\nu^4 = (abu)^4,\\
H &= H_x^2 u_\eta^4 = (\nu\nu_1 x)^2 u_\nu^2 u_{\nu_1}^2,\\
L &= L_x^3 u_l^6 = (\nu\eta x) H_x^2 u_\nu^3 u_\eta^3,\\
g &= g_x^6 = (\nu\eta x)^4 H_x^2,\\
\sigma &= u_\sigma^6 = H_\nu^2 u_\nu^2 u_\eta^4,\\
s &= s_x^4 = (\nu\nu_1 x)^4,\\
Z &= Z_x^4 u_\zeta^2 = (ss'u)^2 s_x^2 s_x'^2,\\
\varrho &= u_\varrho^2 = \theta_\nu^4 u_\vartheta^2,\\
t &= t_x^2 = a_\eta^4 H_x^2,\\
i &= a_\nu^4,\\
J &= s_\nu^4.
\end{aligned}$$

Chapter I. General Theorems on Ternary Forms.

§ 1. The folding process. form sequences.

The fundamental process for producing forms is the folding process, which in the ternary domain can be defined as follows. Let there be given a symbolic product

$$s_x^m t_x^n u_\sigma^\mu u_\tau^\nu.$$

If one replaces the pairs of factors

$$s_x t_x \qquad u_\sigma u_\tau \qquad s_x u_\sigma \text{ or } s_x u_\tau \qquad t_x u_\tau \text{ or } t_x u_\sigma$$

respectively by

$$(stu) \qquad (\sigma\tau x) \qquad s_\sigma \text{ or } s_\tau \qquad t_\tau \text{ or } t_\sigma,$$

folding

$$\text{I} \qquad \text{II} \qquad \text{III} \qquad \text{IV},$$

then the resulting forms have arisen from the original one by folding[13].

[12]Cf. Clebsch–Lindemann, *Vorlesungen über Geometrie.* (Vol. I, p. 288 ff.) System of two cogredient forms. Gordan, "Über Büschel of Kegelschnitten." (*Math. Ann.* vol. XIX, p. 530.) System of two contragredient forms.

[13]Gordan, *Math. Annalen* vol. XVII, p. 219.

Here the expression $s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$ can be regarded either as an actual product of two forms $S \cdot T$, or as a single form with several rows of symbols:

$$s_x^m t_x^n u_\sigma^\mu u_\tau^\nu = A_x^{m+n} u_x^{\mu+\nu}.$$

In the first case we speak of the folding of the form S with T, in the second case of the "folding of the form A in itself." For brevity we always assume in what follows, as indeed is the case for all forms later to be considered, that

$$s_\sigma^\lambda = 0, \qquad t_\tau^\lambda = 0 \tag{a}$$

for any λ and x distinct from 0, and in conjunction with any other foldings. This says that the form $s_x^m t_y^n u_\sigma^\mu u_\tau^\nu$ is a so-called "normal form," that is, it vanishes under application of the Ω-process, both with respect to the variables x and u and with respect to the variables y and v.

Thus condition (a) represents no restriction, but can always be achieved[14].

For the connection among the individual foldings the following theorem holds:

Theorem I. Foldings I and II are fundamental foldings from which, independently of the order of composition, foldings III and IV can be composed. In other words: in order to form all forms arising by folding from a given expression, one has to apply only foldings I and II.

Proof: By the identity theorem, respectively the product theorem for matrices, taking (a) into account, one obtains

$$(\widehat{st}\sigma x) = -t_\sigma, \qquad (\widehat{\sigma\tau} su) = -s_\tau,$$
$$(\widehat{st}\tau x) = s_\tau, \qquad (\widehat{\sigma\tau} tu) = t_\sigma,$$
$$(stu)(\sigma\tau x) = s_\tau + t_\sigma - u_x\, s_\tau t_\sigma.$$

Thus by composing foldings I and II one obtains foldings III and IV, and this holds both when applying folding II to folding I, i.e. to the factors $(stu)u_\sigma u_\tau$, and when applying folding I to folding II, to the factors $(\sigma\tau x)s_x t_x$.

By a *form sequence*[15] we mean an initial form together with all forms that have arisen from it by folding in itself, and we denote the form sequence by the initial form. A "higher form" is to mean here any form with more folding in itself, while among the equally entitled foldings s_τ and t_σ one has to be normalized as higher. According to the law of formation of the form sequence it follows that:

1) Each form of the form sequence is linear in the symbols of the initial form.

2) The form sequence is uniquely determined for initial forms with two rows each of contragredient symbols; for initial forms with more rows of symbols it is to be uniquely normalized by distinguishing special foldings among those connected by the identity theorem.

By Theorem I, a form sequence $s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$ $(m > n;\ \mu > \nu)$ can be arranged according to the following rectangular scheme:

$$\begin{array}{c|c|c|c}
s_x^m t_x^n u_\sigma^\mu u_\tau^\nu & (stu) & (stu)^2 & \cdots \\
(\sigma\tau x) & t_\sigma,\ s_\tau & t_\sigma(stu),\ s_\tau(stu) & \cdots \\
(\sigma\tau x)^2 & t_\sigma(\sigma\tau x),\ s_\tau(\sigma\tau x) & t_\sigma^2,\ t_\sigma s_\tau,\ s_\tau^2 & \cdots \\
\vdots & \vdots & \vdots & \ddots \\
(\sigma\tau x)^\nu & t_\sigma(\sigma\tau x)^{\nu-1},\ s_\tau(\sigma\tau x)^{\nu-1} & t_\sigma^2(\sigma\tau x)^{\nu-2},\ t_\sigma s_\tau(\sigma\tau x)^{\nu-2},\ s_\tau^2(\sigma\tau x)^{\nu-2} & \cdots \\
\vdots & t_\sigma(\sigma\tau x)^\nu & t_\sigma^2(\sigma\tau x)^{\nu-1},\ t_\sigma s_\tau(\sigma\tau x)^{\nu-1} & \cdots
\end{array}$$

and the correspondingly continued lower part[16]

$$\begin{array}{c|c|c}
(stu)^n & \cdots & \cdots \\
t_\sigma(stu)^{n-1},\ s_\tau(stu)^{n-1} & s_\tau(stu)^n & \cdots \\
t_\sigma^2(stu)^{n-2},\ t_\sigma s_\tau(stu)^{n-2},\ s_\tau^2(stu)^{n-2} & t_\sigma s_\tau(stu)^{n-1},\ s_\tau^2(stu)^{n-1} & s_\tau^2(stu)^n.
\end{array}$$

Here one obtains, when one moves

1) one column to the right, all forms arising by folding I from the adjacent forms,

2) one row downward, all forms arising by folding II from the forms above,

[14] Cf. Gordan, *Math. Annalen* vol. V, p. 104.

[15] Cf. Clebsch, "Über eine Fundamentalaufgabe of the Invariantentheorie" (Abh. of the Gött. Ges. d. Wiss. vol. XVII): § 17 and end of § 18. The "form sequence" differs from the "properly reduced equivalent system" introduced there by the principle of ordering according to higher forms.

[16] Here, and similarly in what follows, the lower part marked with an asterisk is to be set at the correspondingly marked place at the upper right of the scheme.

and hence all forms arising by folding.

An arbitrary form of the total form sequence, $t_\sigma^\kappa s_\tau^\lambda (stu)^\mu$ or $t_\sigma^\kappa s_\tau^\lambda (\sigma\tau x)^\nu$, forms the beginning of a new form sequence, consisting of all those forms of the rectangular scheme with $t_\sigma^\kappa s_\tau^\lambda (stu)^\mu$ or $t_\sigma^\kappa s_\tau^\lambda (\sigma\tau x)^\nu$ as left upper corner, which have the factor $t_\sigma^\kappa s_\tau^\lambda$.

Supplementary remark. Theorem I has an exception in the case where the numbers n and μ (respectively m and ν) become equal to zero, so that foldings I, II, and IV no longer exist, while folding III (respectively IV) still does. In this case we designate as the form sequence the diagonal member of the scheme:

$$\begin{array}{cccccc} s_x^m u_\tau^\nu & & & & t_x^n u_\sigma^\mu \\ & s_\tau & & t_\sigma & \\ & s_\tau^2 & & t_\sigma^2 & \\ & & \ddots & & \\ & & s_\tau^\nu & & t_\sigma^\mu. \end{array}$$

In accordance with the absence of foldings I and II, the series expansion according to polars of the form sequence in this case always reduces to the initial member; however, the theorems on reducers remain valid.

§ 2. Series expansions according to polars of the form sequence.

For forms with n variables two kinds of series expansions have been set up explicitly[17]:

1) for forms with one row each of contragredient variables, $s_x^m u_\tau^\nu$, a series progressing by powers of u_x, whose coefficients are "normal forms";

2) for forms with two rows of cogredient variables, $s_x^m t_x^n$, an expansion according to polars of the elementary covariants (polars of the form sequence $s_x^m t_x^n$), which in the ternary case reads as follows[18]:

$$s_x^m t_x^{n-\lambda} t_y^\lambda = \sum_\varrho \frac{\binom{m}{\varrho}\binom{\lambda}{\varrho}}{\binom{m+n-\varrho+1}{\varrho}} \left[(st\widehat{xy})^\varrho s_x^{m-\varrho} t_x^{n-\varrho}\right]_{y^{\lambda-\varrho}}. \tag{I}$$

We combine the two expansions by setting up, for forms with two rows each of contragredient variables,

$$s_x^m t_x^{n-\lambda} t_y^\lambda u_\sigma^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa,$$

expansions according to powers of u_x, whose coefficients are composite polars of the form sequence $s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$, taken with respect to the variables x and u.

It is enough to set up the series for two contragredient rows of symbols (respectively variables), since, by combining each two rows of symbols (variables) into a single new row, forms with more rows of symbols (variables) can be reduced to this case.

In order to ensure that all forms of the form sequence contain only two rows of symbols, respectively variables, we specialize:

$$\begin{aligned} &\text{a) } \sigma = \widehat{st}, \quad \text{a') } y = \widehat{uv}, \\ &\text{b) } s = \widehat{\sigma\tau}, \quad \text{b') } v = \widehat{xy}, \end{aligned}$$

and carry out the expansion for the specialization a), a').

By direct transfer of expansion I one obtains composite polars for forms $(stu)^\mu s_x^m t_x^{n-\lambda} t_y^\lambda$, whereas for forms $(stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa s_x^m t_x^{n-\lambda} t_y^\lambda$, instead of composite polars, terms of such a kind arise whose evaluation makes necessary the introduction of ever new auxiliary variables that are to be set equal only at the end, thereby complicating the calculation unnecessarily.

We therefore take an indirect route, by representing the composite polars of all forms of the form sequence, that is, expressions

$$\left[s_\tau^\varrho (stu)^{\mu-\varrho+r}\right]_{y^\lambda}^{v^\kappa},$$

[17] Cf. Gordan, "Über Kombinanten," §§ 2 and 5 (*Math. Ann.* vol. V).

[18] Cf. Study, *Methoden to the Theorie of the ternären Formen* (Teubner 1889) II. § 7. In distinction to the derivation there, ours gives the method for the rapid computational determination of the numerical coefficients $C_{pr\varrho}$. The Clebsch–Study expansion, upon specialization a), a'), would read

$$(stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa s_x^m t_x^{n-\lambda} (tuv)^\lambda = \sum_{p,r,\varrho} C_{pr\varrho} \left[s_\tau^\varrho (stu)^{\mu+r-\varrho}\right]_{\widehat{uv}^{\lambda-r+\varrho} v^{\nu-\varrho},\, u^p,\, (vw=\widehat{xw})},$$

and therefore does not permit the simplification that occurs already in the course of our calculation, once one knows that all terms with the factor u_x are to be omitted.

by the initial members of these polars, and by inversion of the resulting recursive system of equations obtaining the desired expansion according to polars.

By the transfer principle, for the λ-th polar of a form $s_x^m t_x^n : [s_x^m t_x^n]_{y^\lambda}$ ($\lambda \leq n$) the following expansion holds (the case $\lambda > n$ is reduced to the first by the expansion of $[s_y^m t_y^n]_{x^{m+n-\lambda}}$)[19]:

$$\left[s_x^m t_x^n\right]_{y^\lambda} = \sum_r (-1)^r \frac{\binom{m}{r}\binom{\lambda}{r}}{\binom{m+n}{r}} (st\widehat{xy})^r s_x^{m-r} t_x^{n-\lambda} t_y^{\lambda-r},$$

and correspondingly for the κ-th polar of $u_\sigma^\mu u_\tau^\nu : [u_\sigma^\mu u_\tau^\nu]_{v^\kappa}$

$$\left[u_\sigma^\mu u_\tau^\nu\right]_{v^\kappa} = \sum_\varrho (-1)^\varrho \frac{\binom{\mu}{\varrho}\binom{\kappa}{\varrho}}{\binom{\mu+\nu}{\varrho}} (\sigma\tau\widehat{uv})^\varrho u_\sigma^{\mu-\varrho} u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho}.$$

By multiplication it follows, with introduction of the specialization a) and a'),

$$\left[s_x^m t_x^n (stu)^\mu u_\tau^\nu\right]_{\widehat{uv}^\lambda}^{v^\kappa} = \sum_{r,\varrho} c_{r\varrho}\, (st\widehat{uv})^r (st\widehat{uv})^\varrho s_x^{m-r} t_x^{n-\lambda} (tuv)^{\lambda-r} (stu)^{\mu-\varrho} u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho},$$

and from this, by expansion according to powers of u_x with the first member distinguished ($\sum'_{r,\varrho}$ means that the member $r = 0$, $\varrho = 0$ is to be omitted) and taking account of (a) on p. 26,

$$\begin{aligned} & s_x^m t_x^{n-\lambda} (tuv)^\lambda (stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa \\ & \quad = \left[s_x^m t_x^n (stu)^\mu u_\tau^\nu\right]_{\widehat{uv}^\lambda}^{v^\kappa} \\ & \qquad - \sum_{r,\varrho}{}' c_{r\varrho}\, s_\tau^\varrho (stu)^{\mu-\varrho+r} u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho} s_x^{m-r} t_x^{n-\lambda} (tuv)^{\lambda-r+\varrho} v_x^r \\ & \qquad + u_x \sum_\varrho \sum_{r=1}^{\lambda} c_{r\varrho}\, s_\tau^\varrho (stu)^{\mu-\varrho+r-1} (stv) u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho} s_x^{m-r} t_x^{n-\lambda} (tuv)^{\lambda-r+\varrho} v_x^{r-1} \\ & \qquad \vdots \\ & \qquad - (-1)^\lambda u_x^\lambda \sum_\varrho c_{\lambda\varrho}\, s_\tau^\varrho (stu)^{\mu-\varrho} (stv)^\lambda u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho} s_x^{m-\lambda} t_x^{n-\lambda} (tuv)^\varrho, \end{aligned} \tag{II}$$

where

$$c_{r\varrho} = (-1)^{r+\varrho} \cdot \frac{\binom{m}{r}\binom{\lambda}{r}}{\binom{m+n}{r}} \cdot \frac{\binom{\mu}{\varrho}\binom{\kappa}{\varrho}}{\binom{\mu+\nu}{\varrho}} \qquad \text{for } \lambda \leq n, \ \kappa \leq \nu.$$

and correspondingly for the remaining combinations of the numbers $\lambda, n; \kappa, \nu$. Thus the expression $s_x^m t_x^{n-\lambda} (tuv)^\lambda (stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa$ is reduced to a composite polar, and, by applying the identity

$$(stv) u_\tau = (stu) v_\tau - s_\tau (tuv),$$

to a sum of analogously formed expressions corresponding to higher forms of the form sequence.

By iteration one arrives at the expansion

$$s_x^m t_x^{n-\lambda} (tuv)^\lambda (stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa = \sum_{p,r,\varrho} u_x^p\, C_{pr\varrho} \cdot \left[s_\tau^\varrho (stu)^{\mu-\varrho+r}\right]_{\widehat{uv}^{\lambda-r+\varrho} v^{\kappa-\varrho+p},\, v_x^{r-p}}. \tag{III}$$

The numerical coefficients $C_{pr\varrho}$ are computed uniquely and recursively from the coefficients $c_{r\varrho}$, $c'_{r\varrho}$ of the system of equations arising by iteration of (II) (cf. the example on p. 42). Analogous expansions arise for the other specializations.

§ 3. Reduction theorems.

[19] Gordan, *The Resultante binärer Formen* (Chap. I, § 5). Rend. Circ. Mat. di Palermo XXII. 1906.

The known theorems on the reduction of forms and form systems shall now be brought into connection with §§ 1 and 2, for which some definitions taken over from the theory of binary forms are necessary.

a) We call a system of forms a *relatively complete system* modulo a prescribed sequence of forms if it has the property that all formations arising by folding an arbitrary product of these forms can be expressed as integral rational functions of the forms of the system, up to an additive term consisting of expressions that have arisen by folding the system forms with the system of the modulus[20].

b) We call a form *reducible* when it can be expressed by forms having invariants as factors or by "higher forms," that is, by higher forms of the total form sequence to which the form belongs, or by forms containing the symbols of the modulus in higher order.

The theorems on reducers[21] can now be stated in their most general form as follows:

Definition: *A reducer is a reducible form sequence.*

Theorem II. If the initial form of a form sequence is reducible because one of its members has arisen by folding with a reducer (has a reducer as factor), and if the final form of the form sequence has arisen from exactly this member by folding, then the total form sequence is reducible[22].

Proof: The form sequence arises from the initial member by folding in itself, hence by higher folding with the reducer on the one hand, and by folding the reducer in itself on the other hand. In both cases, by § 2, all forms of the form sequence can be represented as polars (transvections) over the form sequence of the reducer.

The inference from the initial form to the total form sequence no longer holds for the remaining reduction methods. These are:

1) so-called "double reduction."

By this we mean the reduction, in two different ways, of an expression that has two mutually independent reducers as factors, whereby relations arise between the higher forms to which one has reduced.

By systematic application of "double reduction" one must, on the one hand, obtain all relations[23]; on the other hand, if "double reduction" applied to different expressions supplies no new relations, one has both a criterion for the independence of the higher forms and a check on the calculation.

2) *Folding with decomposable forms.*

By "decomposable forms" are to be understood – with a slight generalization of the usual concept – forms that can be expressed by products of forms of lower degree and by "higher" forms. Products of forms pass into products under one folding; likewise products of nonlinear forms under twofold folding in the specializations a') and b') (§ 2), according to the product theorem:

$$a_y b_y a_x b_x = \frac{1}{2} a_y^2 b_x^2 + \frac{1}{2} b_y^2 a_x^2 - \frac{1}{2} (ab\widehat{yx})^2.$$

First polars (transvections) and certain second polars of decomposable forms are therefore again decomposable forms. It follows that:

A member of a onefold (respectively twofold) transvection over a decomposable form, arising by onefold (respectively twofold) folding with a decomposable form, itself becomes a decomposable form:

a) if the next-higher forms in the form sequence of the decomposable form decompose or become reducible, or also if, under onefold folding, the specializations $y = \widehat{uv}$ or $v = \widehat{xy}$ occur;

b) if the remaining onefold foldings lead to higher forms (cf. § 12, B. H_k and $H_k(k\eta x)$).

Such a member of a transvection can also decompose:

c) if the remaining onefold foldings lead to lower forms that decompose independently of the folding with the decomposable form, that is, that can be expressed by products and by the form to be reduced, although in each case one must first verify by calculation that the numerical coefficient of the member concerned does not vanish identically. This is nothing other than "double reduction" of the lower form (cf. § 23, C. $H_q(\theta H u)(\theta s u)$).

[20] Gordan–Kerschensteiner, *Vorlesungen über Invariantentheorie.* Vol. II. p. 227.

[21] Gordan, *Math. Annalen* vol. XVII, p. 222.

[22] The simplification that results from Theorem II can be seen in the following example. For the special form $f = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1$, $a_y^2 a_x^2 u_y^2$ is a reducer. It follows by Theorem II that the form sequence

$$\theta_\nu(\vartheta\nu x)\theta_x^3 u_\vartheta u_\nu^2 = a_\nu^2(abu) - a_\nu b_\nu(aba),$$

is reduced, since $\theta_\nu^4 = a_\nu^2 b_\nu^2 (abu)^2$ has arisen from the member $a_\nu^2(abu)$ by folding. In Gordan, loc. cit., p. 231, by contrast, the reduction of the forms

$$\theta_\nu(\vartheta\nu x), \quad \theta_\nu(\vartheta\nu x)^2, \quad \theta_\nu^2, \quad \theta_\nu^2(\vartheta\nu x), \quad \theta_\nu^2(\vartheta\nu x)^2$$

is carried out individually.

[23] Thus, for example, the following were foand by "double reduction": the reduction of the form a_η^4 (§ 10), the only form included superfluously in Maisano's system; and also the relations mentioned in the note to the Introduction between the three covariants and the three invariants.

Decomposable forms arise by all onefold foldings with functional determinants[24], and moreover by certain higher foldings. One has

$$(st\widehat{yx})s_y = \frac{1}{2}\,t\,s_y^2 - \frac{1}{2}\,s\,t_y^2 + \frac{1}{2}(st\widehat{yx})^2,$$

$$(st\widehat{yx})^2 s_y^2 = \frac{1}{3}\,t\,s_y^3 - \frac{1}{3}\,s\,t_y^3 + s_y(st\widehat{yx})^2 - \frac{1}{3}(st\widehat{yx})^3.$$

Analogous formulas hold for the dualistic forms

$$(\sigma\tau\widehat{\nu u})v_\sigma \quad \text{and} \quad (\sigma\tau\widehat{\nu u})v_\sigma^2.$$

From the theorems of this section the following rule follows for setting up all irreducible forms that arise from the folding of S with T:

Beginning with the lowest foldings, proceed along any previously fixed path (for example row by row, or symmetrically with respect to the diagonal member) in the formation of the form sequence ST until one reaches a form that has a reducer as factor. The form sequence defined by this form is to be omitted as soon as the final form satisfies the condition stated in Theorem II. After all reducible form sequences have been eliminated, all remaining reducible forms, and likewise all decomposable forms, are to be removed by applying double reduction. Places in the total form sequence that are covered twice by independently reducible form sequences give rise to the reduction of higher forms (cf. § 11, D).

[24] cf. Gordan, loc. cit. § 3.

Theorem III. *The forms taken over from the binary domain, that is, those forms which arise from the system forms of the corresponding binary form by bordering according to Clebsch's transfer principle, form a relatively complete system modulo* (abc).

Proof. To a binary relation asserting that the forms $Q'_1, \ldots, Q'_n$ form a complete system of a binary ground form,

$$Q' - F(Q') = 0 = \sum_{\lambda} \big\{(ab)c_x + (bc)a_x + (ca)b_x\big\}\varphi^*_\lambda,$$

there corresponds, by bordering, the ternary relation

$$Q - F(Q_i) = \sum_{\lambda} u_x \cdot (abc)\varphi_\lambda.$$

This is precisely the defining equation for a relatively complete system modulo (abc). For the biquadratic form, the system of the forms taken over consists of the following forms:

$$f = a_x^4, \qquad \theta = \theta_x^4 u_\vartheta^2 = (abu)^2 a_x^2 b_x^2, \qquad K = K_x^6 u_k^3 = (a\theta u)u_\vartheta^2 a_x^3 \theta_x^3,$$
$$j = u_j^6 = (a\theta u)^4 u_\vartheta^2, \qquad \nu = u_\nu^4 = (abu)^4,$$

among which the first four form a relatively complete system modulo $((abc), \nu)$.

§ 5. Reduction of the modulus (abc) to the moduli Δ and ν.

The modulus (abc), given only by a bracket factor, is to be reduced, in agreement with the definition (§ 3, a), to forms of the system. In other words, the form sequence (abc) is to be normalized uniquely (cf. § 1).

$$a_s^2 a_x u_x = \frac{1}{3} a_s^3 u_x = \frac{1}{3} S \cdot u_x. \tag{2}$$

$$\left.\begin{aligned} (a\Delta u)^2 &= a_s - \frac{S}{6}u_x^2, \\ (a\Delta u)^3 &= 0. \end{aligned}\right\} \tag{3}$$

$$\left.\begin{aligned} \theta(s) &= (ss,x)^2 = 2a_t + \frac{1}{3}S\cdot\theta - \frac{1}{3}Tu_x^2, \\ \theta(t) &= (tt,x)^2 = -\frac{1}{3}S\cdot a_t + \frac{2}{3}T\cdot a_s + \frac{1}{18}S^2\cdot\theta - \frac{1}{18}u_x^2\cdot S\cdot T. \end{aligned}\right\} \tag{4}$$

Sequence of moduli: (abc); Δ; s; t (the system of the modulus t consists of the single form t).[25]

It will be shown that the forms

$$\Delta = (abc)^2 a_x^2 b_x^2 c_x^2, \qquad \nu = (abu)^4$$

suffice as moduli, that is, that the forms

$$\Delta = (abc)^2 a_x^2 b_x^2 c_x^2, \qquad a_\nu = (abc)(bcu)^3 a_x^3,$$
$$a_\nu^2 = (abc)^2(bcu)^2 u_x^2, \qquad i = a_\nu^4 = (abc)^4$$

define the form sequence (abc). In the form sequence a_ν, the form a_ν^3 is missing according to the identity

$$a_\nu^3 = (abc)^3 a_x (bcu) = \frac{1}{3} u_x \cdot (abc)^4 = \frac{1}{3} i \cdot u_x.$$

For reducing a modulus to a higher one, according to the definition, we have to reduce the most general foldings, or all special foldings coming into consideration, with the modulus to foldings with the higher modulus.

In the present case the most general foldings arise from the expressions

$$(abc)a_x^3 b_y^3 c_z^3, \qquad (abc)a_x^2 b_y^2 c_z^2 a_z b_x c_x$$

(taken over from the cubic forms),

$$(abc)a_x b_y c_z a_x^2 b_x^2 c_x^2$$

[25] Gordan–Kerschensteiner, loc. cit., p. 134.

(taken over from the quadratic forms), and from the polarisation of these expressions.

For the calculation of these expressions by the polars of Δ and of the form sequence a_ν, respectively of their initial terms, we take the indirect route, as in § 2. We expand the polar terms

$$(abc)^2 a_x^2 b_y^2 c_z^2, \qquad a_\nu(\nu xy)(\nu yz)(\nu zx)a_x a_y a_z = (abc)(bc\widehat{x}u)(bc\widehat{y}z)(bc\widehat{z}x)a_x a_y a_z \quad \text{etc.}$$

as linear aggregates of expressions with the symbolic factor (abc) and obtain, by inverting the system of equations, the desired relations, as well as a relation between the polars taken according to different combinations of the variables. The identity theorem and the product theorem for determinants are used in the evaluation.

The system of equations is, for $(abc)a_x^3 b_y^3 c_z^3$:

$$1) \quad \sum_3 a_\nu(\nu yz)^3 a_x^3 = 6(abc)a_x^3 b_y^3 c_z^3 - 6\sum_3 (abc)a_x^3 b_y^2 b_z c_z^2 c_y^*,$$

$$\begin{aligned} 2) \quad 3(abc)^2 a_x^2 b_y^2 c_z^2 \cdot (xyz) - \frac{1}{2}\sum_3 a_\nu^2(\nu yz)^2 \nu_x^2 (xyz) &= 2\sum_3 (abc)a_x^3 b_y^2 b_z c_z^2 c_y \\ &\quad - 4\sum_3 (abc)a_x a_y a_z b_y^2 b_z c_z^2 c_x, \end{aligned}$$

$$3) \quad a_\nu(\nu xy)(\nu yz)(\nu zx)a_x a_y a_z = 2\sum_3 (abc)a_x a_y a_z b_y^2 b_z c_z^2 c_x,$$

$$4) \quad \sum_3 a_\nu(\nu yz)^3 u_x^3 - \frac{3}{2}\sum_3 a_\nu^2(\nu yz)^2 \nu_x^2 (xyz) + \frac{1}{6} i \cdot (xyz)^3 = 6\sum_3 (abc)a_x a_y a_z b_y^2 b_z c_z^2 c_x.$$

It follows for

$$(abc)a_x^3 b_y^3 c_z^3,$$

and by analogous calculation for

$$(abc)a_x^2 b_y^2 c_z^2 a_z b_x c_x, \qquad (abc)a_x b_y c_z a_z^2 b_x^2 c_x^2 :$$

$$\begin{aligned} (abc)a_x^3 b_y^3 c_z^3 &= \frac{3}{2}(abc)^2 a_x^2 b_y^2 c_z^2 (xyz) + \frac{3}{2} a_\nu(\nu xy)(\nu yz)(\nu zx)a_x a_y a_z \\ &\quad - \frac{1}{12} i \cdot (xyz)^3, \\ \text{(IV.)} \quad (abc)a_x^2 b_y^2 c_z^2 a_z b_x c_x &= \frac{2}{3}(abc)^2 a_x^2 b_y^2 c_z^2 c_x \cdot (xyz) \\ &\quad + \frac{1}{3} a_\nu(\nu xy)(\nu yz)(\nu zx)a_x^3 - \frac{1}{6} a_\nu^2(\nu xy)(\nu zx)a_x^2 \cdot (xyz), \\ (abc)a_x b_y c_z a_z^2 b_x^2 c_x^2 &= \frac{1}{6}(abc)^2 a_x^2 b_z^2 c_z^2 \cdot (xyz). \end{aligned}$$

We have therefore obtained a relatively complete system modulo (Δ, ν), consisting of the four forms f, θ, K, j. It follows from this that all transvections of f over θ not taken over from the binary domain, as well as the transvection $(a\theta u)^2$ reducible in the binary case to $(ab)^4$, can be expressed by the symbols Δ and ν.[26]

We obtain the reduction formulas fundamental for what follows by specializing expansion IV, or more shortly by direct calculation (interchanging the symbols), for $a_\vartheta(a\theta u)^2$, $a_\vartheta^2((a\theta u)^2$, $(a\theta u)^2$, according to the following ansatz:

$$\begin{aligned} a_\vartheta(a\theta u)^2 &= (abc)(bcu)(abu)^2 - \frac{1}{3}(abc)(bcu)\{(bcu)a_x - u_x(abc)\}^2, \\ a_\vartheta^2(a\theta u)^2 &= (abc)^2(abu)^2 - \frac{1}{3}(abc)^2\{(bcu)a_x - u_x(abc)\}^2, \end{aligned}$$

for $((a\theta u)^2)^*$:

$$\begin{aligned} (abu)a_y^3 b_x^3 &= \frac{3}{2} u_\vartheta(\vartheta yx)\theta_y^2 + \frac{1}{4}(\nu yx)^3, \\ ((abu)(acu))b_x^3 &= \frac{3}{2} u_\vartheta(\vartheta\widehat{a}ux)(\theta au)^2 + \frac{1}{4}(\nu\widehat{a}ux)^3. \end{aligned}$$

[26] $\sum_3$ refers to the sum of three terms obtained from the initial term by cyclic interchange of the variables. The equations also hold individually for each term of the sum.

$$\left.\begin{aligned}
(a\theta u)^2 &= \frac{1}{6}\nu\cdot f - \frac{2}{3}u_x\cdot a_\nu + \frac{2}{3}u_x^2\cdot a_\nu^2 - \frac{i}{18}u_x^4,\\
a_\vartheta &= \frac{1}{3}u_x\cdot\Delta,\\
a_\vartheta(a\theta u) &= 0,\\
a_\vartheta^2 &= \Delta,\\
(a\theta u)^3 &= 0,\\
a_\vartheta(a\theta u)^2 &= -\frac{5}{6}a_\nu + \frac{7}{6}u_x\cdot a_\nu^2 - \frac{i}{9}u_x^3,\\
a_\vartheta^2(a\theta u) &= 0,\\
(a\theta u)^4 &= j,\\
a_\vartheta(a\theta u)^3 &= 0,\\
a_\vartheta^2(a\theta u)^2 &= \frac{2}{3}a_\nu^2 - \frac{i}{9}u_x^2.
\end{aligned}\right\} \tag{1}$$

We add the formula, likewise obtained from the reduction of the modulus (abc):

$$a_\nu^3 a_x u_\nu = \frac{1}{3}i\cdot u_x. \tag{2}$$

From formulas (1.) and (2.) and from the formulas formed analogously for the ground form ν, all later reduction formulas are derived by polarisation, except for the relations for the decomposed forms. From formulas (1.) we see:

The form sequence leads:

$$\begin{array}{rcl}
a_\vartheta(a\theta u)^2 & \text{to} & \text{symbols } \nu \text{ alone},\\
a_\vartheta & \text{to} & \text{symbols } \Delta \text{ and } \nu,\\
(a\theta u)^2 & \text{to} & \text{symbols } j, \Delta \text{ and } \nu,\\
(a\theta u)^2 \text{ under folding I} & \text{to} & \text{symbols } j \text{ and } \nu,\\
(a\theta u)^2 \text{ under folding II} & \text{to} & \text{symbols } \Delta \text{ and } \nu.
\end{array}$$

We can therefore replace:

1. modulo (Δ,ν): foldings with j are replaced by corresponding foldings with $(a\theta u)^2$ or $(a\theta u)^3$;

2. modulo (ν): foldings with Δ are replaced by corresponding foldings with a_ϑ or $a_\vartheta(a\theta u)$, or also by special foldings with $(a\theta u)^2$ (which lead only to a single folding I).

In particular:

$$\begin{aligned}
(a\theta u)^2\theta_y^2\theta_x^2 &= \frac{1}{3}(jyx)^2 \pmod{\nu}, & (a\theta u)^2 u_y\theta_y &= -\frac{1}{6}(jyx)^2 \pmod{\nu},\\
(a\theta u)^2\nu_y^2 &= \frac{1}{3}(\Delta u\nu)^2 \pmod{\nu}, & (a\theta u)(a\theta\nu)u_\vartheta\nu_\vartheta &= -\frac{1}{6}(\Delta u\nu)^2 \pmod{\nu},\\
(a\theta u)^2 u_\nu^2\vartheta_\nu^2 &= \frac{1}{3}(\Delta u\nu)^2\Delta_\nu \pmod{\nu}.
\end{aligned}$$

§ 6. Reduction of the modulus (Δ,ν) to the modulus (ν).

The reduction formulas of the preceding paragraph give us the means to set up the relatively complete system modulo ν directly; we shall see that to the system of the transferred forms one has only to add the forms Δ and $(a\Delta u) = N$ (analogously as for the cubic form, and indeed valid in general).

For the reduction of the modulus Δ we consider all special foldings coming into consideration, that is, the foldings of the relatively complete system modulo (Δ,ν) with the system of Δ, and begin with the forms lowest in the coefficients, the foldings of f with Δ.

For the reduction of the forms $(a\Delta u)^2$, $(a\Delta u)^3$, $(a\Delta u)^4$, we replace the symbols Δ by lower forms of the form sequence according to § 5; that is, we develop the expressions

$$a_\vartheta b_\vartheta(b\theta u)^2, \qquad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^2, \qquad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^3$$

1) by polars of the form sequence a_ϑ (symbols Δ and ν),

2) by polars of the form sequence $b_\vartheta(b\theta u)^2$ (symbols ν),

and obtain the reduction formulas (calculation below):

$$
\begin{aligned}
\text{a)}\quad (a\Delta u)^2 &= \frac{1}{2}(\vartheta\nu x)^2 - \frac{2}{5}f\cdot a_\nu^2 - \frac{4}{5}u_x\cdot\theta_\nu(\vartheta\nu x)^2 \\
&\quad + u_x^2\left\{\frac{1}{6}i\cdot f - \frac{1}{40}S\right\}, \\
\text{b)}\quad (a\Delta u)^3 &= -\frac{3}{10}\theta_\nu(\vartheta\nu x), \\
\text{c)}\quad (a\Delta u)^4 &= \frac{21}{10}\theta_\nu^2 - \frac{3}{10}\Pi - \frac{6}{5}u_x\cdot\theta_\nu^3 + \frac{1}{10}u_x^2\cdot\theta.
\end{aligned}
\tag{3}
$$

(The same results are also reached by the double reduction of the expressions $a_\vartheta^2(b\theta u)^2$, $a_\vartheta^2(b\theta u)^3$, $a_\vartheta^2((a\theta u)^2(b\theta u)^2$ and $a_\vartheta^2((a\theta u)(b\theta u))^3$ after elimination of a_ϑ^2.)

From formulas (3.) it follows that $(a\Delta u)^2$ is a reducer with respect to the modulus ν. Hence:

1) The reduction of the system of Δ: the form sequence $(\Delta\Delta' u)^2 = a_\vartheta^2((a\Delta u)^2$ has the reducer $(a\Delta u)^2$ as a factor.

2) The reduction of the system arising by folding with the modulus Δ:

The form sequence $(\theta\Delta u)^2$ has the reducer $(a\Delta u)^2$ as a factor.

For the forms $(\theta\Delta u)$ and Δ_ϑ one obtains:

$$(a\Delta u)^2(abu) = (\theta\Delta u) - u_x\cdot\Delta_\vartheta(\theta\Delta u),$$

$$(a\Delta u)(a\Delta b) = -\frac{1}{2}\Delta_\vartheta + \frac{1}{2}u_x\cdot\Delta_\vartheta^2.$$

From the reducer $(\theta\Delta u)$, however, follows the reduction of the system arising by folding with the modulus Δ.

The relatively complete system modulo ν therefore consists of the six forms:

$$f,\ \theta,\ K,\ j,\ \Delta,\ N.$$

As an example of the sequence expansion in § 2 we give the derivation of formula (3.)a in full; in later calculations the final formula III will be set down directly. (Polars of vanishing forms have already been omitted during the calculation; the first line gives the values inserted according to Expansion II, and the second line gives the final values found recursively, beginning with the last equation of the system.) It follows, according to Expansion II:

$$1)\quad m=3,\quad n=4,\quad \lambda=2,\quad \mu=0,\quad \nu=\chi=1,$$

$$
\begin{aligned}
a_\vartheta b_\vartheta(b\theta u)^2 &= [a_\vartheta]_{b^2}b + \frac{6}{7}\{a_\vartheta b_\vartheta(a\theta u)(b\theta u) - u_x a_\vartheta b_\vartheta(a\theta b)(b\theta u)\} \\
&\quad - \frac{1}{7}\{a_\vartheta(a\theta u)^2 b_\vartheta - 2u_x a_\vartheta(a\theta u)(a\theta b)b_\vartheta + u_x^2 a_\vartheta(a\theta b)^2 b_\vartheta\} \\
&= \frac{2}{3}(a\Delta u)^2 + \frac{1}{5}[a_\vartheta(a\theta u)^2]b + \frac{1}{30}f[a_\vartheta^2(a\theta u)^2] - \frac{2}{5}u_x[a_\vartheta(a\theta u)^2]b^2 \\
&\quad - \frac{1}{15}u_x[a_\vartheta^2(a\theta u)^2]b + \frac{1}{5}u_x^2[a_\vartheta(a\theta u)^2]b^3 + \frac{1}{30}u_x^2[a_\vartheta^2(a\theta u)^2]b^2;
\end{aligned}
$$

$$2)\quad m=2,\quad n=3,\quad \lambda=1,\quad \mu=1,\quad \nu=\chi=1,$$

$$
\begin{aligned}
a_\vartheta(a\theta u)b_\vartheta(b\theta u) &= \frac{2}{5}\{a_\vartheta(a\theta u)^2 b_\vartheta - u_x a_\vartheta(a\theta u)(a\theta b)b_\vartheta\} + \frac{1}{2}a_\vartheta^2(b\theta u)^2 \\
&\quad - \frac{1}{5}\{a_\vartheta^2(b\theta u)(a\theta u) - u_x a_\vartheta^2(a\theta b)(b\theta u)\} \\
&= \frac{1}{2}(a\Delta u)^2 + \frac{2}{5}[a_\vartheta(a\theta u)^2]b + \frac{1}{15}[a_\vartheta^2(a\theta u)^2]f \\
&\quad - \frac{2}{5}u_x[a_\vartheta(a\theta u)^2]b^2 - \frac{1}{10}u_x[a_\vartheta^2(a\theta u)^2]b + \frac{1}{30}u_x^2[a_\vartheta^2(a\theta u)^2]b^2;
\end{aligned}
$$

$$3)\quad m=2,\quad n=3,\quad \lambda=1,\quad \mu=1,\quad \nu=1,\quad \chi=2,$$

$$\begin{aligned}
a_\vartheta(a\theta b)b_\vartheta(b\theta u) &= \frac{2}{5}\{a_\vartheta(a\theta b)(a\theta u)b_\vartheta - u_x a_\vartheta(a\theta b)^2 b_\vartheta\}\\
&= \frac{2}{5}[a_\vartheta(a\theta u)^2]b^2 + \frac{1}{30}[a_\vartheta^2(a\theta u)^2]b - \frac{2}{5}u_x[a_\vartheta(a\theta u)^2]b^3 - \frac{1}{30}u_x[a_\vartheta^2(a\theta u)^2]b^2;
\end{aligned}$$

$$4)\quad m=1,\quad n=2,\quad \lambda=0,\quad \mu=2,\quad \nu=\chi=1,$$

$$\begin{aligned}
a_\vartheta(a\theta u)^2 b_\vartheta &= [a_\vartheta(a\theta u)^2]b + \frac{2}{3}a_\vartheta^2(a\theta u)(b\theta u)\\
&= [a_\vartheta(a\theta u)^2]b + \frac{1}{6}[a_\vartheta^2(a\theta u)^2]f - \frac{1}{6}u_x[a_\vartheta^2(a\theta u)^2]b;
\end{aligned}$$

$$5)\quad m=1,\quad n=2,\quad \lambda=0,\quad \mu=2,\quad \nu=1,\quad \chi=2,$$

$$\begin{aligned}
a_\vartheta(a\theta u)(a\theta b)b_\vartheta &= [a_\vartheta(a\theta u)^2]b^2 + \frac{1}{3}a_\vartheta^2(a\theta b)(b\theta u)\\
&= [a_\vartheta(a\theta u)^2]b^2 + \frac{1}{12}[a_\vartheta^2(a\theta u)^2]b - \frac{1}{12}u_x[a_\vartheta^2(a\theta u)^2]b^2;
\end{aligned}$$

$$6)\quad m=1,\quad n=2,\quad \lambda=0,\quad \mu=2,\quad \nu=1,\quad \chi=3,$$

$$a_\vartheta(a\theta b)^2 b_\vartheta = [a_\vartheta(a\theta u)^2]b^3;$$

$$7)\quad m=2,\quad n=4,\quad \lambda=2,\quad \mu=\nu=\chi=0,\quad \text{(according to Expansion I)},$$

$$a_\vartheta^2(b\theta u)^2 = [a_\vartheta^2]_{ub^2} + \frac{1}{10}\{[a_\vartheta^2(a\theta u)^2]f - 2u_x[a_\vartheta^2(a\theta u)^2]b + u_x^2[a_\vartheta^2(a\theta u)^2]b^2\};$$

$$8)\quad m=1,\quad n=3,\quad \lambda=1,\quad \mu=1,\quad \nu=\chi=0,\quad \text{(Expansion I)},$$

$$a_\vartheta^2(a\theta u)(b\theta u) = \frac{1}{4}[a_\vartheta^2(a\theta u)^2]f - \frac{1}{4}u_x[a_\vartheta^2(a\theta u)^2]b;$$

$$9)\quad m=1,\quad n=3,\quad \lambda=1,\quad \mu=\chi=1,\quad \nu=0,\quad \text{(Expansion I)},$$

$$a_\vartheta^2(a\theta b)(b\theta u) = \frac{1}{4}[a_\vartheta^2(a\theta u)^2]b - \frac{1}{4}u_x[a_\vartheta^2(a\theta u)^2]b^2.$$

From 1) and 4) it follows:

$$\begin{aligned}
(a\Delta u)^2 &= \frac{6}{5}[a_\vartheta(a\theta u)^2]b + \frac{1}{5}f[a_\vartheta^2(a\theta u)^2] + \frac{3}{5}u_x[a_\vartheta(a\theta u)^2]b^2 - \frac{3}{20}u_x[a_\vartheta^2(a\theta u)^2]b\\
&\quad - \frac{3}{10}u_x^2[a_\vartheta(a\theta u)^2]b^3 - \frac{1}{20}u_x^2[a_\vartheta^2(a\theta u)^2]b^2,\\
(a\Delta u)^2 &= -a_\nu b_\nu + \frac{3}{5}fa_\nu^2 + \frac{4}{5}u_x a_\nu^2 b_\nu - \frac{3}{20}u_x^2 a_\nu^2 b_\nu^2 - \frac{1}{12}u_x^2 if.
\end{aligned}$$

(For conversion into the symbols θ cf. § 8 and § 9.) Formula (3.)a can be computed still more briefly by direct transfer of Expansion I, introducing the auxiliary variable $w = x\widehat{u}b$ whereas already for formulas (3.)b and (3.)c the path taken here becomes the simpler one; for (3.)b one knows in advance, by § 9, that all terms with factor u_x are to be omitted. For the calculation of (3.)b and (3.)c one obtains:

$$\begin{aligned}
&(3.)\text{b}\quad 1)\quad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^2 = \frac{1}{2}(a\Delta u)^3 + \frac{4}{5}[a_\vartheta(a\theta u)^2]_{ub}b + \frac{7}{360}[a_\vartheta^2(a\theta u)^2]_{ub^2},\\
&\qquad\quad\; 2)\quad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^2 = [a_\vartheta(a\theta u)^2]_{ub}b + \frac{13}{36}[a_\vartheta^2(a\theta u)^2]_{ub^2};
\end{aligned}$$

$$\begin{aligned}
&(3.)\text{c}\quad 1)\quad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^3 = \frac{1}{2}(a\Delta u)^4 + \frac{6}{5}[a_\vartheta(a\theta u)^2]_{ub^2}b + \frac{53}{120}[a_\vartheta^2(a\theta u)^2]_{ub^3}\\
&\qquad\qquad - \frac{6}{5}u_x[a_\vartheta(a\theta u)^2]_{ub^3}b^2 - \frac{3}{5}u_x[a_\vartheta^2(a\theta u)^2]_{ub^2}b + \frac{19}{120}u_x^2[a_\vartheta^2(a\theta u)^2]_{ub^3}b^2,\\
&\qquad\quad\; 2)\quad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^3 = \frac{3}{4}[a_\vartheta^2(a\theta u)^2]_{ub^2}.
\end{aligned}$$

Supplementary remark: the transvectants of f over j that are reducible according to § 5 are computed analogously. One obtains

$$\begin{aligned} g_\nu &= -\theta_\nu + u_x\left(\frac{9}{2}\theta_\nu^2 - \frac{1}{2}H\right) - 3u_x^2\theta_\nu^3 + \frac{1}{2}u_x^3\theta, \\ g_\nu^2 &= \frac{21}{10}\theta_\nu^2 - \frac{3}{10}H - 2u_x\theta_\nu^3 + \frac{1}{2}u_x^2\theta, \\ g_\nu^3 &= -\frac{7}{10}\theta_\nu^3 + \frac{1}{2}u_x^2\theta, \qquad g_\nu^4 = \frac{3}{5}\theta. \end{aligned} \tag{4}$$

$$g_\nu^3 = -\frac{7}{10}\theta_\nu^3 + \frac{1}{2}u_x^2\theta, \qquad g_\nu^4 = \frac{3}{5}\theta. \tag{4}$$

From (3.) and (4.) it follows, by transfer from the binary domain, that

$$(\theta\theta' u)^2 = \frac{1}{3}\, j \cdot f \pmod{\nu}.$$ [27]

The remaining forms of the form sequence $(\theta\theta' u)^2$ and the form θ'_ν are reducible to symbols ν. We can therefore replace:

mod ν : foldings with $f \cdot j$ by the corresponding foldings with $(\theta\theta' u)^2$.

Furthermore,

$$(\vartheta\vartheta' x)^2 = \frac{4}{3} f \cdot \Delta, \qquad \theta_{g\nu}(\vartheta\vartheta' x) = -N.$$

Chapter III. The relatively complete system modulo (s, ϱ, t).

§ 7. Overview of the formation of the system.

In order to pass from the relatively complete system modulo ν to the relatively complete system with the next higher modulus, one has, by Definition § 3, to form the system of ν with respect to this higher modulus and to fold it with the forms of the relatively complete system modulo ν. But $\nu = u_\nu^4$, as a contravariant of the fourth degree, is dual to the form f. If therefore we regard ν as the ground form, we obtain a relatively complete system of six forms modulo $\nu(\nu) = (\nu\nu_1 x)^4 = s_x^4$.

Thus, for the formation of the relatively complete system modulo s (System III), we have to transvect

System I: $f,\ \theta,\ K,\ j,\ \Delta,\ N$ over

System II: $\nu,\ H = (\nu\nu_1 x)^2 u_\nu^2 u_{\nu_1}^2,\ L = (\nu\eta x) H_x^2 u_\nu^3 u_\eta^3,$
$g = (\nu\eta x)^4 H_x^2,\ \sigma = H_\eta^2 u_\nu^2 u_\eta^4,\ (\nu\sigma x).$

It will be shown (formula (13.)) that the form g is reducible, and hence that System II consists of only five forms.

In the course of the calculation the quadratic forms

$$\varrho = u_\varrho^2 = \theta_\nu^4 u_\vartheta^2, \qquad t = t_x^2 = a_\eta^4 H_x^2$$

are adjoined as moduli, so that one obtains a relatively complete system modulo (s, ϱ, t); and the modulus (ϱ, t) is to be regarded as a higher modulus than the modulus s.

The arrangement of the individual forms is to follow the order in the coefficients. We normalize the folding

$$\theta_\nu(K_\nu, \theta_\nu, \ldots)$$

as higher than the folding

$$H_y(H_y, L_y, \ldots).$$

Consequently, by § 1 and § 3b, the order of the individual forms of the same order is uniquely determined.

The formation of the forms of the same order is carried out in tabular form:

I. Indication of which forms of Systems I and II are to be folded with each other in order to obtain the prescribed order in the coefficients.

[27] Gordan–Kerschensteiner, loc. cit., p. 181.

II. Listing of the irreducible forms according to the scheme of the form sequence.

III. Reduction of the reducible or decomposable form sequences or forms, that is, of those places where the total form sequence breaks off, thereby proving that all irreducible constructions have been exhausted by those indicated.

IV. Consequences: 1) indication of newly arising reducers; 2) indication of those forms which do not enter, in products with forms of the same system, into folding with forms of the other system.

It will be shown that only the following occur:

1) foldings of one form of System I with one form of System II;

2) foldings of powers of θ, respectively H, or of two forms of one system with one form of the second system.

We shall obtain the following scheme for folding:

System I	folded with	System II
1. order f		2. order ν
2. ,, θ		4. ,, H
3. ,, K, j, Δ		6. ,, L, σ
4. ,, $N, \theta^2, f \cdot j$		8. ,, $(\nu\sigma x), H^2$
5. ,, $\theta \cdot K$		10. ,, $H \cdot L$
6. ,, $\theta^3, \Delta \cdot j$		12. ,, H^3.

For checking the calculation, besides the "double reduction", the two special forms serve:

$$\text{I.}\quad \begin{array}{lll} f = \sum\limits^{3} x_1^3 x_2, & u_\nu^2 = \frac{i}{6} u_x^2, & s = \frac{i}{3} f, \\ a_\eta^2 = -\frac{1}{9} i\theta, & H_\nu^2 = \frac{i}{6}\theta, & \sigma = -\frac{i}{6} j, \\ g = -\frac{2}{3} i\Delta, & \varrho = 0, & t = 0, \end{array}$$

$$\text{II.}\quad \begin{array}{lll} f = \sum\limits^{3} x_1^4, & s = \frac{4}{3} i f, & a_\eta^2 = \frac{2}{9} i\theta, \\ \Delta_\nu^2 = \frac{1}{15} i\theta, & \sigma = \frac{4}{3} i j, & g = \frac{4}{3} i\Delta, \\ \varrho = 0, & t = 0. & \end{array}$$

Supplementary remark: for the ground form ν, formulas (1.), (2.), (3.), (4.) correspond to the following:

$$\begin{array}{r|c|c} (\nu\eta x)^2 = A & H_\nu(\nu\eta x) = 0 & H_\nu^2 = \sigma \\ (\nu\eta x)^3 = 0 & H_\nu(\nu\eta x)^2 = B & H_\nu^2(\nu\eta x) = 0 \\ (\nu\eta x)^4 = g & H_\nu(\nu\eta x)^3 = 0 & H_\nu^2(\nu\eta x)^2 = C, \end{array} \tag{5}$$

$$A = \frac{1}{6} s\nu - \frac{2}{3} u_x s_\nu + \frac{2}{3} u_x^2 s_\nu^2 - \frac{J}{18} u_x^4, \quad B = -\frac{5}{6} s_\nu + \frac{7}{6} u_x s_\nu^2 - \frac{J}{9} u_x^3, \quad C = \frac{2}{3} s_\nu^2 - \frac{J}{9} u_x^2.$$

$$s_\nu^3 u_x s_\nu = \frac{1}{3} u_x s_\nu^4 = \frac{1}{3} J u_x. \tag{6}$$

$$\begin{aligned} &\text{a)}\quad (\nu\sigma x)^2 = \frac{1}{2}(Hsu)^2 - \frac{2}{5}\nu \cdot s_\nu^2 - \frac{4}{5} u_x\, s_\eta (Hsu)^2 + u_x^2 \left\{ \frac{1}{6} J \cdot \nu - \frac{1}{40}(ss'u)^4 \right\}, \\ &\text{b)}\quad (\nu\sigma x)^3 = -\frac{3}{10} s_\eta (Hsu), \\ &\text{c)}\quad (\nu\sigma x)^4 = \frac{21}{10} s_\eta^2 - \frac{3}{10} Z - \frac{6}{5} u_x s_\eta^3 + \frac{1}{10} u_x^2 s_\eta^4. \end{aligned} \tag{7}$$

$$\begin{aligned} &\text{a)}\quad g_\nu = -s_\eta + u_x \left(\frac{9}{2} s_\eta^2 - \frac{1}{2} Z \right) - 3u_x^2 s_\eta^3 + \frac{1}{2} u_x^3 s_\eta^4, \\ &\text{b)}\quad g_\nu^2 = \frac{21}{10} s_\eta^2 - \frac{3}{10} Z - 2u_x s_\eta^3 + \frac{1}{2} u_x^2 s_\eta^4, \\ &\text{c)}\quad g_\nu^3 = -\frac{7}{10} s_\eta^3 + \frac{1}{2} u_x s_\eta^4, \\ &\text{d)}\quad g_\nu^4 = \frac{3}{5} s_\eta^4. \end{aligned} \tag{8}$$

§ 8. Forms of the third order (System III).

Folding of f with ν.
Irreducible forms:

$$\begin{array}{cccc} a_\nu & \cdot & \cdot & \cdot \\ \cdot & a_\nu^2 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & a_\nu^4 = i. \end{array}$$

Reduction:

$$a_\nu^3 = \frac{1}{3} i u_x \qquad \text{(formula (2.))}.$$

Consequences: 1) a_ν^3 is a reducer. 2) f enters into folding with ν only in products with contragredient forms (j). The same holds for ν with respect to f.

§ 9. Forms of the fourth order (System III).

Folding of θ with ν.
Irreducible forms:

$$\begin{array}{cccc} (\theta\nu x) & \theta_\nu & \cdot & \cdot \\ (\theta\nu x)^2 & \theta_\nu(\theta\nu x) & \theta_\nu^2 & \cdot \\ \cdot & \theta_\nu(\theta\nu x)^2 & \cdot & \theta_\nu^3. \end{array}$$

reductions: From the reducer a_ν^3, by double reduction of the expressions $a_\nu^3(abu)$, $a_\nu^3 b_\nu$, $a_\nu^3 b_\nu(abu)$ according to the ansatz:

$$\begin{aligned} \theta_\nu^2(\vartheta\nu x) &= a_\nu^2(\widehat{a}b\nu x)(abu) - \frac{1}{3}(\nu\nu_1 x)^3 = a_\nu b_\nu(\widehat{a}b\nu x)(abu) + \frac{1}{6}(\nu\nu_1 x)^3 \\ &= a_\nu^3(abu) - a_\nu^2 b_\nu(abu) = 2a_\nu^2 b_\nu(abu) = \frac{2}{3}a_\nu^3(abu), \\ \theta_\nu^2(\vartheta\nu x)^2 &= a_\nu^2(\widehat{a}b\nu x)^2 - \frac{1}{3}(\nu\nu_1 x)^4 = a_\nu b_\nu(\widehat{a}b\nu x)^2 + \frac{1}{6}(\nu\nu_1 x)^4 \\ &= if - 2a_\nu^3 b_\nu + a_\nu^2 b_\nu^2 - \frac{1}{3}s = 2a_\nu^3 b_\nu - 2a_\nu^2 b_\nu^2 + \frac{1}{6}s = \frac{2}{3}if - \frac{2}{3}a_\nu^3 b_\nu - \frac{1}{6}s, \\ \theta_\nu^3(\vartheta\nu x) &= a_\nu^2 b_\nu(\widehat{a}b\nu x)(abu) = a_\nu^3 b_\nu(abu). \end{aligned}$$

The formulas obtained are:

$$\left.\begin{array}{r} \theta_\nu^2(\theta\nu x) = 0 \\ \theta_\nu^2(\theta\nu x)^2 = \dfrac{4}{9}if - \dfrac{1}{6}s \end{array}\right| \left.\begin{array}{l} \theta_\nu^3(\theta\nu x) = 0 \\ \\ \end{array}\right| \begin{array}{l} \theta_\nu^4 u_x^2 = u_\varrho^2 = \varrho \quad \text{(adjoined modulus, § 7)}, \\ \\ \end{array} \tag{9}$$

Consequences: 1) $\theta_\nu^2(\theta\nu x)^2$ is a reducer. 2) ν enters into folding with θ only in products with contragredient forms (g).

§ 10. Forms of the fifth order (System III).

$$\text{Folding of} \left\{ \begin{array}{l} K, j, \Delta \text{ with } \nu, \\ f \text{ with } H. \end{array} \right.$$

Irreducible forms:

$$\text{A)} \quad \begin{array}{cccc} \cdot & K_\nu & \cdot & \cdot \\ (k\nu x)^2 & \cdot & K_\nu^2 & \cdot \\ (k\nu x)^3 & K_\nu(k\nu x)^2 & \cdot & \cdot \\ \cdot & K_\nu(k\nu x)^3 & \cdot & \cdot \end{array} \qquad \text{B)} \quad \begin{array}{c} (j\nu x) \\ (j\nu x)^2 \\ (j\nu x)^3 \\ \cdot \end{array} \qquad \text{C)} \quad \begin{array}{c} \Delta_\nu \\ \cdot \\ \cdot \\ \cdot \end{array}$$

$$\text{D)} \quad \begin{array}{cccc} (aHu) & (aHu)^2 & \cdot & \cdot \\ a_\eta & a_\eta(aHu) & a_\eta(aHu)^2 & \cdot \\ \cdot & a_\eta^2 & \cdot & \cdot \end{array}$$

reductions:

A) form sequence K_ν^3: reducer a_ν^3.
form $K_\nu^2(k\nu x)^2$: reducer $\theta_\nu^2(\vartheta\nu x)^2$.

$$(k\nu x), \quad K_\nu(k\nu x), \quad K_\nu^2(k\nu x)$$

are decomposable forms according to § 3.
B) form

$$(j\nu x)^4 = (\widehat{a}\theta\nu x)^4 = a_\nu^4\theta + \theta_\nu^4 f - 4a_\nu^3\theta_\nu - 4a_\nu\{a_\nu\theta_x - (\widehat{a}\theta\nu x)\}^3$$
$$+ 6a_\nu^2\{a_\nu\theta_x - (\widehat{a}\theta\nu x)\}^2 = \varrho f + \frac{5}{3}i\theta - 6a_\nu^2(\widehat{a}\theta\nu x)^2 + 4a_\nu(\widehat{a}\theta\nu x)^3,$$

and therefore

$$(j\nu x)^4 = \varrho f + \frac{5}{3}i\theta \ \mathrm{mod}(\Delta, \nu), \qquad \text{(cf. § 5 end).}$$

C) form Δ_ν^4: reducer θ_ν^4.
forms Δ_ν^2 and Δ_ν^3: reducer a_ν^3 or θ_ν^4 by replacing the form Δ by lower forms of the form sequence (§ 5 end), i.e. by double reduction of the expressions

$$a_\vartheta a_\nu^3, \qquad a_\vartheta a_\nu^3\theta_\nu, \qquad a_\vartheta\theta_\nu^4.$$

The double calculation of Δ_ν^3 gives rise to the reduction of the higher form a_η^3.
Reduction formulas:

$$\begin{aligned} &\text{a)} \quad \Delta_\nu^2 = \frac{3}{5}a_\eta^2 - \frac{3}{10}(asu)^2 + \frac{1}{3}i\theta + \frac{1}{5}u_x^2(2a_\varrho^2 - t), \\ &\text{b)} \quad \Delta_\nu^3 = -\frac{3}{10}a_\varrho + \frac{3}{5}u_x\left(\frac{13}{10}a_\varrho^2 - \frac{2}{5}t\right), \\ &\text{c)} \quad \Delta_\nu^4 = a_\varrho^2 - \frac{2}{5}t. \end{aligned} \tag{10}$$

Derivation of the formulas:

$$\begin{aligned} \text{a)} \quad a_\vartheta a_\nu^3 = \frac{1}{3}i\theta &= [a_\vartheta]_{\nu^3} + \frac{6}{7}[a_\vartheta^2]_{\nu^2} + \frac{6}{5}[a_\vartheta(a\theta u)^2]_{\nu^2}\, ux^2 + \frac{11}{30}[a_\vartheta^2(a\theta u)^2]_\nu\, \nu x^2 \\ &- \frac{6}{7}u_x[a_\vartheta^2]_{\nu^3} - \frac{1}{6}u_x[a_\vartheta^2(a\theta u)^2]_{\nu^2}\, \nu x^2, \\ \frac{1}{3}i\theta &= \Delta_\nu^2 - \frac{2}{3}u_x\Delta_\nu^3 - a_\nu a_{\nu_1}(\nu\nu_1 x)^2 + \frac{2}{5}a_\nu^2(\nu\nu_1 x)^2 + \frac{1}{5}u_x^2 a_\nu^2 a_{\nu_1}(\nu\nu_1 x)^2; \end{aligned}$$

$$\begin{aligned} \text{b)} \quad a_\vartheta a_\nu^3\theta_\nu = 0 &= [a_\vartheta]_{\nu^4} + \frac{9}{14}[a_\vartheta^2]_{\nu^3} + \frac{3}{5}[a_\vartheta(a\theta u)^2]_{\nu^2}\, \nu x^2 + \frac{17}{120}[a_\vartheta^2(a\theta u)^2]_\nu\, \nu x^2 \\ &- \frac{9}{14}u_x[a_\vartheta^2]_{\nu^4} - \frac{1}{24}u_x[a_\vartheta^2(a\theta u)^2]_{\nu^2}\, \nu x^2, \\ 0 &= \frac{5}{6}\Delta_\nu^3 - \frac{1}{2}u_x\Delta_\nu^4 - \frac{1}{4}a_\eta^3 + \frac{1}{20}u_x a_\eta^4; \end{aligned}$$

$$\begin{aligned} \text{b')} \quad a_\vartheta\theta_\nu^4 = a_\varrho &= a_\vartheta\theta_\nu\{a_\nu - (a\theta\nu x)\}^3 \\ &= -\frac{3}{2}[a_\vartheta]_{\nu^3} + \frac{3}{5}[a_\vartheta(a\theta u)^2]_{\nu^2}\, \nu x^2 - \frac{37}{120}[a_\vartheta^2(a\theta u)^2]_\nu\, \nu x^2 \\ &+ \frac{3}{2}u_x[a_\vartheta^2]_{\nu^4} + \frac{49}{120}u_x[a_\vartheta^2(a\theta u)^2]_{\nu^2}\, \nu x^2, \\ a_\varrho &= -\frac{3}{2}\Delta_\nu^3 + \frac{3}{2}u_x\Delta_\nu^4 - \frac{11}{20}a_\eta^3 + \frac{7}{20}u_x a_\eta^4; \end{aligned}$$

$$\begin{aligned} \text{c)} \quad a_\vartheta^2\theta_\nu^4 = a_\varrho^2 &= [a_\vartheta^2]_{\nu^4} + \frac{3}{5}[a_\vartheta^2(a\theta u)^2]_{\nu^2}\, \nu x^2, \\ a_\varrho^2 &= \Delta_\nu^4 + \frac{2}{5}a_\eta^4. \end{aligned}$$

D) reduction of a_η^3 by double reduction of Δ_ν^3 (C).
The remaining reductions are obtained by a computation dualistically opposite to that of § 9.

Reduction formulas:

$$\begin{array}{rl|rl}
a_\eta^2(aHu) & = -\frac{1}{6}(asu)^3, & a_\eta^2(aHu)^2 & = \frac{4}{9}i\nu - \frac{1}{6}(asu)^4, \\
a_\eta^3 & = -a_\varrho + \frac{3}{5}u_x a_\varrho^2 + \frac{1}{5}u_x t, & a_\eta^3(aHu) & = 0, \\
 & & a_\eta^4 & = t \quad \text{(adjoined as a modulus).}
\end{array} \tag{11}$$

Consequences:

1) $(j\nu x)^4$, Δ_ν^2, $a_\eta^2(aHu)$ are reducers.

2) ν enters only in products with contragredient forms in foldings with K, j, Δ; the same holds for f in relation to H, and for j in relation to ν, since $\theta_\nu(\theta \cdot j, \nu, x^3)$ becomes reducible according to § 11 B.

§ 11. Forms of the sixth order (System III).

$$\text{Folding of } \left\{ \begin{array}{l} \theta^2, f \cdot j, N \text{ with } \nu, \\ \theta \text{ with } H. \end{array} \right.$$

Irreducible forms:

$$\text{A) } (\vartheta^2\nu x)^3, \quad \theta_\nu(\vartheta^2\nu x)^3 \quad \text{B) } a_\nu(j\nu x), \quad a_\nu^2(j\nu x) \quad \text{C) } N_\nu$$

$$\text{D)} \quad \begin{array}{c|c|c|c}
 & (\theta Hu) & (\theta Hu)^2 & \cdot \\
(\vartheta\eta x) & \theta_\eta & H_\vartheta(\theta Hu),\ \theta_\eta(\theta Hu) & \theta_\eta(\theta Hu)^2 \\
(\vartheta\eta x)^2 & H_\vartheta(\vartheta\eta x),\ \theta_\eta(\vartheta\eta x) & \theta_\eta^2 & \theta_\eta^2(\theta Hu) \\
\cdot & \theta_\eta(\vartheta\eta x)^2 & \cdot & \cdot
\end{array}$$

reductions:

A) $(\vartheta^2\nu x)^4$ decomposes according to formula (3.)a:

$$(a\Delta\vartheta x)^4 = \frac{1}{2}(\vartheta^2\nu x)^4 - \frac{3}{5}f \cdot a_\nu^2(\nu\vartheta x)^2 = \frac{1}{3}\Delta^2 + f\Delta_\vartheta^2.$$

B) $a_\nu(j\nu x)^2$ decomposes according to formula (9.)a, indem wir according to § 6 end $f \cdot j$ by $(\theta\theta' u)^2$ ersetzen, and the reducibility of θ'_ϑ take account of:

$$\frac{1}{3}a_\nu(j\nu x)^2 = (\theta\theta'\nu x)^2\theta_\nu = \theta_\nu^3\theta - \theta_\nu^2\theta'_\nu = \theta_\nu^3\theta + \theta_\nu^2(\widehat{j\nu\theta' u}) = \theta_\nu^3\theta - \frac{2}{3}\theta_\nu^3\theta.$$

forms $a_\nu(j\nu x)^3$ and $a_\nu^2(j\nu x)^2$: reducer a_j and $(j\nu x)^4$:

$$a_\nu(j\nu x)^3 = a_j(j\nu x)^3 - (j\nu x)^3(j\nu\widehat{a}u),$$

$$a_\nu^2(j\nu x)^2 = a_j^2(j\nu x)^2 - 2a_j(j\nu x)^2(j\nu\widehat{a}u) + (j\nu x)^2(j\nu\widehat{a}u)^2.$$

C) form sequence N_ν^2; reducer Δ_ν^2. $(n\nu x)$ and $N_\nu(n\nu x)$ decompose as a transvection over functional determinants according to § 3.

D) *Overview of the reductions:*

a) form sequence $\theta_\eta^2 H_\vartheta$: reducer $a_\eta^2(aHu)$. (Leads to forms with higher symbols.)

b) form sequence $\theta_\eta H_\vartheta$: reducer $a_\eta^2(aHu)$ by double reduction. (Leads to forms with higher symbols and to higher forms of the total form sequence, i.e. to the form sequence θ_η^2.) Instead of $\theta_\eta H_\vartheta$ we consider the form sequence

$$\theta_\eta(\vartheta\eta x)(\theta Hu) = \theta_\eta H_\vartheta + \theta_\eta^2,$$

ersetzen in it the symbols θ by the lower form (abu), gelangen so to the double reduction of the form sequence

$$a_\eta^2(aHu)(abu) = \theta_\eta H_\vartheta + \frac{3}{2}\theta_\eta^2 \mod s$$

up to the terminal form

$$a_\eta^3 b_\eta(bHu)(abu) = \theta_\eta^3 H_\vartheta + \frac{1}{2}\theta_\eta^4.$$

c) form sequence θ_η^3: By double reduction of those forms which simultaneously belong to the form sequences $\theta_\eta H_\vartheta$ and $\theta_\eta^2 H_\vartheta$ - i.e. the forms of the form sequence $\theta_\eta^2 H_\vartheta$ - to forms with higher symbols on the one hand, and to forms with higher symbols and to the higher form sequence θ_η^3 on the other hand.

d) forms $\theta_\eta^2(\vartheta\eta x)$, $\theta_\eta^2(\vartheta\eta x)^2$, $\theta_\eta^3(\vartheta\eta x)$: By double reduction by means of the reducers a analogous to the computation of § 9.

e) forms $\theta_\eta^2(\theta H u)^2$ and $\theta_\eta^2(\vartheta\eta x)^2$: By double reduction of the expressions $\Delta_\nu^2(a\Delta u)^4$ and $(a\Delta\nu x)^4$ by means of the reducers Δ_ν^2 and $(a\Delta u)^4$.

f) forms H_ϑ and H_ϑ^2: Decomposition of forms. Is obtained for H_ϑ^2 by double reduction of the expression $\Delta_\nu^2(a\Delta u)^2$ or also by direct calculation of the products $(a_\nu^2)^2$, $a_\nu^2 b_{\nu_1}$ by forms of the system. (The analogous calculation of $(a_\nu)^2$ gives a relation between decomposable forms.)

g) Reduction of higher forms by the fact that forms of the form sequence $\theta\cdot H$ admit two reduction possibilities (belong to two reducible form sequences), namely:

g: by double reduction of $\theta_\eta^2(\vartheta\eta x)^2$; allows reduction d) and e) zu.

$s_\vartheta^2(\theta su)$: by double reduction of $\theta_\eta^3(\vartheta\eta x)$; allows reduction c) and d) zu.

$s_\vartheta^2(\theta su)^2$: by double reduction of $\theta_\eta^2 H_\vartheta^2$; allows reduction a) and has the reducers $\theta_\nu^2(\vartheta\nu x)^2$ as a factor.

s_ν^2: by double reduction of $\theta_\eta^3 H_\vartheta$; allows reduction a) and c) zu.

The proof of the independence of the remaining forms is obtained by double reduction of the form sequence $\Delta_\nu^2(a\Delta u)^2 = \theta_\eta^2$.

Execution of the reductions:

The existence of reductions a), b), c), d) is clear a priori, since according to the stated law of formation the relations arising from double reduction are independent of one another. For reductions e), f), g), it must be shown by computation that no identities arise.

Proceeding symmetrically with the diagonal term in the form sequence $\theta\cdot H$ up to the form θ_η, and partly indicating the calculation, we also give the formulas obtained by reductions a), b), c), d), since they are used partly in reductions e), f), g), and partly later.

1) H_ϑ and H_ϑ^2 according to f). According to the ansatz[28]

$$\begin{aligned} a_\nu b_{\nu_1}^2 &= \nu a_\nu b_\nu^2 - (aHu)(bHu)b_\eta \sim \nu a_\nu b_\nu^2 - f a_\eta (aHu)^2 - (abu)(aHu)b_\eta + (abu)^2 b_\eta, \\ a_\nu^2 b_{\nu_1}^2 &= b_\nu^2\{a_\nu u_\nu - (a\nu\nu_1 u)\}^2 \sim \nu\{a_\nu^2 b_\nu^2 - 2a_\nu b_\nu (aHu)(bHu) - \frac{1}{6}(asu)^2(bsu)^2\} \\ &\sim \nu a_\nu^2 b_\nu^2 - 2a^2(aHu)(abu) - \frac{1}{6}(asu)^2(bsu)^2 + (\vartheta\eta x)^2(\theta Hu)^2 \end{aligned}$$

after substitution of the value of $\theta_\eta H_\vartheta$, the formulas arise:

$$\begin{aligned} &\text{a)}\quad H_\vartheta \sim 2a_\nu a_\varrho^2 + 2\nu b_\nu(\vartheta\eta x)^2 + 2f a_\eta (aHu)^2 - 2\theta_\eta, \\ &\text{b)}\quad H_\vartheta^2 \sim (a^2)^2 + 2\theta_\eta^2 + \frac{2}{3}(\theta su)^2 + \frac{1}{12}s\nu - \frac{1}{9}if\nu - \frac{1}{6}f(asu)^4. \end{aligned} \tag{12}$$

2) $\theta_\eta H_\vartheta$ according to b):

$$\begin{aligned} a_\eta^2(aHu)(abu) &\sim [a_\eta^2(aHu)]_{bu} + \frac{1}{2}f[a_\eta^2(aHu)^2] = a_\eta b_\eta (aHu)(abu) \\ &\quad + a_\eta(\widehat{ab}\eta x)(abu)(aHu) \sim \theta_\eta(\vartheta\eta x)(\theta Hu) + \frac{1}{2}\theta_\eta^2 + \frac{1}{4}(\nu\eta x)^2. \end{aligned}$$

According to the formulas (5.) and (11.) follows:

$$\theta_\eta H_\vartheta \sim -\frac{3}{2}\theta_\eta^2 - \frac{1}{4}(\theta su)^2 - \frac{1}{12}s\nu + \frac{2}{9}if\nu.$$

3) $\theta_\eta H_\vartheta(\theta Hu)$ is computed according to b), analogously to $\theta_\eta H_\vartheta$:

$$\theta_\eta H_\vartheta(\theta Hu) \sim -\theta_\eta^2(\theta Hu) - \frac{1}{6}(\theta su)^3.$$

4) $\theta_\eta H_\vartheta(\vartheta\eta x)$ according to b); $\theta_\eta^2(\vartheta\eta x)$ according to d) (cf. § 9):

$$\theta_\eta H_\vartheta(\vartheta\eta x) \sim -\theta_\eta^2(\vartheta\eta x) - \frac{1}{6}s_\vartheta(\theta su) \sim -\frac{1}{3}(\vartheta\varrho x) - \frac{1}{6}s_\vartheta(\theta su),$$

[28] The sign $\sim$ in place of the equality sign means that products arising from u_x with higher forms, other than those arising by foldings III and IV, are omitted.

$$\theta_\eta^2(\vartheta\eta x) \sim \frac{2}{3}a_\eta^3(abu) \sim -\frac{1}{3}(\vartheta\varrho x).$$

5)

$\theta_\eta H_\vartheta^2$ according to b)	$\theta_\eta^2 H_\vartheta$ according to a)	$\theta_\eta^2 H_\vartheta$ according to b) and thereby θ_η^3 according to c)
from $a_\eta^2(aHb)(bHu)$	from $a_\eta^3(Hab)(abu)$	from $a_\eta^3(bHu)(abu)$

$$\theta_\eta H_\vartheta^2 \sim -\frac{3}{2}\theta_\eta^2 H_\vartheta - \frac{1}{4}s_\vartheta(\theta su)^2 + \frac{1}{6}s_\nu - \frac{4}{9}ia_\nu \sim -\theta_\varrho - \frac{1}{6}s_\nu - \frac{1}{9}ia_\nu,$$

$$\theta_\eta^2 H_\vartheta \sim \frac{2}{3}\theta_\varrho - \frac{1}{6}s_\vartheta(\theta su)^2 + \frac{2}{9}s_\nu - \frac{2}{9}ia_\nu,$$

$$\theta_\eta^3 H_\vartheta \sim \frac{2}{3}\theta_\varrho - \frac{2}{3}\theta_\eta^3 + \frac{5}{36}s_\nu, \qquad \theta_\eta^3 \sim \frac{1}{4}s_\vartheta(\theta su)^2 - \frac{1}{8}s_\nu + \frac{1}{3}ia_\nu.$$

6) $\theta_\eta^2(\theta Hu)^2$ according to e):

$$\Delta_\nu^2(a\Delta u)^2 = [(a\Delta u)^4]_{\nu^2} = -\frac{12}{5}\theta_\eta^2(\theta Hu)^2 - \frac{3}{10}\sigma - \frac{1}{20}(\theta su)^4 + \nu\varrho \quad \text{(according to formula (3.)c)},$$

$$\Delta_\nu^2(a\Delta u)^4 = [\Delta_\nu^2]_{aa}^4 = -\frac{3}{5}\theta_\eta^2(\theta Hu)^2 + \frac{1}{5}\sigma - \frac{3}{10}(\theta su)^4 + \frac{1}{3}ij \quad \text{(according to formula (10.)a)},$$

$$\theta_\eta^2(\theta Hu)^2 = -\frac{1}{6}\sigma + \frac{1}{12}(\theta su)^4 - \frac{1}{9}ij + \frac{1}{3}\nu\varrho.$$

7) $\theta_\eta^2(\vartheta\eta x)^2$ according to d) and e). From this reduction of the higher form g.
By a computation analogous to that in § 9, it follows:

$$\begin{aligned}\theta_\eta^2(\vartheta\eta x)^2 &= \frac{1}{2}if - \frac{1}{2}a_\eta^2 b_\eta - \frac{1}{12}g = \frac{2}{3}tf - \frac{2}{3}a_\eta^3 b_\eta - \frac{1}{6}g\\ &= -\frac{1}{6}g - \frac{1}{3}(\vartheta\varrho x)^2 + \frac{4}{15}fa_\varrho^2 + \frac{8}{15}ft \quad \text{(according to formula (11.))}.\end{aligned}$$

By the corresponding computation as above follows:

$$((a\Delta\nu x)^4) = [(a\Delta u)^4]_{\nu x^4} = \frac{21}{10}\theta_\eta^2(\vartheta\eta x)^2 + \frac{7}{10}s_\vartheta^2 - \frac{3}{10}g \quad \text{(according to formula (3.)c)},$$

$$\begin{aligned}(a\Delta\nu x)^4 &= -\frac{1}{3}i\Delta + 6[\Delta_\nu^2]a^2 - 4[\Delta_\nu^3]a + \Delta_\nu^4 f\\ &= -\frac{36}{5}\theta_\eta^2(\vartheta\eta x)^2 - \frac{9}{5}s_\vartheta^2 - \frac{3}{5}g - \frac{3}{5}(\vartheta\varrho x)^2\\ &\quad + \frac{5}{3}i\Delta + \frac{37}{25}fa_\varrho^2 + \frac{74}{25}ft \quad \text{(according to formula (10.))}.\end{aligned}$$

Thus

$$\begin{aligned}\frac{93}{10}\theta_\eta^2(\vartheta\eta x)^2 &= -\frac{3}{10}g - \frac{5}{2}s_\vartheta^2 - \frac{3}{5}(\vartheta\varrho x)^2\\ &\quad + \frac{5}{3}i\Delta + \frac{37}{25}fa_\varrho^2 + \frac{74}{25}ft,\end{aligned}$$

and by comparing the two values of $\theta_\eta^2(\vartheta\eta x)^2$ according to g):

$$0 = g - 2s_\vartheta^2 + 2(\vartheta\varrho x)^2 + \frac{4}{3}i\Delta - \frac{4}{5}fa_\varrho^2 - \frac{8}{5}ft. \tag{13}$$

8) $\theta_\eta^2 H_\vartheta(\theta Hu)$ according to a) and b), from this $\theta_\eta^3(\theta Hu)$ according to c) (forms with the Faktor u_x vanish):

$$\theta_\eta^2 H_\vartheta(\theta Hu) = -\frac{1}{6}s_\vartheta(\theta su)^3 - \frac{1}{3}(\nu\varrho x),$$

$$\theta_\eta^2 H_\vartheta(\theta Hu) = -\frac{1}{3}\theta_\eta^3(\theta Hu) - \frac{1}{3}(\nu\varrho x),$$

$$\theta_\eta^3(\theta Hu) = \frac{1}{2}s_\vartheta(\theta su)^3.$$

9) $\theta_\eta^2 H_\vartheta(\vartheta\eta x)$ according to a) and b), from this $\theta_\eta^3(\vartheta\eta x)$ according to c). $\theta_\eta^3(\vartheta\eta x)$ according to d), from this reduction of the higher form $s_\vartheta^2(\theta su)$ according to g) (forms with the Faktor u_x vanish):

$$\theta_\eta^2 H_\vartheta(\vartheta\eta x) = \frac{1}{3}(atu),$$

$$\theta_\eta^2 H_\vartheta(\vartheta\eta x) = \frac{1}{3}(atu) + \frac{1}{3}\theta_\eta^3(\vartheta\eta x) - \frac{1}{6}s_\vartheta^2(\theta su),$$

$$\theta_\eta^3(\vartheta\eta x) = \frac{1}{2}s_\vartheta^2(\theta su),$$

$$\theta_\eta^3(\vartheta\eta x) = \frac{2}{5}\theta_\varrho(\vartheta\varrho x) + \frac{1}{5}(atu),$$

$$s_\vartheta^2(\theta su) = \frac{4}{5}\theta_\varrho(\vartheta\varrho x) + \frac{2}{5}(atu). \tag{14}$$

10) $\theta_\eta^2 H_\vartheta^2$ according to a) and by reducer $\theta_\nu^2(\vartheta\nu x)^2$, $\theta_\eta^3 H_\vartheta$ according to a) and c), θ_η^4 according to c), from this reduction of the higher forms $s_\vartheta^2(\theta su)^2$ and s_ν^2 according to g):

$$\begin{aligned}\text{According to a)}\quad \theta_\eta^2 H_\vartheta^2 &= [a_\eta^2(aHu)^2]b^2 + \frac{1}{3}[a_\eta^4]_{bu^2} - \frac{1}{3}H_\nu^2(\nu\eta x)^2\\ &= \frac{4}{9}ia_\nu^2 - \frac{1}{6}s_\vartheta^2(\theta su)^2 - \frac{5}{18}s_\nu^2 + \frac{1}{3}(atu)^2 + \frac{1}{54}Ju_x^2.\end{aligned}$$

$$\begin{aligned}\text{By } \theta_\nu^2(\vartheta\nu x)^2:\quad \theta_\eta^2 H_\vartheta^2 &= [\theta_\nu^2(\vartheta\nu x)^2]_{\nu_1} + \frac{1}{3}[\theta_\nu^4]_{\nu_1}\hat{\nu}_1 x^2 - \frac{1}{3}s_\vartheta^2(\theta su)^2\\ &= \frac{4}{9}ia_\nu^2 - \frac{1}{6}s_\nu^2 - \frac{1}{3}s_\vartheta^2(\theta su)^2 + \frac{1}{3}(\nu\varrho x)^2.\end{aligned}$$

$$\begin{aligned}\text{According to a)}\quad \theta_\eta^3 H_\vartheta = &-[a_\eta^2(aHu)]_{bu}b^2 - \frac{1}{2}[a_\eta^2(aHu)^2]b^2 - \frac{1}{3}[a_\eta^3]b + \frac{1}{12}[a_\eta^4]_{bu^2}\\ &+ \frac{1}{2}u_x[a_\eta^2(aHu)^2]b^3\\ = &-\frac{2}{9}ia_\nu^2 + \frac{1}{12}s_\nu^2 + \frac{4}{15}\theta_\varrho^2 - \frac{7}{90}(\nu\varrho x)^2 + \frac{1}{30}(atu)^2 + \frac{2}{27}i^2u_x^2 - \frac{1}{36}Ju_x^2.\end{aligned}$$

$$\begin{aligned}\text{According to c)}\quad \theta_\eta^3 H_\vartheta : a_\eta^3(Hab)(bHu) &= \frac{1}{2}(atu)^2 = \theta_\eta^2 H_\vartheta(\theta Hu)(\vartheta\eta x) + \frac{1}{4}H_\nu^2(\nu\eta x)^2\\ &+ \frac{1}{2}\theta_\eta^2 H_\vartheta^2 = \frac{3}{2}\theta_\eta^2 H_\vartheta^2 + \theta_\eta^3 H_\vartheta - u_x[a_\eta^2(aHu)^2]b^3\\ &+ \frac{1}{4}H_\nu^2(\nu\eta x)^2,\end{aligned}$$

and therefore

$$\begin{aligned}\theta_\eta^3 H_\vartheta = &-\frac{2}{3}ia_\nu^2 + \frac{5}{12}s_\nu^2 + \frac{1}{2}(\nu\varrho x)^2 - \frac{1}{2}(atu)^2\\ &+ \frac{4}{27}i^2u_x^2 - \frac{1}{12}Ju_x^2.\end{aligned}$$

According to c)

$$\begin{aligned}\theta_\eta^4 = &\frac{4}{9}ia_\nu^2 - \frac{1}{6}s_\nu^2 + \frac{16}{15}\theta_\varrho^2\\ &- \frac{14}{45}(\nu\varrho x)^2 + \frac{2}{15}(atu)^2.\end{aligned}$$

According to g), from the two values for $\theta_\eta^2 H_\vartheta^2$:

$$\begin{aligned}s_\vartheta^2(\theta su)^2 &= \frac{2}{3}s_\nu^2 - 2(atu)^2 + 2(\nu\varrho x)^2 - \frac{1}{9}Ju_x^2\\ &= \frac{8}{9}ia_\nu^2 + \frac{8}{15}\theta_\varrho^2 - \frac{14}{15}(atu)^2 + \frac{38}{45}(\nu\varrho x)^2 - \frac{4}{27}i^2u_x^2.\end{aligned} \tag{15}$$

According to g), from the two values for $\theta_\eta^3 H_\vartheta$:

$$s_\nu^2 = \frac{4}{3}ia_\nu^2 + \frac{4}{5}\theta_\varrho^2 + \frac{8}{5}(atu)^2 - \frac{26}{15}(\nu\varrho x)^2 + \frac{1}{6}Ju_x^2 - \frac{2}{9}i^2u_x^2. \tag{16}$$

Formula (16.) gives the basis for the reduction of the modulus $(ss'u)^4$, and formula (13.) gives the basis for the reduction of the system of s. (§ 17.)

Consequences:

1) $a_\nu(j\nu x)^3$, $\theta_\eta H_\vartheta$, $\theta_\eta^2(\vartheta\eta x)$, $\theta_\eta^2(\theta Hu)^2$ are reducers, $a_\nu(j\nu x)^2$, H_ϑ and H_ϑ^2 decomposable forms.

2) The form of the system II: $j(\nu) = g = (\nu\eta x)^4 H_x^2$ is reducible.

3) Since the only contragredient form g becomes reducible, ν does not enter at all in powers or in products with forms of the same system in folding with System I.

θ enters into foldings only as a contravariant in products or powers; correspondingly, H enters only as a covariant (cf. § 13 B).

§ 12. Forms of the seventh order.

$$\text{Folding of } \left\{ \begin{array}{l} \theta\cdot K \text{ with } \nu, \\ K, j, \Delta \text{ with } H, \\ f \text{ with } L, \sigma. \end{array} \right.$$

Irreducible forms:

$$\text{B)} \quad \begin{array}{c|c|c|c|c} \cdot & \cdot & (KHu)^2 & \cdot & \cdot \\ \cdot & K_\eta & \cdot & K_\eta(KHu)^2 & \cdot \\ (k\eta x)^2 & \cdot & K_\eta^2 & \cdot & \cdot \\ (k\eta x)^3 & H_k(k\eta x)^2,\ K_\eta(k\eta x)^2 & \cdot & \cdot & \cdot \\ \cdot & K_\eta(k\eta x)^3 & \cdot & \cdot & \cdot \end{array}$$

$$\text{C)} \quad \begin{array}{cc} (j\eta x) & H_j \\ \cdot & H_j(j\eta x) \\ \cdot & H_j^2 \end{array} \qquad \text{D)} \quad \begin{array}{c} (\Delta Hu) \\ \Delta_\eta \end{array}$$

$$\text{E)} \quad \begin{array}{cccc} a_i & (aLu)^2 & (aLu)^3 & \cdot \\ \cdot & \cdot & a_i(aLu)^2 & a_i(aLu)^3 \\ \cdot & a_i^2 & \cdot & \cdot \end{array}$$

reductions:

A) $(\vartheta\cdot k, \nu, x)^4$ decomposes as single Überschiebung over the decomposable form $(\vartheta^2\nu x)^4$.

B) form sequence $H_k K_\eta$: reducer $H_\vartheta\theta_\eta$.

form sequence $K_\eta^2(KHu)$: reducer $a_\eta^2(aHu)$.

form sequence $K_\eta^2(k\eta x)$: reducer $\theta_\eta^2(\vartheta\eta x)$.

The from this arising double reduction of the form sequence K_η^3 gives rise to the reduction of the higher form sequence $(j\eta x)^3$.

(KHu), $(k\eta x)$, $K_\eta(k\eta x)$ decomposable forms according to § 3.

$H_k(KHu)$, $K_\eta(KHu)$ decompose as arising by single folding with the decomposable form $K_\nu(k\nu x)$ and the functional determinant

$$K_\nu^2 = \theta_\nu^2(a\theta u) \equiv a_\nu^2(a\theta u) \mod \nu^2.$$

A relation between decomposable forms corresponds to the double representation of K_ν^2. One obtains:

$$H_k(KHu) = 2K_\nu(\nu\nu_1 k) = 2\theta_{\nu_1}(\nu\nu_1 a\theta) = 2\theta_\nu^2 a_\nu - a_\nu^2\theta_\nu + f\theta_\eta(\theta Hu)^2 - f\nu\theta_\vartheta^3 \mod \nu^3,$$

$$\begin{aligned} K_\eta(KHu) &= -K_\nu^2(\nu\nu_1 x) = -a_\nu^2(a\theta u)(\nu\nu_1 x) \\ &= a_\eta(aHu)^2\theta + a_\nu^2\theta_\nu = -\theta_\nu(\theta Hu)^2 f - \theta_\nu^2 a_\nu. \end{aligned}$$

Therefore

$$0 = a_\nu^2\theta_\nu + \theta_\nu^2 a_{\nu_1} + f\theta_\eta(\theta Hu)^2 + \theta a_\eta(aHu)^2 \mod \nu^3. \tag{17}$$

The forms H_k, $H_k(k\eta x)$, H_k^2, $H_k^2(k\eta x)$ decompose as arising by single folding with the decomposable forms H_ϑ and H_ϑ^2 (cf. § 3. 2), a) and b)).

Es is

$$H_k = H_\vartheta(a\theta u) \sim [H_\vartheta]_{ua} - fH_\vartheta(\theta Hu)$$

(specialization $y = uv$ of 2)a).

$$H_k(k\eta x) = H_\vartheta a_\eta - H_\vartheta\theta_\eta f, \qquad H_\vartheta a_\eta = \frac{5}{4}[H_\vartheta]a - \frac{1}{4}H_\vartheta a_\vartheta \quad (\S\ 3.2)b),$$

$$H_\vartheta^2(a\theta u) = [H_\vartheta^2]_{ua}, \qquad H_\vartheta^2 a_\eta = [H_\vartheta^2]a = H_k^2(k\eta x) + H_\vartheta^2\theta_\eta f.$$

C) form sequence $(j\eta x)^3$: Reducible by double reduction of the form sequence K_η^3 according to B); according to the ansatz:

$$0 = a_\eta^3(a\theta u) = \theta_\eta^3(a\theta u) = \theta_\eta + (a\theta\eta x)^3(a\theta u) - \theta_\eta^3(a\theta u) = \frac{1}{2}(j\eta x)^3 \mod(\nu^3, s, \varrho, t, i, J).$$

The analogous ansatz holds up to the terminal form of the form sequence:

$$0 = H_\vartheta^2(a\theta\eta x)a_\eta^3 = H_\vartheta^2(a\theta\eta x)\theta_\eta^3 - \frac{1}{2}H_\vartheta^2(j\eta x)^4 \mod(\nu^3, s, \varrho, t, i, J).$$

form $(j\eta x)^2$: decomposes 1) by twofold polarization of the relation for the decomposable form H_ϑ^2 ((12.) b);

2) by direct calculation of the products $a_\nu\theta_\nu^3$; thereby Decomposition of the higher form $(\Delta Hu)^2$. One obtains:

$$\left.\begin{aligned} &\text{a)} \quad \frac{1}{3}(\Delta Hu)^2 + \frac{1}{3}(j\eta x)^2 = \theta_\nu^2 a_{\nu_1}^2, \\ &\text{b)} \quad (j\eta x)^2 = \frac{4}{3}\theta_\nu^3 a_{\nu_1}, \end{aligned}\right\} \mod(\nu^3, s, \varrho, t, i, J). \tag{18}$$

according to the ansatz:

$$H_\vartheta^2(a\theta u)^2 - \frac{1}{3}(\Delta Hu)^2 = 2\theta_\eta^2(a\theta u)(aHu) + \theta_\nu^2 a_{\nu_1}^2 = (a\theta u)(aHu)\{a_\eta - (a\theta\eta x)^2\} + \theta_\nu^2 a_{\nu_1}^2,$$

$$\begin{aligned} a_{\nu_1}\theta_\nu^3 &= -(a\widehat{\nu}\nu_1 u)\theta_\nu^2(\theta\widehat{\nu}\nu_1 u) - \frac{1}{2}(u\nu\nu_1 u)\theta_\nu\theta_{\nu_1}(\vartheta\nu\nu_1 u) \\ &= -\frac{3}{2}\theta_\eta^2(\theta Hu)(aHu) = -\frac{3}{2}\theta_\eta^2(\theta Hu)(a\theta u) \\ &= -\frac{3}{2}\{a_\eta - (a\theta\eta x)^2\}\{(aHu) - (a\theta u)\}(a\theta u). \end{aligned}$$

D) form sequence $\Delta_\eta(\Delta Hu)$: reducer Δ_ν^2.

$(\Delta Hu)^2$ decomposable form according to C).

E) form sequence $a_i^2(aLu)$: reducer $a_\eta^2(aHu)$. forms (aLu) and $a_i(aLu)$: Decomposition as a transvectionen over functional determinants according to § 3.

F) form sequence a_σ^3: reducer a_ϱ^3.

forms a_σ^2 and a_σ^3: reducer a_ν^3 and a_η^3 by double reduction of the corresponding expressions

$$H_\nu a_\nu^3 \quad \text{and} \quad H_\nu a_\eta^3, \qquad H_\nu a_\nu^3 a_\nu \quad \text{and} \quad H_\nu a_\eta^4$$

(cf. § 10. C) reduction of Δ_ν^2 and Δ_ν^3).

From this reduction for the higher forms $a_\nu s_\nu(asu)^2$, $a_\nu^2 s_\nu(asu)^2$ and for forms in symbolsn $\varrho, t(a_\nu, a_\eta^2)$ taking account of (16.):

$$\left.\begin{aligned} a_\nu s_\nu(asu)^2 &= \frac{1}{3}i\theta_\nu^2 - \frac{1}{18}iH, \\ a_\nu^2 s_\nu(asu)^2 &= 0, \end{aligned}\right\} \mod(\varrho, t). \tag{19}$$

form a_σ: decomposes by polarization of (12.)b or, more briefly, by direct expansion of the products $a_\nu^2\theta_\nu^3$ according to the ansatz:

$$a_\nu^2\theta_{\nu_1}^3 = (a_\nu^2\theta_{\nu_1})a_{\nu_1} - (a\theta\nu_1 x)^2 = -2a_\eta(aHu)(a\theta H)(a\theta\eta x) + (a\theta\nu_1 x)^2 a_\nu^2\theta_{\nu_1}$$

and taking account of the reduction of $H_j(j\eta x)^2$ (C):

$$a_\sigma = 2a_\nu^2\theta_{\nu_1}^3 \mod(s, \varrho, t, i). \tag{20}$$

Consequences:

1) $(j\eta x)^3$, $\Delta_\eta(\Delta Hu)$, a_σ^2 are reducers, $(j\eta x)^2$, $(\Delta Hu)^2$, a_σ decomposable forms.

2) f enters into folding with L only in products with contragredient forms; therefore f, and likewise K and N, do not enter at all into folding with σ and consequently with $(\nu\sigma x)$. The form j enters only in products with contragredient forms; Δ does not enter at all in products in folding with H and L (this follows from $\Delta_\eta(\Delta Hu)$ and $(\Delta Hu)^2$). Likewise σ, and consequently $(\nu\sigma x)$, do not enter in products in folding with any form of System I. Thus, among products in System II, taking account of the consequences of § 11, there remain only powers of H and products of H with L.

§ 13. Forms of the eighth order (System III).

$$\text{Folding of}\begin{cases}\Delta\cdot j \text{ with } \nu,\\ \theta^2, f\cdot j, N \text{ with } H,\\ \theta \text{ with } L,\sigma.\end{cases}$$

Irreducible forms:

A) $\Delta_\nu(j\nu x)$ B) $\begin{matrix}(\vartheta^2\eta x)^3\\(\vartheta^2\eta x)^4\end{matrix}$ $H_\vartheta(\vartheta^2\eta x)^3,\ \theta_\eta(\vartheta^2\eta x)^3,$

C) $(aHu)H_j,\quad a_\eta(aHu)H_j$ D) $H_n,\ N_\eta,$

$$\text{E)}\quad \begin{array}{c|c|c|c} \cdot & (\theta Lu)^2 & (\theta Lu)^3 & \cdot \\ \theta_i & \cdot & L_\vartheta(\theta Lu)^2,\ \theta_i(\theta Lu)^2 & \theta_i(\theta Lu)^3 \\ (\vartheta lx)^2 & \theta_i^2 & \cdot & \cdot \\ \cdot & \theta_i(\vartheta lx)^2 & \cdot & \cdot \end{array}$$

reductions:

A) $\Delta_\nu(j\nu x)^3$: reducer $a_\nu(j\nu x)^3$.

$\Delta_\nu(j\nu x)^2$ is reducible by the decomposable form $a_\nu(j\nu x)^2$ according to § 3. 2)b, taking account of the reducer $\theta_{\vartheta j}$. Indeed, according to § 11 B), one obtains:

$$\frac{3}{10}\Delta_\nu(j\nu x)^2+\frac{3}{5}a_\nu(j\nu x)a_\vartheta(j\nu x)+\frac{1}{10}a_\nu(j\nu x)^2=\frac{3}{5}\theta_\nu^3\theta_{\vartheta j}+\frac{2}{5}\theta_\nu^3\theta_{\vartheta j}\theta_{\vartheta j},$$

(reducer a_j and $\theta_{\vartheta j}$).

B) The forms $H_\vartheta(\vartheta'\eta x)^2$, $H_\vartheta^2(\vartheta'\eta x)$, $H_\vartheta^2(\vartheta'\eta x)^2$ decompose as arising by successive single folding with the decomposable forms H_ϑ and H_ϑ^2, respectively $H_\vartheta a_\eta$ and $H_\vartheta^2 a_\eta$ according to § 3. 2)b) and according to the relation:

$$H_\vartheta(\vartheta'\eta x)^2=2H_\vartheta a_\eta^2 f-2H_\vartheta a_\eta b_\eta.$$

C) form sequence $a_\eta(j\eta x)^2$: reducers a_j and $(j\eta x)^3$.

form $a_\eta(j\eta x)$ decomposes: reducer a_j and single folding with the decomposable form $(j\eta x)^2$ according to § 3, 2)a (specialization $y=uv$).

form $a_\eta H_j$ decomposes: 1) by single folding with the decomposable form $a_\nu(j\nu x)^2$;

2) by a special twofold folding with the decomposable form $(j\eta x)^2$; from this one obtains a relation between decomposable forms.

From $a_\nu(j\nu x)^2$:

$$0=\theta_\nu'\theta_\nu+2a_\eta H_j \mod(s,\varrho,t,i,J).$$

From $(j\eta x)^2$:

$$0=\theta_\nu'\theta_\nu-\frac{3}{2}a_\eta H_j \mod(s,\varrho,t,i,J)$$

(products with contravariants are omitted), according to the ansatz:

$$0=\frac{1}{2}a_\nu(abu)^2\theta_\nu^3+\frac{1}{2}u_x(ubu)\theta_{\nu_1}^3(\theta bu)-\frac{3}{4}(\widehat{j}\eta bu)^2+\frac{3}{4}H_j(\widehat{j}\eta bu)-\frac{1}{8}fH_j^2$$

and by interchanging the symbols in $a_\nu(ubu)(\theta bu)\theta_{\nu_1}^3$.

D) form sequence $N_\eta(NHu)$: reducer Δ_ν^2.

(NHu), $(n\eta x)$, $N_\eta(n\eta x)$, $H_\eta(NHu)$ decomposable forms (cf. § 12 B), $(NHu)^2$ decomposes by single folding with the form $(\Delta Hu)^2$.

E) form sequences $\theta_i L_\vartheta$, $\theta_i^2(\theta Lu)^2$, $\theta_i^2(\vartheta lx)$:

reducers: $\theta_\eta H_\vartheta$, $\theta_\eta^2(\theta Hu)^2$, $\theta_\eta^2(\vartheta\eta x)$.

forms (θLu), (ϑlx), L_ϑ, $L_\vartheta(\theta Lu)$, $\theta_i(\theta Lu)$, $L_\vartheta(\vartheta lx)$, $\theta_i(\vartheta lx)$, L_ϑ^2, $L_\vartheta^2(\theta Lu)$, $\theta_i^2(\theta Lu)$ decompose. (Dualistically opposite to the decomposable forms of § 12 B).

F) form sequence $\theta_\sigma(\vartheta\sigma x)$: reducer a_σ^2.

forms $(\vartheta\sigma x)$ and $(\vartheta\sigma x)^2$ decompose, as arising by single folding with the decomposable form a_σ; θ_σ, by twofold folding arising, decomposes, since the form sequence

$$a_{\nu_2}^2(a\theta u)\theta_{\nu_1}^3\equiv H_j^2(j\eta x)\equiv 0 \mod u_x(s,\varrho,t,i,J):$$

$$\theta_\sigma = a_\sigma(abu)^2 = a_\nu^2(abu)^2\theta_\nu^3.$$

Consequences:

θ^3 or $\theta^2 K$ does not enter into folding with H; θ enters into folding with L only as a contravariant in powers or products, and not at all into folding with σ and $(\nu\sigma x)$.

N does not enter in products in folding with irgend einer form of System II enters, ebensowenig wie with powers or products of forms of System II. There remain an products in System I, taking account of the consequences of the last paragraphs, only products $f\cdot j$, $\Delta\cdot j$, powers of θ and products of $\theta\cdot K$.

§ 14. Forms of the ninth order (System III).

$$\text{Folding of}\begin{cases}\theta\cdot K \text{ with } H,\\ K, j, \Delta \text{ with } L,\\ j, \Delta \text{ with } \sigma,\\ f \text{ with } H^2.\end{cases}$$

Irreducible forms:

A) $(\vartheta\cdot k,\eta,x)^4,\ H_k(\vartheta\cdot k,\eta,x)^4$ B) $(KLu)^3,\ K_i(KLu)^3,\ K_i^2,\ (klx)^3,\ K_i(Klx)^3,$

C) $L_j,\ L_j^2$ D) $\Delta_i,$

E) $(j\sigma x)$ G) $(aH^2u)^3,\ (aH^2u)^4,\ a_\eta(aH^2u)^3.$

reductions:

A) the forms $H_\vartheta(k\eta x)^3$, $H_\vartheta^2(k\eta x)^2$, $H_\vartheta^2(k\eta x)^3$ decompose as arising by successive single folding with decomposable forms.

B) form sequences K_iL_k, $K_i^2(KLu)$, $K_i^2(klx)$:

reducers: $\theta_\eta H_\vartheta$, $a_\eta^2(aHu)$, $\theta_\eta^2(\vartheta\eta x)$.

forms L_i, $K_i(KLu)$, $K_i(klx)$, $(KLu)^2$, $(klx)^2$ decompose as a transvectionen over functional determinants (§ 3).

forms L_k, $L_k(KLu)$, $L_k(KLu)^2$, $L_k(klx)$, $L_k(klx)^2$, L_k^2, $L_k^2(KLu)$, $L_k^2(klx)$, L_k^3, $K_i(KLu)^2$, $K_i(klx)^2$ decompose as arising by single folding with the decomposable forms:

$$L_\vartheta,\ H_k(KHu),\ L_\vartheta(\vartheta lx),\ H_k^2,\ L_\vartheta^2,\ L_\vartheta^2(\theta Lu),\ K_\eta(KHu),\ \theta_i(\vartheta lx)$$

according to § 3. 2), a) and b).

C) form sequence $(jlx)^3$: reducer $(j\eta x)^3$.

forms $(jlx)^2$ and $L_j(jlx)$: arising by single folding with the decomposable form $(j\eta x)^2$.

D) form sequence $\Delta_i(\Delta Lu)$: reducer Δ_ν^2.

forms $(\Delta Lu)^2$, $(\Delta Lu)^3$: Single folding with the decomposable form $(\Delta Hu)^2$.

E) form sequence $(j\sigma x)^2$: reducer a_σ^2 by replacing j by lower forms of the form sequence (§ 5):

$$a_\sigma^2(a\theta u)^2 = \frac{1}{3}(j\sigma x)^2, \qquad a_\sigma^2(a\theta\sigma x)^2 = \frac{1}{3}(j\sigma x)^4.$$

F) form sequence Δ_σ^2: reducer a_σ^2.

form Δ_σ: reducer a_σ^2 by replacing Δ by lower forms of the form sequence:

$$a_\sigma a_\varrho^2 = \frac{2}{3}\Delta_\sigma \mod \nu.$$

Consequences: Only the form j enters in folding with σ and $(\nu\sigma x)$ enters.

N does not enter in folding with L enters, since the Δ_i correspondinglye form N_i decomposes.

§ 15. Forms of the tenth order (System III).

$$\text{Folding of}\begin{cases}\theta^2,\ f\cdot j \text{ with } L,\\ \theta \text{ with } H^2.\end{cases}$$

Irreducible forms:

A) $(\vartheta^2lx)^4,\ \theta_i(\vartheta^2lx)^4$ B) $(aLu)^2L_j,\ a_i(aLu)^2L_j,$

C) $(\theta H^2u)^3,\ (\theta H^2u)^4,\ H_\vartheta(\theta H^2u),\ \theta_\eta(\theta H^2u)^3.$

reductions:

A) and B): Decomposition of the remaining forms by single folding with decomposable forms.

C) Decomposition of the remaining forms according to § 13 B) by the dualistically opposite consideration.

Consequences:

$\theta^2 K$ does not enter into folding with L; correspondingly, $H^2 L$ does not enter into folding with K. The forms H^3 and $H^2 L$ do not enter into folding with θ. The folding scheme of § 7 is therefore proved.

§ 16. Forms of the 11th, 12th, 13th, and 15th orders (System III).

11th order.

$$\text{Folding of} \left\{ \begin{array}{l} \theta \cdot K \text{ with } L, \\ K, j \text{ with } H^2, \\ j \text{ with } (\nu\sigma x), \\ f \text{ with } H \cdot L. \end{array} \right.$$

Irreducible forms:

$$\begin{array}{llll} \text{A)} & (\vartheta \cdot k, l, x)^5,\ \theta_i(\vartheta \cdot k, l, x)^5 & \text{B)} & (KH^2u)^4,\ K_\eta(KH^2u)^4, \\ \text{C)} & (\nu\sigma j), & \text{D)} & (a_\eta H \cdot L, u)^4. \end{array}$$

12th order.

$$\text{Folding of} \left\{ \begin{array}{l} \theta^3 \text{ with } L, \\ \theta \text{ with } H \cdot L. \end{array} \right.$$

Irreducible forms:

$$\text{E)}\ (\vartheta^3 lx)^6, \qquad \text{F)}\ (\theta, H \cdot L, u)^4, \qquad L_\vartheta(\theta, H \cdot L, u^4).$$

13th order.

$$\text{Folding of } K \text{ with } H \cdot L.$$

Irreducible forms:

$$\text{G)}\ (K, H \cdot L, u)^5, \qquad K_\eta(K, H \cdot L, u)^5.$$

15th order.

$$\text{Folding of } K \text{ with } H^3.$$

Irreducible form:

$$\text{H)}\ (KH^3u)^6.$$

reductions:

The forms H_j^2, H_j^2 have reducer L_j^3, from the relation:

$$(\nu l x) L_x^3 = -\frac{1}{2} H^2 \mod(\sigma, s).$$

Decomposition or reducibility of the remaining forms follows from the considerations of the earlier paragraphs.

Consequences:

According to the folding scheme of § 7 and the consequences of §§ 8–15, the irreducible forms of System III (117 forms) are thereby exhausted.

Chapter IV. The relatively complete system mod (ϱ, t).

§ 17. Reduction of the modulus $(ss'u)^4$ and of the system of s.

For the formation of the next higher relatively complete system, one must, in analogy with § 7, form the system of s with respect to the next modulus

$$\nu(s) = (ss'u)^4$$

and fold it with the forms of the relatively complete system mod s. It will be shown that, after introduction of the modulus (ϱ, t) as the higher modulus,

A) the modulus $(ss'u)^4$ becomes reducible,
B) the system of s consists of the single form s.

A) The reduction of the modulus $(ss'u)^4$ follows at once from the reducer s_ν^2 found by formula (16.), § 11. From

$$s_\nu^2 = \frac{4}{3} i a_\nu^2 + \frac{4}{5}\theta_\varrho^2 + \frac{8}{5}(atu)^2 - \frac{26}{15}(\nu\varrho x)^2 + \frac{1}{6} J \cdot u_x^2 - \frac{2}{9} i^2 \cdot u_x^2$$

there follows, by polarization from x to ν_1,

$$\begin{aligned}(ss'u)^4 &= -\frac{2}{3} J \cdot \nu + 6 s_\nu^2 s_{\nu_1}^2 \\ &= \frac{24}{5}\theta_\nu^2\theta_\varrho^2 + \frac{48}{5} a_\nu^2 (atu)^2 - \frac{52}{5} H_\varrho^2 + \frac{4}{3} i \cdot (asu)^4 + \frac{1}{3} J \cdot \nu - \frac{4}{9} i^2 \cdot \nu,\end{aligned} \tag{21}$$

and with this the reduction of the modulus $(ss'u)^4$.

B) The reduction of

$$Z = (ss'u)^2 = \theta(s)$$

and hence the reduction of the system of s is obtained by replacing the form Z by the lower form g_ν^2, according to formula (8.)b, § 7, taking into account the relation

$$s_\eta^2 = s_\nu^2 (\nu\nu_1 x)^2 - \frac{1}{3} Z.$$

The form g_ν^2 has, by formula (13.), § 11, the reducer s_ν^2 as a factor. By formula (8.) and (16.) one obtains

$$g_\nu^2 = -Z + \frac{14}{5} i a_\eta^2 + \frac{14}{15} i (asu)^2 \ \operatorname{mod}(\varrho, t),$$

and by formulas (13.), (14.), (15.), (16.),

$$\begin{aligned} g_\nu^2 &= 2[s_\vartheta^2]_{\nu^2} - \frac{4}{3} i \Delta_\nu^2 = 2 s_\vartheta^2 s_\nu^2 - \frac{6}{5}[s_\vartheta^2(\theta s u)^2]\widehat{\nu x}^{\,2} - \frac{4}{3} i \Delta_\nu^2 \\ &= \frac{4}{5} i \cdot a_\eta^2 - \frac{2}{5} i \cdot (asu)^2 + \frac{1}{3} J \cdot \theta \ \operatorname{mod}(\varrho, t).\end{aligned}$$

Thus

$$Z = 2i \cdot a_\eta^2 + \frac{4}{3} i \cdot (asu)^2 - \frac{1}{3} J \cdot \theta \ \operatorname{mod}(\varrho, t). \tag{23}$$

The relatively complete system mod $((ss'u)^4, \varrho, t)$ therefore passes into a relatively complete system mod (ϱ, t), and from this, by transvection over the known system of two quadratic forms, the absolutely complete system arises. In order to form the relatively complete system mod (ϱ, t), since by formula (23.) the system of s reduces to the form s, we must transvect the relatively complete system mod (s, ϱ, t) over s and over powers of s. It will be shown that powers of s enter into folding only with System I.

The announced relation is

$$\begin{aligned}(ass')^4 &= \frac{24}{5}\theta_\nu^2\theta_\varrho^2 a_\nu^2 a_\varrho^2 + \frac{48}{5} a_\nu^2 b_\nu^2 (tab)^2 - \frac{52}{5} H_\varrho^2 a_\nu^4 + \frac{5}{3} J \cdot i - \frac{4}{9} i^3, \\ (ass')^4 &= \frac{24}{5} a_\varrho^2 a_{\varrho'}^2 - \frac{12}{5} t_\varrho^2 + \frac{5}{3} J \cdot i - \frac{4}{9} i^3.\end{aligned} \tag{22}$$

[28] From formula (21.) there follows the relation, mentioned in the note to the introduction, between the three invariants of order 9, with use of formula (10.)c.

Among forms of the same order and the same degree, we define as "higher forms" those which have arisen from higher forms of the relatively complete system mod (s, ϱ, t) by folding with s, regarding the folding with s merely as a polarization of the original forms. Thus the order of the individual forms is uniquely determined (cf. § 1, form sequences 2). For instance,

$$(\theta_\eta(\theta H u), s, u)^2 \text{ is higher than } s_\eta((\theta H u)^2, s, u),$$

$$(\theta_\eta(\theta H u)^2, s, u) \text{ is higher than } (\theta_\eta(\theta H u), s, u)^2.$$

We divide the resulting system into the four subsystems

System s_I :	obtained by folding s and powers of s with System I;
System s_{II} :	obtained by folding s with System II;
System s_{III} :	obtained by folding s with the individual forms (without products) of System III and with products of one form each from Systems I and II;
System s_{IV} :	obtained by folding s with products of one form each from Systems I and III, II and III, III and III.

It should also be noted that from now on all formulas are to be understood mod (ϱ, t); from formula (27.) on, they are to be understood mod $(\varrho, t, i, J,$ higher forms), without this being expressly stated each time.

For the reduction we collect the formulas obtained earlier:

$$\begin{aligned}
s_\nu^2 &= \frac{4}{3} i a_\nu^2 + \frac{1}{6} J \cdot u_x^2 - \frac{2}{9} i^2 \cdot u_x^2, & s_\nu^3 &= \frac{1}{3} J \cdot u_x, \quad s_\nu^4 = J, \\
g &= 2 s_\vartheta^2 - \frac{4}{3} i \Delta, & s_\vartheta^2(\theta s u) &= 0, \\
s_\vartheta^2(\theta s u)^2 &= \frac{8}{9} i \cdot a_\nu^3 - \frac{4}{27} i^2 \cdot u_x^2, \\
Z &= 2i \cdot a_\eta^2 + \frac{4}{3} i (a s u)^2 - \frac{1}{3} J \cdot \theta, \\
g_\nu &= -s_\eta + u_x \left\{ 2 i a_\eta^2 - \frac{2}{3} i (a s u)^2 + \frac{2}{3} J \cdot \theta \right\}, \\
g_\nu^2 &= \frac{4}{5} i a_\eta^2 - \frac{2}{5} i (a s u)^2 + \frac{1}{3} J \cdot \theta, & g_\nu^3 &= 0, \qquad g_\nu^4 = 0.
\end{aligned} \tag{24}$$

Consequences:

$$s_\nu^3, \quad s_\vartheta^2(\theta s u), \quad s_\vartheta^2 s_\nu, \quad s_\vartheta^2 \theta_\nu$$

are reducers.

§ 18. System s_I.

$$\text{Folding of} \quad \left\{ \begin{array}{l} s \text{ with } f;\ \theta;\ K, j, \Delta;\ f \cdot j, N;\ \Delta \cdot j, \\ s^3 \text{ with } \theta, K. \end{array} \right.$$

Irreducible forms.

Order 5:

$$(asu), \qquad (asu)^2, \qquad (asu)^3, \qquad (asu)^4.$$

Order 6:

$$\begin{array}{cccc}
(\theta s u) & (\theta s u)^2 & (\theta s u)^3 & (\theta s u)^4 \\
s_\vartheta & s_\vartheta(\theta s u) & s_\vartheta(\theta s u)^2 & s_\vartheta(\theta s u)^3 \\
\cdot & s_\vartheta^2 & \cdot & \cdot
\end{array}$$

Order 7:

$$\begin{array}{ccccl}
\cdot & (Ksu)^2 & (Ksu)^3 & (Ksu)^4 & \text{B) } s_j, \\
\text{A) } s_k & s_k(Ksu) & s_k(Ksu)^2 & s_k(Ksu)^3 & \\
\cdot & s_k^2 & \cdot & \cdot & \text{C) } (\Delta s u),\ (\Delta s u)^2.
\end{array}$$

Order 8:

$$\text{A)}\quad s_j(asu),\ s_j(asu)^2,\ s_j(asu)^3,$$
$$\text{B)}\quad (Nsu)^2,\quad s_n,\quad s_n(Nsu).$$

Order 10:

$$\text{A) } s_j(\Delta su),\qquad \text{B) } s_\vartheta(\theta s'u)^4.$$

Order 11:

$$(Ks^2u)^6,\qquad s_k(Ks^2u)^5,\qquad s_k(Ks^2u)^6.$$

Reductions:

The form sequences

$$s_\vartheta^2(\theta su),\quad s_k^2(Ksu),\quad s_j^2,\quad (\Delta su)^3,\quad (Nsu)^3$$

have the reducer $s_\vartheta^2(\theta su)$ as a factor; s_j^2 and $(\Delta su)^3$ are obtained by going back from j and Δ to lower forms of the form sequence:

$$\begin{aligned} s_\vartheta^2(\theta su)(a\theta u)^3 &= -\frac{1}{2}s_j^2,\\ s_\vartheta^2(\theta su)(a\theta u)^2 &= \frac{1}{3}(\Delta su)^3,\\ s_\vartheta^2(\theta su)^2(a\theta u)^2 &= \frac{1}{3}(\Delta su)^4+\frac{1}{3}s_j^2. \end{aligned}$$

Forms $s_\vartheta^2 s_{\vartheta'}$ and $s_\vartheta^2 s_{\vartheta'}^2$: reducers $s_\vartheta^2(\theta su)$ and $\theta_{\vartheta'}$:

$$\begin{aligned} s_\vartheta^2 s_{\vartheta'} &= s_\vartheta^2\theta_{\vartheta'} - s_\vartheta^2(\theta s\widehat{\vartheta'}x),\\ s_\vartheta^2 s_{\vartheta'}^2 &= s_\vartheta^2 s_{\vartheta'}\theta_{\vartheta'} - s_\vartheta^2 s_{\vartheta'}(\theta s\widehat{\vartheta'}x). \end{aligned}$$

Form sequence $s_j(\Delta su)^2$: reducers Δ_j and $(\Delta su)^3$.

Consequences:

s does not enter into powers in folding with f, j, Δ, N. The forms f, j, Δ, N enter into folding only in products with contragredient forms; however, in products with f those forms may still be quadratic in x, and in products with Δ they may still be linear in x. Thus the following products of forms are to be considered:

$$f\cdot(0,\lambda),\quad f\cdot(1,\lambda),\quad f\cdot(2,\lambda),\quad j\cdot(\mu,0),\quad N,\quad \Delta\cdot(0,\lambda),\quad \Delta\cdot(1,\lambda)\quad (\lambda>0),$$

where (μ,λ) denotes a form of degree μ in x and of degree λ in u.

Neither θ nor K enters into powers or products with arbitrary forms in folding with s or s^2. For products with the functional determinant K, $K\cdot\varphi_x$ ($\varphi\neq f$ or θ), the relation between decomposing forms is

$$K\cdot\varphi_x = (a\theta u)\varphi_x = f\cdot(\varphi\theta u) - \theta\cdot(\varphi a u).$$

The above scheme for folding System s_I is thereby proved.

§ 19. System s_{II}.

Folding of s with ν; H; L; H^2.

Irreducible forms:

Order 6	Order 8	Order 10	Order 12
s_ν	$(Hsu),\ (Hsu)^2$	$(Lsu)^2,\ (Lsu)^3$	$(H^2su)^3$
.	s_η	s_l	.
.	$s_\nu^4 = J$	.	.

Reductions:

Form sequences $s_\eta(Hsu)$ and $s_l(Lsu)$: reducer s_ν^2.

Form sequence s_σ: reducer s_ν^2 from H, $s_\nu^2 = \dfrac{2}{3}s_\sigma$.

$(H^2su)^4$ decomposes by formula (7.)a (cf. § 11 A). Hence $(H\cdot L, s, u)^4$ also decomposes.

Consequences:

s^2 does not enter into folding with System II. The form ν enters only in products with contragredient forms. The form H, regarded as a covariant, enters into folding in products with forms $(1,\lambda)$, $(2,\lambda)$, $(3,\lambda)$, with arbitrary λ. The forms σ and $(\nu\sigma x)$ do not enter at all, and L, as a functional determinant, does not enter into products in folding; the full transvections over covariants of System III become reducible. The products of Systems I and II that remain are

$$f\cdot\nu, \qquad \Delta\cdot\nu.$$

§ 20. Forms of order 7 (System s_{III}).

Folding of s with $f\cdot\nu$, a_ν, a_ν^2.
Irreducible forms:

$$\begin{gathered} s_\nu(asu),\quad a_\nu(asu),\quad s_\nu(asu)^2,\quad a_\nu(asu)^2,\quad s_\nu(asu)^3,\quad a_\nu(asu)^3,\\ a_\nu s_\nu,\quad a_\nu s_\nu(asu),\quad a_\nu^2(asu),\quad a_\nu^2(asu)^2. \end{gathered}$$

Reductions:

The evident reductions that arise by direct folding with the reducers s_ν^2 and $s_\eta(Hsu)$ will not be mentioned, nor will the decomposition of transvections with functional determinants.

For the forms $a_\nu s_\nu(asu)^2$ and $a_\nu^2 s_\nu(asu)^2$, see formula (19.), § 12; moreover,

$$a_\nu s_\nu(asu)^3 = a_\nu(a\widehat{\nu}_1\nu_2 u)^3(\nu_1\nu_2\nu) = 0.$$

Form sequence $a_\nu^2 s_\nu$: reducer $s_\vartheta^2(\theta su)$, obtained by going back from a_ν^2 to lower forms, i.e. by double reduction of the expressions

$$s_\vartheta^2(a\theta s)(a\theta u), \qquad s_\vartheta^2(a\theta s)(a\theta u)^2$$

according to the Ansatz

$$\begin{aligned} s_\vartheta^2(a\theta s)(a\theta u) &= [(a\theta u)^2]s^3 + \frac{1}{5}[a_\vartheta(a\theta u)^2]s^2 + \frac{1}{60}[a_\vartheta^2(a\theta u)^2]s\\ &= \frac{1}{2}a_\nu^2 s_\nu - \frac{2}{3}a_\nu^2 s_\nu^2,\\ s_\vartheta^2(a\theta s)(a\theta u)^2 &= \frac{4}{5}[a_\vartheta(a\theta u)^2]_{us}s^2 + \frac{23}{180}[a_\vartheta^2(a\theta u)^2]_{us}s\\ &= -\frac{1}{2}a_\nu^2 s_\nu(asu). \end{aligned}$$

Here $a_\nu^2 b_\nu(abu) = 0$.

The formulas obtained are

$$\begin{aligned} a_\nu^2 s_\nu \;&=\; \frac{4}{9}i\,a_\nu^2 b_\nu, & a_\nu s_\nu(asu)^2 \;&=\; \frac{1}{3}i\theta_\nu^2 - \frac{1}{18}iH,\\ a_\nu s_\nu(asu)^3 \;&=\; 0, & a_\nu^2 s_\nu(asu) \;&=\; 0,\\ a_\nu^2 s_\nu(asu)^2 \;&=\; 0. \end{aligned} \tag{25}$$

Consequences:

1) $a_\nu s_\nu(asu)^2$ and $a_\nu^2 s_\nu$ are reducers.

2) The values of the forms $(\Delta su)^3$, $(\Delta su)^4$, already recognized as reducible in § 19, are thereby obtained; likewise for the corresponding form sequence $(agu)^2$, which, when folded with ν, gives rise to double reduction through the reducer g_ν:

$$\begin{aligned} &\text{a)}\quad (\Delta su)^3 = -\frac{6}{5}a_\nu^2(asu) + 3a_\nu s_\nu(asu) + 2i\theta_\nu(\theta\nu x),\\ &\text{b)}\quad (\Delta su)^4 = -\frac{2}{5}a_\nu^2(asu)^2 + \frac{4}{3}i\theta_\nu^2 + \frac{1}{9}iH. \end{aligned} \tag{26}$$

From formula (24.) and (26.), mod (i, J) (cf. p. 71),

$$\begin{aligned} &\text{a)}\quad (agu)^2 = \frac{2}{3}(\Delta su)^2 - \frac{4}{3}a_\nu s_\nu + \frac{3}{5}s\cdot a_\nu^2,\\ &\text{b)}\quad (agu)^3 = 2a_\nu s_\nu(asu) - a_\nu^2(asu)^2,\\ &\text{c)}\quad (agu)^4 = -a_\nu^2(asu)^2. \end{aligned} \tag{27}$$

§ 21. Forms of 8th order (System s_{III}).

Folding of s with the forms of 4th order (System III). (§ 9.)

Irreducible forms:

$$\begin{array}{ll} (\vartheta\nu x)s_\nu, & ((\vartheta\nu x), s, u)^2, \\ ((\vartheta\nu x)^2, s, u), & ((\vartheta\nu x)^2, s, u)^2,\ \theta s_\nu, \\ (\vartheta\nu x)^2 s_\nu, & ((\vartheta\nu x)^2, s, u)s_\nu, \end{array}$$

$$\begin{array}{ll} ((\vartheta\nu x), s, u)^3,\ (\theta_\nu su)^2, & ((\vartheta\nu x), s, u)^4,\ (\theta_\nu su)^3, \\ ((\vartheta\nu x)^2, s, u)^3,\ (\theta_\nu(\vartheta\nu x), s, u)^2,\ (\theta_\nu^2 su), & ((\vartheta\nu x), s, u)^3,\ (\theta_\nu^2 su)^2,\ (\theta_\nu(\vartheta\nu x), s, u)^4. \end{array}$$

Reductions:

$((\vartheta\nu x), s, u)s_\nu$ decomposes as a transvection over functional determinants, by § 3.

Form sequence $((\vartheta\nu x), s, u)^2 s_\nu$: reducers s_ν^2 and $s_\vartheta^2\theta_\nu$:

$$(\widehat{\vartheta}\nu su)s_\nu(\theta su) = s_\nu s_\vartheta(\theta su) = -\frac{1}{2}(\widehat{\vartheta}\nu su)^2(\theta su),$$

$$\theta_\nu(\widehat{\vartheta}\nu su)s_\nu = \theta_\nu s_\nu s_\vartheta = -\frac{1}{2}\theta_\nu(\widehat{\vartheta}\nu su)^2.$$

Form sequence $\theta_\nu s_\nu(\theta su)$: reducer $a_\nu s_\nu(asu)^2$ by double reduction of the expressions

$$a_\nu s_\nu(asu)^2(abu) \quad \text{and} \quad a_\nu s_\nu(asu)^2(abu)(sbu);$$

$\theta_\nu s_\nu(\theta su)^3$ by double reduction of $(agu)^3(abu)(bgu)^3$.

Form sequence $\theta_\nu(\vartheta\nu x)s_\nu$: reducer $a_\nu^2 s_\nu$ from $\theta_\nu(\vartheta\nu x)s_\nu = a_\nu^2(abu)s_\nu$.

Form sequence $(\theta_\nu(\vartheta\nu x)^2, s, u)$: double reduction of the forms of the sequence $\theta_\nu(\widehat{\vartheta}\nu su)s_\nu$, which at the same time belong to the form sequences $((\vartheta\nu x)su)^2 s_\nu$ and $\theta_\nu(\vartheta\nu x)s_\nu$.

The form $(\theta_\nu(\vartheta\nu x), s, u)$ decomposes as a single transvection over the functional determinant $\theta_\nu(\vartheta\nu x) = a_\nu^2(abu)$.

The form $((\vartheta\nu x)^2, s, u)^4$: double reduction of the expression $(a\Delta u)^2(\Delta su)^4$, or also of $(\theta gu)^4$, from formulas (27.) and analogously to the reduction of $(j\eta x)^4$ (§ 12 C)).

Double reduction gives the formulas:

$$\begin{aligned} 0 &= \theta_\nu s_\nu(\theta su) - \frac{1}{6}(\widehat{\vartheta}\nu su)^2(\theta su) - \frac{1}{12}(Hsu), \\ 0 &= \theta_\nu s(\theta su)^2 - \frac{1}{4}(\widehat{\vartheta}\nu su)^2(\theta su)^2 - \frac{1}{24}(Hsu)^2, \\ 0 &= (\widehat{\vartheta}\nu su)^2(\theta su)^2 + 2\theta_\nu(\widehat{\vartheta}\nu su)(\theta su)^2 + 3\theta_\nu^2(\theta su)^2 - \frac{1}{3}H(su)^2, \\ 0 &= \theta_\nu s_\nu(\theta su)^3 + \frac{1}{2}\theta_\nu(\widehat{\vartheta}\nu su)(\theta su)^3, \\ 0 &= \theta_\nu s_\nu s_\vartheta - \frac{1}{4}s_\eta, \quad 0 = \theta_\nu s_\nu s_\vartheta(\theta su), \quad 0 = \theta_\nu s_\nu s_\vartheta(\theta su)^2. \end{aligned} \tag{28}$$

The last group corresponds to $((\theta_\nu(\vartheta\nu x)^2, s, u)^2, \ldots)$.

Consequences:

1) $\theta(\vartheta\nu x)s_\nu$, $(\widehat{\vartheta}\nu su)(\theta su)s_\nu$, $\theta_\nu(\widehat{\vartheta}\nu su)^2$, $(\widehat{\vartheta}\nu su)^2(\theta su)^2$ are reducers.

2) The forms of 4th order $(5, \lambda)$, $(6, \lambda)$ etc. do not enter into folding with s^2.

3) For the form sequence $(\theta gu)^2$, formulas (27.), with the reductions of this paragraph taken into account, give the following formulas:

$$\begin{aligned} &\text{a)} \quad (\theta gu)^2 = \frac{4}{3}\theta_\nu s_\nu + \frac{2}{3}s_\nu s_\vartheta - \frac{1}{3}s\theta_\nu^2 - \frac{1}{9}sH - \frac{1}{3}f\, a_\nu^2(asu)^2 + \frac{1}{3}a_\nu^2(asu)^2, \\ &\text{b)} \quad (\theta gu)^3 = -\frac{1}{3}(\widehat{\vartheta}\nu su)^2(\theta su) - \theta_\nu(\widehat{\vartheta}\nu su)(\theta su) - \theta_\nu^2(\theta su), \\ &\text{c)} \quad (\theta gu)^4 = 2\theta_\nu^2(\theta su)^2 - \frac{1}{3}(Hsu)^2, \\ &\quad g\vartheta(\theta gu) = (\widehat{\vartheta}\nu su)(\vartheta\nu x)s_\nu - \theta_\nu(\vartheta\nu x)(\widehat{\vartheta}\nu su), \\ &\quad g\vartheta(\theta gu)^2 = \frac{2}{3}s\theta_\nu^3, \quad g\vartheta(\theta gu)^3 = \frac{1}{3}\theta_\nu^3(\theta su), \quad g\vartheta(\theta gu)^4 = 0. \end{aligned} \tag{29}$$

§ 22. Forms of 9th order (System s_{III}).

Folding of s with the forms of 5th order (System III) and the product $\Delta \cdot \nu$ (§ 10).
Irreducible forms:

A) $(k\nu x)^2 s_\nu, ((k\nu x)^3, s, u), (k\nu x)^3 s_\nu, ((k\nu x)^2, s, u)^2, K_\nu s_\nu,$
$((k\nu x)^3, s, u)^2, ((k\nu x)^2, s, u)s_\nu, (K_\nu (k\nu x)^3, s, u),$
$(K_\nu su)^2,\ (K_\nu su)^3,\ (K_\nu su)^4, ((k\nu x)^2, s, u)^3, (K_\nu^2 su)^4,$

B) $(j\nu x)s_\nu, ((j\nu x)^2, s, u), (j\nu x)^3, s, u, ((j\nu x)^2, s, u)^2,$

C) $s_\nu(\Delta su),\quad (\Delta_\nu su),$

D) $(aHu)s_\eta, (a_\eta su), ((aHu), s, u)^2, ((aHu)^2, s, u),$
$(aHu)^2 s_\eta, (a_\eta su)^2, (a_\eta^2 su), ((aHu), s, u)^3, ((aHu)^2, s, u)^2,$
$((aHu), s, u)^4, (a_\eta su)^3, (a_\eta (aHu), s, u)^2, (a_\eta (aHu)^2, s, u), (a_\eta (aHu), s, u)^3.$

Reductions:

A) The reductions follow directly from the reductions of § 21 by single folding. The form sequence $K_\nu s_\nu (Ksu)^2$ has the reducers $a_\nu s_\nu (asu)^2$ and $\theta_\nu s_\nu (\theta su)^2$ as factors. Hence the higher forms $K_\nu^2 (Ksu)^2$, $K_\nu^2 (Ksu)^3$ decompose according to the Ansatz:

$$0 = a_\nu s_\nu (asu)^2 (a\theta u) = K_\nu s_\nu (Ksu)^2 - \theta_\nu s_\nu (\theta su)^2 (a\theta u).$$

B) Form sequence $(j\nu x)^2 s_\nu$: reducer $a_\nu^2 s_\nu$ from

$$(j\nu x)^2 s_\nu = 3a_\nu^2 s_\nu (a\theta u)^2.$$

$((j\nu x)^3, s, u)^2$: reducer $a_\nu^2 s_\nu$ from

$$((j\nu x)^3, s, u)^2 = -6a_\nu^2 s_\nu (a\theta u)^2 (\theta su).$$

C) Forms $\Delta_\nu (\Delta su)^2$ and $s_\nu (\Delta su)^2$: reducer g_ν from formula (27.)a.

Form $\Delta_\nu s_\nu$: 1) reducer $a_\nu^2 s_\nu$ from $a_g a_\nu^2 s_\nu = \frac{2}{3}\Delta_\nu s_\nu$; 2) decomposes by formula (27.)a through double reduction of the expression $(\Delta s\widehat{\nu} x)^2$.

Hence the higher form $a_\eta s_\eta$ decomposes.

D) Form sequence $(aHu)^2 s_\eta (asu)$: reducer $a_\nu s_\nu (asu)^2$ by double reduction of the form sequence $[a_\nu s_\nu (asu)^2]_{\nu_1 x^2}$; reduction of $(aHu)^2 s_\eta (asu)^2$ from $a_\nu s_\nu (asu)^2$ and $a_\nu s_\nu (asu)^3$. Hence the reduction of $(a_\eta, s, u)^4$.

Form sequence $a_\eta (aHu) s_\eta$: reducers $a_\nu^2 s_\nu$, $a_\eta^2 s_\eta$ by double reduction of the expression $(\Delta s\widehat{\nu} x)^3$, since $a_\eta^2 s_\eta$ does not arise by folding from the reducible term $a_\nu^2 s_\nu (\nu\nu_1 x)$ of the form $a_\eta (aHu) s_\eta$.

Form sequence $(a_\eta^2 su)^2$: reducer $a_\nu^2 s_\nu$ from

$$a_\nu^2 s_\nu s_{\nu_1} = -\frac{1}{2}(a_\eta^2 (Hsu))^2.$$

The form $(a_\eta (aHu), s, u)$ decomposes as a transvection over a functional determinant.

The reduction formulas obtained are:

$$\begin{aligned}
&\text{a)}\quad 0 = (aHu)^2 (asu) s_\eta - a_\eta (asu)(Hsu)^2 - a_\eta (aHu)(asu)(Hsu),\\
&\text{b)}\quad 0 = (aHu)^2 (asu)^2 s_\eta - a_\eta (aHu)(asu)^2 (Hsu),\\
&\text{c)}\quad 0 = a_\eta (asu)^2 (Hsu)^2,\\
&\qquad 0 = a_\eta s_\eta + \frac{1}{4} a_\nu^2 g - \frac{1}{4} s \cdot a_\eta^2,\\
&\qquad 0 = a_\eta (aHu) s_\eta - \frac{1}{2} a_\eta^2 (Hsu), \quad 0 = a_\eta^2 (Hsu)^2, \ldots, \quad 0 = a_\eta^2 s_\eta.
\end{aligned} \tag{30}$$

Consequences:

1) $(j\nu x)^2 s_\nu$, $\Delta_\nu s_\nu$, $\Delta_\nu (\Delta su)^2$, and consequently $N_\nu (Nsu)^2$, $(aHu)^2 (asu) s_\eta$, $a_\eta (aHu) s_\eta$, $a_\eta^2 (Hsu)^2$, are reducers; $a_\eta s_\eta$ is a decomposable form that passes into decomposable forms under all single and double foldings.

2) The forms of 5th order $(5, \lambda)$, $(6, \lambda)$ etc. do not enter into folding with s^2.

§ 23. Forms of the tenth order (System s_{III}).

Folding of s with the forms of the sixth order (System III) (§ 11).
Irreducible forms:

A) $(\vartheta^2\nu x)^3 s_\nu, ((\vartheta^2\nu x)^3, s, u), ((\vartheta^2\nu x)^3, s, u)^2, (\vartheta^2\nu x)^3 s_\nu s_\vartheta, \theta_\nu(\vartheta^2\nu x)^3 s_\vartheta,$

B) $a_\nu(j\nu x)s_\nu, (a_\nu(j\nu x), s, u)^2, (a_\nu(j\nu x), s, u)^3, (a_\nu(j\nu x), s, u)^4, (a_\nu^2(j\nu x), s, u)^2, (a_\nu^2(j\nu x), s, u)^3,$

C) $((\vartheta\eta x)^2, s, u), ((\vartheta\eta x), s, u)^2, (\theta Hu)s_\eta, (\theta_\eta su), ((\theta Hu), s, u)^2, ((\theta Hu)^2, s, u),$
$((\theta\eta x), s, u)^3, (\theta_\eta su)^2, (\theta_\eta^2 su), ((\theta Hu), s, u)^3, ((\theta Hu)^2, s, u)^2, (H_\vartheta(\theta Hu), s, u)^2,$
$(\theta_\eta(\theta Hu), s, u)^2, (\theta_\eta(\theta Hu)^2, s, u), ((\theta Hu), s, u)^4, (\theta_\eta su)^4, (\theta_\eta(\theta Hu), s, u)^3, (\theta_\eta^2(\theta Hu), s, u)^2.$

Reductions: A) The forms $((\vartheta^2\nu x)^3, s, u)^2$, $((\vartheta^2\nu x)^3, s, u)^3$ decompose:

$$(\vartheta\nu x)(\widehat{\vartheta\nu su})(\widehat{\vartheta\nu su}) = (\vartheta\nu x)s_\vartheta s_\nu - (\vartheta\nu x)s_\nu s_\vartheta = -2\theta(\vartheta\nu x)s_\nu s_\vartheta + (\vartheta\nu x)s_\vartheta^2.$$

The remaining forms either have reducers as factors or are single foldings with decomposed forms.

B) Reducer $a_\nu^2 s_\nu$ and $(\widehat{j\nu su})s_\nu$.

C) 1. The form sequence $(\theta Hu)^2 s_\eta$: a) reducer $(aHu)^2(asu)s_\eta$; b) double reduction of the form sequences $(\theta gu)^3 g_\nu$ and $g_\nu(agu)^3(bgu)^3(abu)$. Hence reduction of the higher forms $(\theta, s, u)^3$, $(H_\vartheta(\theta Hu), s, u)^3$ and $(a_\nu b_{\nu_1}^2, s, u)^4$ [the latter form replacing $(H_\vartheta su)^4$].

2. $(\vartheta\eta x, s, u)^4$: reducer $(a_\eta su)^4$.

3. $((\vartheta\eta x)^2, s, u)^3$: reducer $s_\vartheta^2 s_\nu$ from $(\widehat{\vartheta\eta su})^2(Hsu)$. The forms $(\vartheta\eta x)s_\eta$, $(\vartheta\eta x)^2 s_\eta$, $((\vartheta\eta x)^2, s, u)^2$, $\theta_\eta s_\eta$ decompose by folding with the decomposed form $a_\eta s_\eta$.

4. Form sequences $H_\vartheta(\theta Hu)s_\eta$, $\theta_\eta(\theta Hu)s_\eta$, $H_\vartheta(\vartheta\eta x)s_\eta$, $\theta_\eta(\vartheta\eta x)s_\eta$: reducers $a_\eta(aHu)s_\eta$ and $a_\eta^2 s_\eta$.

The form sequence $(\theta_\eta(\vartheta\eta x), s, u)^2$: reducer $a_\eta^2(Hsu)^2$ [except for $(\theta_\eta^2(\theta Hu), s, u)^2$, which arises from the term $a_\eta^2(\theta su)(Hsu)$ by folding].

5. $(H_\vartheta(\vartheta\eta x), s, u)$, $(H_\vartheta(\vartheta\eta x), s, u)^2$: double reduction of $a_\eta(aHu)s_\eta(abu)$ and $g_\vartheta(\theta gu)g_\nu$.

The form sequence $(H_\vartheta(\vartheta\eta x), s, u)^3$: reducer $((\vartheta\nu x)^2, s, u)^2 s_\nu$.

6. $(H_\vartheta(\theta Hu), s, u)$ decomposes by single folding with the decomposed form $(\theta_\nu(\vartheta\nu x), s, u)$ according to § 3. 2), c), that is, by double reduction of the lower decomposed form $(H_\vartheta su)^2$:[29]

$$\begin{aligned}
0 &= \theta_\nu(\vartheta\nu_1)(\theta su) + \frac{2}{3}\theta_\nu^2(\vartheta\nu_1 x)(\theta su) + \theta_\nu(\vartheta\nu x)(\theta su)s_{\nu_1}\\
&\equiv -\frac{1}{2}H_\vartheta(\theta Hu)(\theta su) + \theta_\eta(\theta su)(Hsu) + \frac{1}{2}H_\vartheta(\theta su)(Hsu)\\
&\equiv -\frac{1}{2}H_\vartheta(\theta Hu)(\theta su) + \theta_\eta(\theta su)(Hsu) + \frac{1}{2}[H_\vartheta]_{sx}^2 - \frac{1}{2}\frac{3}{5}H_\vartheta(\theta Hu)(\theta su)\\
&\equiv -\frac{3}{5}H_\vartheta(\theta Hu)(\theta su).
\end{aligned}$$

By double reduction, according to 1. and 5., the following formulae arise:

$$\begin{aligned}
0 &= \theta_\eta(\theta su)(Hsu)^2 + \frac{1}{4}H_\vartheta(\theta Hu)(\theta su)^2\\
&\quad - \frac{3}{4}\theta_\eta(\theta Hu)(\theta su)(Hsu) - \frac{5}{4}\theta_\eta(\theta Hu)^2(\theta su)^3\\
&\quad - (asu)^3 b_\eta(\theta Hu)^2 - a_\nu(asu)^3 b_\nu^2,\\
0 &= H_\vartheta(\theta Hu)(\theta su)^3 - 7\theta_\eta(\theta Hu)(\theta su)^2(Hsu),\\
0 &= a_\nu(asu)^2 b_{\nu_1}^2(bsu)^2 - \theta_\eta(\theta su)^2(Hsu)^2\\
&\quad + 6\theta_\eta(\theta Hu)(\theta su)^2(Hsu),\\
0 &\equiv (H_\vartheta(\vartheta\eta x), s, u), \qquad 0 \equiv H_\vartheta(\widehat{\vartheta\eta su})(Hsu) - 4\theta_\eta^2(Hsu).
\end{aligned} \tag{31}$$

Consequences:

1. $(\theta Hu)^2 s_\eta$, $H_\vartheta(\theta Hu)s_\eta$, $\theta_\eta(\theta Hu)s_\eta$, $H_\vartheta(\vartheta\eta x)s_\eta$, $\theta_\eta(\vartheta\eta x)s_\eta$, $(H_\vartheta(\vartheta\eta x), s, u)$, $(\theta_\eta(\vartheta\eta x), s, u)^2$ are reducers.

2. s^2 does not enter into folding with the forms $(5, \lambda)$, etc., of order 6.

[29]The sign $\equiv$ means that products of forms of lower degree have been omitted.

§ 24. Forms of the 11th order (System s_{III}).

Folding of s with the forms of the 7th order (System III) (§ 12).
Irreducible forms:

A) $((k\eta x)^3, s, u)$, $(K_\eta(k\eta x)^3, s, u)$, $(K_\eta su)^2$, $((KHu)^2 su)^2$,
$((KHu)^2, s, u)^3$, $((KHu)^2, s, u)^4$, $(K_\eta(KHu)^2, s, u)^3$,
B) $((j\eta x), s, u)^2$, $(H_j su)$, $((j\eta x), s, u)^3$,
C) $(\Delta_\eta su)$,
D) $(aLu)^2 s_\eta$, $(a_\eta su)^2$, $((aLu)^2, s, u)^2$, $((aLu)^3, s, u)^2$, $((aLu)^3, s, u)^3$, $(a_l(aLu)^2, s, u)^2$.

Reductions:

A) Reduction or decomposition by single folding with the corresponding forms in the symbols $\theta - H$.
$(K_\eta(KHu)^2, s, u)$ and $(K_\eta(KHu)^2, s, u)^2$: decomposition according to § 3. 2), a), by folding with $K_\nu^2(Ksu)$, $K_\nu^2(Ksu)^2$.
$(K_\eta(KHu), s, u)^4 \equiv (a_\nu^2\theta_{\nu_1}, s, u)^4$: decomposition according to § 3. 2), b), from the decomposed form $K_\nu^2(Ksu)^3$.
B) Form sequence $(j\eta x)s_\eta$: reducer $(j\nu x)^2 s_\nu$.
Form $H_j(\widehat{j\eta su})(Hsu)$: reducer $(j\nu x)(\widehat{j\nu su})^2$.
Form $H_j(j\eta x)(Hsu)$: decomposes by reducing the decomposed form $((j\eta x)^2, s, u)^2$ by the reducer $(j\nu x)^2 s_\nu$ (formula (18.)a):

$$0 = -2(j\nu x)^2 s_\nu s_{\nu_1} = [(j\eta x)^2]_{su^2} + H_j(j\eta x)(Hsu).$$

C) Reducers $\Delta_\nu s_\nu$ and $\Delta_\nu(\Delta su)^2$.
D) Reduction and decomposition by single folding with the corresponding forms in the symbols $f - H$.
E) From (30.)c one obtains the reduction of the decomposed form sequence $(a_\sigma su)^2$:

$$0 = a_\eta(Hsu)^2(as\widehat{\nu x})^2 = a_\eta(Hsu)^2(a\widehat{\nu\eta}u)^2 + 2a_\eta(a\widehat{\nu\eta}u)(\widehat{\eta\sigma}u)(Hsu)^2 = \frac{1}{3}a_\sigma(asu)^2,$$
$$0 = a_\eta a_\nu(Hsu)^2(as\widehat{\nu x})(asu) = a_\eta(a\widehat{\nu\eta}u)^2(Hsu)^2(asu) + a_\eta(a\widehat{\nu\eta}u)(\widehat{\eta\sigma}u)(Hsu)^2(asu) = \frac{1}{3}a_\sigma(asu)^3.$$

Consequence: s^2 does not enter into folding with the forms of order 7.

§ 25. Forms of the 12th, 13th, 14th, and 15th orders (System s_{III}).

Folding of s with the forms of orders 8, 9, 10, 11 (System III) (§§ 13–16).
Irreducible forms:
12th order.

A) $((\vartheta^2\eta x)^4, s, u)$, $\theta_\eta(\vartheta^2\eta x)^3 s_\vartheta$,
B) $((aHu)H_j, s, u)^2$, $((aH\cdot u)H_j, s, u)^3$,
C) $(\theta_\nu su)^2$, $((\theta Lu)^2, s, u)^2$, $((\theta Lu)^3, s, u)$, $((\theta Lu)^2, s, u)^3$, $(\theta_l(\theta Lu)^2, s, u)^2$.

Reductions: The reductions arising by direct folding with reducers or decomposed forms are no longer to be mentioned.

B) Decomposition of $((aHu)H_j, s, u)$ and $(a_\eta(aHu)H_j, s, u)$ as transvections over functional determinants.
Decomposition of $(a_\eta(aHu)H_j, s, u)^2$: double reduction of the decomposed form $[\theta_\nu\theta_{\nu_1}^3]_{su^3}$ (§ 13. C):

$$0 \equiv [\theta_{\nu_1}^3\theta_\nu]_{su^3} \equiv -\frac{3}{4}\theta_\nu(\theta su)^2(\theta\theta' u)\theta_{\nu_1}^3$$
$$\equiv -\frac{9}{8}(\theta\theta'\eta su)(\theta\theta' u)(\theta Hu)(\theta' Hu)\theta'_\nu(\theta su) \equiv -\frac{3}{8}a_\eta(aHu)H_j(asu)^2.$$

C) From formulae (31.) follows the reduction of the decomposed forms

$$[L_\vartheta(\theta Lu)]_{su^3} \quad \text{and} \quad [\theta_l(\theta Lu)]_{s^2u^3},$$

that is, of the products $[\theta_\nu^2 H]_{su^3}$ and $[a_\nu^2(bHu)^2]_{su^3}$:

$$\begin{aligned}
&0 \equiv \theta_\nu^2(\theta su)^2(Hsu), 0 \qquad\qquad \equiv [L_\vartheta(\theta Lu)]_{su^4} - 7[\theta_l(\theta Lu)]_{su^4},\\
&0 \equiv a_\nu^2(asu)^2(bHu)^2(bsu)\\
&\quad + 6\theta_l(\theta Lu)^2(\theta su)(Lsu),\\
&0 \equiv \theta_\nu^3(\theta su)(Hsu)^2\\
&\quad \text{from } 0 \equiv \theta_l^2(\theta Lu)(\theta su)(Lsu)^2\\
&\quad (\text{reducer } a_\eta^2(Hsu)^2).
\end{aligned} \tag{32}$$

13th order.

$$\text{A) } ((KLu)^3, s, u)^2,\ ((KLu)^3, s, u)^3, \qquad \text{B) } (L_j su)^2, \qquad \text{C) } ((aH^2u)^3, s, u)^2.$$

14th order.

$$\text{A) } ((aLu)^2 L_j, s, u)^2, \qquad \text{B) } ((\theta H^2 u)^3, s, u)^2.$$

15th order.

$$((KH^2u)^4, s, u)^2.$$

Reductions: Decomposition of $(K_l(KLu)^3, s, u)^2$, $(H_\vartheta(\theta H^2u)^3, s, u)$, $((K, H \cdot L, u)^5, s, u)$ according to § 3. 2), c); that is, with simultaneous consideration of the decomposed forms $(K_l(KLu)^2, s, u)^3$, $(H_\vartheta(\theta H'u)^2, s, u)^2$, $((K, H \cdot L, u)^4, s, u)^2$.
Consequence: s^2 does not enter into folding with individual forms of System III.

§ 26. System s_{IV}.

1) *Folding of s with System I–III.*
Reductions: According to the reductions of the last paragraphs ($a_\nu^2 s_\nu (aHu)^2(asu)s_\eta$, etc.), the forms $(0, \lambda)$, $(1, \lambda)$, $(2, \lambda)$ do not permit foldings I and III simultaneously; hence, by § 18, the forms f, Δ, N do not enter in products in folding to form System s_{IV}.

The form j does not enter into folding, since, by the same reductions, the foldings contragredient to j,

$$(K_\nu(k\nu x)^3, s, u), \qquad ((\vartheta^2\eta x)^4, s, u) \quad \text{etc.},$$

correspond to the foldings cogredient to j:

$$K_\nu(k\nu x)^2 s_k, \qquad (\vartheta^2\eta x)^3 s_\vartheta \quad \text{etc.}$$

2) *Folding of s with System II–III*, that is, of s with $H \cdot (1, \lambda)$, $H \cdot (2, \lambda)$, $H \cdot (3, \lambda)$.
Irreducible forms:

$$\begin{aligned}
\text{A) } & (a_\nu H, s, u)^4, ((aHu)^2 H, s, u)^3, ((aHu)^2 H, s, u)^4,\\
& ((aLu)^2 H, s, u)^4, ((aLu)^3 H, s, u)^3, ((aH^2u)^3 H, s, u)^4,\\
\text{B) } & (a_\nu^2 H, s, u)^3, (a_\eta(aHu)^2 H, s, u)^3, (a_l(aLu)^2 H, s, u)^4,\\
\text{C) } & (\theta_\nu H, s, u)^4, ((\theta Hu)^2 H, s, u)^3, ((\theta Hu)^2 H, s, u)^4,\\
& ((\theta Lu)^2 H, s, u)^4, ((\theta Lu)^3 H, s, u)^3, ((\theta H^2u)^3 H, s, u)^4,\\
\text{D) } & (\theta_\eta(\theta Hu)^2 H, s, u)^3, (\theta_l(\theta Lu)^2 H, s, u)^4,\\
\text{E) } & ((KLu)^3 H, s, u)^4, ((KH^2u)^4 H, s, u)^4,\\
\text{F) } & ((j\nu x)^2 H, s, u)^3, ((j\nu x)^2 H, s, u)^4, ((j\eta x) H, s, u)^4,\\
& (H_j H, s, u)^3, (L_j H, s, u)^4.
\end{aligned}$$

Reductions: ν does not enter into folding, analogously to j.

B) and D). $(a_\nu^2 H, s, u)^4$ and $(\theta_\nu^2 H, s, u)^4$ decompose by formula (27.)c) and (29.)c), taking into account $(gHu)^2 = -\frac{1}{3}s\sigma$:

$$(a_\nu^2 H, s, u)^4 = \frac{1}{3}(asu)^4\sigma, \qquad (\theta_\nu^2 H, s, u)^4 = -\frac{1}{6}(\theta su)^4\sigma;$$

hence decomposition of $(a_\eta(aHu)H, s, u)^4$, $(\theta_\eta(\theta Hu)H, s, u)^4$.

$(\theta_\nu^2 H, s, u)^3$, $(\theta_\nu^3 H, s, u)^3$, and hence $(\theta_\eta^2(\theta Hu)H, s, u)^4$, decompose according to § 25, forms of order 12, C).

3) *Folding of s with System III–III*, that is, of s with forms of System III:

$$(3,\lambda)(3,\lambda),\quad (3,\lambda)(2,\lambda),\quad (3,\lambda)(1,\lambda),\quad (2,\lambda)(2,\lambda),\quad (2,\lambda)(1,\lambda),\quad (1,\lambda)(1,\lambda).$$

Irreducible forms:

$$\begin{aligned}
&\text{A)}\quad (a_\nu^2 a_\nu^2, s, u)^3, (a_\nu^2 a_\nu^2, s, u)^4, (a^2 a_\eta (aHu)^2, s, u)^3,\\
&\text{B)}\quad (a_\nu H_j, s, u)^4, ((aHu)^2 H_j, s, u)^3, ((aLu)^2 H_j, s, u)^4, ((aLu)^3 H_j, s, u)^2, ((aH^2u)^3 H_j, s, u)^3.
\end{aligned}$$

Reductions:

Products II–III, for the same order in the symbols ν and f, are to be considered higher than products III–III.

The reductions arise partly from relations between products (syzygies), partly by replacing the products by the corresponding decomposed forms and reducing the folding of these forms with s. Products containing contravariants are always omitted, since they pass into products.

A) f quadratic, symbols ν of arbitrary order. By § 11, formulae (12.), and by analogous computation, one has:

$$\begin{aligned}
&1)\quad 0 = a_\nu b_{\nu_1} + \frac{1}{2} f (aHu)^2 - \frac{1}{4}\theta H + \frac{1}{2} u_x H_\vartheta - \frac{1}{4} u_x^2 H_\vartheta^2,\\
&2)\quad 0 = a_\nu b_{\nu_1}^2 + f a_\eta (aHu)^2 - \theta_\eta - \frac{1}{2} H_\vartheta,\\
&3)\quad 0 = a_\nu^2 b_{\nu_1}^2 - H_\vartheta^2 + 2\theta_\eta^2.
\end{aligned}$$

It follows that

$$\begin{aligned}
4)&\quad 0 = a_\nu^2 b_{\nu_1}^2 (j\nu_1 x),\\
5)&\quad L_\vartheta(\theta L u) = -a_\nu b_\eta (bHu)^2 + \frac{3}{4} a_\nu^2 (bHu)^2 + \frac{3}{4}\theta_\nu^2 H,\\
6)&\quad L_\vartheta(\theta L u) = -6 a_\nu b_\eta (bHu)^2 + 2 a_\nu^2 (bHu)^2 - \frac{1}{2}\theta_\nu^2 H,\\
7)&\quad 0 = a_\nu (bHu)^2 + f (aLu)^3 + \theta_\nu H,\\
8)&\quad 0 = a_\nu (bLu)^3 + \frac{3}{4} (\theta H u)^2 H,\\
9)&\quad 0 = (aHu)^2 (bHu)^2 + 2 (\theta H u)^2 H,\\
10)&\quad 0 = a_\eta (aHu)^2 b_\eta (bHu)^2,\\
11)&\quad 0 \equiv [a_\nu^2 b_\eta (bHu)]_{\widehat{su^4}}.
\end{aligned}$$

The reductions of

$$(H_\vartheta s u)^4, \qquad (L_\vartheta(\theta L u), s, u)^3, \qquad (L_\vartheta(\theta L u), s, u)^4$$

(formulae (31.) and (32.)) together with formulae 1) to 11) give the reduction of all foldings of s with products that are quadratic in the symbols f and arbitrary in ν.

B) Products of two of the symbols f, θ, j; ν in arbitrary order. By formulae (17.), (18.), (20.), and by direct computation, the relations are:

$$\begin{aligned}
1)&\quad 0 = a_\nu \theta_{\nu_1} + \frac{1}{4}\theta (aHu)^2 + \frac{1}{4} f (\theta H u)^2,\\
2)&\quad 0 = \theta_\nu \theta_{\nu_1} + \frac{1}{2}\theta (\theta H u)^2.
\end{aligned}$$

Therefore:

$$\begin{aligned}
&3)\quad 0 = 3 a_\nu (\theta H u)^2 + \theta (aLu)^3 + 2 f (\theta L u)^3,\\
&4)\quad 0 = 3\theta_\nu (aHu)^2 + f (\theta L u)^3 + 2\theta (aLu)^3,\\
&5)\quad 0 = a_\nu (\theta L u)^3, \qquad\qquad 0 = \theta_\nu (aLu)^3, \qquad 0 = (aHu)^2 (\theta H u)^2,\\
&6)\quad 0 = a_\nu \theta_\nu^2 + \theta_\nu a_\nu^2 + f \theta_\eta (\theta H u)^2 + \theta a_\eta (aHu)^2,\\
&7)\quad 0 = \theta_\nu \theta_\nu^2 + \theta\theta_\eta (\theta H u)^2 + \frac{1}{6} f H_j,
\end{aligned}$$

8) $0 = a_\nu (j\nu x)^2 + fH_j,$

9) $0 = (aHu)^2 (j\nu x)^2 - 2a_\nu H_j,$

10) $0 = \theta_\nu (j\nu x)^2,$ $\qquad 0 = \theta H_j,$

11) $0 = a_\nu \theta_\nu^3 - \dfrac{3}{4}(j\eta x)^2,$

12) $0 = a_\nu^2 \theta_\nu^2 - \dfrac{1}{3}(j\eta x)^2 - \dfrac{1}{3}(\Delta Hu)^2,$

13) $0 = a_\nu^2 \theta_\nu^3 - \dfrac{1}{2} a_\sigma,$

14) $0 = \theta_\nu \theta_\nu^3,$ $\qquad 0 = a_\eta H_j.$

The formulae have been carried far enough for the products to become polarizable: that is, by folding only one new product arises, because a) the folding of the product into itself becomes reducible, and b) the remaining products arising by folding become reducible or lead to contravariants (cf. § 3. 2), a) and b)).

Formulae 1) to 5) give the reduction of all foldings of s with products arising from $a_\nu \theta_\nu$ by folding. Formulae 6) to 10) give the reduction of those products arising from $a_\nu \theta_\nu^2$ by folding. By § 24 A), taking formulae 6) to 10) into account, the foldings of s with products arising from $a_\nu^2 \theta_\nu$ decompose.

One has:

$$0 \equiv a_\nu^2 (asu)^2 \theta_{\nu_1} (\theta su)^2, \qquad 0 \equiv a_\nu^2 (asu) \theta_{\nu_1} (\theta su)^3 - K_\eta (KHu)^2 (Ksu)^3,$$

$$0 \equiv a_\nu^2 (asu)^2 \theta_{\nu_1} (\theta su), \qquad 0 \equiv a_\nu^2 (asu) \theta_{\nu_1} (\theta su)^2,$$

$$0 \equiv a_\nu^2 (asu) \theta_{\nu_1} (\theta su).$$

The reducers

$$((j\eta x)^2, s, u)^3 \text{ [from } (\widehat{\vartheta \eta su})^2 (Hsu), \text{ § 23. } C3],$$

$$(H_j (j\eta x), s, u)^2 \text{ [§ 24, } B], \quad ((\Delta Hu)^2, s, u)^2 \text{ [§ 24, } C], \quad (a_\sigma su)^2 \text{ [§ 24, } E]$$

together with formulae 11) to 14) give the reduction of all products arising from $a_\nu \theta_{\nu_1}^3$, $a_\nu^2 \theta_{\nu_1}^2$, $a_\nu^2 \theta_{\nu_1}^3$.

Our preceding computations can be summarized in the following *conclusion*:

The relatively complete system modulo (ϱ, t) consists of the following 331 forms and subsystems, which are also reproduced below in tabular order.

Table I (cf. p. 90)

u \ x	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
0	i				f		Δ		$K_\nu(k\nu x)^3$		$K_\eta(k\eta x)^3$				$(\vartheta^3\eta x)^4$	$H_k(k\vartheta, \eta x)^4$		$\theta_l(\vartheta k, lx)^5$				$(\vartheta^2 lx)^6$
1		u_x				$\theta_\nu(\vartheta\nu x)^2$		$\theta_\eta(\vartheta\eta x)^2$	N	$(k\nu x)^3$	$\theta_\nu(\vartheta^2\nu x)^3$	$(k\eta x)^3$	$H_\vartheta(\vartheta^2\eta x)^3$ $\theta_\eta(\vartheta^2\eta x)^3$		$\theta_l(\vartheta^2 lx)^4$		$(\vartheta k, \eta, x)^4$		$(\vartheta k, l, x)^5$			
2			a_ν^2		θ a_η^2		$(\vartheta\nu x)^2$	$K_\nu(k\nu x)^2$	$(\vartheta\eta x)^2$	$H_k(k\eta x)^2$ $K_\eta(k\eta x)^2$		$(\vartheta^2\nu x)^3$ $K_l(klx)^3$		$(\vartheta^2\eta x)^3$		$(\vartheta^2 lx)^4$						
3		θ_ν^3		a_ν	$\theta_\nu(\vartheta\nu x)$	Δ_ν a_η	K $H_\vartheta(\vartheta\eta x)$ $\theta_\eta(\vartheta\eta x)$	Δ_η	$(k\nu x)^2$ $\theta_l(\vartheta lx)^2$		$(k\eta x)^2$		$(klx)^3$									
4	ν		H θ_ν^2	$(j\nu x)^3$ $a_\eta(aHu)$	θ_η^2	$(\vartheta\nu x)$ a_l^2		N_ν $(\vartheta\eta x)$		H_η N_η $(\vartheta lx)^2$												
5		$a_\eta(aHu)^2$	$\theta_\eta^2(\theta Hu)$	θ_ν	$\frac{K_\nu^2}{(aHu)}$	θ_η	K_η^2 (ΔHu) a_l		Δ_l													
6	j σ		$(j\nu x)^2$ $(aHu)^2$	L $a_\nu^2(j\nu x)$ $H_\vartheta(\theta Hu)$ $\theta_\eta(\theta Hu)$	$\frac{K_\nu}{\theta_l^2}$			K_η														
7		$\theta_\eta(\theta Hu)^2$	$H_j(j\eta x)$ $a_l(aLu)^2$		$a_\nu(j\nu x)$ (θHu)		$\Delta_\nu(j\nu x)$ θ_l	K_l^2														
8	H_j^2 $a_l(aLu)^3$	$(\nu\sigma x)$ $(j\nu x)$	$(\theta Hu)^2$	$K_\eta(KHu)^2$ $(j\eta x)$ $(aLu)^2$																		
9		H_j $(aLu)^3$	$a_\eta(aHu)H_j$ $L_\vartheta(\theta Lu)^2$ $\theta_l(\theta Lu)^2$		(KHu)																	
10	$\theta_l(\theta Lu)^3$	L_j^2 $(j\sigma x)$ $a_\eta(aH^2u)^3$		$H_j(aHu)$ $(\theta Lu)^2$																		
11		$(\theta Lu)^3$	$K_l(KLu)^3$ L_j $(aH^3u)^3$																			
12	$(aH^2u)^4$	$a_\eta(aLu)^2L_j$ $H_\vartheta(\theta H^2u)^3$ $\theta_\eta(\theta H^2u)^3$		$(KLu)^3$																		
13	$(\nu\sigma j)$		$(aLu)^2L_j$ $(\theta H^3u)^3$																			
14	$(\theta H^2u)^4$	$K_\eta(KH^2u)^4$ $(a_\eta H\cdot L, u)^4$																				
15	$L_9(\theta, H\cdot L, u)^4$		$(KH^2u)^4$																			
16	$(\theta, H\cdot L, u)^5$																					
17	$K_\eta(K, H\cdot L, u)^5$																					
18		$(K, H\cdot L, u)^5$																				
19																						
20																						
21	$(KH^3u)^6$																					

(The numbers in the uppermost horizontal row denote the degree in the x; the numbers in the first vertical row denote the degree in the u.)

Table II (cf. p. 90)

$u \backslash x$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0	J				s		s_j^3					s_ν	$(k\nu x)^3 s_\nu$	$(\vartheta^3\nu x)^2 s_\vartheta$ $(\vartheta^2\nu x)^2 s_\vartheta$		$\theta_\eta(\vartheta^2\eta x)^2 s_\eta$	
1							(asu)	s_η	s_l^2 (Δsu)	$s_\eta(Nsu)$ $(\vartheta\nu x)^2 s, u$	$((k\nu x)^3, s, u)_s$ $(K_\nu(k\nu x)^3, s, u)$		$K_\eta(k\eta x)^3, s, u$		$(\vartheta^2\nu x)^2 s_\nu$		$(\vartheta^2\eta x)^4, s, u$
2					$(asu)^2$	$s_\theta(\theta su)$	$(\Delta su)^2/a_\nu s_\nu$	$(\vartheta\nu x)^2, s, u$ $(\theta_l(\vartheta\nu x)^2, s, u)$		$\theta_\eta(\vartheta^2\eta x)^2, s, u$		$(k\eta x)^2 s_\nu$ $((k\nu x)^2, s, u)$		$(k\eta x)^3, s, u$			
3			$(asu)^3$	$s_\vartheta(\theta su)^2$ s_ν	$a_\nu s_\nu(asu)$ $a_l^2(asu)$	s_η	(θsu) $a_l^2 su$		$(Nsu)^2$ $(\vartheta\eta x)s_\nu$ $((\vartheta\nu x)^2, s, u)$	$((k\nu x)^2, s, u)^2$	$(\vartheta^2\eta x)^2, s, u$						
4	$(asu)^4$	$s_\vartheta(\theta su)^3$	$a_\nu^2(asu)^2$	$(\theta_\nu^2 su)$	$(\theta su)^2$	$s_k(Ksu)^3$ $a_\nu(asu)$ $s_\nu(asu)$	$((\vartheta\nu x)^2, s, u)^2$	$s_\nu(\Delta su)$ (Δsu) $(aHu)s_\eta$ $(a_\nu su)$		$(\Delta_\eta su)$							
5			$(\theta su)^3$	$s_k(Ksu)^3, s_j$ $s_\vartheta(\theta s' u)^4$ $s_\nu(asu)^2$ $a_\nu(asu)^2$	(Hsu) $((\vartheta\nu x)^2, s, u)$ $(\theta_\nu(\vartheta\nu x)^2, s, u)$ $(\theta_\nu^2 su)$	$((j\nu x)^2, s, u)$ $(aHu)^2 s_\nu$ $(a_\eta su)^2$	$(Ksu)^2$ s_η $(\theta_\eta^2 su)$		$((k\nu x)^2, s, u)^2$ $K_\nu s_\nu$								
6	$(\theta su)^4$	$s_\nu(asu)^3$ $a_\nu(asu)^3$	$(Hsu)^2$ $(\theta_\nu(\vartheta\nu x), s, u)^3$ $(\theta_\nu^2 su)^2$	$a_\eta s_\nu^2$ $(a_\eta(aHu), s, u)^2$ $(a_\eta(aH^2u), s, u)$	$(Ksu)^3$	$s_\eta(asu)$ $((\vartheta\nu x), s, u)^3$ $(\theta_\eta su)$	$((k\nu x)^2, s, u)^3$	$s_\nu(\Delta su)$ $a_\nu(j\nu x)s_\nu$ $(\theta Hu)s_\eta$ $(\theta_\eta su)$									
7	$\theta_\nu(\vartheta\nu x), s, u)^4$	$(a_\eta(aHu), s, u)^3$	$(Ksu)^4$ $(\theta_\eta^2(\theta Hu), s, u)^2$ $(a_\eta^2 - a_l^2, s, u)^3$	$s_j(asu)^3$ $s_k(K^2u)^3$ $((j\nu x), s, u)^2$ $(\theta_\eta su)^2$	$((j\nu x), s, u)$ $((j\eta x), s, u)$ $(aHu), s, u)$ $(aH^2u), s, u)$	$((\vartheta\eta x), s, u)^3$	$(aLu)^3 s_l/(asu)^3$										
8	$(a^2 - a_l^2, s, u)^4$	$s_j(asu)^3$ $s_k(K^2u)^3$ $((j\nu x), s, u)^4$ $(\theta_\eta su)^3$	$((j\nu x)^2, s, u)^2$ $(aHu), s, u)^3$ $((aHu)^2, s, u)^3$	$(Lsu)^2$ $(a_l^2(j\nu x), s, u)^3$ $H_\vartheta(\theta Hu), s, u)^3$ $\theta_l(\theta Hu), s, u)^3$	$(asu)^3$	$(K, su)^3$		$(K_\vartheta su)^2$									
9	$K_j^2 su^4$ $((aHu), s, u)^4$	$(Lsu)^4$ $(a_l^2(j\nu x), s, u)^4$ $(\theta_\nu(\theta Hu), s, u)^2$	$(K_\nu^3 u)^6$ $(a_\eta(aLu)^2, s, u)^3$ $(a_l^2 H, s, u)^3$	$(K, su)^3$	$(a_\nu(j\nu x), s, u)^3$ $(\theta Hu), s, u$ $(\theta Hu)^2, s, u$		$(\theta_l su)^3$										
10			$(K, su)^4$ $(a_l^2 a_\eta(aHu)^2, s, u)^3$	$((j\eta x), s, u)^3$ (H_j, su) $((aLu)^2, s, u)^2$ $((aLu)^3, s, u)$													
11	$(a_\nu(j\nu x), s, u)^4$ $((\theta Hu), s, u)^4$	$(K_l(KHu)^2, s, u)^3$ $((j\eta x), s, u)^3$ $((aLu)^2, s, u)^4$ $(a_\eta H, s, u)^3$	$(H^2 su)^3$ $\theta_l(\theta Lu)^2, s, u)^3$		$((KHu)^2, s, u)^3$												
12		$(a_\eta(aHu)^2 H, s, u)^3$	$((KHu)^3, s, u)^3$	$((aHu)H_j, s, u)^2$ $((\theta Lu)^2, s, u)^3$													
13	$((KHu)^3, s, u)^4$	$((aHu)H_j, s, u)^3$ $((\theta Lu)^2, s, u)^4$ $(\nu \cdot H, s, u)^3$	$(L_j su)^2$ $((aH^3u)^3, s, u)^3$ $((j\eta x)H, s, u)^3$ $((aHu)^2 H, s, u)^3$														
14	$((j\nu x)^2 H, s, u)^4$ $((aHu)^2 H, s, u)^4$	$(\theta_\eta(\theta Hu)^2 H, s, u)^3$		$((KLu)^3, s, u)^2$													
15	$(a_l(aLu)^3 H, s, u)^4$	$((KLu)^3, s, u)^3$	$((aLu)^2 L_j, s, u)^2$ $((\theta Hu)^3 H, s, u)^3$ $((\theta Hu)^2 H, s, u)^2$														
16	$((\theta Hu)^2 H, s, u)^4$ $(\sigma \cdot H_j, s, u)^4$	$((j\eta x)H, s, u)^4$ $(H_j H, s, u)^3$ $((aLu)^2 H, s, u)^4$ $((aLu)^3 H, s, u)^3$															
17	$(\theta(\theta Lu)^2 H, s, u)^4$		$((KH^3u)^3, s, u)^3$														
18		$((\theta Lu)^2 H, s, u)^4$ $((\theta Lu)^3 H, s, u)^3$ $(aHu)^2 H, s, u)^4$															
19	$((aH^3u)^3 H, s, u)^4$ $(L_j H, s, u)^5$																
20		$((KLu)^3 H, s, u)^5$	$((aLu)^3 H, s, u)^5$														
21	$((\theta H^3u)^3 H, s, u)^4$ $((aLu)^2 H_j, s, u)^4$																
22																	
23	$((KH^3u)^4 H, s, u)^4$	$((aH^3u)^4 H, s, u)^3$															

(The numbers in the uppermost horizontal row denote the degree in the x; the numbers in the first vertical row denote the degree in the u.)

3. On the Invariant Theory of Forms in n Variables[30]

Jahresbericht der DMV 19 (1910), pp. 101–104

In the projective invariant theory of forms in n variables the main problem has been solved: the finiteness of the system of forms has been proved. A series of further questions, however, have not been treated, namely those concerned with methods for investigating the connection among forms. I would like briefly to communicate the results that I have obtained with respect to such questions, but for the sake of clarity I shall first sketch the questions in the ternary domain, where they are known.

I have first in mind the two fundamental theorems of the symbolic method: the theorem that all invariant formations can be represented symbolically, and the second theorem that all relations existing among invariants are obtained by successive application of a small number of symbolic identities; thus the symbolic method gives all relations without leaving the domain of invariants.[31] Built upon this are the processes for generating the forms, the symbolic folding process and the nonsymbolic differentiation processes, the theory of series expansions, and the like. In this connection I should also mention a theorem proved in my dissertation,[32] namely that all foldings can be generated by successive application of the two possible "cogredient" foldings; by this, for a symbolic product $s_x t_x u_\sigma u_\tau$, I mean the two foldings (stu) and $(\sigma\tau x)$. On this theorem one can build the theory of reducers and of the reduction of systems of forms, in analogy with the binary domain, as I showed in my dissertation.

When we now pass to the n-ary domain, even the first theorem of the symbolic method holds only in a conditional measure. It is well known that already for quaternary forms there occur, besides the point and plane coordinates x and u, line coordinates p_{ik}, the subdeterminants from two rows of point coordinates. Analogously, in the n-ary domain we have rows of variables p_ρ, whose individual elements are the subdeterminants from ρ rows x, and rows of symbols S^ρ, whose elements behave like subdeterminants from ρ rows s cogredient to the u. The symbolic representation by determinants is known to hold only when the symbol rows S^ρ are resolved into the rows s and these are distributed among different determinants; only such aggregates of determinants as depend on the S^ρ then have real meaning. But which aggregates of determinants accomplish this cannot be surveyed, and thereby all the other theorems mentioned in the ternary domain are lost.

I would now like to state a simple principle by which one obtains a symbolic representation explicit in the rows S^ρ, and thereby also recovers the remaining theorems. This is an application of the familiar theorem on corresponding matrices: "To each row of variables p_ρ there corresponds one-to-one another row $q_{n-\rho}$ whose individual elements are subdeterminants from $n-\rho$ rows u connected with the ρ rows x by equations $(u \mid x) = 0$." Correspondingly, to each symbol row S^ρ there is assigned another $S_{n-\rho}$, behaving as if it were composed from $n-\rho$ rows cogredient to the x.

The theorem also permits a second formulation, suited for applications: "To every matrix product $(v^{(1)} \cdots v^{(\rho)} \mid y^{(1)} \cdots y^{(\rho)})$,[33] where the rows of the second matrix are contragredient to those of the first, there is assigned a determinant; and conversely each determinant can be transformed into a matrix product with a prescribed number ρ of rows." Thus determinant and matrix product appear as fully equivalent, and the first theorem of the symbolic method assumes the form: "All invariant formations can be represented symbolically by matrix products." From two symbol rows S^σ, T^τ one can now, with the aid of rows of variables, very easily form a matrix product explicitly containing these symbol rows and hence representing an invariant formation of the form $(S^\sigma \mid p_\sigma)(T^\tau \mid p_\tau)$, namely the following:

$$(q_{n-\sigma-\lambda} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda}), \qquad (\lambda = 1, 2, \ldots, \tau \text{ or } n-\sigma). \tag{1}$$

More important is the converse: that all invariant formations can be represented explicitly by matrix products, so that the above process is the extension of the binary and ternary folding process. To prove this converse it is necessary to transfer the second theorem of the symbolic method to the n-ary domain, namely to set up the totality of the independent indecomposable identities in which all rows S^ρ, p_ρ are regarded as indecomposable. These identities arise from the known ones containing only rows x and u,[34] by replacing the product theorem for determinants by a system of product theorems for matrices, and by contracting different

[30] Lecture delivered at the Salzburg meeting of natural scientists, 1909.

[31] Study: *Methoden zur Theorie der ternären Formen.* II. § 6. (Leipzig, Teubner, 1889.)

[32] *Über die Bildung des Formensystems der ternären biquadratischen Form.* § 1. (Crelle's Journal Bd. 134. 1908.)

[33] That is, the sum over the products of corresponding determinants, formed from the matrix of the ρ rows v and that of the ρ rows y.

[34] E. Pascal in *Memorie d. R. Acc. d. Lincei.* (4) 5 (1888), p. 375.

matrix products into a single one by means of precisely these product theorems. One obtains in this way two dualistic formulas, of which only one is given:

$$\begin{gathered}(S_1^{\rho_1} S_2^{\rho_2} \cdots S_k^{\rho_k} \,|\, x p_{\rho-1}) = \sum_{i=1}^{k} \varepsilon \,(S_1^{\rho_1} \cdots S_{i-1}^{\rho_{i-1}} S_{i+1}^{\rho_{i+1}} \cdots S_k^{\rho_k} q_{n-\rho_i+1} \,|\, S_{n-\rho_i} x), \\ \rho = \sum_{i=1}^{k} \rho_i < n, \qquad \varepsilon = \pm 1.\end{gathered} \tag{2}$$

The only specialization contained in these formulas, namely that a row x enters, cannot be avoided; for completely general symbol and variable rows we arrive at a "decomposition identity," an identity in which a row p_ρ is decomposed into the individual rows x. The simplest special case of the decomposition identity was already used by Clebsch in his "fundamental problem of invariant theory" to derive the elementary covariants in the theory of series expansions. With the aid of these identities, which mediate the transition from the non-explicit determinant expressions to the matrix product, one can in fact show that the desired explicit symbolic representation is obtained with the matrix representation.

For the folding process, however, a theorem entirely analogous to that in the ternary domain holds, and the reduction theorems can likewise be built upon it analogously. If in (1) one denotes the number λ as the defect of the folding, and calls foldings for which $\sigma = \tau$ cogredient, then all foldings can be generated by successive application of the cogredient foldings of defect 1. The proof of this theorem rests essentially on the fact that the most general forms can be replaced by "normal forms," that is, by forms that vanish identically under the application of all invariant differential operations. In the quaternary domain this latter fact was proved by Mertens;[35] our knowledge of the most general differential operations–which, as is well known, arise from the folding processes by replacing the symbol rows by differential symbols–allows us, using the decomposition identity, to transfer Mertens's proof almost verbatim to the n-ary domain.

[35] Über invariante Gebilde quaternärer Formen. Wiener Berichte. Bd. 98. 1889.

4. On the Invariant Theory of Forms in n Variables

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Introduction.

In the projective invariant theory of forms in n variables, the questions adjoining the main problem–the proof of finiteness of the system of forms–have scarcely been treated when one goes beyond the binary and ternary domains. These are the questions concerning the mutual dependence of the forms and the methods for controlling this connection among forms. These methods rest essentially–as has been completely proved in the binary and ternary domains–on two theorems designated by Study as the "fundamental theorems of the symbolic method," namely the theorem on the symbolic representability of the forms and the theorem on symbolic identities.[36] The latter theorem says that all relations existing among integral invariant formations are obtained by successive application of a small number of symbolic identities; thus the symbolic method, and only it, gives knowledge of the connection among forms without leaving the domain of invariants.

Gordan, in his Erlangen program,[37] already pointed to the reason for the difficulty in passing to n variables: it lies in the fact that the first theorem of the symbolic method is no longer valid in its general form. In the n-ary domain there occur, as is well known, variable rows of higher level p_ρ[38] and analogously symbol rows of higher level S^ρ; one arrives at these rows both when, as originally with Clebsch,[39] one starts from forms with arbitrarily many rows p_ρ, and when one starts, following F. Deruyts and A. Capelli,[40] from forms containing only arbitrarily many cogredient rows x. A symbolic representation that would contain the rows p_ρ, S^ρ explicitly[41] does not exist; it is necessary to resolve the p_ρ, S^ρ into the individual rows x and the s contragredient to them, and to distribute these among different determinants. It is indeed possible, by means of differential conditions (cf. formula (19.)), to verify that a given aggregate of determinants has the property of depending only on the rows p_ρ, S^ρ; but in this way one obtains no overview of the totality of invariant formations and of the processes for their generation.[42]

We shall now show in §§ 2 and 6 how, using the "theorem on corresponding matrices,"[43] the first fundamental theorem is recovered in its generality by replacing representation by determinants with representation by "matrix products." In § 2 it is shown that every matrix product explicitly containing the rows S^ρ, p_ρ is an invariant formation of the ground form; the converse given in § 6, that every invariant formation can be represented explicitly by matrix products, succeeds only through the transfer of the second fundamental theorem (§§ 3–5).

This theorem is known for forms that contain only variable rows x.[44] The general problem, to set up the totality of the indecomposable identities in which all rows S^ρ, p_ρ are regarded as indecomposable, was formulated by Study;[45] at the same time he sees in the number of identities that occur a measure of the difficulty of the methods in the n-ary domain. Since it is now possible to combine the totality of the identities in a single formula (36.), this difficulty is at least pressed down to a minimum. In applications, however, certain distinguished special cases of (36.) will often occur, such as (20.), (21.), characterized by the fact that they admit explicit representation without substitution of new rows; or formula (31.), designated as the "decomposition identity," since a row p_ρ appears to be decomposed into the rows y.

Upon the two fundamental theorems the further theory can now easily be built. The first theorem gives directly the processes for generating the forms, the symbolic folding process and the nonsymbolic differentiation processes,[46] while the decomposition identity eliminates, among these processes, all equivalent ones (§ 6).

[36] E. Study, *Methoden zur Theorie der ternären Formen* (Leipzig, Teubner, 1889). II. § 6. For the occurrence of these fundamental theorems also in the invariant theory of special transformation groups, cf. Study, *Das Apollonische Problem* (Math. Ann. Bd. 49 [1897]) and *Zur Differentialgeometrie der analytischen Kurven* (Transactions of the American Math. Society, Bd. 1 [1909]).

[37] P. Gordan, *Über das Formensystem binärer Formen.* S. 46. (Leipzig, Teubner, 1875).

[38] For the notation see § 1.

[39] A. Clebsch, über eine Fundamentalaufgabe der Invariantentheorie. (Abh. der Ges. d. Wiss. zu Göttingen, Bd. XVII, 1872).

[40] Cf. A. Capelli, *Lezioni sulla teoria delle forme algebriche* (Napoli 1902), § 22–24, and the literature cited there.

[41] More precise definition of "explicit representation" in § 2.

[42] A computationally very valuable development of the symbolism due to Study (communicated by F. Engel, Ber. d. K. Sächs. Ges. d. Wiss., math.-phys. Kl., 1900, p. 126, and for the quaternary domain by E. Study, *Geometrie der Dynamen*, p. 135) is also not an explicit representation in our sense. It would, for example, hardly lead to the nonsymbolic differentiation processes for generating the forms.

[43] Apart from Grassmann's *Ausdehnungslehre* of 1862, no. 112, first in A. Brill, über zwei Berührungsprobleme, § 2 (Math. Ann. Bd. 4. 1871).

[44] E. Pascal, Giorn. di mat. 26 (1888), pp. 33, 102; Rendic. d. Accad. d. Linc. (4) 4 (1888), p. 119; ib. Mem. (4) 5 (1888), p. 375.

[45] "Methoden ...", p. 204, note 17).

[46] In the paper "Invariante Gebilde von Nullsystemen" (Sitzungsber. d. Ak. d. Wiss. Wien, Bd. 97, Abt. IIa, 1888), Mertens arrived by another path, not directly generalizable, at the quaternary folding process. The alternating invariant of 6 rows S^2 occurring there differs from the one arising with us by a single application of substitution (11.) by an aggregate of even invariants. For a special case the quaternary folding process is also found

Knowledge of the differentiation processes further yields the transfer of the concept of "normal form" to the n-ary domain and, by means of a proof procedure given by Mertens[47] in the quaternary domain, the theorem that a system of invariant formations can be replaced by an equivalent system of normal forms (§ 7). Under this latter assumption I showed in my dissertation[48] that the ternary folding process can be generated by successive application of two fundamental foldings. On the basis of this theorem and the theory of series expansions, I then developed, by introducing the concept of the "form row," the principal reduction theorems for systems of forms. Entirely analogously, in the n-ary domain the folding process can be generated from $n-1$ fundamental foldings (§ 8), and the reduction theorems can be set up (§ 9).

Afterword. On the occasion of the Salzburg communication, Prof. E. Müller of Vienna drew my attention to the fact that the calculation with matrix products underlying this paper is, in principle, identical with the calculus of Grassmann's theory of extension.[49] Indeed, Grassmann's "combinatorial product" $[S^{\rho_1} S^{\rho_2} \cdots S^{\rho_i}]$ can be regarded as a matrix given by these rows (cf. § 2), specifically the "progressive product" as the matrix of the rows themselves, the "regressive" as the matrix of the assigned rows $S_{n-\rho_1}, S_{n-\rho_2}, \ldots, S_{n-\rho_i}$, while the matrix corresponding to the "mixed product" arises when several rows S are combined by substitution (9.) into new rows P, Q. Our "assigned row" corresponds to Grassmann's "complement," our "matrix product" to a "mixed product of level number 0."

Under this correspondence, the development given in the *Ausdehnungslehre* of 1862, no. 113, becomes identical with a formula dualistic to our formula (32.). In the special case $\rho = 1$, (32.) goes over into (17.), the dualistic formula into (16.). From these special cases of Grassmann's development, Müller[50] has recently derived the identities (20.), (21.) mentioned above.

In contrast to the Grassmann–Müller derivation, our derivation at the same time gives the proof of completeness of the identities, which appears to be the essential point for the questions considered here, and it proceeds from (20.), (21.) further to the most general indecomposable identities (36.), which do not seem to have been noticed before now.

§ 1. Notations and definitions. Summary of known results.

We denote, as usual, by the variable row x the row of elements $x_1, x_2, \ldots, x_n$, and analogously by the variable row p_ρ $(\rho = 1, 2, \ldots, n-1)$ the row of the $\binom{n}{\rho}$ elements $p_{i_1 i_2 \ldots i_\rho}$, where $p_{i_1 i_2 \ldots i_\rho}$ denotes the $\binom{n}{\rho}$ determinants formed from the matrix of the ρ rows $x^{(1)}, x^{(2)}, \ldots, x^{(\rho)}$,

$$\begin{vmatrix} x_{i_1}^{(1)} & x_{i_2}^{(1)} & \cdots & x_{i_\rho}^{(1)} \\ x_{i_1}^{(2)} & x_{i_2}^{(2)} & \cdots & x_{i_\rho}^{(2)} \\ \vdots & \vdots & & \vdots \\ x_{i_1}^{(\rho)} & x_{i_2}^{(\rho)} & \cdots & x_{i_\rho}^{(\rho)} \end{vmatrix}, \qquad (i_1 < i_2 < \cdots < i_\rho = 1, 2, \ldots, n).$$

According to the theorem on corresponding matrices,[51] to every row p_ρ there is assigned one-to-one a second row $q_{n-\rho}$ by the relation, valid for each element of the row,

$$p_{i_1 i_2 \ldots i_\rho} = (-1)^{\frac{\rho(\rho+1)}{2} + i_1 + i_2 + \cdots + i_\rho} q_{j_1 j_2 \ldots j_{n-\rho}}, \tag{1}$$

where the i and j are each in good order and together form a complete permutation of the numbers 1 through n. The row $q_{n-\rho}$ consists of the $\binom{n}{\rho}$ determinants of degree $n-\rho$, $q_{j_1 j_2 \ldots j_{n-\rho}}$, formed from the matrix of $n-\rho$ rows $u^{(1)}, u^{(2)}, \ldots, u^{(n-\rho)}$ contragredient to the x. These rows satisfy the $\rho(n-\rho)$ condition equations

$$(u^{(k)} \mid x^{(l)}) = \sum_{\alpha=1}^{n} u_\alpha^{(k)} x_\alpha^{(l)} = 0; \qquad (k = 1, 2, \ldots, n-\rho;\ l = 1, 2, \ldots, \rho). \tag{2}$$

in P. Gordan, *Das simultane System von zwei quadratischen quaternären Formen* (Math. Ann. Bd. 56. 1903).

[47] F. Mertens, Über invariante Gebilde quaternärer Formen. (Sitzber. d. Ak. d. Wiss. Wien. Bd. 98. Abt. IIa. 1889.)

[48] Über die Bildung des Formensystems der ternären biquadratischen Form. (Dieses Journal, Bd. 134. 1908.)

[49] Hermann Graßmanns gesammelte mathematische und physikalische Werke. Ersten Bandes zweiter Teil: Die Ausdehnungslehre von 1862. In Gemeinschaft mit H. Graßmann d. Jüngeren herausgegeben v. Fr. Engel (Leipzig, Teubner 1896).

[50] E. Müller, Beiträge zur Graßmannschen Ausdehnungslehre. 1. Mitteilung: Einige allgemeine Sätze. (Sitzungsber. d. Ak. d. Wiss. Wien; Bd. 118. Abt. IIa. Juli 1909), Satz 16 und 18.

[51] Cf. for example Gordan–Kerschensteiner, *Vorlesungen über Invariantentheorie*, Bd. I, p. 99.

We shall assume them normalized (with Clebsch[52]) so that the proportionality factor otherwise entering in (1.) is set equal to one.

According to Clebsch[53] (and likewise according to the later investigations of F. Deruyts and A. Capelli, cf. the Introduction), the most general forms can be reduced to those that contain at most one of each of the $n-1$ rows p_ρ, and hence can be represented symbolically as follows:

$$(S^1 \mid p_1)^{m_1}(S^2 \mid p_2)^{m_2}\cdots(S^{n-1} \mid p_{n-1})^{m_{n-1}}, \tag{3}$$

where, in analogy with (2.), we have set

$$(S^\rho \mid p_\rho) = \sum S^{i_1 i_2 \dots i_\rho} p_{i_1 i_2 \dots i_\rho}.$$

In consequence of the nonlinear identical relations existing between the elements of a row p_ρ, the symbol rows S^ρ can be normalized so that they satisfy the very same identities, hence so that they behave like the determinants of a ρ-rowed matrix whose rows $s^{(1)}, s^{(2)}, \dots, s^{(\rho)}$ are contragredient to the x.[54] Thus to each row S^ρ there can be assigned one-to-one another $S_{n-\rho}$ through the relation

$$S_{i_1 i_2 \dots i_{n-\rho}} = (-1)^{\frac{(n-\rho)(n-\rho+1)}{2}+i_1+i_2+\dots+i_{n-\rho}} S^{j_1 j_2 \dots j_\rho}, \tag{4}$$

where again the i and j are each in good order and together exactly exhaust the numbers 1 through n. The elements $S_{i_1 i_2 \dots i_{n-\rho}}$ of the row $S_{n-\rho}$ behave like the determinants formed from $(n-\rho)$ rows $\sigma^{(1)}, \sigma^{(2)}, \dots, \sigma^{(n-\rho)}$ cogredient to the x, and the rows s and σ are linked by the $\rho(n-\rho)$ conditions

$$(s^{(k)} \mid \sigma^{(l)}) = 0, \qquad (k = 1, 2, \dots, \rho;\ l = 1, 2, \dots, n-\rho). \tag{5}$$

In consequence of the relations (1.) and (4.), each factor of (3.) is capable of the equivalent fourfold symbolic representation

$$(S^\rho \mid p_\rho) = (S^\rho q_{n-\rho}) = (S_{n-\rho} p_\rho) = (-1)^{\rho(n-\rho)}(q_{n-\rho} \mid S_{n-\rho}), \tag{6}$$

where, as always in what follows, $(S^\rho q_{n-\rho})$ and $(S_{n-\rho} p_\rho)$ are abbreviations for the determinants $(s^{(1)} \cdots s^{(\rho)} u^{(1)} \cdots u^{(n-\rho)})$ and $(\sigma^{(1)} \cdots \sigma^{(n-\rho)} x^{(1)} \cdots x^{(\rho)})$; by Laplace expansion these prove to depend only on the rows $S^\rho, q_{n-\rho}$ and respectively $S_{n-\rho}, p_\rho$, as the above notation is meant to indicate. The expressions $(S^\rho \mid p_\rho)$ and $(q_{n-\rho} \mid S_{n-\rho})$ mean, according to the preceding discussion, the products of the matrices of the rows s and x, respectively of the rows u and σ, where by matrix product is meant the sum over the products of corresponding determinants, that is, the value of the determinant of the product matrix.

The characteristic number $\rho(n-\rho)$ belonging to a symbol row (respectively variable row) will be called, following Clebsch, the weight of the row;[55] the number ρ, which gives the number of rows s, the upper weight index; the number $n-\rho$, which gives the number of rows σ, the lower weight index. It follows from the relations (4.) that a symbol row can occur in one and the same relation once with upper and once with lower weight index; the same holds for the occurrence of the corresponding rows p_ρ and $q_{n-\rho}$. It remains to note that we generally denote variable rows of higher level whose associated matrix consists of rows x by p, those whose matrix consists of rows u by q; symbol rows cogredient to the x by small Greek letters $\sigma, \tau, \alpha, \beta$; symbol rows cogredient to the u by small Latin letters s, t, a, b; all other symbol rows by capital Latin letters with weight index: S^ρ, T^τ, A^ρ or $S_{n-\sigma}$, $T_{n-\tau}$, $A_{n-\rho}$. Corresponding to an upper or lower weight index, we also write the individual elements of a row with upper or lower indices, as for instance in (4.).

§ 2. Representation by matrix products.

In what follows we always understand by "matrix product" the sum over the products of corresponding determinants of two matrices with the same number of rows, where the rows of the second matrix are contragredient to those of the first. The rows of each matrix may be: 1. individually given; 2. combined, as in (6.), into a row S^ρ; 3. united into different rows $S^\rho, S^\sigma \dots$. The matrix product is to be denoted by the symbol (|) used in (6.). The representation (6.) is a special application of a second formulation of the theorem on corresponding matrices: "To every ρ-rowed matrix product $(v^{(1)} v^{(2)} \cdots v^{(\rho)} \mid y^{(1)} \cdots y^{(\rho)})$ there is

[52] a. a. O. § 5.
[53] a. a. O. § 12.
[54] Clebsch, a. a. O. § 4.
[55] a. a. O. § 11.

assigned a determinant; and conversely to every determinant there can be assigned a matrix product with prescribed number ρ of rows ($\rho = 1, 2, \ldots, n-1$)."[56] Indeed, one has only to combine the rows y into a row p_ρ (respectively the rows v into a row q_ρ) and to carry out the substitutions given by (1.) in order to pass from a given matrix product to the determinant and conversely. The value of the determinant is, however, not given by its individual elements, but first results from Laplace expansion. Since determinant and matrix product therefore prove to be equivalent, the theorem on the symbolic representability of the forms (the first fundamental theorem) is expressed also as follows:

Theorem I. "All invariant formations can be represented symbolically by matrix products, and conversely all matrix products are invariant formations."[57]

Representation by matrix products allows us to overcome the essential difficulty of the non-explicit representation caused by the determinant representation.[58] By "explicit representation" we mean a representation into which only the rows $S^\rho, T^\tau, \ldots$ themselves enter, or rows R^ρ uniquely derivable from these, but in which the individual component rows $s, t, \ldots$ no longer occur. We shall obtain in § 6, for all invariant formations that depend only on the symbol rows $S^\rho, T^\tau, \ldots$, a representation explicit in these symbol rows, and thereby the transfer to the n-ary domain of the generating processes, the folding process and the differentiation processes. The possibility of explicit representation is evident from the fact that only the simplest matrix products can be assigned precisely to the determinants occurring in the determinant representation, so that all other matrix products are combinations of different determinants into a single expression. For if one resolves the symbol and variable rows into the individual rows s, u, x, then, according to the first fundamental theorem in its usual form, an aggregate of determinants must necessarily arise. The proof that, conversely, all determinant aggregates that represent invariant formations of prescribed ground forms can be combined into explicit matrix products follows by using the second fundamental theorem (§ 6).

In order to obtain from two symbol rows S^σ, T^τ, with the help of variable rows, an invariant formation of the ground form $(S^\sigma \mid p_\sigma)(T^\tau \mid p_\tau)$ that explicitly contains all rows (the folding process), we need only form a matrix product of the third kind according to the definition given at the beginning of the paragraph:

$$(q_{n-\sigma-\lambda} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda}), \qquad (\lambda = 1, 2, \ldots, \tau, n-\sigma). \tag{7}$$

Expanded, (7.) gives:

$$\begin{aligned}&(v^{(1)} \cdots v^{(n-\sigma-\lambda)} s^{(1)} \cdots s^{(\sigma)} \,|\, \alpha^{(1)} \cdots \alpha^{(n-\tau)} y^{(1)} \cdots y^{(\tau-\lambda)})\\ &\qquad = \sum_{i,k,l} \varepsilon\, q_{k_1 \ldots k_{n-\sigma-\lambda}} S^{k_{n-\sigma-\lambda+1} \ldots k_{n-\lambda}} T_{l_1 \ldots l_{n-\tau}} p^{l_{n-\tau+1} \ldots l_{n-\lambda}},\end{aligned} \tag{8}$$

that is, an expression depending only on the rows S, T, q, p. Here $\varepsilon = \pm 1$[59] and the sum is to be understood in such a way that the indices of each row $S \ldots$ are in good order, and that for each combination $i_1, i_2, \ldots, i_{n-\lambda}$ the k as well as the l individually run through all then possible permutations of the numbers i. The number λ entering into (7.) will be called the defect of the folding in the case $\sigma > \tau$; the reason for this latter restriction appears in § 6.[60]

The determinants of the two matrices entering in (7.) can be regarded as elements of new rows $Q^{n-\lambda}, P_{n-\lambda}$ whose associated rows are Q_λ, P^λ. According to (8.), these rows Q, P can be expressed through the original S and q, respectively T and p. We indicate this by the equations

$$\begin{aligned} q_{n-\sigma-\lambda} S^\sigma &= Q^{n-\lambda} \sim Q_\lambda, &\qquad &\text{or also} \quad Q_\lambda = [q_{n-\sigma-\lambda} S^\sigma]_\lambda,\\ T_{n-\tau} p_{\tau-\lambda} &= P_{n-\lambda} \sim P^\lambda, &\qquad &\text{or also} \quad P^\lambda = [T_{n-\tau} p_{\tau-\lambda}]^\lambda. \end{aligned} \tag{9}$$

Taking (4.) into account, one then has, analogously to (6.), the equivalent fourfold symbolic representation for the matrix product (7.):

$$(q_{n-\sigma-\lambda} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda}) = (q_{n-\sigma-\lambda} S^\sigma P^\lambda) = (Q_\lambda T_{n-\tau} p_{\tau-\lambda}) = (-1)^{\lambda(n-\lambda)} (P^\lambda \,|\, Q_\lambda). \tag{10}$$

A further substitution of new rows can be performed on (7.) by setting

$$(q_{n-\sigma-\lambda} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda}) = (R^{\sigma+\lambda} \,|\, p_{\sigma+\lambda})(R^{\tau-\lambda} p_{\tau-\lambda}). \tag{11}$$

[56] For $\rho > n$ the matrix product is known to vanish; for $\rho = n$ it goes into the product of two determinants.

[57] Invariant formations of prescribed ground forms are, of course, only those aggregates of matrix products that depend on the symbol rows $S^\rho \ldots$.

[58] Cf. the Introduction.

[59] This notation is to be retained throughout.

[60] The number λ always runs up to the smaller of the two last indicated numbers.

Here too (8.) shows that the products of the elements of the rows R can be expressed uniquely through the rows S and T. The substitutions (9.) and (11.) will be used especially when a further folding with other rows is performed with (7.).

The matrix representation enables us to state invariantly the quadratic relations between the elements of a row p_ρ (from which, as is known, all the others follow), hence, according to § 1, the relations linear in the coefficients of the ground form that the rows S^ρ satisfy. These are the identities

$$(q_{n-\rho-\lambda}S^\rho \mid S_{n-\rho}p_{\rho-\lambda}) = 0, \qquad (\lambda = 1, 2, \ldots, \rho, n-\rho), \tag{12}$$

where the p and q are to be regarded as arbitrary quantities. We shall see in § 5 that the identities with odd defect number λ are consequences of those with higher defect. The relations (12.) are a direct consequence of the relations (5.); for one has

$$(q_{n-\rho-\lambda}S^\rho \mid S_{n-\rho}p_{\rho-\lambda}) = (v^{(1)} \cdots v^{(n-\rho-\lambda)}s^{(1)} \cdots s^{(\rho)} \mid \sigma^{(1)} \cdots \sigma^{(n-\rho)}y^{(1)} \cdots y^{(\rho-\lambda)}),$$

and the application of the product theorem gives, in consequence of (5.), a vanishing $(n-\lambda)$-rowed determinant. The identities (12.) say that in order to find all types of foldings it suffices to fold the product of forms linear in each variable row; or also, by § 6, that the normalized form $(S^\rho \mid p_\rho)^m$ possesses no invariant formation linear in the coefficients.[61]

The conditions (12.) are also sufficient to characterize the rows p_ρ. For the property that the individual elements of p_ρ are determinants of a matrix is an invariant property, and therefore it must be expressible invariantly; but by Theorem VI, the conditions (12.) give the greatest possible invariant restriction for the p_ρ, namely that the linear form formed from the row p_ρ as coefficients possesses no invariant formations at all.

§ 3. The symbolic identities.

We now have to transfer the second fundamental theorem of the symbolic method, namely to set up the totality of the irreducible indecomposable identities in which all rows S^ρ, p_ρ are regarded as indecomposable. We also stipulate the following. Corresponding to the meaning of symbol and variable rows, those identities are to be regarded as composite that arise when one specializes the variable row p_ρ to $p_{\rho-1}y$ and applies one of the identities of the system to the resulting expressions; while, on the other hand, identities containing k symbol rows are to count as indecomposable even if they arise from identities with fewer symbol rows by the above process. The rows S^ρ, p_ρ need not necessarily enter the identities explicitly; for their interpretation as indecomposable quantities it suffices to show that the matrix products that occur depend only on these rows. We shall show, however, that in this case, partly by applying substitution (11.), an explicit representation can always be obtained. The general identities are obtained, by means of the principle of matrix representation, from the special identities given by Pascal, which contain only rows x and u and constitute a direct transfer of those given by Study for the ternary domain:[62]

$$\sum_{i=1}^{n}(-1)^{i+1}(a^{(i)} \mid x)(a^{(1)} \cdots a^{(i-1)}a^{(i+1)} \cdots a^{(n)}u) = (-1)^{n+1}(u \mid x)(a^{(1)} \cdots a^{(n)}), \tag{13}$$

$$\sum \pm (a^{(1)} \mid x^{(1)})(a^{(2)} \mid x^{(2)}) \cdots (a^{(n)} \mid x^{(n)}) = (a^{(1)} \cdots a^{(n)})(x^{(1)} \cdots x^{(n)}), \tag{14}$$

$$\sum_{i=1}^{n}(-1)^{i+1}(u \mid a^{(i)})(a^{(1)} \cdots a^{(i-1)}a^{(i+1)} \cdots a^{(n)}x) = (-1)^{n+1}(u \mid x)(a^{(1)} \cdots a^{(n)}). \tag{15}$$

Two further identities arise from (13.) and (15.) through the substitutions $(a^{(i)} \mid x) = (a^{(i)}q_{n-1})$ and respectively $(u \mid a^{(i)}) = (p_{n-1}a^{(i)})$, and so by our interpretation they are not to be regarded as essentially different. (14.) represents the product theorem for determinants; we shall show how (13.) and (14.) (and, dualistically, (15.) and (14.)) can be replaced by the system of suitably formed product theorems for matrices. The product theorem, formed once for ρ-rowed matrices and once for $(\rho-1)$-rowed matrices, reads

$$(a^{(1)}a^{(2)} \cdots a^{(\rho)} \mid xp_{\rho-1}) = (a^{(1)}a^{(2)} \cdots a^{(\rho)} \mid xy^{(2)} \cdots y^{(\rho)}) = \sum \pm (a^{(1)} \mid x)(a^{(2)} \mid y^{(2)}) \cdots (a^{(\rho)} \mid y^{(\rho)}),$$

$$\sum \pm (a^{(2)} \mid y^{(2)}) \cdots (a^{(\rho)} \mid y^{(\rho)}) = (a^{(2)} \cdots a^{(\rho)} \mid y^{(2)} \cdots y^{(\rho)}) = (a^{(2)} \cdots a^{(\rho)}p_{\rho-1}) = (a^{(2)} \cdots a^{(\rho)}q_{n-\rho+1});$$

[61] Cf. an addendum by Study in Grassmann's works, first volume, second part, p. 510 (condition that a quantity S^ρ is simple).

[62] For formulation and literature of the problem, cf. the Introduction.

and from this follows the system of product theorems, where $\rho = 2, 3, \dots, n$:

$$\begin{aligned}(a^{(1)}a^{(2)}\cdots a^{(\rho)} \,|\, xp_{\rho-1}) &= \sum_{i=1}^{\rho}(-1)^{i+1}(a^{(i)} \,|\, x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(\rho)} \mid p_{\rho-1}) \\ &= \sum_{i=1}^{\rho}(-1)^{i+1}(a^{(i)} \,|\, x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(\rho)}q_{n-\rho+1}).\end{aligned} \tag{16}$$

For $\rho = n$, (16.) goes over into (13.); if we further specialize $p_{n-1} = y^{(2)}p_{n-2}$, $p_{n-2} = y^{(3)}p_{n-3}$, etc., and apply the identities (16.) to the expressions $(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(n)} \mid y^{(2)}p_{n-2})$, $(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(h-1)}a^{(h+1)}\cdots a^{(n)} \mid y^{(3)}p_{n-3})$, etc., then we pass from (13.) to (14.). Thus the identity (14.) is, by our definition, composite from the identities (16.); equivalently, the system (16.) is equivalent to the identities (13.) and (14.). The system (16.) satisfies one of the conditions required for the general identities: it contains the row $p_{\rho-1}$ as an indecomposable quantity and at the same time in explicit form. It is the only system satisfying this condition and only this condition. For we can form no further determinants or matrix products from the rows a, x, p; and between these no relations independent of (16.) can exist, since otherwise, by eliminating $(a^{(1)}\cdots a^{(\rho)} \mid xp_{\rho-1})$, there would arise a relation between ρ $(\rho < n)$ expressions $(a^{(i)} \mid x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}a^{(\rho)}q_{n-\rho+1})$, whereas by (13.) at least $n+1$ such expressions must enter into a relation. From analogous considerations one obtains the system equivalent to (15.) and (14.):

$$\begin{aligned}(uq_{\rho-1} \,|\, a^{(1)}\cdots a^{(\rho)}) &= \sum_{i=1}^{\rho}(-1)^{i+1}(u \,|\, a^{(i)})(q_{\rho-1} \mid a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(\rho)}) \\ &= \sum_{i=1}^{\rho}(-1)^{i+1}(u \,|\, a^{(i)})(p_{n-\rho+1}a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(\rho)}).\end{aligned} \tag{17}$$

In order to pass from (16.) to identities between arbitrary rows $A^{\rho_1}, B^{\rho_2}, \dots, x, p$, we observe that these identities must become identical with (16.) as soon as we resolve $A^{\rho_1} = a^{(1)}a^{(2)}\cdots a^{(\rho_1)}$, $B^{\rho_2} = b^{(1)}b^{(2)}\cdots b^{(\rho_2)}$, etc., and that they can also be combined into final expressions only by the operations determined by (16.) as soon as the individual expressions are of the type entering into (16.). Thus we obtain, if we also set

$$\rho = \rho_1 + \rho_2 + \cdots + \rho_k \leq n, \tag{18}$$

$$\begin{aligned}&(a^{(1)}\cdots a^{(\rho_1)}b^{(1)}\cdots b^{(\rho_2)}\cdots k^{(1)}\cdots k^{(\rho_k)} \,|\, xp_{\rho-1}) = (A^{\rho_1}B^{\rho_2}\cdots K^{\rho_k} \,|\, xp_{\rho-1}) \\ &\quad= \sum_{i=1}^{\rho_1}(-1)^{i+1}(a^{(i)} \,|\, x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(\rho_1)}B^{\rho_2}\cdots K^{\rho_k}q_{n-\rho+1}) \\ &\qquad+ (-1)^{\rho_1\rho_2}\sum_{i=1}^{\rho_2}(-1)^{i+1}(b^{(i)} \,|\, x)(b^{(1)}\cdots b^{(i-1)}b^{(i+1)}\cdots b^{(\rho_2)}A^{\rho_1}\cdots K^{\rho_k}q_{n-\rho+1}) \\ &\qquad+ \cdots \\ &\qquad+ (-1)^{(\rho_1+\cdots+\rho_{k-1})\rho_k}\sum_{i=1}^{\rho_k}(-1)^{i+1}(k^{(i)} \,|\, x)(k^{(1)}\cdots k^{(i-1)}k^{(i+1)}\cdots k^{(\rho_k)}A^{\rho_1}\cdots K^{\rho_{k-1}}q_{n-\rho+1}).\end{aligned}$$

Each individual sum (but not already a part of the sum) does in fact contain the rows $A^{\rho_1}, B^{\rho_2}, \dots$ as indecomposable quantities; for it satisfies the necessary and sufficient conditions[63]

$$\left(a^{(i+1)}\middle|\frac{\partial}{\partial a^{(i)}}\right) = 0;\ \dots\ ;\left(k^{(j+1)}\middle|\frac{\partial}{\partial k^{(j)}}\right) = 0. \quad (i = 1, 2, \dots, \rho_1 - 1;\ j = 1, 2, \dots, \rho_k - 1) \tag{19}$$

We show that it also contains all rows explicitly. For if we set $B^{\rho_2}\cdots K^{\rho_k}q_{n-\rho+1} = q_{n-\rho_1+1}$, then by (16.) and (10.) we obtain

$$\sum(-1)^{i+1}(a^{(i)} \,|\, x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(\rho_1)}q_{n-\rho+1}) = (A^{\rho_1} \,|\, xp_{\rho_1-1})$$

[63]Capelli, a. a. O. § 23, gives sufficient conditions: $\left(a^{(k)} \mid \frac{\partial}{\partial a^{(i)}}\right) = 0$, $(k > i,\ i = 1, 2, \dots, \rho_1 - 1)$. These most simply imply the necessary and sufficient conditions by application of the Poisson bracket process after Wellstein, Von den Differentialgleichungen der projektiven Invarianten, Math. Ann. 67 (1909) § 3.

$$= (A_{n-\rho_1} x p_{\rho_1 - 1}) = (-1)^{(\rho_1+1)(n+1)} (q_{n-\rho_1+1} \,|\, A_{n-\rho_1} x)$$
$$= (-1)^{(\rho_1+1)(n+1)} (B^{\rho_2} \cdots K^{\rho_k} q_{n-\rho+1} \,|\, A_{n-\rho_1} x).$$

If we compute the remaining sums analogously and subsequently set, since the rows are already characterized as distinct by the weight indices, $B^{\rho_2} = A^{\rho_2}$, $C^{\rho_3} = A^{\rho_3}, \ldots$, then we obtain the desired identities:

$$\begin{gathered}(A^{\rho_1} A^{\rho_2} \cdots A^{\rho_k} \,|\, x p_{\rho-1}) = \sum_{i=1}^{k} (-1)^{\delta_i} (A^{\rho_1} \cdots A^{\rho_{i-1}} A^{\rho_{i+1}} \cdots A^{\rho_k} q_{n-\rho+1} \,|\, A_{n-\rho_i} x), \\ \delta_i = (\rho_1 + \rho_2 + \cdots + \rho_{i-1}) \rho_i + (\rho_i + 1)(n+1),\end{gathered} \tag{20}$$

and from (17.) the dualistic identities:

$$\begin{gathered}(u q_{\sigma-1} \,|\, A_{\sigma_1} A_{\sigma_2} \cdots A_{\sigma_k}) = \sum_{i=1}^{k} (-1)^{\varepsilon_i} (u A^{n-\sigma_i} \,|\, p_{n-\sigma+1} A_{\sigma_1} \cdots A_{\sigma_{i-1}} A_{\sigma_{i+1}} \cdots A_{\sigma_k}), \\ \varepsilon_i = (\sigma_1 + \sigma_2 + \cdots + \sigma_{i-1}) \sigma_i + (\sigma_i + 1)(n+\sigma).\end{gathered} \tag{21}$$

Theorem II. With (20.) and (21.) the totality of the indecomposable identities between the rows A^{ρ_i}, x, p, respectively A_{σ_i}, u, q, is exhausted, and at the same time the totality of those indecomposable identities that admit explicit representation without carrying out substitution (11.).

The first part of the theorem follows from the considerations leading to (20.) and (21.); the second part from the impossibility of an identity between two symbol rows:

$$(q_\rho A^\sigma \,|\, B_{\rho+\sigma-\tau} p_\tau) = \varepsilon_1 (q_\rho B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} p_\tau) + \varepsilon_2 (q_\rho q_{n-\tau} \,|\, B_{\rho+\sigma-\tau} A_{n-\sigma}),$$

where $\rho \le \tau \le \sigma$ and none of the rows has weight $1 \cdot (n-1)$. (Other assumptions about ρ, τ, σ can be reduced to this one by interchanging the row designations.) This impossibility follows, for example, by expanding (8.) under the specialization $q_\rho = l M^{\rho-1}$, $q_{n-\tau} = l N^{n-\tau-1}$, if one sets $l_1 = 1$, $M^{2,3,\ldots,\rho} = 1$, $A^{\rho+1,\ldots,\rho+\sigma} = 1$, all other elements of the rows l, M, A equal to zero, while N and B are arbitrary. Since identities between arbitrary rows, by resolving a row p_τ into $y^{(1)} y^{(2)} \cdots y^{(\tau)}$, pass into identities composite from (20.), the second part of the assertion is proved as soon as (20.) can be generated from identities with only two symbol rows. In fact, from

$$(A^{\rho_1} A^{\rho_2} \,|\, x p_{\rho-1}) = \varepsilon (A^{\rho_1} q_{n-\rho+1} \,|\, A_{n-\rho_2} x) + (A^{\rho_1} \,|\, x p_{\rho_1-1}), \quad \text{where } p_{\rho_1-1} \sim A^{\rho_2} q_{n-\rho+1}, \tag{20a}$$

by specializing $A^{\rho_1} = B^{\sigma_1} B^{\sigma_2}$, that is, by

$$(B^{\sigma_1} B^{\sigma_2} \,|\, x p_{\rho_1-1}) = \varepsilon (B^{\sigma_1} q_{n-\rho_1+1} \,|\, B_{n-\sigma_2} x) + (B^{\sigma_1} \,|\, x p_{\sigma_1-1}), \quad \text{where } p_{\sigma_1-1} \sim B^{\sigma_2} q_{n-\rho_1+1},$$

one obtains an identity (20.) in three symbol rows; and by further specializations of $B^{\sigma_1} = C^{\tau_1} C^{\tau_2}$, etc., the general identities (20.). The nonexistence of the explicit representation of an identity between two symbol rows and two arbitrary variable rows thus implies the nonexistence in the case of arbitrarily many symbol rows, provided that among the variable rows no row x or u is contained.

§ 4. The decomposition identities.

We now consider the most general case of identities among arbitrary symbol and variable rows in whose final expressions, without performing substitution (11.), a row p_τ occurs resolved into its individual rows $y^{(1)}\cdots y^{(\tau)}$; we call these identities "decomposition identities." Following the remark at the end of the preceding paragraph, we first determine them for the case of only two symbol rows, that is, we expand the expression $(q_\rho A^\sigma \mid B_{\rho+\sigma-\tau}p_\tau)$. We set:

$$q_{\rho-\alpha}^{(i_1\ldots i_\alpha)} \sim p_{n-\rho}\, y^{(i_1)}\cdots y^{(i_\alpha)}, \qquad p_{\tau-\alpha}^{(i_1\ldots i_\alpha)} = y^{(i_{\alpha+1})}\cdots y^{(i_\tau)}, \qquad p_\tau = y^{(1)}\cdots y^{(\tau)}. \tag{22}$$

With respect to the numbers ρ, σ, τ four cases must be distinguished:

$$\begin{aligned}
&1)\quad \rho+\sigma\leq n,\quad \tau\leq\rho,\quad \tau\leq\sigma;\\
&2)\quad \rho+\sigma\leq n,\quad \tau\geq\rho,\quad \tau\leq\sigma;\\
&3)\quad \rho+\sigma\leq n,\quad \tau\leq\rho,\quad \tau>\sigma;\\
&4)\quad \rho+\sigma\leq n,\quad \tau\geq\rho,\quad \tau\geq\sigma.
\end{aligned}$$

We shall single out the first case, and then briefly characterize the others from it. By (20.), (10.) and (9.) we obtain, when we decompose the row $y^{(1)}y^{(2)}\cdots y^{(\tau)}B_{\rho+\sigma-\tau}$ into $y^{(1)}$ and $y^{(2)}\cdots y^{(\tau)}B_{\rho+\sigma-\tau}$:

$$\begin{aligned}
(q_\rho A^\sigma \mid y^{(1)}y^{(2)}\cdots y^{(\tau)}B_{\rho+\sigma-\tau}) &= (q_\rho^{(1)}A^\sigma \mid y^{(2)}\cdots y^{(\tau)}B_{\rho+\sigma-\tau})\\
&\quad + (-1)^\rho (q_\rho[A_{n-\sigma}y^{(1)}]^{\sigma-1} \mid y^{(2)}\cdots y^{(\tau)}B_{\rho+\sigma-\tau}).
\end{aligned} \tag{23}$$

The last term in (23.) now has exactly the form of the original expression–only σ is replaced by $\sigma-1$, and τ by $\tau-1$–and therefore again admits equation (23.). Thus, since $\tau\leq\sigma$, we can apply operation (23.) τ times and, taking account of (10.), obtain the relation:

$$\begin{aligned}
(q_\rho A^\sigma \mid p_\tau B_{\rho+\sigma-\tau}) &= (-1)^{\rho\sigma+(\sigma+\tau)(n+1)}(q_\rho B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma}p_\tau)\\
&\quad + \sum_{i_1=1}^{\tau}(-1)^{\rho(i_1-1)}(q_{\rho-1}^{(i_1)}[A_{n-\sigma}y^{(1)}\cdots y^{(i_1-1)}]^{\sigma-i_1+1} \mid y^{(i_1+1)}\cdots y^{(\tau)}B_{\rho+\sigma-\tau}).
\end{aligned} \tag{24}$$

Every term under the summation sign is again of the type of the original expression; ρ is replaced by $\rho-1$, σ by $\sigma-i_1+1$, and τ by $\tau-i_1$, so that condition 1) is even more strongly satisfied. Since $\tau\leq\rho$, we can therefore apply operation (24.) τ times and obtain the desired decomposition identity:

$$\begin{aligned}
(q_\rho A^\sigma \mid p_\tau B_{\rho+\sigma-\tau}) &= \varepsilon_{i_0}(q_\rho B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma}p_\tau)\\
&\quad + \sum_{i_1=1}^{\tau}\varepsilon_{i_1}(q_{\rho-1}^{(i_1)}B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma}p_{\tau-1}^{(i_1)})\\
&\quad + \sum_{i_2=i_1+1}^{\tau}\varepsilon_{i_2}(q_{\rho-2}^{(i_1i_2)}B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma}p_{\tau-2}^{(i_1i_2)}) + \cdots\\
&\quad + \varepsilon_{i_\tau}(q_{\rho-\tau}^{(i_1\ldots i_\tau)}B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma})\\
&= \sum_{\alpha=0}^{\tau}\sum_{i_\alpha=i_{\alpha-1}+1}^{\tau}\varepsilon_{i_\alpha}(q_{\rho-\alpha}^{(i_1\ldots i_\alpha)}B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma}p_{\tau-\alpha}^{(i_1\ldots i_\alpha)}).
\end{aligned} \tag{25}$$

For the sign factor ε one obtains from (24.):

$$\begin{aligned}
\varepsilon_{i_\alpha} = (-1)^{\delta_{i_\alpha}}, \qquad \delta_{i_\alpha} &= \sum_{\beta=1}^{\alpha}(\rho-\beta+1)(i_{\beta-1}+i_\beta+1)\\
&\quad + (\rho-\alpha)(\sigma+i_\alpha+\alpha) + (\sigma+\tau+\alpha)(n+1),
\end{aligned} \tag{26}$$

where $i_0=0$ is to be put. The summation sign in (25.) is to be understood as follows: to each combination $i_1i_2\ldots i_{\alpha-1}$, where $i_1<i_2<\cdots<i_{\alpha-1}$, all values i_α for which $i_{\alpha-1}<i_\alpha\leq\tau$ are adjoined. Taking (26.) into account, one easily verifies that the individual sums in (25.) satisfy the differential equations analogous to (19.):

$$\left(\frac{\partial}{\partial y^{(i)}}\middle| y^{(i+1)}\right) = 0, \qquad (i=1,2,\ldots,\tau-1), \tag{27}$$

and hence depend only on the row p_τ. Thus the decomposition identity is an indecomposable identity among the rows A, B, q, p; we shall return to its explicit representation in the next paragraph.

In case 2), operation (23.) can likewise be applied τ times, but operation (24.) stops after the ρ-th application. Thus in the final expression of (25.) the first summation sign must be replaced by $\sum_{\alpha=0}^{\rho}$. In cases 3) and 4), operation (23.) can be carried out only σ times; in place of (24.) one then has the relation:

$$(q_\rho A^\sigma \,|\, p_\tau B_{\rho+\sigma-\tau}) = (-1)^{\rho\sigma}(q_\rho^{(\sigma+1\ldots\tau)} B^{n-\rho+\tau-\sigma} \,|\, A_{n-\sigma} p_{\tau-\sigma}^{(\sigma+1\ldots\tau)}) \\ + \sum_{i_1=1}^{\sigma} (-1)^{\rho(i_1-1)}(q_{\rho-1}^{(i_1)}[A_{n-\sigma} y^{(1)} \cdots y^{(i_1-1)}]^{\sigma-i_1+1} \,|\, y^{(i_1+1)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau}); \tag{28}$$

repeating (28.) $\tau-\sigma$ times, where the summation sign becomes $\sum_{i_\beta=i_{\beta-1}+1}^{\sigma+\beta-1}$–as the $(\sigma-i_1+1)$-fold application of (23.) to the terms under the summation sign in (28.) shows–gives all terms of the individual sum in (25.) corresponding to the index $\alpha = \tau-\sigma$, with the corresponding sign. Then cases 3) and 4) pass over into cases 1) and 2), for the numbers corresponding to σ, τ become $\sigma' = \sigma - i_{\tau-\sigma} + \tau - \sigma$, $\tau' = \tau - i_{\tau-\sigma} = \sigma'$; and as the procedure continues, $\tau' < \sigma'$. Thus in the final expression of (25.) the first summation sign must be replaced by $\sum_{\alpha=\tau-\sigma}^{\tau,\rho}$.

Analogously, from (21.) one obtains the dual decomposition identity, in which a row q_ρ is resolved into the individual rows $v^{(1)}, \ldots, v^{(\rho)}$. If we put

$$\dot{p}_{\tau-\beta}^{(i_1\ldots i_\beta)} \sim v^{(i_1)} \cdots v^{(i_\beta)} q_{n-\tau}; \qquad \dot{q}_{\rho-\beta}^{(i_1\ldots i_\beta)} = v^{(i_{\beta+1})} \cdots v^{(i_\rho)}; \qquad q_\rho = v^{(1)} \cdots v^{(\rho)}, \tag{29}$$

then

$$(q_\rho A^\sigma \,|\, p_\tau B_{\rho+\sigma-\tau}) = \sum_{\beta=\max(0,\tau-\sigma)}^{\min(\tau,\rho)} \sum_{i_\beta=i_{\beta-1}+1}^{\rho} \varepsilon(\dot{q}_{\rho-\beta}^{(i_1\ldots i_\beta)} B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} \dot{p}_{\tau-\beta}^{(i_1\ldots i_\beta)}), \tag{30}$$

where the summation index β runs from the larger of the lower values to the smaller of the upper values. All terms of an individual sum in (25.) and (30.) are polars of a single form, produced by the polar processes

$$\left(\frac{\partial}{\partial p_{\tau-\alpha}} \middle| p_{\tau-\alpha}^{(i_1\ldots i_\alpha)}\right), \qquad \left(q_{\rho-\alpha}^{(i_1\ldots i_\alpha)} \middle| \frac{\partial}{\partial q_{\rho-\alpha}}\right).$$

To express this, we denote by Π an aggregate of such polar operations, multiplied by constants, and obtain, in place of (25.) and (30.), the relation

$$(q_\rho A^\sigma \,|\, p_\tau B_{\rho+\sigma-\tau}) = \sum_{\alpha=\max(0,\tau-\sigma)}^{\min(\tau,\rho)} \Pi(q_{\rho-\alpha} B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} p_{\tau-\alpha}). \tag{31}$$

§ 5. The decomposition identities in explicit form.

If in (25), in case 4), we specialize τ to $\sigma+\rho$, we obtain

$$(q_\rho A^\sigma \,|\, y^{(1)} \cdots y^{(\rho+\sigma)}) = \sum_{1 \le i_1 < \cdots < i_\rho \le \rho+\sigma} \varepsilon_{i_\rho}(q_\rho \,|\, y^{(i_1)} \cdots y^{(i_\rho)})(A^\sigma \,|\, y^{(i_{\rho+1})} \cdots y^{(i_{\rho+\sigma})}), \tag{32}$$

where $\varepsilon_{i_\rho} = (-1)^{\delta_{i_\rho}}$ and $\delta_{i_\rho} = \rho(\rho+1)/2 + i_1 + \cdots + i_\rho$. This is a more general form of the product theorem for matrices than (16.) and (17.), and follows by Laplace expansion of the determinant of the product matrix

$$\sum \pm (v^{(1)} y^{(1)}) \cdots (v^{(\rho)} y^{(\rho)})(a^{(1)} y^{(\rho+1)}) \cdots (a^{(\sigma)} y^{(\rho+\sigma)}).$$

Thus the more general product theorem is composed out of the identities (20.) and (21.), respectively, and the selection of the special form (16.), (17.) is thereby explained.

Relation (32.) now gives the means for explicitly representing the decomposition identities. For this purpose we regard the forms occurring in (31.) as ground forms; that is, we perform substitution (11.), which changes nothing in the explicit representation in the rows A, B, since by (8.) the newly introduced rows R can be expressed through the rows A, B themselves and not through their component rows $a^{(1)} a^{(2)} \ldots$, $b^{(1)} b^{(2)} \ldots$. Thus let

$$(q_{\rho-\alpha} B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} p_{\tau-\alpha}) = (R_{\rho-\alpha} p_{n-\rho+\alpha} \,|\, R^{\tau-\alpha} p_{\tau-\alpha}). \tag{33}$$

Then, for the individual sum of (25.) corresponding to the index α, we get

$$\begin{aligned}&\sum \varepsilon_{i_\alpha}(R_{\rho-\alpha}p_{n-\rho}y^{(i_1)}\cdots y^{(i_\alpha)}\,|\,R^{\tau-\alpha}y^{(i_{\alpha+1})}\cdots y^{(i_\tau)})\\ &\quad=\sum \varepsilon_{i_\alpha}([R_{\rho-\alpha}p_{n-\rho}]^\alpha y^{(i_1)}\cdots y^{(i_\alpha)}\,|\,R^{\tau-\alpha}y^{(i_{\alpha+1})}\cdots y^{(i_\tau)})\\ &\quad=\varepsilon([R_{\rho-\alpha}p_{n-\rho}]^\alpha R^{\tau-\alpha}\,|\,p_\tau)=\varepsilon(R^{\tau-\alpha}q_{n-\tau}\,|\,R_{\rho-\alpha}p_{n-\rho}).\end{aligned}\tag{34}$$

Thus an explicit final expression is in fact obtained, and the symmetric form in which q_ρ and p_τ occur shows that (30.) leads to the same final expression, which is therefore independent of the original decomposition. There is a difference between (25.) and (30.) only as long as the rows y and v have independent meaning, so that the decomposition identity passes, by our definition, into a decomposable identity. To arrive at identities with more symbol rows, one need only, according to § 3 (end), resolve a row q_ρ into $q_{\rho_1}C^{\sigma_1}$, or p_τ into $p_{\tau_1}C_{\tau-\tau_1}$. The resulting expressions

$$(q_{\rho_1}C^{\sigma_1}\,|\,[R^{\tau-\alpha}q_{n-\tau}]_\lambda R_{\rho+\sigma_1-\alpha-\lambda}),\qquad ([R_{\rho-\alpha}p_{n-\rho}]^\lambda R^{\tau-\alpha}\,|\,p_{\tau_1}C_{\tau-\tau_1})\tag{35}$$

are exactly of the type that occurs for two symbol rows, and therefore again admit explicit representation after a substitution corresponding to (33.). Repetition therefore gives explicit representation for arbitrarily many rows. It may also be noted that the first and last sums in (25.) and (30.), respectively, give the explicit representation without applying (33), solely by formula (32), or else reduce altogether to a single term containing all rows explicitly. The relation obtained from (32.) by resolving q_ρ into $q_{\rho_1}C^{\sigma_1}$ and so forth can be regarded as the product theorem in its most general form.

In summary we shall show:

Theorem III: With the identities

$$(q_\rho A^\sigma\,|\,p_\tau B_{\rho+\sigma-\tau})=\sum_{\alpha=\max(0,\tau-\sigma)}^{\min(\tau,\rho)}\varepsilon(R^{\tau-\alpha}q_{n-\tau}\,|\,R_{\rho-\alpha}p_{n-\rho}),\tag{36}$$

where the rows R are determined by (33), together with those obtained from them by resolving q_ρ, p_τ into $q_{\rho_1}C^{\sigma_1}$, etc., the totality of indecomposable identities is exhausted.

Indeed, the resulting identities are the most general of their kind, since the individual rows are no longer subject to any restriction as to weight or number, as was still the case in (20.) and (21.). Nor do any further identities among these rows exist; for when a row p_τ is resolved into $y^{(1)}\cdots y^{(\tau)}$, the most general identities are composed out of (20.), hence by § 3 (end) out of (20a.); and the composition discussed following (16.) is precisely the one carried out in § 4. The transition to arbitrarily many rows is supplied, as the discussion of (20a.) shows, by applying always the same operation to the expressions that arise from resolving q_ρ into $q_{\rho_1}C^{\sigma_1}$, and so on. It also follows that the direct transition from (20.) instead of from (20a.) gives nothing new. This is easily verified by calculation, by specializing once a row q_ρ in (25.), and another time carrying out the transition from (20.) by the method of § 4. In both cases the same sums arise, which pass by application of the general product theorem (see above) into the identities obtained from (36.). The identities (20.), (21.) correspond to the special cases $\tau = 1$, $\rho = 1$; only two individual sums then occur, and hence, in accordance with an earlier remark, explicit representation without the substitution (33.), agreeing with what was found before.

At the end let us briefly mention two special cases of the decomposition identity. Put in (31): $\tau=\sigma$, $\rho=n-\sigma$, and denote the row p_τ by $p'_\sigma\sim q'_{n-\sigma}$. Then

$$(A^\sigma\,|\,p_\sigma)(B^\sigma\,|\,p'_\sigma)-(A^\sigma\,|\,p'_\sigma)(B^\sigma\,|\,p_\sigma)=\sum_{\alpha=1}^{\min(\sigma,n-\sigma)}\Pi(q_{n-\sigma-\alpha}B^\sigma\,|\,A_{n-\sigma}p_{\sigma-\alpha}).\tag{37}$$

This is Clebsch's formula[64] for expanding a form with two cogredient rows p_σ, p'_σ into elementary covariants. The elementary covariants belonging to a form $(A^\sigma\,|\,p_\sigma)^m(B^\sigma\,|\,p'_\sigma)^n$ are then the following:

$$\left\{\sum_{\alpha=1}^{\min(\sigma,n-\sigma)}\varepsilon(q_{n-\sigma-\alpha}B^\sigma\,|\,A_{n-\sigma}p_{\sigma-\alpha})\right\}^\rho(A^\sigma\,|\,p_\sigma)^{m-\rho}(B^\sigma\,|\,p'_\sigma)^{n-\rho},\qquad(\rho=0,1,\dots,m,n).\tag{38}$$

[64] Clebsch, loc. cit., pp. 25–32. Clebsch proves that the individual terms on the right-hand sides depend only on the rows A, B; the explicit representation would be obtained by applying (32.) to his expressions Φ_h.

They arise, as we shall briefly say, from the ground form by "cogredient folding of defect ρ" (cf. § 2).

Second, put $\tau = \sigma - \lambda$, $\rho = n - \sigma - \lambda$, $B^\sigma = A^\sigma$. Then

$$
\begin{aligned}
(q_{n-\sigma-\lambda}A^\sigma \mid A_{n-\sigma}p_{\sigma-\lambda}) &= (-1)^\lambda (q_{n-\sigma-\lambda}A^\sigma \mid A_{n-\sigma}p_{\sigma-\lambda}) \\
&\quad + (-1)^{(n-\sigma)(\sigma-\lambda)} \sum_{\alpha=1}^{\min(\sigma-\lambda, n-\sigma-\lambda)} \Pi(q_{n-\sigma-\lambda-\alpha}A^\sigma \mid A_{n-\sigma}p_{\sigma-\lambda}).
\end{aligned}
\tag{39}
$$

Thus, as noted in § 2, the conditions (12.) characterizing the symbol rows for odd defect λ are consequences of the conditions of higher defect. For a linear form $(A^\sigma \mid p_\sigma) = (B^\sigma \mid p_\sigma)$, (39.) implies that its irreducible invariant formations that are quadratic in the coefficients reduce to ones of even defect; this agrees with the known fact that a linear form in the binary and ternary domains has no invariant formations.

It should be noted that the general identities were originally derived under the assumption that the individual rows satisfy conditions (12.). Since, however, these individual rows enter the identities only linearly, the latter also hold for the more general rows that do not satisfy conditions (12.).

§ 6. Explicit representation of the invariant formations. Folding and differentiation processes.

The results of the preceding paragraphs allow us to complete the theorem stated in § 2 on symbolic representability (the first fundamental theorem) by adding explicit representability; namely, we show:

Theorem IV: "All invariant formations of arbitrary ground forms can be represented explicitly by matrix products; that is, aside from variable rows p_ρ, only the rows $S^\sigma, \ldots, T^\tau$ of the original forms, or the rows P, Q, R derived from them by substitution (9.) or (11.), enter the final expressions."

For the proof, suppose that an invariant formation depends, besides variable rows, on the rows $S^\sigma, \ldots, T^\tau$; and suppose that these rows have been resolved sufficiently far into individual rows $S^\sigma = S^{\sigma_1} \cdots S^{\sigma_i}, \ldots, T^\tau = T^{\tau_1} \cdots T^{\tau_k}$ to make representation by determinants possible. Let the rows be ordered in a sequence $S^\sigma, \ldots, T^\tau$ that is arbitrary in itself but fixed once and for all. We pick out an aggregate of determinants that depends on the first total row $S^\sigma = S^{\sigma_1} \cdots S^{\sigma_i}$, not merely on individual component rows. This dependence, however, is by § 3–§ 5 a consequence of the general identities. If we now resolve a variable row p_ρ into the rows y, v, or one of the later rows T into the individual rows t, respectively τ, then the most general identities are composed out of (20.) and (21.). But these latter identities always lead to an expression containing the row $S^\sigma = S^{\sigma_1} \cdots S^{\sigma_i}$ explicitly, namely the left-hand side of (20.) and (21.); for by (19.) one verifies that only the complete sum on the right–not a part of it corresponding to a term of (20a.)–has the property of depending only on S^σ. As the procedure continues, all rows that have once entered explicitly remain explicit; application of the identities can at most require the combining of several rows into a single one, i.e. substitution (9.). If finally only a single row T^τ remains to be brought to explicit form, two cases are possible. There may still occur a free variable row, that is, one not connected with other rows by substitution (9.); resolving it into component rows y or v completes the procedure and gives, as in (31.), polars of one or more forms containing all rows explicitly. If no free variable row remains, then from the manner in which they arose we recognize, in the totality of the expressions that contain the individual rows t or τ but depend on the row T^τ, the terms of a decomposition identity; by § 5, however, these always admit explicit representation in all rows after applying substitution (11.). This proves the theorem in full generality. It should also be noted that when only two symbol rows occur, substitution (11.) never appears. For, since $\sigma, n-\sigma, \tau, n-\tau < n$, no variable rows can enter only when the two symbol rows are united in a single determinant and thus occur explicitly; but as soon as variable rows enter, (34.) and (33.) show that substitution (11.) becomes unnecessary.

It follows from what has just been said that, in order to generate all invariant formations, it suffices to unite the symbol rows, after adjoining variable rows, into all possible matrix products. The most general matrix product is now of the form

$$
(P^{\mu_1} \cdots P^{\mu_i} \mid Q_{\nu_1} \cdots Q_{\nu_k}), \qquad \sum \mu = \sum \nu,
\tag{40}
$$

where the P and Q may be rows of the original forms, or may have arisen from the original rows by a sequence of substitutions (9.) or (11.), or finally may be variable rows. But the expressions (40.) can be generated by successively uniting two symbol rows at a time into a matrix product, with the help of variable rows; for from

$$
(q_{n-\mu-\lambda}P^\mu \mid Q_{\nu_1}p_{n-\mu-\nu_1-\lambda}) = (R_\lambda Q_{\nu_1} \mid p_{n-\mu-\nu_1-\lambda})
$$

one obtains, by uniting with the row Q_{ν_2}, an expression

$$([R_\lambda Q_{\nu_1}]^{n-\lambda-\nu_1}\,|\,Q_{\nu_2}p_{n-\mu-\nu_1-\nu_2-\lambda}) = (q_{n-\mu-\lambda}P^\mu\,|\,Q_{\nu_1}Q_{\nu_2}p_{n-\mu-\nu_1-\nu_2-\lambda}),$$

and hence, by continuing the procedure, an expression (40.). If P and Q are not rows of the original forms, then (9.) and (11.), in conjunction with what was just shown, imply that they too arose through a sequence of unions of two symbol rows at a time. Thus:

Theorem V: The process for generating all invariant formations, the folding process, consists in successively uniting two symbol rows at a time into a matrix product, with the help of variable rows.

From two symbol rows S^σ, T^τ, where $\sigma \geq \tau$, the following matrix products can be formed:

$$(q_{n-\sigma-\lambda}S^\sigma\,|\,T_{n-\tau}p_{\tau-\lambda}), \qquad (\lambda = 1, 2, \ldots, \min(\tau, n-\sigma)), \tag{41}$$

$$(q_{n-\tau-\mu}T^\tau\,|\,S_{n-\sigma}p_{\sigma-\mu}), \qquad (\mu = \sigma - \tau + \lambda), \tag{42}$$

$$(q_{n-\tau-\nu}T^\tau\,|\,S_{n-\sigma}p_{\sigma-\nu}), \qquad (\nu = 1, 2, \ldots, \sigma - \tau). \tag{43}$$

From (31.), however, the following values are obtained for (42.) and (43.):

$$(q_{n-\sigma-\lambda}S^\sigma\,|\,T_{n-\tau}p_{\tau-\lambda}) + \sum_\alpha \Pi(q_{n-\sigma-\lambda-\alpha}S^\sigma\,|\,T_{n-\tau}p_{\tau-\lambda-\alpha}), \tag{42a}$$

$$\Pi(q_{n-\sigma}S^\sigma)(T_{n-\tau}p_\tau) + \sum_\beta \Pi(q_{n-\sigma-\beta}S^\sigma\,|\,T_{n-\tau}p_{\tau-\beta}). \tag{43a}$$

Thus the forms (42.) can be expressed linearly by (41.) and polars of (41.), and the forms (43.) by polars of the ground form and of the forms (41.). Likewise (31.) gives the linear dependence of the forms (41.) on (42.) and polars of (42.). Hence, according to Clebsch's definition,[65] the forms (42.) are equivalent to the forms (41.), while (43.) leads back to the ground form. The generating process is therefore given with equal right by (41.) or (42.); we shall define the process (41.), which is simpler with respect to the number λ, as the folding process. The number λ, the defect of the folding (cf. § 2), gives the number of rows missing from the determinant product, namely in that matrix product which, for the given folding, contains the maximal number of rows. Since λ runs only up to the smaller of the two values $\tau, n-\sigma$, where $\sigma \geq \tau$, we have

$$\lambda \leq \frac{n}{2},\ n \text{ even}; \qquad \lambda \leq \frac{n-1}{2},\ n \text{ odd}. \tag{44}$$

The reduction of (42.) and (43.) to (41.) remains valid even when, through further folding, the rows p, q have passed into symbol rows; for by (36.) one has

$$(A^{n-\tau-\varkappa}T^\tau\,|\,S_{n-\sigma}B_{\sigma-\varkappa}) = \sum_{\alpha=\max(0,\varkappa-\sigma+\tau)}^{\min(\tau,n-\sigma)} \varepsilon(R^{\tau-\alpha}B^{n-\sigma+\varkappa}\,|\,A_{\tau+\varkappa}R_{n-\sigma-\alpha}),$$

where the rows R have arisen from S and T according to (33.). Thus these forms containing symbol rows only are obtained by folding $(q_\tau A^{n-\tau-\varkappa}\,|\,B_{\sigma-\varkappa}p_{n-\sigma})$ or $(q_{\tau-\alpha}B^{n-\sigma+\varkappa}\,|\,A_{\tau+\varkappa}p_{n-\sigma-\alpha})$–according as $\sigma-\tau \gtreqless 2\varkappa$–with the ground form and the forms (41.).

From the symbolic folding process one obtains, in the usual way, the invariant differentiation processes if one merely replaces the symbol rows $S^\sigma, T_{n-\tau}$ by the rows of differentiation symbols $\frac{\partial}{\partial p_\sigma}$, $\frac{\partial}{\partial q_{n-\tau}}$; as is well known, the product $\frac{\partial}{\partial p_{i_1\ldots i_\sigma}}\frac{\partial}{\partial q_{j_1\ldots j_{n-\tau}}}$ then has the real meaning $\frac{\partial^2}{\partial p_{i_1\ldots i_\sigma}\partial q_{j_1\ldots j_{n-\tau}}}$. Since the rows $\frac{\partial}{\partial p_\sigma}$ and $\frac{\partial}{\partial q_{n-\tau}}$ are cogredient with the rows $S^\sigma, T_{n-\tau}$, they satisfy the same identities as these, in particular also the ones existing between (42.) and (42a.), and between (43.) and (43a.). In summary we obtain:

Theorem VI: All invariant formations of arbitrary ground forms arise from these by repeated application of the symbolic folding process (41.), or also of the invariant differentiation processes

$$\left(q_{n-\sigma-\lambda}\frac{\partial}{\partial p_\sigma}\middle|\frac{\partial}{\partial q_{n-\tau}}p_{\tau-\lambda}\right), \qquad (\sigma \geq \tau,\ \lambda = 1, 2, \ldots, \min(\tau, n-\sigma)). \tag{45}$$

It remains to note that in every folding (41.) the total weight of the form, that is, the sum of the weights of the individual variable rows (cf. § 1), decreases by the positive number $2(\lambda^2 + \lambda(\sigma-\tau))$. This implies,

[65] Clebsch, loc. cit., § 7.

independently of the general finiteness proof, the finiteness of the form row, that is, of the totality of invariant formations of a given ground form that are linear in the coefficients.[66]

It should further be noted that Theorem VI still holds for forms that do not satisfy conditions (12.). For each factor in (3), $(S^\rho \,|\, p_\rho)^{m_\rho}$, can be replaced by a product of m_ρ linear factors $(A^\rho \,|\, p_\rho)(B^\rho \,|\, p_\rho)\cdots(M^\rho \,|\, p_\rho)$; the invariant formations thereby pass into those of a system of linear forms, which need only afterwards be set equal to one another. For every single linear form, however, conditions (12.)–which, when differential symbols are introduced, contain only second-order differential quotients–are always satisfied.

§ 7. Expansion of the general forms according to normal forms.

In what follows (§ 8) we shall derive a principal property of the folding process (41.), namely its composability from fundamental foldings. For this we need the transfer to the n-ary domain of an important theorem given by Mertens[67] in the quaternary domain. This theorem says that any finite system of forms can be replaced by an equivalent system of normal forms. By a "normal form" we mean–generalizing the binary and ternary definition[68]–a form all of whose invariant formations linear in the coefficients vanish identically, with the exception of the ground form itself. Thus by Theorem VI (§ 6) a normal form φ satisfies the system of differential equations

$$\left(q_{n-\sigma-\lambda}\frac{\partial}{\partial p_\sigma}\,\middle|\,\frac{\partial}{\partial q_{n-\tau}}p_{\tau-\lambda}\right)\varphi = 0, \qquad (\sigma \geq \tau,\ \lambda = 1, 2, \ldots, \min(\tau, n-\sigma)), \tag{46}$$

where the rows $q_{n-\sigma-\lambda}, p_{\tau-\lambda}$ are to be regarded as arbitrary quantities. In the quaternary domain–taking (39.) into account for $\sigma = \tau = 2$–these differential equations agree completely with the ten differential equations given by Mertens without introduction of the arbitrary rows p, q (loc. cit., formula 28). Their knowledge, together with Theorem VI, allows us to transfer his proof almost word for word to n variables. The only thing to add is the verification, obtained by the decomposition identity (30.), that the constructed forms are normal forms; in the quaternary domain this verification still follows without further transformation. It is therefore enough to give a short exposition of the proof, and for simplicity in the case relevant below, namely that of a single ground form f and its form row (cf. § 6, end); for invariant formations of arbitrary degree in the coefficients of arbitrarily many ground forms, the proof remains the same.

The basic idea of the proof is to replace, by introducing the transformed coefficients, a form with $n-1$ rows p_ρ by forms with n rows ξ cogredient to the x, namely the rows of the transformation quantities. From these forms the sought normal forms are obtained directly by invariant differentiation processes. The expansion of the general forms according to normal forms is mediated by a series expansion for forms with ρ ($\rho < n$) cogredient rows ξ; invariant differentiation processes lead back to the forms in the original rows p_ρ.

Let $\xi^{(1)}, \ldots, \xi^{(n)}$ be the rows of transformation quantities, so that

$$\begin{gathered} x_i = \xi_i^{(1)}x_1' + \xi_i^{(2)}x_2' + \cdots + \xi_i^{(n)}x_n', \\ p_{i_1\ldots i_\rho} = \sum_j (\xi_{i_1}^{(j_1)}\cdots\xi_{i_\rho}^{(j_\rho)})p'_{j_1\ldots j_\rho}. \end{gathered} \tag{47}$$

The transformed coefficients of the forms $f, \varphi, \ldots$ will be denoted by $(f), (\varphi), \ldots$, and let

$$f = (S^1 \,|\, p_1)^{m_1}(S^2 \,|\, p_2)^{m_2}\cdots(S^{n-1} \,|\, p_{n-1})^{m_{n-1}}.$$

Then

$$\begin{gathered} (f) = (S^1 \,|\, \xi^{(1)})^{m_{11}}\cdots(S^1 \,|\, \xi^{(n)})^{m_{1n}}(S^2 \,|\, \xi^{(1)}\xi^{(2)})^{m_{21}}\cdots(S^{n-1} \,|\, \xi^{(2)}\cdots\xi^{(n)})^{m_{n-1,n}} \\ \cdot(s^{(1)} \,|\, \xi^{(1)})^{k_1}(s^{(2)} \,|\, \xi^{(2)})^{k_2}\cdots(s^{(n)} \,|\, \xi^{(n)})^{k_n}, \end{gathered} \tag{48}$$

where $m_{11} + \cdots + m_{1n} = m_1$, and so on. From each coefficient (f) for which

$$k_1 \geq k_2 \geq k_3 \geq \cdots \geq k_n, \tag{49}$$

the differentiation process exhausting exactly the rows ξ,

$$\left(\frac{\partial}{\partial\xi^{(1)}}\middle|p_1\right)^{k_1-k_2}\left(\frac{\partial}{\partial\xi^{(1)}}\frac{\partial}{\partial\xi^{(2)}}\middle|p_2\right)^{k_2-k_3}\cdots\left(\frac{\partial}{\partial\xi^{(1)}}\cdots\frac{\partial}{\partial\xi^{(n-1)}}\middle|p_{n-1}\right)^{k_{n-1}-k_n}\left(\frac{\partial}{\partial\xi^{(1)}}\cdots\frac{\partial}{\partial\xi^{(n)}}\right)^{k_n} \tag{50}$$

[66] Cf. Clebsch, loc. cit., § 11 and 12: finiteness of the system of elementary covariants.

[67] F. Mertens, *Über invariante Gebilde quaternärer Formen* (see the Introduction), cited as "Mertens."

[68] Study, loc. cit., p. 54.

gives the form

$$\varphi = \big(s^{(1)} \mid p_1\big)^{k_1-k_2}\big(s^{(1)}s^{(2)} \mid p_2\big)^{k_2-k_3}\cdots\big(s^{(1)}\cdots s^{(n-1)} \mid p_{n-1}\big)^{k_{n-1}-k_n}\big(s^{(1)}s^{(2)}\cdots s^{(n)}\big)^{k_n}. \tag{51}$$

The forms φ are the sought normal forms. Indeed, they have the following properties defining normal forms.

1. They are invariant formations of the ground form f that are linear in the coefficients. This follows from the fact that they arise from f by the invariant differentiation processes $\left(\frac{\partial}{\partial p_\rho}\middle|\xi^{(i_1)}\xi^{(i_2)}\cdots\xi^{(i_\rho)}\right)$ and (50.). Thus (by § 6, end) they can be expressed linearly by polars of the finite form row of f. These polars are obtained by regarding the variables p_ρ entering φ as coefficients of a form decomposed into linear forms with the variable rows p_ρ:

$$H = (q_{n-1} \mid p_{n-1})^{k_1-k_2}(q_{n-2} \mid p_{n-2})^{k_2-k_3}\cdots(q_1 \mid p_1)^{k_{n-1}-k_n}. \tag{52}$$

The simultaneous invariants of the form rows f and H give all relevant polars, since the latter must be of the same dimension in the individual rows p_ρ as φ itself.

2. They satisfy the differential equations (46.). If the differential process (45.) is applied to φ, then

$$\varphi' = \big(q_{n-\sigma-\lambda}s^{(1)}\cdots s^{(\sigma)} \mid [s^{(1)}\cdots s^{(\tau)}]_{n-\tau}p_{\tau-\lambda}\big), \qquad (\sigma > \tau).$$

From the decomposition identity (30.) it follows, if the rows $q_\sigma = s^{(1)}\cdots s^{(\sigma)}$, $p_{n-\tau} = [s^{(1)}\cdots s^{(\tau)}]_{n-\tau}$ formed out of the rows s are identified with the q_ρ, p_τ occurring there, that

$$\varphi' = \sum_{\beta=\sigma-\tau+\lambda}^{n-\tau,\sigma}\sum_{i_\beta}\varepsilon\,(q_{\sigma-\beta}^{(i_1\ldots i_\beta)}q_{n-\tau+\lambda}\,|\,p_{\sigma+\lambda}p_{n-\tau-\beta}^{(i_1\ldots i_\beta)}).$$

Here

$$p_{n-\tau-\beta}^{(i_1\ldots i_\beta)} \sim s^{(1)}\cdots s^{(\tau)}s^{(i_1)}\cdots s^{(i_\beta)},$$

where $i_1,\ldots,i_\beta$ denote any β distinct numbers chosen from the numbers 1 to σ. Since $\beta > \sigma-\tau$, it follows that among these numbers $i_1,\ldots,i_\beta$ there must be some between 1 and τ; hence $p_{n-\tau-\beta}^{(i_1\ldots i_\beta)}$ vanishes identically, and therefore every individual matrix product, and with it φ', vanishes.

For the equivalence of the system of normal forms with the form row f, it remains only to show that all forms of the form row can be expanded in polars of the normal forms in the sense fixed under 1. Let Π be an aggregate of polar operations

$$\left(\frac{\partial}{\partial\xi^{(i)}}\middle|\xi^{(k)}\right), \qquad (i \neq k;\ i = 1,2,\ldots,\rho;\ k = 1,2,\ldots,\rho), \tag{53}$$

and let Π^σ be an aggregate of the special operations (53.) for which $i=\sigma$ and $k<\sigma$. For a form F of the ρ rows $\xi^{(1)},\ldots,\xi^{(\rho)}$ that has degree k_ρ in the row $\xi^{(\rho)}$, the expansion[69] holds:

$$F = \sum_{\alpha=0}^{k_\rho}\Pi F_\alpha, \qquad (\rho \leq n), \tag{54}$$

where F_α has degree α in the last row $\xi^{(\rho)}$ and arises from F by the invariant differential operations

$$\left(\frac{\partial}{\partial\xi^{(1)}}\cdots\frac{\partial}{\partial\xi^{(\rho)}}\middle|\xi^{(1)}\cdots\xi^{(\rho)}\right)^\alpha \tag{55}$$

and Π^ρ. If, in constructing F_α, one replaces process (55.) by

$$\left(\frac{\partial}{\partial\xi^{(1)}}\cdots\frac{\partial}{\partial\xi^{(\rho)}}\middle|p_\rho\right)^\alpha, \tag{56}$$

then a form $F_\alpha(p_\rho)$ with only $\rho-1$ rows $\xi^{(1)},\ldots,\xi^{(\rho-1)}$ is obtained, to which (54.) can again be applied. Continuing the procedure leads to forms $G(p_\rho,p_{\rho-1},\ldots,p_1)$. In order to obtain from these the expansion of F according to (54.), one replaces the individual rows p_ρ by $\xi^{(1)},\ldots,\xi^{(\rho)}$ and applies to the resulting forms the

[69] F. Mertens, *Über eine Formel der Determinantentheorie.* Sitzungsberichte der Akademie der Wissenschaften, Vienna, Math.-Naturw. Kl., vol. 91, 2nd section (1885), formula 20).

polar process Π (53.); this gives (cf. (48.)) a sum of transformed coefficients (G). For $\rho = n$, the process (55.) remains at the first step. If c denotes numerical constants and we put, for brevity,

$$(\xi^{(1)}\xi^{(2)}\cdots\xi^{(n)}) \sim \Delta; \qquad \left(\frac{\partial}{\partial\xi^{(1)}}\frac{\partial}{\partial\xi^{(2)}}\cdots\frac{\partial}{\partial\xi^{(n)}}\right) = \nabla, \tag{57}$$

then, in the case $\rho = n$,

$$F = \sum c\Delta^{\alpha}(G), \tag{58}$$

whereas for $\rho < n$ only $\alpha = 0$, and hence $\sum c(G)$, appears on the right.

Applying (58.) to a transformed coefficient (f) (48.), and using the commutativity of the polar processes Π^{σ} with all operations (56.) for which $\rho \geq \sigma$, as well as the fact that polars of transformed coefficients again give transformed coefficients, one obtains

$$G = \left(\frac{\partial}{\partial\xi^{(1)}}\middle| p_1\right)^{\beta_1}\left(\frac{\partial}{\partial\xi^{(1)}}\frac{\partial}{\partial\xi^{(2)}}\middle| p_2\right)^{\beta_2}\cdots\left(\frac{\partial}{\partial\xi^{(1)}}\cdots\frac{\partial}{\partial\xi^{(n-1)}}\middle| p_{n-1}\right)^{\beta_{n-1}}\nabla^{\beta_n}\Pi(f) = \sum c\varphi,$$

and from (58.) it follows that

$$(f) = \sum c\Delta^{\alpha}(\varphi) \qquad \text{(Mertens, formula 23)).} \tag{59}$$

To pass from (59.) to the expansion of f itself, we again regard the variables p_ρ as coefficients of a form H (52.) with the degrees $m_1, \ldots, m_{n-1}$ corresponding to f, and put $m = m_1 + \cdots + m_{n-1}$. Then, by the definition of invariants,

$$f \cdot \Delta^m = \sum (f)(H) = \sum c\Delta^{\alpha}(\varphi)(H),$$

and application of the process ∇^m that exactly exhausts the rows ξ gives

$$f = \sum c\Phi, \tag{60}$$

where the forms Φ are derived from φ and H by invariant differentiation processes with respect to the variables, and hence are simultaneous invariants of the form rows φ and H.[70] But since φ is a normal form, the form row φ reduces to the form φ itself; the form row H contains all possible invariant variable combinations. Thus the simultaneous invariants of φ and H are the polars of φ in the sense indicated under 1.; in other words, (60.) gives the expansion of f in polars of the normal forms. It remains to verify the same expansion for an arbitrary form θ of the form row f. Let Γ be the normal forms derived from (θ) by (50.), so that

$$(\theta) = \sum c\Delta^{\beta}(\Gamma). \tag{61}$$

The forms θ are characterized by the relation, valid for every transformed coefficient,

$$(\theta)\Delta^{\sigma} = \sum c(f) \qquad \text{(Mertens, formula 9)).}$$

But every form F of the n rows ξ which contains the factor Δ^{σ} also contains the second $\nabla^{\sigma}\Pi F$ (Mertens, formula 20)); hence

$$(\theta) = \sum c\nabla^{\sigma}(f), \qquad \text{therefore} \qquad \Gamma = \sum c\varphi,$$

and from (61.) we obtain the relation corresponding to (59.),

$$(\theta) = \sum c\Delta^{\beta}(\varphi),$$

which implies for θ the expansion (60.) in polars of the normal forms φ. Thus:

Theorem VII. Every form row f can be replaced by an equivalent system of normal forms.

[70] Instead of the direct conclusion, Mertens, § 7, introduces a series of further differential processes that essentially serve to form the form row H; this is done by our Theorem VI.

§ 8. Reduction of the folding process to fundamental foldings.

Following Theorem VII, we now carry out the reduction of the folding process to fundamental foldings mentioned at the beginning of § 7. We call (cf. § 5) every folding of two cogredient rows S^ρ, T^ρ a "cogredient folding", and specifically the cogredient folding of defect 1 of two rows S^ρ, T^ρ a "fundamental folding of type ρ". We then prove, completing Theorem VI:

Theorem VIII: The general folding process (41.) can be composed from the $(n-1)$ fundamental foldings of type ρ, where $\rho = 1, 2, \ldots, n-1$. In other words, to generate all invariant formations it suffices to apply successively the $n-1$ fundamental foldings.

In the binary domain the folding process (41.) coincides directly with the only possible fundamental folding; in the ternary domain I proved Theorem VIII in § 1 of my dissertation (cf. the Introduction). There, because of (44.), the general cogredient foldings are still identical with the fundamental foldings, and it is noteworthy that Theorem VIII for n variables gives the reduction to fundamental foldings, not merely the much weaker reduction to cogredient foldings. The proof is modeled in principle on the ternary proof: the most general foldings are constructed from the fundamental foldings, using the symbolic identities. We shall generate:

1. arbitrary foldings of defect 1 by cogredient foldings of defect 1, that is, by fundamental foldings (the analogue of the ternary case),
2. arbitrary foldings of defect λ by cogredient foldings of defect λ and arbitrary foldings of defect 1,
3. cogredient foldings of defect λ by cogredient foldings of defect $\lambda - 1$ and arbitrary foldings of defect 1.

As is essential, we assume by Theorem VII that the two forms S and T to be folded are normal forms. Thus, by (46.), the relations hold:

$$\begin{aligned} (q_{n-\sigma_1-\lambda} S^{\sigma_1} \,|\, S_{n-\sigma_2} p_{\sigma_2-\lambda}) = 0, \quad (\sigma_1 \geq \sigma_2,\ \lambda = 1, 2, \ldots, \sigma_2, n-\sigma_1), \\ (q_{n-\tau_1-\lambda} T^{\tau_1} \,|\, T_{n-\tau_2} p_{\tau_2-\lambda}) = 0, \quad (\tau_1 \geq \tau_2,\ \lambda = 1, 2, \ldots, \tau_2, n-\tau_1). \end{aligned} \tag{62}$$

It is further sufficient, by § 2 (end), to suppose that the forms S and T are linear in all variable rows; that is, the constructed forms belong to the form row

$$f = (S^1 \,|\, p_1)(S^2 \,|\, p_2) \cdots (S^{n-1} \,|\, p_{n-1})(T^1 \,|\, p_1)(T^2 \,|\, p_2) \cdots (T^{n-1} \,|\, p_{n-1}).$$

1) Generation of $(q_{n-\sigma-1} S^\sigma \,|\, T_{n-\tau} p_{\tau-1})$, where $\sigma > \tau$, from fundamental foldings of type $\tau + \alpha$, $\alpha = 0, 1, \ldots, \sigma - \tau$. From the fundamental folding of type τ

$$(q_{n-\tau-1} S^\tau \,|\, T_{n-\tau} p_{\tau-1})$$

we obtain, by the substitution $w \sim T_{n-\tau} p_{\tau-1}$, a form of the form row f with three rows of upper weight index (cf. § 1) $\tau + 1$:

$$(w S^\tau \,|\, p_{\tau+1})(S^{\tau+1} \,|\, p_{\tau+1})(T^{\tau+1} \,|\, p_{\tau+1}). \tag{63}$$

A fundamental folding of type $\tau + 1$ applied to (63.) gives

$$(q_{n-\tau-2} w S^\tau \,|\, S_{n-\tau-1} p_\tau);$$

from (31.), by the substitution $q_{n-\tau-2} w = q'_{n-\tau-1}$, this has the value

$$\varepsilon (q_{n-\tau-2} w S^{\tau+1} \,|\, S^\tau p_\tau) + \sum_{\alpha=1}^{\tau, n-\tau-1} \Pi (q'_{n-\tau-1-\alpha} S^{\tau+1} \,|\, S_{n-\tau} p_{\tau-\alpha}).$$

The expression under the summation sign vanishes identically by (62.); thus we have reached a form

$$(w S^{\tau+1} \,|\, p_{\tau+2})(S^{\tau+2} \,|\, p_{\tau+2})(T^{\tau+2} \,|\, p_{\tau+2}),$$

which is obtained from (63.) by changing τ into $\tau + 1$. Repeating the process $\sigma - \tau$ times therefore leads to the desired folding:

$$(w S^\sigma \,|\, p_{\sigma+1}) = \varepsilon (q_{n-\sigma-1} S^\sigma \,|\, T_{n-\tau} p_{\tau-1}).$$

We indicate the process just carried out by the formula

$$T^\tau S^\tau, \quad S^{\tau+1}, \quad S^{\tau+2}, \ldots, S^\sigma; \tag{64}$$

and see that only the normal-form property of S, not that of T, is used. A process dual to (64.) can be performed, using only the normal-form property of T, in the reverse order, namely:

$$S^\sigma T^\sigma, \quad T^{\sigma-1}, \quad T^{\sigma-2}, \ldots, T^\tau, \tag{65}$$

according to the relations:

$$\begin{aligned}
(q_{n-\sigma-1}S^\sigma \,|\, T_{1-\sigma}p_{\sigma-1}) &= \varepsilon(T_{n-\sigma}y \,|\, p_{\sigma-1}), \quad y \sim q_{n-\sigma-1}S^\sigma,\\
(q_{n-\sigma}T^{\sigma-1} \,|\, T_{n-\sigma}yp_{\sigma-2}) &= \varepsilon(T_{n-\sigma+1}y \,|\, p_{\sigma-2}),\\
&\vdots\\
(q_{n-\tau-1}T^\tau \,|\, T_{n-\tau-1}yp_{\tau-1}) &= \varepsilon(q_{n-\sigma-1}S^\sigma \,|\, T_{n-\tau}p_{\tau-1}).
\end{aligned}$$

We should also point out the connection with the decrease of total weight given in § 6 (end): for foldings of defect 1 this decrease has the value $2(1+(\sigma-\tau))$, and for fundamental foldings the value $2\cdot 1$. Therefore, if composition out of fundamental foldings is possible at all, exactly $\sigma-\tau+1$ such foldings are necessarily required, as was carried out above.

2) Generation of general foldings from cogredient and fundamental foldings. The weight decrease for a general folding (41) is $2(\lambda^2+\lambda(\sigma-\tau)) = 2[\lambda^2+(\sigma-\tau)(1+(\lambda-1))]$. This gives the possibility of decomposing it into a cogredient folding of defect λ (weight decrease $2\lambda^2$) and $\sigma-\tau$ foldings of defect 1 with difference of weight indices $\lambda-1$ (weight decrease $2\{1+(\lambda-1)\}$). In the case $\lambda=1$ this decomposition agrees with the one given under 1); we show that it can indeed be realized.

From the cogredient folding of defect λ

$$(q_{n-\tau-\lambda}S^\tau \,|\, T_{n-\tau}p_{\tau-\lambda})$$

we obtain, by the substitution $R^\lambda \sim T_{n-\tau}p_{\tau-\lambda}$, a form with the two rows of upper weight indices $\tau+\lambda$ and $\tau+1$:

$$(R^\lambda S^\tau \,|\, p_{\tau+\lambda})(S^{\tau+1} \,|\, p_{\tau+1}). \tag{66}$$

The row with the lower weight index $\tau+1$ belongs to a normal form; hence (66.) admits a folding of defect 1 composed out of λ fundamental foldings, in the order (65):

$$(R^\lambda S^\tau)S^{\tau+\lambda}, \quad S^{\tau+\lambda-1}, \quad S^{\tau+\lambda-2}, \ldots, S^{\tau+1}.$$

Taking account of (31.) and (62.), this gives

$$(q_{n-\tau-\lambda-1}R^\lambda S^\tau \,|\, S_{n-\tau-1}p_\tau) = \varepsilon(R^\lambda S^{\tau+1} \,|\, p_{\tau+\lambda+1}),$$

that is, a form obtained from (66.) by replacing τ by $\tau+1$, just as in 1) for (63.). Repeating the process $\sigma-\tau$ times leads to the desired folding:

$$(R^\lambda S^\sigma \,|\, p_{\sigma+\lambda}) = \varepsilon(q_{n-\sigma-\lambda}S^\sigma \,|\, T_{n-\tau}p_{\tau-\lambda}).$$

In the whole process only the normal-form property of S was used. Viewing T as a normal form gives the dual process, with complete reversal of the order, according to the formulas

$$\begin{aligned}
(q_{n-\sigma-\lambda}S^\sigma \,|\, T_{n-\sigma}p_{\sigma-\lambda}) &= \varepsilon(T_{n-\sigma}R_\lambda \,|\, p_{\sigma-\lambda}), \quad R_\lambda \sim q_{n-\sigma-\lambda}S^\sigma,\\
(q_{n-\sigma}T^{\sigma-1} \,|\, T_{n-\sigma}R_\lambda p_{\sigma-\lambda-1}) &= \varepsilon(T_{n-\sigma+1}R_\lambda \,|\, p_{\sigma-\lambda-1}),\\
&\vdots\\
(q_{n-\tau-1}T^\tau \,|\, T_{n-\tau-1}R_\lambda p_{\tau-\lambda}) &= \varepsilon(q_{n-\sigma-\lambda}S^\sigma \,|\, T_{n-\tau}p_{\tau-\lambda}).
\end{aligned}$$

3) Generation of cogredient foldings of defect λ from cogredient foldings of defect $\lambda-1$ and fundamental foldings. The weight decrease for cogredient foldings of defect λ is $2\lambda^2 = 2((\lambda-1)^2+2\lambda-1)$, giving the possibility of a decomposition into a cogredient folding of defect $\lambda-1$ and $2\lambda-1$ fundamental foldings. This is most

easily found by first putting $n = 2\lambda$. The only possible cogredient folding of defect λ is $(S^\lambda \,|\, T_\lambda) = (S^\lambda \,|\, T_{n/2})$; it contains one row u and one row x fewer than the folding of defect $\lambda - 1$ of the same rows, $(uS^\lambda \,|\, T_\lambda x)$. Conversely, we compose the folding of defect λ out of fundamental foldings which effect the disappearance of a row u and a row x (weight decrease $2(n-1) = 2(2\lambda - 1)$) and at the same time generate a new symbol row R^λ, together with a cogredient folding of defect $\lambda - 1$. The simplest folding that does this is the following one of defect 1:

$$(sT^\lambda \,|\, \sigma p_\lambda) = \varepsilon(S^{2\lambda-1} \,|\, R_{\lambda-1} p_\lambda) = \varepsilon(R^\lambda \,|\, p_\lambda),$$

where

$$R^{\lambda+1} = sT^\lambda, \quad s = S^1, \quad \sigma = S_1, \quad R^\lambda \sim \sigma R_{\lambda-1}.$$

By (65.) and (64.) it is composed out of the $2\lambda - 1$ fundamental foldings

$$T^\lambda S^\lambda, S^{\lambda-1}, \ldots, S^1; \qquad R^{\lambda+1} S^{\lambda+1}, S^{\lambda+2}, \ldots, S^{2\lambda-1}. \tag{67}$$

A folding of defect $\lambda - 1$, taking account of (21.) and (62.), gives

$$(uS^\lambda \,|\, \sigma R_{\lambda-1} x) = \varepsilon(R^{\lambda+1} \,|\, S_\lambda x) = \varepsilon(sT^\lambda \,|\, S^\lambda x) = \varepsilon(S^\lambda \,|\, T_\lambda).$$

The general case, for two rows S^ρ, T^ρ and arbitrary n, is obtained directly from the special one. In place of (67.) one has the following $2\lambda - 1$ fundamental foldings:

$$T^\rho S^\rho, S^{\rho-1}, \ldots, S^{\rho-\lambda+1}; \qquad R^{\rho+1} S^{\rho+1}, S^{\rho+2}, \ldots, S^{\rho+\lambda-1}. \tag{68}$$

The first half of the foldings gives the form

$$(q_{n-\rho-1} T^\rho \,|\, S_{n-\rho+\lambda-1} p_{\rho-\lambda}),$$

from which, by the substitutions

$$S_{n-\rho+\lambda-1} p_{\rho-\lambda} \sim w, \qquad T^\rho w = R^{\rho+1},$$

and application of the second half of the foldings, one obtains

$$(q_{n-\rho-\lambda} S^{\rho+\lambda-1} \,|\, R_{n-\rho-1} p_\rho).$$

Substituting

$$q_{n-\rho-\lambda} S^{\rho+\lambda-1} \sim y,$$

one obtains, by a cogredient folding of defect $\lambda - 1$,

$$(q_{n-\rho-\lambda+1} S^\rho \,|\, y R_{n-\rho-1} p_{\rho-\lambda+1}). \tag{69}$$

We show that this is the desired folding of defect λ. Substitute

$$R_{n-\rho-1} p_{\rho-\lambda+1} = R_{n-\lambda}$$

and observe that resolving y gives

$$(q_{n-\rho-\lambda+1} R^\lambda \,|\, S_{n-\rho} y) = \varepsilon(q_{n-\rho-\lambda} S^{\rho+\lambda-1} \,|\, S_{n-\rho} p'_{\rho-1}) = 0.$$

Thus by (20.) the value of (69.) is

$$\varepsilon(S^\rho R^\lambda \,|\, p_{\rho+\lambda-1} y) = \varepsilon(q_{n-\rho-\lambda} S^{\rho+\lambda-1} \,|\, [S^\rho R^\lambda]_{n-\rho-\lambda} p_{\rho+\lambda-1}).$$

This folding is of type (43.), hence equivalent to

$$(q_{n-\rho-\lambda} S^\rho \,|\, R_{n-\rho-1} p_{\rho-\lambda+1}) = \varepsilon(wT^\rho \,|\, R_\lambda p_{\rho-\lambda+1}), \qquad R_\lambda \sim q_{n-\rho-\lambda} S^\rho.$$

Here R_λ is already of the desired final form; the expression corresponds to the form $(sT^\lambda \,|\, S_\lambda x)$ in the special case. Now, observing that resolving w and R_λ gives

$$(wR^{n-\lambda} \,|\, T_{n-\rho} p_{\rho-\lambda+1}) = \varepsilon(q'_{n-1} R^{n-\lambda} \,|\, S_{n-\rho+\lambda-1} p_{\rho-\lambda}) = \varepsilon(q''_{n-\sigma-1} S^\rho \,|\, S_{n-\rho+\lambda-1} p_{\rho-\lambda}) = 0,$$

we get by (21.) the value

$$(wq_{n-\rho+\lambda-1} \,|\, T_{n-\rho}R_\lambda) = \varepsilon(q_{n-\rho+\lambda-1}[T_{n-\rho}R_\lambda]^{n-\lambda} \,|\, S_{n-\rho+\lambda-1}p_{\rho-\lambda}) \quad \text{equivalent to} \quad (q_{n-\rho-\lambda}S^\rho \,|\, T_{n-\rho}p_{\rho-\lambda}).$$

In the whole process only the normal-form property of S, not that of T, was used. Thus the cogredient folding of defect $\lambda-1$ used to generate (69.) can be replaced by $2\lambda-3$ fundamental foldings and a cogredient folding of defect $\lambda-2$, which in turn admits a decomposition into fundamental foldings, and so on. Analogously, the generation is obtained by using the normal-form property of T and applying the dualistic process, that is, the fundamental foldings

$$S^\rho T^\rho, \quad T^{\rho-1}, \dots, T^{\rho-\lambda+1}; \qquad R^{\rho-1}T^{\rho-1}, \quad T^{\rho-2}, \dots, T^{\rho-\lambda+1},$$

and a folding dual to (69.). The folding

$$(q_{n-\rho-\lambda}T^\rho \,|\, S_{n-\rho}p_{\rho-\lambda}),$$

equivalent to the preceding one, is then obtained.

Adding the decomposition obtained under 2), we have indeed obtained a generation of the most general foldings from fundamental foldings. It remains to show that this generation is unique, apart from permutation of the order, i.e. that to each folding there corresponds one and only one number α_ρ of fundamental foldings of type ρ, where $\rho = 1, 2, \dots, n-1$.

Let m_ρ be the degrees in the variable rows p_ρ of the initial form f, and l_ρ the degrees of the form produced by folding. Each fundamental folding of type ρ transforms the degrees $m_{\rho-1}, m_\rho, m_{\rho+1}$ respectively into $m_{\rho-1}+1, m_\rho-2, m_{\rho+1}+1$. Thus the α satisfy the system of equations

$$m_\rho - l_\rho = -\alpha_{\rho-1} + 2\alpha_\rho - \alpha_{\rho+1}, \qquad (\rho = 1, 2, \dots, n-1), \tag{70}$$

where α_0 and $\alpha_n = 0$ are to be put, and whose determinant has value n.[71] The α are thereby uniquely determined, and indeed, by Theorem VIII, as positive integers; this gives conditions for the differences $m_\rho - l_\rho$.

The results of this paragraph hold by virtue of relations (62.); however, if the relations (62.) do not hold, the final expressions are by no means equivalent to the initial expressions. The definition of equivalence given in § 6 also admits the formulation:[72] "Two forms are equivalent if and only if they differ by forms of higher folding, i.e. by forms that exhibit a greater weight decrease." This greater weight decrease always occurs when the two rows $q_{n-\sigma-i}, p_{\tau-\lambda}$ entering the folding are really variable rows, as happens in (41.), (42.), in the foldings appearing in this paragraph as equivalent, and also in the equivalent forms of Theorem VII. But it need not occur when these rows are derived from symbol rows by substitution, as was the case for the forms of this paragraph which could be expressed in terms of one another by virtue of (62.). One easily sees that the vanishing forms even have, in part, a higher total weight.

§ 9. Form rows. Reduction theorems.

We now define the form row introduced in § 6 (end) somewhat more sharply as "the totality of invariant formations of a given initial form that are linear in the coefficients, i.e. the totality of forms that arise from the initial form by folding in itself, arranged according to higher forms."

To explain the higher forms, suppose by Theorem VII that the initial form is given as a product of two normal forms $S \cdot T$. By a higher form we mean forms with higher folding in itself, and among forms with the same folding in itself, specially normalized ones. More precisely: suppose the form A has arisen by α_ρ fundamental foldings of type ρ, and the form B by β_ρ such foldings. Then B is higher than A:

1. if, in the system of inequalities $\beta_\rho \geq \alpha_\rho$, $\rho = 1, 2, \dots, n-1$, the inequality sign holds for at least one value of ρ;[73]

2. if the equality sign holds for all values ρ, but fewer symbol rows have been combined into a single matrix product. This normalization proves necessary because of the reductions in § 8. Accordingly, for example, $(S^{n-1} \,|\, T_{n-1})$ is higher than $(S^{n-1} \,|\, [S^1T^\rho]_{n-\rho-1}p_\rho)$;

[71] If Δ_n denotes the n-rowed determinant, then the recurrence formula holds: $\Delta_n = 2\Delta_{n-1} - \Delta_{n-2}$; that is, $\Delta_n - \Delta_{n-1} = \Delta_{n-1} - \Delta_{n-2} = \cdots = \Delta_2 - \Delta_1 = 1$, since $\Delta_2 = 3$, $\Delta_1 = 2$.

[72] Cf. the consideration at the end of the next paragraph.

[73] If in the system of inequalities the sign $>$ holds in part and the sign $<$ in part, then the forms are not comparable, since a form arising from another by folding always has a positive value system of fundamental foldings, hence in our case positive or vanishing values $\beta_\rho - \alpha_\rho$.

3. if the equality sign holds for all values ρ and the number of symbol rows combined into one matrix product is the same, then B is made higher than A by a normalization arbitrary in itself but fixed once and for all. This normalization is always possible because of the finite number of forms belonging to a value system α; it can be achieved, for instance, by privileging the one form S (larger number of symbol rows S, larger sum of the upper weight indices of the rows S, and so on).

It is useful to think of the form row as arranged as an $(n-1)$-dimensional manifold of lattice points, where the $n-1$ directions correspond to the $n-1$ fundamental foldings. To every form there then corresponds one and only one value system α, i.e. a positive integral lattice point, while the different forms belonging to a value system α are uniquely determined in their order by the above normalization. An arbitrary form of the form row gives the initial form of a new form row, contained as a part of the total form row; indeed it is contained as the part consisting of the forms obtained from the new initial form by proceeding in the $n-1$ positive directions.

By virtue of Theorem VIII and the general theory of series expansions, exactly the reduction theorems of the binary and ternary domains (cf. dissertation, § 3) hold for form rows. We shall briefly derive the principal theorem. We first state the definition:

"In a given system of forms, suppose a certain finite number of forms has been combined into a module. We call a form reducible if it can be expressed by forms that have invariants as factors, or by 'higher forms'; that is, by higher forms of the total form row to which the form belongs, or by forms that contain the symbols of the module in higher order. A reducible form row is called a reducer." Then:

Theorem IX: If the initial form of a form row is reducible by having arisen through folding with a reducer, then the entire form row arising from this initial form is reducible.

For the proof, observe that a form h arising by folding a form f with a second form g can, because of (45.), be regarded as a form f containing several cogredient variable rows p_ρ, p'_ρ. Such forms can, by the general theory of series expansion, be represented as polars of the elementary covariants of f, by applying the series expansion successively to each pair of cogredient variable rows p_ρ, p'_ρ. By (38.), the elementary covariants that occur are the forms generated by cogredient folding. Since, however, the order in which the series expansion is applied is still arbitrary, we may in particular choose an order corresponding to the generation of the general foldings from fundamental foldings. The resulting system of elementary covariants is then identical with the form row f; the forms that arise by cogredient folding of higher defect are arranged among those that arise by fundamental foldings. But one also obtains polars of the form row f for any order of the series expansion, to which a second system of elementary covariants may correspond. Since the form row exhausts the totality of invariant formations, every form of the second system can be expressed linearly by polars of the form row, and indeed by polars of those forms that show the same or a greater weight decrease. The latter follows because (cf. § 7) the polars can be regarded as simultaneous invariants of a form row H (52.), with the degree numbers corresponding to the form h, and of the form row f. Every folding of H in itself corresponds to a weight decrease which, after substitution (11.) is performed, is transferred to the symbol rows and hence to the forms of f.[74]

An arbitrary form of the form row h has arisen by folding h in itself; that is, on the one hand by a higher folding of g with f, and on the other by folding f or g in itself. In both cases the series expansion leads to polars of the form row f, just as was shown above for the initial form h. If in particular f is a reducer, then an expansion in polars of the reducer results; hence the form row h becomes reducible.

Applications of the theorem to the reduction of complete systems of forms follow analogously to the binary and ternary domains.

[74]This consideration also underlies the second formulation of the equivalence concept (§ 8, end). Further explanations of the different systems of elementary covariants in the ternary domain are found in Study, loc. cit., II, § 7, Theorems 7) and 8). By our Theorem VII the same results also hold for n variables.

5. Rational Function Fields

Jahresbericht der DMV 22 (1913), pp. 316–319

The following questions originally go back to conversations with E. Fischer. Some of the questions, moreover, were already raised and settled by E. Steinitz for the special case of one indeterminate — and indeed under more general assumptions on the coefficient field.[75]

By a "rational function field" I understand a field whose elements are rational functions of n indeterminates, with coefficients from a prescribed number field, which in particular may also comprise all complex numbers. Examples of such fields are the field of symmetric functions of n quantities, or more generally the totality of rational functions of n indeterminates that allow the permutations of a certain group (Lagrange's *Gattungsbereiche*); furthermore, the invariant field.

Between the first two fields just mentioned and the latter there is a characteristic difference: the former contain n algebraically independent functions, the elementary symmetric functions; whereas the number of algebraically independent invariants is always smaller than the number of indeterminates. It is therefore important that, in general, one can associate such fields of the second kind one-to-one with fields of the first kind, by replacing some of the indeterminates by numbers. In what follows I shall therefore restrict myself to fields of the first kind, i.e. to fields with n algebraically independent functions; by virtue of the association, the results then hold in general.

The questions group themselves around three basis concepts: rational basis, minimal basis, and integrality basis.

1. By a "rational basis" I understand a finite number of functions of the field such that every function of the field can be represented as a rational combination of this finite number, with coefficients from the prescribed coefficient field. By simple considerations — which in the case of one indeterminate are also found in E. Steinitz, loc. cit. — one proves the existence of a rational basis for every rational function field. For example, by Lagrange's theorem, the elementary symmetric functions and one function belonging to the group form a rational basis for the Lagrange *Gattungsbereiche* mentioned earlier.

2. Every rational basis must contain (for fields of the first kind) at least n functions; if it contains exactly n functions, which then must be algebraically independent, I call it a "minimal basis." The question of the existence of a minimal basis can be answered in part by theorems of Lüroth, Castelnuovo, and Enriques.[76] According to these, for fields of one indeterminate there always exists a minimal basis: the Lüroth function of the field, uniquely determined up to fractional linear transformation (cf. Steinitz § 24). For fields of two indeterminates, a minimal basis exists always, and in general only, if one permits an algebraic extension of the coefficient field, whereas for three and more indeterminates a minimal basis in general does not exist at all. No attempt has yet been made to characterize the special fields with minimal basis.

I have pursued further the question of the minimal basis for Lagrange's *Gattungsbereiche.* Here it acquires the following meaning: "If the Lagrange *Gattungsbereich* belonging to the group G possesses a minimal basis, then one can construct rationally, by a parametric representation, the totality of equations with prescribed group G; and indeed for every number field that contains the coefficient field of the minimal basis — hence in particular for any arbitrary number field if this coefficient field is the field of rational numbers."

The simplest example is provided by the symmetric group; here the elementary symmetric functions form a minimal basis with rational numerical coefficients. From this one obtains as parametric representation simply the equation with indeterminate coefficients. Hilbert's irreducibility theorem shows how one passes from this to arbitrarily many affectless equations over every number field.

Now the existence of the minimal basis for all groups occurring in equations of degrees 3 and 4 follows directly from the theorems of Lüroth and Castelnuovo; at the same time it further appears that this minimal basis has rational numerical coefficients. Thus, for every arbitrary number field, one can rationally construct the totality of equations of degrees 3 and 4 with prescribed group. For the cyclic and dihedral groups there exists a minimal basis with the field of the nth roots of unity as coefficient field; the same holds for a few further metacyclic groups. The question whether the minimal basis is at all characteristic for certain groups must remain undecided.

3. To arrive at the last basis concept, I consider the totality of polynomials contained in the field. By an

[75] E. Steinitz: Algebraische Theorie der Körper, especially § 24. Crelle's Journal 137. 1910.

[76] J. Lüroth: Beweis eines Satzes über rationale Kurven. Math. Ann. 9. 1875. — G. Castelnuovo: Sulla razionalità delle involuzioni piane. Math. Ann. 44. 1893. — F. Enriques: Sopra una involuzione non razionale dello spazio. Rend. Acc. Linc. vol. 21. 21 Jan. 1912.

"integrality basis" I understand a finite number of these polynomials such that every polynomial of the field can be represented as an integral rational combination of this finite number, with coefficients from the prescribed coefficient field. The question of the existence of an integrality basis — which, for example, contains as a special case the finiteness of the invariant system — was posed by Hilbert in his "Mathematical Problems"[77] in a somewhat different formulation, as the problem of relatively integral functions.

For the moment I can indicate only one class of fields for which an integrality basis exists. Namely, if the field contains a system of n polynomials such that the homogeneous resultant of the terms of highest dimension of these polynomials is different from zero, then an integrality basis exists; it is given by the coefficients of all u in the equation, irreducible in the field, for the linear form:

$$u_0 = u_1 x_1 + u_2 x_2 + \cdots + u_n x_n.$$ [78]

If, in particular, the value of the resultant is equal to a unit of the coefficient field, then the integrality of the representation is also secured. An example of this class of fields is furnished by Lagrange's *Gattungsbereiche*, since the resultant of the elementary symmetric functions has the value ± 1. A further example (without integrality) is given by all fields that contain only one algebraically independent polynomial; here the integrality basis is identical with the Lüroth function of the smallest subfield containing all polynomials. Thus, as is well known, all integral rational projective invariants of a quadratic form in n variables are integral functions of the discriminant.

The condition just given is only sufficient, by no means necessary, for the existence of an integrality basis. One can easily construct fields in which the resultant of every n polynomials vanishes, and which nevertheless possess an integrality basis given by the coefficients of that irreducible equation. The question arises whether perhaps even in the most general case the coefficients of that equation furnish the integrality basis.

[77] D. Hilbert: Mathematische Probleme. Lecture, Paris 1900. Problem 14. Göttinger Nachr. 1900.

[78] This irreducible equation is found, in the case of one indeterminate, in E. Steinitz's proof of Lüroth's theorem.

6. Fields and Systems of Rational Functions

Math. Ann. 76 (1915), pp. 161–196

The present paper treats basis questions for arbitrary systems of rational and integral rational functions. The methods of field theory used here make the settlement of these questions for fields of rational functions — rational function fields[79] — appear as the essential matter, while the generalization of the results to arbitrary systems arises as a consequence.

Among basis questions for general systems, up to now only the existence of a module basis, guaranteed by Hilbert's theorem (Math. Ann. 36), has been known. In what follows, the question of rational representability is answered completely by the existence of a rational basis for every arbitrary system (§ 7); the rational basis of fields is already obtained in § 4. This existence of the rational basis permits us throughout to start from the abstractly defined field or system, and thereby to avoid difficulties that are caused only by the special choice of the rational basis and not by the system itself, such as the occurrence of special denominators or of fundamental points of the basis functions.

For fields the question of the minimal basis is also treated, i.e. of a rational basis consisting of algebraically independent functions (§ 6). The methods of field theory further lead to a distinguished rational basis, the involution basis[80] (§ 5), which becomes especially important for integrality domains consisting of polynomials (§ 8). In that setting it gives a representation with a fixed denominator (a power of a function determined by the integrality domain), as is known for example in the special case of the typical representation of invariants.

From rational representability we now draw, as far as possible, consequences for integral rational representability, i.e. for the question of finiteness in the narrower sense.

The finiteness theorems known up to now all rest on the fact that Hilbert's theorem on module bases guarantees finiteness for all systems in which a representation

$$F = A_1 f_1 + A_2 f_2 + \cdots + A_k f_k$$

entails a second representation

$$F = B_1 f_1 + B_2 f_2 + \cdots + B_k f_k,$$

where the B_i belong to the system and are of lower degree than F.

By contrast, the theorem on the rational basis and the methods of field theory permit one to infer finiteness under assumptions of an entirely different kind.

Thus, as a partial answer to a problem of Hilbert (Mathematical Problems 14), one obtains a class of relatively integral functions (relatively integral domains of the first kind) which are finite integrality domains and whose integrality basis is given by the involution basis mentioned above (§ 10). In analogy with Hilbert's proof of Kronecker's theorem on the fundamental system of algebraic integers (Complete invariant systems, § 2; Math. Ann. 42), it can further be shown that all regular systems of polynomials — i.e. all systems that can be mapped to ones without fundamental points — possess an integrality basis (§ 12), whereas nonregular systems without integrality basis can easily be specified. As a simple example of this fact one may mention the theorem: "In every arbitrary system of infinitely many polynomials in one indeterminate, there exists a finite number of these polynomials such that every polynomial of the system can be represented as an integral rational combination of this finite number." Further examples are found in § 13.

Finally, the last paragraphs (§§ 14 and 15) give a sharpening of the basis theorems with respect to integrality.

An essential tool of the investigation is the transfer principle of § 3, according to which it suffices to consider all basis questions raised here for those systems in which the number of indeterminates coincides with the number of algebraically independent functions of the system.

The impetus for the present work came from conversations with Mr. E. Fischer, in particular from a question of his concerning the minimal basis of Lagrange's *Gattungsbereiche*. The investigations relating to this, which led to the construction of equations with prescribed group (cf. the cited communication on "Rational Function Fields," part 2), are reserved for a further publication.

The essential extension of the present paper over the original communication consists in the fact that the basis theorems stated there only for fields or special integrality domains are here transferred to arbitrary

[79]Cf. a preliminary communication on the occasion of the Vienna meeting of natural scientists: Rationale Funktionenkörper, Jahresber. d. D. Math.-Ver. 22 (1913), where an overview is given of the questions and of the results concerning function fields.

[80]The name was chosen because, in its geometric interpretation, the rational function field represents the most general involution in a linear space.

systems. This transfer succeeds in a simple way by means of the concept of the smallest containing field of an arbitrary system, as a generalization of the quotient field of an integrality domain (§ 7). I was led to form this concept by the fact that Mr. K. Hentzelt, after learning the rational basis of fields, was able to prove the existence of the rational basis for arbitrary systems by a detour through Hilbert's module basis. I was also led to the theorem on the integrality basis of regular systems by K. Hentzelt's observation that the arguments I had applied to integrality domains were valid for arbitrary systems subject to the same assumptions; likewise I owe to Mr. Hentzelt a few isolated smaller remarks.

Finally, let it be mentioned that in the case of one indeterminate the rational basis and minimal basis of fields are found in E. Steinitz's "Algebraische Theorie der Körper," § 24 (Crelle's Journal 137); there the coefficient field is taken to be a field in the most general sense. The question how far the theorems given here remain valid under these more general assumptions is not touched on in what follows; in any event the proofs require modification, since, for example, the tools from the theory of functional determinants fail for fields of characteristic p.

§ 1. Fields $\mathfrak{K}_{n\rho}$ and systems $\mathfrak{S}_{n\rho}$.

In what follows we are concerned with the investigation of systems of rational functions in n indeterminates, especially fields of rational functions.

By $f(x_1,\dots,x_n)$, $g(x_1,\dots,x_n)$, ..., or abbreviated $f(x)$, $g(x)$, ..., one should therefore always understand rational functions of $x_1,\dots,x_n$; these are assumed to be in reduced representation — that is, numerator and denominator relatively prime. Integral rational functions (polynomials) will be denoted by capital letters, $F(x)$, $G(x)$, ... The coefficient field Ω is assumed to be some number field; in particular, it may also contain all complex numbers. The field Ω may also contain a finite number of parameters.

The quantities from Ω, hence all functions of degree zero, are always counted as belonging to the system.

With these stipulations, the field consisting of rational functions — the rational function field — can be defined abstractly.

Definition I: A system of rational functions is called a field if it satisfies the conditions:

1. *Together with $f(x)$, and for every quantity c from Ω, it also contains $c \cdot f(x)$.*
2. *Together with $f(x)$ and $g(x)$, it always also contains $f(x)+g(x)$, $f(x)\cdot g(x)$ and — for $g(x)\neq 0$ — the quotient $f(x):g(x)$.*[81]

Special fields are those which arise by adjoining a finite number of rational functions

$$f_1(x),\ f_2(x),\dots,f_k(x)$$

to Ω; they will be denoted by

$$\Omega\big(f_1(x),\dots,f_k(x)\big) \quad \text{or} \quad \Omega(f_1,\dots,f_k).$$

[82] Among these special fields we also mention the field arising by adjoining $x_1,\dots,x_n$ to Ω,

$$\Omega(x_1,\dots,x_n),$$

which is identical with the totality of rational functions of $x_1,\dots,x_n$ with coefficients from Ω.

We also introduce (following E. Steinitz) the concept of an intermediate field:

"A field $\mathfrak{M}$ lies between Ω_1 and Ω_2 if it contains Ω_1 and is contained in Ω_2;"

then Definition I also admits the following formulation:

The field of rational functions of n indeterminates is an intermediate field between Ω and $\Omega(x_1,\dots,x_n)$ which effectively contains the n indeterminates in at least one function.

Beside the number of indeterminates, there arises for systems of rational functions a further characteristic number, the number of algebraically independent functions, or the algebraic rank, by the following definition.

Definition II: A system of rational functions has algebraic rank ρ if ρ functions of the system can be specified that are algebraically independent, but every $(\rho+1)$ functions of the system are algebraically dependent.[83]

[81] Here $f(x)$ and $g(x)$ may also be of degree zero, i.e. quantities from Ω.

[82] That these "special fields" already exhaust the totality of all fields follows in § 4.

[83] ρ functions $f_1(x),\dots,f_\rho(x)$ are called algebraically independent if from every relation

$$F\big(f_1(x),\dots,f_\rho(x)\big)=0 \quad \text{identically in } x_1,\dots,x_n$$

Systems of (finitely or infinitely many) rational functions in n indeterminates and of algebraic rank ρ will be denoted by the double index

$$\mathfrak{S}_{n\rho}.$$

In particular, by

$$\mathfrak{K}_{n\rho}$$

one is to understand a field of n indeterminates and of algebraic rank ρ. The inequality holds:

$$1 \leq \rho \leq n.$$

We give a few more examples of fields $\mathfrak{K}_{nn}$ and $\mathfrak{K}_{n\rho}$:

1. Fields $\mathfrak{K}_{nn}$ are Lagrange's *Gattungsbereiche*, which can be defined as

> "the totality of rational (and integral rational) functions of $x_1, \dots, x_n$ that allow the permutations of a permutation group G of $x_1, \dots, x_n$."

They always contain the field of symmetric functions, hence also the n algebraically independent elementary symmetric functions.

2. Fields $\mathfrak{K}_{n\rho}$ are the projective invariant fields, which are formed from the totality of the rational (and integral rational) invariants of one or more ground forms.[84] Here $\rho < n$, since the number of algebraically independent invariants is always smaller than the number of coefficients.

The corresponding statement holds for the invariant fields with respect to subgroups of the projective group.

§ 2. Auxiliary theorem on algebraically dependent functions.

The auxiliary theorem of this paragraph will show in § 3 that the algebraic rank of a system is its properly characteristic number. Namely, the auxiliary theorem shows that by a successive numerical specialization of some indeterminates, such that ρ algebraically independent functions remain algebraically independent, an arbitrary function algebraically dependent on these ρ functions can neither vanish identically nor become indeterminate.

Let therefore the system

$$g_1(x),\ g_2(x), \dots, g_\rho(x) \tag{1}$$

have algebraic rank ρ, where $\rho < n$, and let $h(x)$ be a rational function algebraically dependent on the functions (1). By known theorems, the rank of the functional matrix of (1) is equal to ρ. The indeterminates may therefore be named, for instance, so that

$$\frac{\partial(g_1, g_2, \dots, g_\rho)}{\partial(x_1, x_2, \dots, x_\rho)} = \frac{\Gamma(x)}{G(x)^{\rho+1}} \neq 0, \tag{2}$$

where $G(x)$, as usual, denotes the common denominator of the functions (1).

Now choose the number a so that the product $G(x) \cdot \Gamma(x)$ is not divisible by $(x_i - a)$, where i is understood to be one of the numbers $\rho+1, \rho+2, \dots, n$. Thus, for $x_i = a$, the common denominator of the functions (1) cannot vanish, and the ρ functions

$$[g_1(x)]_{x_i=a}, \dots, [g_\rho(x)]_{x_i=a} \tag{3}$$

are likewise algebraically independent. As we shall say, the algebraic rank of the system (1) is preserved under the substitution $x_i = a$.

Let the irreducible equation satisfied by $h(x)$ with respect to the functions (1) be given by

$$\begin{aligned} A_0(g_1, \dots, g_\rho)\, h(x)^\tau + A_1(g_1, \dots, g_\rho)\, h(x)^{\tau-1} + \cdots \\ + A_\tau(g_1, \dots, g_\rho) = 0, \end{aligned} \tag{4}$$

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it follows that

$$F(\lambda_1, \dots, \lambda_\rho) = 0 \quad \text{identically in } \lambda_1, \dots, \lambda_\rho.$$

In the opposite case they are called algebraically dependent.

[84] It is expedient to consider only transformations of determinant 1; then every rational combination of arbitrarily many invariants again admits the transformation, and one does indeed obtain a field. The homogeneous, isobaric invariants taken by themselves do not form a field, but they do form a system $\mathfrak{S}_{n\rho}$.

where $A_0(g_1, \dots, g_\rho)$ and $A_\tau(g_1, \dots, g_\rho)$ do not vanish identically. In order to pass from this to an identity between polynomials, we set

$$g_i(x) = G_i(x) : G(x); \qquad h(x) = H_1(x) : H_2(x), \tag{5}$$

and note that — because of the assumed reduced representation of $h(x)$ — $H_1(x)$ and $H_2(x)$ are relatively prime. By virtue of (5), (4) becomes

$$\begin{aligned} &B_0(G, G_1, \dots, G_\rho)\, G(x)^{\sigma_0} H_1(x)^\tau \\ &\quad + B_1(G, G_1, \dots, G_\rho)\, G(x)^{\sigma_1} H_1(x)^{\tau-1} H_2(x) + \cdots \\ &\quad + B_\tau(G, G_1, \dots, G_\rho)\, G(x)^{\sigma_\tau} H_2(x)^\tau = 0, \end{aligned} \tag{6}$$

where the B_k are homogeneous forms in G, G_i, and at least one of the nonnegative exponents σ_k must be zero.

Suppose now that $H_1(x)$ were divisible by $(x_i - a)$. Then, by (6),

$$B_\tau(G, G_1, \dots, G_\rho)\, G(x)^{\sigma_\tau} H_2(x)^\tau$$

would also be divisible by $(x_i - a)$. This is excluded by assumption for $G(x)$; hence from the divisibility of B_τ there would also follow the divisibility of

$$A_\tau = B_\tau : G^\lambda,$$

and between the functions (3) there would exist the relation $A_\tau = 0$, contrary to the assumed algebraic independence. Finally $H_2(x)$, being relatively prime to $H_1(x)$, also cannot be divisible by $(x_i - a)$;[85] thus the assumption that $H_1(x)$ is divisible by $(x_i - a)$ leads to a contradiction. In the same way one shows that $H_2(x)$ also cannot be divisible by $(x_i - a)$. Hence the function $[h(x)]_{x_i=a} = h'(x)$ can neither vanish identically nor become indeterminate.

The above conclusion can now be applied again to the system (3) and the function $h'(x)$, again assumed to be in reduced representation, and continuation of the procedure leads finally to the following auxiliary theorem.

Auxiliary theorem: The two systems

$$g_1(x),\ g_2(x), \dots, g_\rho(x),$$

$$g_1(x),\ g_2(x), \dots, g_\rho(x),\ h(x)$$

shall each have algebraic rank ρ, where $\rho < n$, and suppose, for instance, that

$$\frac{\partial(g_1, \dots, g_\rho)}{\partial(x_1, \dots, x_\rho)} = \frac{\Gamma(x)}{G(x)^{\rho+1}} \neq 0.$$

The numbers

$$a_n,\ a_{n-1}, \dots, a_{\rho+1}$$

are to be chosen so that, under the substitution

$$x_n = a_n, \quad x_{n-1} = a_{n-1}, \dots, \quad x_{\rho+1} = a_{\rho+1},$$

the product $G(x) \cdot \Gamma(x)$ does not vanish — that is, the algebraic rank of the system (1) is preserved. In the successive substitutions

$$[h(x)]_{x_n=a_n} = h^{(n-1)}(x), \qquad [h^{(n-1)}(x)]_{x_{n-1}=a_{n-1}} = h^{(n-2)}(x), \dots,$$
$$[h^{(\rho+1)}(x)]_{x_{\rho+1}=a_{\rho+1}} = h^*(x_1, x_2, \dots, x_\rho),$$

let the functions $h(x)$, $h^{(n-1)}(x)$, $\dots$, $h^{(\rho+1)}(x)$ each be put into reduced representation. Under these assumptions, in the function $h^*(x_1, \dots, x_\rho)$ neither numerator nor denominator can vanish identically, and the function also cannot become $\frac{0}{0}$.[86]

[85] This last conclusion, resting on the reduced representation, would no longer be possible for a substitution $x_i = a$, $x_k = b$ while the remaining assumptions are preserved. Then $H_1(x)$ and $H_2(x)$ could vanish simultaneously, and the quotient would become indeterminate under the substitution, equal to $\frac{0}{0}$; for example

$$h(x) = \frac{x_3(\alpha x_1 + \beta x_2) + x_4(\gamma x_1 + \delta x_2)}{x_3(\alpha' x_1 + \beta' x_2) + x_4(\gamma' x_1 + \delta' x_2)}$$

under the substitution $x_3 = 0$, $x_4 = 0$.

[86] For example, in the function $h(x)$ of the preceding note, the successive substitution gives

$$[h(x)]_{x_4=0} = h^{(3)}(x) = \frac{\alpha x_1 + \beta x_2}{\alpha' x_1 + \beta' x_2}, \qquad h^*(x_1, x_2) = \frac{\alpha x_1 + \beta x_2}{\alpha' x_1 + \beta' x_2}.$$

§ 3. One-to-one mapping of systems $\mathfrak{S}_{n\rho}$ onto systems $\mathfrak{S}^*_{\rho\rho}$.

From the auxiliary theorem we shall immediately obtain a mapping of the systems $\mathfrak{S}_{n\rho}$ onto systems $\mathfrak{S}^*_{\rho\rho}$, by replacing some of the indeterminates by numbers; the algebraic rank ρ thereby presents itself as the properly characteristic number.

Since $\mathfrak{S}_{n\rho}$ has algebraic rank ρ, there exist ρ functions from $\mathfrak{S}_{n\rho}$,

$$g_1(x), \ldots, g_\rho(x), \tag{1}$$

which are algebraically independent. Further, if $f(x)$ denotes an arbitrary function from $\mathfrak{S}_{n\rho}$, then the system

$$g_1(x), \ldots, g_\rho(x),\ f(x) \tag{2}$$

also has algebraic rank ρ, and the systems (1) and (2) satisfy the conditions of the auxiliary theorem.[87]

If, therefore, one determines the numbers

$$a_n,\ a_{n-1}, \ldots, a_{\rho+1}$$

according to the auxiliary theorem in such a way that the algebraic rank ρ of the system (1) is preserved by the substitution — which is possible in arbitrarily many ways — then in the function defined by

$$[f(x)]_{x_n=a_n} = f^{(n-1)}(x), \qquad [f^{(n-1)}(x)]_{x_{n-1}=a_{n-1}} = f^{(n-2)}(x); \ldots,$$
$$[f^{(\rho+1)}(x)]_{x_{\rho+1}=a_{\rho+1}} = f^*(x_1, \ldots, x_\rho),$$

namely

$$f^*(x_1, \ldots, x_\rho), \tag{3}$$

neither numerator nor denominator can vanish identically.

Now let $f(x)$ run through all (finitely or infinitely many) functions from $\mathfrak{S}_{n\rho}$. We consider the system consisting of the totality of all functions (3); it has the following properties:

1. It has algebraic rank ρ; for it contains the functions arising from (1),

$$g_1^*(x_1, \ldots, x_\rho), \ldots, g_\rho^*(x_1, \ldots, x_\rho),$$

which, because of the choice of $a_n, a_{n-1}, \ldots, a_{\rho+1}$, are algebraically independent. The system is therefore to be denoted by

$$\mathfrak{S}^*_{\rho\rho}.$$

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2. The correspondence of the functions $f(x)$ and $f^*(x)$ is without exception one-to-one.

For if $f_1(x)$ and $f_2(x)$ belong to $\mathfrak{S}_{n\rho}$, then their difference

$$d(x) = f_1(x) - f_2(x)$$

is also algebraically dependent on the system (1); moreover

$$d^*(x) = f_1^*(x) - f_2^*(x).$$

Since $d(x)$, together with the system (1), satisfies the conditions of the auxiliary theorem, it follows from

$$d(x) \neq 0$$

necessarily that

$$d^*(x) \neq 0;$$

in other words, different functions from $\mathfrak{S}_{n\rho}$ correspond to different functions from $\mathfrak{S}^*_{\rho\rho}$.

3. From every relation between functions from $\mathfrak{S}^*_{\rho\rho}$,

$$F\big(f_1^*(x), f_2^*(x), \ldots, f_k^*(x)\big) = 0 \quad \text{identically in } x_1, \ldots, x_\rho,$$

[87] If necessary, the indeterminates are to be renumbered.

[88] Correspondingly, by $f^*(x)$, $g^*(x)$, ... one should throughout understand mapping functions, i.e. functions of $x_1, \ldots, x_\rho$.

it follows, for the uniquely corresponding functions from $\mathfrak{S}_{n\rho}$, that

$$F\big(f_1(x), f_2(x), \ldots, f_k(x)\big) = 0 \quad \text{identically in } x_1, \ldots, x_n.$$

For with $f_1(x), f_2(x), \ldots, f_k(x)$, every rational combination of these functions is also algebraically dependent on the system (1).

It further follows from

$$\begin{aligned} h(x) &= F\big(f_1(x), \ldots, f_k(x)\big) : \\ h^*(x) &= F\big(f_1^*(x), \ldots, f_k^*(x)\big). \end{aligned} \tag{4}$$

By the auxiliary theorem, therefore,

$$h(x) \neq 0 \quad \text{implies} \quad h^*(x) \neq 0;$$

q.e.d.

In summary, we obtain

Theorem I: To every system $\mathfrak{S}_{n\rho}$ of (finitely or infinitely many) rational functions there can be assigned — in arbitrarily many ways — a one-to-one system $\mathfrak{S}^*_{\rho\rho}$, in such a way that all relations existing between functions from $\mathfrak{S}^*_{\rho\rho}$ are also preserved in $\mathfrak{S}_{n\rho}$ and conversely. In particular, by (4), a field $\mathfrak{K}_{n\rho}$ again corresponds to a field $\mathfrak{K}^*_{\rho\rho}$.

From Theorem I we draw the consequence that simplifies all further proofs: *properties of systems $\mathfrak{S}_{n\rho}$ which are characterized by finitely or infinitely many relations between functions of the system hold for the systems $\mathfrak{S}_{n\rho}$ as soon as they hold for a mapping system $\mathfrak{S}^*_{\rho\rho}$.*

We call this consequence briefly the "transfer principle."

The simplest example of this mapping is furnished by the theory of algebraic invariants. In fact, by a linear transformation one can always numerically fix some coefficients of the ground form in such a way that all relations valid for the special form also remain valid in general.

§ 4. Rational basis of the fields $\mathfrak{K}_{n\rho}$.

In the following investigations, grouped around basis concepts, we shall consistently use the "transfer principle" of § 3. We first prove the existence of the basis for fields, but state the definitions generally:

Definition III: A finite number of functions from $\mathfrak{S}_{n\rho}$ is called a rational basis if every function from $\mathfrak{S}_{n\rho}$ can be represented as a rational combination of this finite number, with coefficients from Ω.

The rational basis must contain ρ algebraically independent functions, since the algebraic rank of a system derived from a rational basis is equal to the algebraic rank of the rational basis.

We first show that the existence proof for the rational basis admits the transfer principle. Namely, Definition III can be formulated as follows:

The functions from $\mathfrak{S}_{n\rho}$

$$g_1(x),\ g_2(x), \dots, g_k(x)$$

form a rational basis if and only if the infinitely many relations

$$f(x) = \gamma\big(g_1(x), \dots, g_k(x)\big) \tag{1}$$

are satisfied, where $f(x)$ runs through all functions from $\mathfrak{S}_{n\rho}$ and γ denotes a rational function of k arguments, varying with f.

Thus, if in the mapping system $\mathfrak{S}^*_{\rho\rho}$ a rational basis is given by

$$g^*_1(x),\ g^*_2(x), \dots, g^*_k(x),$$

which entails the relations

$$f^*(x) = \gamma\big(g^*_1(x), \dots, g^*_k(x)\big),$$

then by Theorem I the relations (1) are fulfilled for $\mathfrak{S}_{n\rho}$. Thus the *transfer principle for the rational basis* says:

From the existence of a rational basis for a mapping system $\mathfrak{S}^*_{\rho\rho}$ follows the rational basis for the original system $\mathfrak{S}_{n\rho}$.[89]

To prove the existence of the rational basis of all fields $\mathfrak{K}_{n\rho}$, we may therefore restrict ourselves to fields $\mathfrak{K}^*_{\rho\rho}$; here a direct generalization of an argument used by E. Steinitz leads to the goal.[90]

Let

$$g^*_1(x_1, \dots, x_\rho), \dots, g^*_\rho(x_1, \dots, x_\rho) \tag{2}$$

be a system of ρ algebraically independent functions from $\mathfrak{K}^*_{\rho\rho}$. By eliminating $x_1, \dots, x_\rho$ one obtains ρ relations:

$$\begin{aligned} &F_1\big(g^*_1(x), \dots, g^*_\rho(x), x_1\big) = 0, \dots, \\ &F_\rho\big(g^*_1(x), \dots, g^*_\rho(x), x_\rho\big) = 0, \end{aligned} \tag{3}$$

where in each relation $F_i = 0$ the indeterminate x_i actually occurs; otherwise, contrary to the assumption, there would be an algebraic dependence among the functions (2). The relations (3) say that

$$x_1, x_2, \dots, x_\rho$$

are algebraic elements with respect to

$$\Omega\big(g^*_1(x), \dots, g^*_\rho(x)\big).$$

Hence the field arising by adjoining $x_1, \dots, x_\rho$,

$$\Omega\big(g^*_1, \dots, g^*_\rho, x_1, \dots, x_\rho\big) = \Omega(x_1, \dots, x_\rho),$$

is a finite algebraic extension of $\Omega(g^*_1, \dots, g^*_\rho)$. By a known theorem,[91] $\mathfrak{K}^*_{\rho\rho}$, as an intermediate field between $\Omega(g^*_1, \dots, g^*_\rho)$ and $\Omega(x_1, \dots, x_\rho)$, is itself an algebraic extension of $\Omega(g^*_1, \dots, g^*_\rho)$, while $\Omega(x_1, \dots, x_\rho)$ is an

[89] The transfer principle is formulated correspondingly for every basis question treated here.

[90] E. Steinitz, Algebraische Theorie der Körper, § 24, Crelle's Journal 137 (1909).

[91] Compare, for example, Weber, Lehrbuch d. Algebra 1, § 144 (1st ed.) or § 55 (small edition).

algebraic extension of $\mathfrak{K}^*_{\rho\rho}$. Thus $\mathfrak{K}^*_{\rho\rho}$ arises from $\Omega(g^*_1,\dots,g^*_\rho)$ by adjoining a finite number of elements from $\mathfrak{K}^*_{\rho\rho}$, or also by adjoining one suitably chosen function; i.e. one has

$$\mathfrak{K}^*_{\rho\rho} = \Omega(g^*_1,\dots,g^*_\rho,g^*_{\rho+1},\dots,g^*_k) = \Omega(g^*_1,\dots,g^*_\rho,g^*_0).$$

By the transfer principle we therefore have

Theorem II: Every field $\mathfrak{K}_{n\rho}$ possesses a rational basis of algebraic rank ρ; the rational basis can in particular be chosen so that it contains at most $\rho+1$ functions.

Let an arbitrary system of ρ algebraically independent functions from $\mathfrak{K}_{n\rho}$ be given by $h_1(x),\dots,h_\rho(x)$, and suppose it can be extended by the above method to a rational basis

$$h_1(x),\dots,h_\rho(x);h_{\rho+1}(x),\dots,h_\sigma(x).$$

By the defining property, relations exist:

$$\begin{aligned} h_i(x) &= \chi_i\big(g_1(x),\dots,g_k(x)\big) \quad (i=1,2,\dots,\sigma),\\ g_j(x) &= \psi_j\big(h_1(x),\dots,h_\sigma(x)\big) \quad (j=1,2,\dots,k), \end{aligned}$$

where χ_i and ψ_j denote rational functions of their arguments. Thus:

From any rational basis one obtains every further rational basis by a birational transformation; and:

If

$$h_1(x),\dots,h_\rho(x)$$

is any system of ρ algebraically independent functions from $\mathfrak{K}_{n\rho}$, then $\mathfrak{K}_{n\rho}$ is a finite algebraic extension of

$$\Omega\big(h_1(x),\dots,h_\rho(x)\big).$$

§ 5. Involution form and involution basis of the fields $\mathfrak{K}_{n\rho}$.

Whereas every rational basis constructed in § 4 had the disadvantage of depending on the arbitrarily chosen starting functions, we now show the existence of a *rational basis uniquely determined by the field.*

By § 4, $\Omega(x_1,\dots,x_\rho)$ is an algebraic extension — say of degree i — of $\mathfrak{K}^*_{\rho\rho}$, and, because

$$\mathfrak{K}^*_{\rho\rho}(x_1,\dots,x_\rho) = \Omega(x_1,\dots,x_\rho),$$

arises by adjoining the elements $x_1,\dots,x_\rho$, which are algebraic with respect to $\mathfrak{K}^*_{\rho\rho}$. We therefore consider the integral function $\Phi^*(z,u)$ irreducible with respect to $\mathfrak{K}^*_{\rho\rho}$, with the zero

$$z = u_1x_1+u_2x_2+\dots+u_\rho x_\rho = u(x),$$

which is of ith degree in z. Further, as the product of the quantities conjugate to $[z-u(x)]$, $\Phi^*(z,u)$ is a homogeneous form of dimension i in $z,u_1,\dots,u_\rho$; hence as coefficient of z^i it contains a function from $\mathfrak{K}^*_{\rho\rho}$ that is free of u. If we make this highest coefficient equal to 1 by division, then the irreducible function $\Phi^*(z,u)$ is uniquely determined. The homogeneous form so defined,

$$\begin{aligned} \Phi^*(z,u) = z^i + {\textstyle\sum}' g^*_{\alpha\alpha_1\dots\alpha_\rho}(x_1,\dots,x_\rho) z^\alpha u_1^{\alpha_1}\cdots u_\rho^{\alpha_\rho},\\ (\alpha+\alpha_1+\dots+\alpha_\rho=i), \end{aligned} \tag{1}$$

is to be called the *involution form*[92] of $\mathfrak{K}^*_{\rho\rho}$; from (1) one obtains

$$\begin{aligned} \Phi(z,u) = z^i + {\textstyle\sum}' g_{\alpha\alpha_1\dots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1}\cdots u_\rho^{\alpha_\rho},\\ (\alpha+\alpha_1+\dots+\alpha_\rho=i), \end{aligned} \tag{2}$$

as an *involution form* of $\mathfrak{K}_{n\rho}$.

The coefficients of the involution form $\Phi^*(z,u)$ will turn out to be a rational basis of $\mathfrak{K}^*_{\rho\rho}$; i.e. the relation

$$\Omega\big(g^*_{\alpha\alpha_1\dots\alpha_\rho}(x_1,\dots,x_\rho)\big) = \mathfrak{K}^*_{\rho\rho} \quad (\alpha+\alpha_1+\dots+\alpha_\rho=i) \tag{3}$$

[92] The reason for the name will be clear at the end of the paragraph; $\sum'$ means that the summation is to extend over all terms except z^i.

holds. For the proof we observe that $\Phi^*(z,u)$ has coefficients from $\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x))$ and, because of its irreducibility with respect to $\mathfrak{K}^*_{\rho\rho}$, is all the more irreducible with respect to this subfield. Hence $\Phi^*(z,u)$ gives the irreducible equation of $u(x)$ with respect to $\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x))$. It follows that $\Omega(x_1,\ldots,x_\rho)$ is an algebraic extension — of degree i — of $\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x))$. Since $\mathfrak{K}^*_{\rho\rho}$ lies between $\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x))$ and $\Omega(x_1,\ldots,x_\rho)$, the identity of degrees is known to imply the identity of the fields, hence relation (3). By the transfer principle, the coefficients of $\Phi(z,u)$ correspondingly yield a rational basis of $\mathfrak{K}_{n\rho}$; i.e. we have

Theorem III: The coefficients of the involution form constitute a rational basis determined by the field, the involution basis; for fields $\mathfrak{K}^*_{\rho\rho}$ this basis is uniquely determined.[93]

We add an irrational interpretation of this fact, which establishes the connection with the geometric theory of involutions.

The factorization of $\Phi^*(z,u)$ into linear factors is given in an extension field by

$$\begin{aligned}\Phi^*(z,u) = (z - u_1x_1 - \cdots - u_\rho x_\rho)(z - u_1x_1^{(1)} - \cdots - u_\rho x_\rho^{(1)})\cdots \\ \cdot(z - u_1x_1^{(i-1)} - \cdots - u_\rho x_\rho^{(i-1)}).\end{aligned} \tag{4}$$

Comparison with (1) shows that the functions

$$g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x_1,\ldots,x_\rho) \quad (\alpha+\alpha_1+\cdots+\alpha_\rho = i)$$

are identical with the elementary symmetric functions of the rows of quantities

$$\begin{matrix} x_1 & x_2 & \cdots & x_\rho \\ x_1^{(1)} & x_2^{(1)} & \cdots & x_\rho^{(1)} \\ \vdots & \vdots & & \vdots \\ x_1^{(i-1)} & x_2^{(i-1)} & \cdots & x_\rho^{(i-1)}. \end{matrix} \tag{5}$$

Thus every function from $\mathfrak{K}^*_{\rho\rho}$ is a rational combination of these elementary functions, symmetric in the rows (5), and conversely every symmetric function of the rows (5) belongs to $\mathfrak{K}^*_{\rho\rho}$.

Thus, as

$$f^*(x) = \frac{F^*(x)}{H^*(x)}$$

runs through all functions from $\mathfrak{K}^*_{\rho\rho}$, there is a system of infinitely many relations:

$$\frac{F^*(x)}{H^*(x)} = \frac{F^*(x^{(1)})}{H^*(x^{(1)})} = \cdots = \frac{F^*(x^{(i-1)})}{H^*(x^{(i-1)})}, \tag{6}$$

or, summarized,

$$F^*(x^{(k)})H^*(x^{(l)}) - F^*(x^{(l)})H^*(x^{(k)}) = 0 \quad (k,l = 0,1,\ldots,i-1),$$

where neither $F^*(x^{(k)})$ nor $H^*(x^{(k)})$ vanishes identically, since they are conjugate to $F^*(x)$ and $H^*(x)$ and formally symmetric.

Conversely, for every row of quantities ξ satisfying the infinitely many relations

$$F^*(x)H^*(\xi) - H^*(x)F^*(\xi) = 0, \qquad H^*(\xi) \neq 0, \tag{7}$$

one obtains membership in the rows of quantities (5).

Indeed, if $G^*(x)$ denotes the common denominator of the coefficients of $\Phi^*(z,u)$, then by assumption (7) the function

$$G^*(\xi)\cdot\Phi^*(z,u;\xi),$$

which is also integral in ξ, does not vanish identically — that is, the coefficients of $\Phi^*(z,u;\xi)$ do not become indeterminate — and has the zero

$$z = u_1\xi_1 + u_2\xi_2 + \cdots + u_\rho\xi_\rho.$$

Because of (7), $\Phi^*(z,u;x)$ therefore also has the zero $z = u(\xi)$; i.e. ξ belongs to the rows of quantities (5). The corresponding assertion holds by (6) if ξ satisfies a system of equations with respect to a conjugate row of quantities:

$$F^*(x^{(k)})H^*(\xi) - H^*(x^{(k)})F^*(\xi) = 0; \qquad H^*(\xi) \neq 0.$$

[93] For fields $\mathfrak{K}_{n\rho}$ the involution form may depend on the mapping; uniqueness can be obtained by a mapping with indeterminate coefficients.

Thus the rows of quantities (5) are defined as the solutions ξ of the system of equations

$$F^*(x)H^*(\xi) - H^*(x)F^*(\xi) = 0; \qquad H^*(\xi) \neq 0,$$

where $F^*(x) : H^*(x)$ runs through all functions from $\mathfrak{K}^*_{\rho\rho}$; here the row x may also be replaced by any conjugate row.

Geometrically this says the following, interpreting each row x as a point: the solutions of (7) represent, in a ρ-dimensional linear space, ∞^ρ groups of i points each, in such a way that any point of the group determines the individual group uniquely. Such a system of point-groups is called an "*involution in a linear space*."[94]

Correspondingly, a field $\mathfrak{K}_{n\rho}$ characterizes an involution in a linear space consisting of curves, surfaces, and so on.

By the preceding, the degree i of $\Omega(x_1, \ldots, x_\rho)$ with respect to $\mathfrak{K}^*_{\rho\rho}$ is equal to the number of solution systems of (7) satisfying the side condition $H^*(\xi) \neq 0$ — which we may call the solution systems of $\mathfrak{K}^*_{\rho\rho}$. Hence the argument used to prove the existence of the involution basis also permits the following irrational formulation:

"If $\mathfrak{L}^*$ is a divisor of $\mathfrak{K}^*_{\rho\rho}$ and every solution system of the field $\mathfrak{L}^*$ is also a solution system of $\mathfrak{K}^*_{\rho\rho}$, then $\mathfrak{L}^*$ and $\mathfrak{K}^*_{\rho\rho}$ are identical."

The corresponding statement holds, by the transfer principle, for divisors $\mathfrak{L}$ of $\mathfrak{K}_{n\rho}$.

§ 6. Minimal basis of the fields $\mathfrak{K}_{n\rho}$.

This paragraph gives an overview of known theorems, but in an expanded formulation made possible by the transfer principle and the existence of the rational basis.

Definition IV: A rational basis of $\mathfrak{K}_{n\rho}$ is called a minimal basis if it contains only the minimum number ρ of functions, which must then be algebraically independent.

From theorems of Lüroth, Castelnuovo, and Enriques[95] one obtains the facts:

I. Every field $\mathfrak{K}_{n1}$ possesses a minimal basis, the Lüroth function of the field, uniquely determined up to linear fractional transformation.

II. For fields $\mathfrak{K}_{n2}$, a minimal basis exists always, and in general only, if one permits an algebraic extension of the coefficient field Ω.

III. For fields $\mathfrak{K}_{n\rho}$ ($\rho \geq 3$), in general no minimal basis exists. A characterization of the special fields $\mathfrak{K}_{n\rho}$ with minimal basis has not yet been attempted.

We briefly indicate the proof of Lüroth's theorem for fields $\mathfrak{K}^*_{11}$ given by E. Steinitz:[96]

Let $\Phi^*(z, u)$ be the involution form of $\mathfrak{K}^*_{11}$, and let $\psi^*(x)$ be any coefficient of $\Phi^*(z, u)$ that actually contains the indeterminate x. It turns out that the degree of $\Omega(x)$ with respect to $\Omega(\psi^*(x))$ is equal to the degree of $\Omega(x)$ with respect to $\mathfrak{K}^*_{11}$, whence the identity of the fields follows. Further, if $\Omega(\chi^*(x))$ is equal to $\Omega(\psi^*(x))$, then the mutual rational dependence of $\chi^*(x)$ and $\psi^*(x)$ implies the linear fractional dependence of the two functions.

By the transfer principle, Lüroth's theorem therefore also admits the formulation:

The minimal basis of the fields $\mathfrak{K}_{n1}$ consists of any function of the involution basis; and any two functions of the involution basis of $\mathfrak{K}_{n1}$ depend on one another by a linear fractional transformation.

G. Castelnuovo proves Fact II, by means of the geometric theory of involutions, for fields $\mathfrak{K}^*_{22}$ derived from a rational basis of three functions. Since the transfer principle also remains valid under finite algebraic extension of the coefficient field Ω, the existence of the rational basis makes the proof complete for all fields $\mathfrak{K}_{n2}$.

Regarding III, it should be noted that Enriques constructs a nonrational involution in three-dimensional space; that is, a field $\mathfrak{K}_{33}$ which has no minimal basis.

§ 7. Arbitrary system $\mathfrak{S}_{n\rho}$, linear family $\mathfrak{L}_{n\rho}$, and integrality domain $\mathfrak{J}_{n\rho}$ of rational functions. Existence of the rational basis.

From the existence of the rational basis of the fields $\mathfrak{K}_{n\rho}$ we shall now obtain the rational basis for arbitrary systems of rational and integral rational functions. We arrange these systems in the order in which they

[94] The excluded solutions of (7), for which $F^*(\xi) = 0$, $H^*(\xi) = 0$, and likewise the excluded solutions of the system of equations corresponding to the involution basis, are called fundamental points of the involution; this also includes those arising by homogenization, the points "lying at infinity."

[95] J. Lüroth: Beweis eines Satzes über rationale Kurven, Math. Ann. 9 (1875). — G. Castelnuovo: Sulla razionalità delle involuzioni piane, Math. Ann. 44 (1893). — F. Enriques: Sopra una involuzione non razionale dello spazio, Rend. Acc. Linc., Vol. 21, 21 Jan. 1912.

[96] Loc. cit., § 24.

successively arise from one another through the elementary operations of addition, multiplication, and division.

I. As always, let $\mathfrak{S}_{n\rho}$ denote an arbitrary system of rational (or integral rational) functions of n indeterminates and of algebraic rank ρ. The quantities from Ω are counted as belonging to the system.

II. The system $\mathfrak{L}_{n\rho}$ is called a *linear family* if it satisfies the conditions:

1. Together with $f(x)$, $\mathfrak{L}_{n\rho}$ always also contains $c \cdot f(x)$, where c is an arbitrary quantity from Ω;
2. Together with $f(x)$ and $g(x)$, $\mathfrak{L}_{n\rho}$ always also contains $f(x) + g(x)$.

III. The linear family $\mathfrak{J}_{n\rho}$ is called an *integrality domain* under the further condition:

3. Together with $f(x)$ and $g(x)$, $\mathfrak{J}_{n\rho}$ always also contains $f(x) \cdot g(x)$.

IV. The integrality domain $\mathfrak{K}_{n\rho}$ is called a *field* under the further condition:

4. Together with $f(x)$ and $g(x)$ — where $g(x) \neq 0$ — $\mathfrak{K}_{n\rho}$ always also contains the quotient $f(x) : g(x)$.

Conditions 1 through 4 are those stated in § 1 for the field.[97]

These domains are now to be derived successively from one another.

a) An arbitrary system $\mathfrak{S}_{n\rho}$ is enlarged to a linear family $\mathfrak{L}_{n\rho}$ by the requirement:

$\mathfrak{L}_{n\rho}$ contains, in addition to $\mathfrak{S}_{n\rho}$, all functions $h(x)$ and only those that can be represented linearly with coefficients from Ω by a finite number of functions from $\mathfrak{S}_{n\rho}$:

$$h(x) = c_1 f_1(x) + c_2 f_2(x) + \cdots + c_k f_k(x),$$

where $f_1(x), \ldots, f_k(x)$ run through all functions from $\mathfrak{S}_{n\rho}$, and $c_1, \ldots, c_k$ through all quantities from Ω.

For adding rational combinations of functions of the system preserves the algebraic rank of a system. $\mathfrak{L}_{n\rho}$ satisfies conditions 1 and 2, and is therefore a linear family. Moreover, every linear family containing $\mathfrak{S}_{n\rho}$ also contains $\mathfrak{L}_{n\rho}$ by 1 and 2; hence $\mathfrak{L}_{n\rho}$ is the smallest linear family containing $\mathfrak{S}_{n\rho}$.

b) Correspondingly, from a linear family $\mathfrak{L}_{n\rho}$ arises the smallest containing integrality domain $\mathfrak{J}_{n\rho}$ by the requirement:

$\mathfrak{J}_{n\rho}$ contains, in addition to $\mathfrak{L}_{n\rho}$, all functions $h(x)$ and only those that can be represented as an *integral rational combination* — with coefficients from Ω — of a finite number of functions from $\mathfrak{L}_{n\rho}$, $f_1(x), \ldots, f_k(x)$, where $f_1(x), \ldots, f_k(x)$ run through all functions from $\mathfrak{L}_{n\rho}$.

c) Finally, from an integrality domain $\mathfrak{J}_{n\rho}$ arises the smallest containing field $\mathfrak{K}_{n\rho}$ by the requirement: $\mathfrak{K}_{n\rho}$ contains, in addition to $\mathfrak{J}_{n\rho}$, all functions $h(x)$ and only those that can be represented as quotients of two functions from $\mathfrak{J}_{n\rho}$.

Combining the three extensions into one step gives:

Every system $\mathfrak{S}_{n\rho}$ can be enlarged to a smallest containing field $\mathfrak{K}_{n\rho}$ by the requirement:

$\mathfrak{K}_{n\rho}$ contains, in addition to $\mathfrak{S}_{n\rho}$, all functions $h(x)$ and only those that can be represented as rational combinations — with coefficients from Ω — of a finite number of functions from $\mathfrak{S}_{n\rho}$, $f_1(x), \ldots, f_k(x)$, where $f_1(x), \ldots, f_k(x)$ run through all functions from $\mathfrak{S}_{n\rho}$.

From the latter fact the existence of the rational basis for arbitrary domains follows immediately:

Let $\mathfrak{K}_{n\rho}$ be the smallest field containing $\mathfrak{S}_{n\rho}$, and let a rational basis of $\mathfrak{K}_{n\rho}$ be given by

$$h_1(x),\ h_2(x), \ldots, h_\sigma(x). \tag{1}$$

By the definition of $\mathfrak{K}_{n\rho}$, relations exist:

$$\begin{aligned} h_1(x) &= r_1\big(g_1(x), \ldots, g_\lambda(x)\big); \ldots; \\ h_\sigma(x) &= r_\sigma\big(g_\mu(x), \ldots, g_\nu(x)\big), \end{aligned} \tag{2}$$

where $r_1, \ldots, r_\sigma$ are rational functions of their arguments, and

$$g_1(x), \ldots, g_\lambda(x), \ldots, g_\mu(x), \ldots, g_\nu(x) \tag{3}$$

all belong to $\mathfrak{S}_{n\rho}$. Because of the relations (2), (3), in addition to (1), also represents a rational basis of $\mathfrak{K}_{n\rho}$; because the functions (3) belong to $\mathfrak{S}_{n\rho}$, this is at the same time a rational basis of $\mathfrak{S}_{n\rho}$. We therefore have

Theorem IV: An arbitrary system $\mathfrak{S}_{n\rho}$ of rational (or integral rational) functions possesses a rational basis, consisting of a finite number of functions of the system, of algebraic rank ρ.

Now let $\mathfrak{L}_{n\rho}$ in particular be a linear family, and let

$$g_1(x), \ldots, g_\rho(x),\ g_{\rho+1}(x), \ldots, g_\nu(x) \tag{4}$$

[97] $f(x)$ and $g(x)$ may also be of degree zero, i.e. quantities from Ω.

be a rational basis of $\mathfrak{L}_{n\rho}$. Suppose, for instance, that

$$g_1(x), \ldots, g_\rho(x)$$

are algebraically independent. If one then sets

$$g(x) = c_1 g_{\rho+1}(x) + c_2 g_{\rho+2}(x) + \cdots + c_{\nu-\rho} g_\nu(x),$$

then the c_i can be chosen as numbers from Ω in such a way that

$$g_1(x), \ldots, g_\rho(x),\ g(x) \tag{5}$$

becomes a rational basis of the smallest containing field $\mathfrak{K}_{n\rho}$. But since $\mathfrak{L}_{n\rho}$ is a linear family, $g(x)$ belongs to $\mathfrak{L}_{n\rho}$, and (5) represents a rational basis of $\mathfrak{L}_{n\rho}$; that is,

Theorem V: The rational basis of every linear family $\mathfrak{L}_{n\rho}$ of rational (or integral rational) functions — hence in particular the rational basis of every integrality domain $\mathfrak{J}_{n\rho}$ and every field $\mathfrak{K}_{n\rho}$ — can be chosen in particular so that it contains at most $\rho + 1$ functions (and at least ρ).

The bound on the number of functions in the rational basis found in § 4 for fields therefore rests on the property of the field of being a linear family, whereas the restriction to ρ functions in the case of the existence of the minimal basis uses all assumptions of the field.

It remains to note that, by § 3 (4), all characteristic properties of the systems and smallest containing systems are preserved under the mapping.

§ 8. Involution basis of the integrality domains $\mathfrak{J}_{n\rho}$ consisting of polynomials.

In the following paragraphs we are concerned with systems $\mathfrak{S}_{n\rho}$ whose elements are integral rational functions (polynomials); first with integrality domains $\mathfrak{J}_{n\rho}$ consisting of polynomials. For these domains $\mathfrak{J}_{n\rho}$, the divisibility notions are first to be fixed.

Let $F(x), G(x)$ be two polynomials from $\mathfrak{J}_{n\rho}$, and let $L(x)$ be a common divisor of these polynomials, so that one has

$$F(x) = L(x) \cdot F_1(x), \qquad G(x) = L(x) \cdot G_1(x).$$

$L(x)$ is called a *common divisor in* $\mathfrak{J}_{n\rho}$ if and only if the three polynomials $L(x), F_1(x), G_1(x)$ belong to $\mathfrak{J}_{n\rho}$.

Two polynomials from $\mathfrak{J}_{n\rho}$ are called *relatively prime in* $\mathfrak{J}_{n\rho}$ if they possess no common divisor in $\mathfrak{J}_{n\rho}$.

Let $\mathfrak{K}_{n\rho}$ be the smallest field containing $\mathfrak{J}_{n\rho}$. Then, from the definition, every function from $\mathfrak{K}_{n\rho}$ has a representation

$$f(x) = F_1(x) : F_2(x),$$

where $F_1(x), F_2(x)$ belong to $\mathfrak{J}_{n\rho}$ and are relatively prime in $\mathfrak{J}_{n\rho}$. Correspondingly, several functions from $\mathfrak{K}_{n\rho}$ can be brought to a common denominator in $\mathfrak{J}_{n\rho}$:

$$f_1(x) = \frac{F_1(x)}{F(x)}, \ldots, f_k(x) = \frac{F_k(x)}{F(x)}.$$

$F(x)$ is called *a least common denominator in* $\mathfrak{J}_{n\rho}$ if and only if the polynomials from $\mathfrak{J}_{n\rho}$,

$$F(x), F_1(x), \ldots, F_k(x),$$

are relatively prime in $\mathfrak{J}_{n\rho}$.

Such a representation can be possible in different ways, since in $\mathfrak{J}_{n\rho}$ the laws of unique factorization into irreducible functions do not hold in general.

We now investigate the significance of the involution basis of $\mathfrak{K}_{n\rho}$ for $\mathfrak{J}_{n\rho}$, by transferring the notions of involution form and involution basis to integrality domains. Let

$$\Phi(z,u) = z^i + \sum{}' g_{\alpha\alpha_1\ldots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \qquad (\alpha + \alpha_1 + \cdots + \alpha_\rho = i)$$

be an involution form of $\mathfrak{K}_{n\rho}$, and let $G(x)$ be a least common denominator in $\mathfrak{J}_{n\rho}$ of the coefficients of $\Phi(z,u)$. By multiplication with $G(x)$, $\Phi(z,u)$ passes into a form $\Psi(z,u)$ whose coefficients are polynomials from $\mathfrak{J}_{n\rho}$ and are relatively prime in $\mathfrak{J}_{n\rho}$:

$$G(x) \cdot \Phi(z,u) = \Psi(z,u) = G(x) z^i + \sum{}' G_{\alpha\alpha_1\ldots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \qquad (\alpha + \alpha_1 + \cdots + \alpha_\rho = i).$$

Every form $\Psi(z,u)$ arising from an involution form $\Phi(z,u)$ of $\mathfrak{K}_{n\rho}$ by multiplication with a polynomial from $\mathfrak{J}_{n\rho}$ in such a way that the coefficients of $\Psi(z,u)$ are polynomials from $\mathfrak{J}_{n\rho}$ and are relatively prime in $\mathfrak{J}_{n\rho}$, shall be called an *involution form* of $\mathfrak{J}_{n\rho}$; and the coefficients of $\Psi(z,u)$ shall be called a corresponding *involution basis.*

It is first clear from the meaning of the involution basis of $\mathfrak{K}_{n\rho}$ that every involution basis of $\mathfrak{J}_{n\rho}$ is at the same time a rational basis. For the further investigation we consider a mapping domain $\mathfrak{J}^*_{\rho\rho}$ for whose smallest containing field $\mathfrak{K}^*_{\rho\rho}$ the irreducible equation with zero

$$z = u_1 x_1 + \cdots + u_\rho x_\rho$$

is given by $\Phi^*(z,u) = 0$.

According to § 5, the coefficients of

$$\Phi^*(z,u) = z^i + \sum{}' g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \qquad (\alpha + \alpha_1 + \cdots + \alpha_\rho = i)$$

represent the elementary symmetric functions of the rows of quantities conjugate with respect to $\mathfrak{K}^*_{\rho\rho}$,

$$x, \quad x^{(1)}, \ldots, x^{(i-1)}.$$

Let $F^*(x)$ be any polynomial from $\mathfrak{K}^*_{\rho\rho}$. Since

$$F^*(x) = \frac{1}{i}\Big(F^*(x) + F^*(x^{(1)}) + \cdots + F^*(x^{(i-1)})\Big),$$

it follows that $F^*(x)$ is an integral symmetric function of these rows of quantities; consequently it can be expressed as an integral rational combination of the elementary symmetric functions, that is, of the functions

$$g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x).$$

If now

$$G^*(x)\cdot\Phi^*(z,u)=\Psi^*(z,u)=G^*(x)z^i+\sum{}' G^*_{\alpha\alpha_1\ldots\alpha_\rho}(x)z^\alpha u_1^{\alpha_1}\cdots u_\rho^{\alpha_\rho},\qquad (\alpha+\alpha_1+\cdots+\alpha_\rho=i)$$

is an involution form of $\mathfrak{J}^*_{\rho\rho}$, then every polynomial from $\mathfrak{K}^*_{\rho\rho}$, and in particular every polynomial from $\mathfrak{J}^*_{\rho\rho}$, is an integral rational combination of the quotients

$$G^*_{\alpha\alpha_1\ldots\alpha_\rho}(x):G^*(x).$$

Taking the transfer principle into account, this gives

Theorem VI: To every integrality domain of polynomials $\mathfrak{J}_{n\rho}$ there exists at least one distinguished rational basis, the involution basis, such that in the rational representation of all polynomials from $\mathfrak{J}_{n\rho}$ by the functions of the involution basis only a power of one fixed function (the highest coefficient of the corresponding involution form) occurs in the denominator.

§ 9. Relatively integral domains $\mathfrak{G}_{n\rho}$ consisting of polynomials.

In what follows we consider special integrality domains consisting of polynomials, which form an intermediate stage between integrality domains and fields.

An integrality domain of polynomials $\mathfrak{G}_{n\rho}$ is called a "relatively integral domain" under the condition:

4a) $\mathfrak{G}_{n\rho}$ contains, along with $F(x)$ and $G(x)$ — where $G(x)\neq 0$ — the quotient $F(x):G(x)$ always and only when this quotient is integral in the indeterminates, that is, is a polynomial.

First the identity of the relatively integral domains with the totality of the polynomials of a field $\mathfrak{K}_{n\sigma}$ $(\sigma\geq\rho)$ of rational functions is to be proved.

Let $\mathfrak{K}_{n\rho}$ be the smallest field containing $\mathfrak{G}_{n\rho}$. For every function $f(x)$ from $\mathfrak{K}_{n\rho}$ there is a representation as quotient of two polynomials from $\mathfrak{G}_{n\rho}$:

$$f(x)=F_1(x):F_2(x),$$

where polynomials of degree zero are also allowed. As soon as $f(x)$ becomes a polynomial, it belongs to $\mathfrak{G}_{n\rho}$ by condition 4a); and since $\mathfrak{G}_{n\rho}$ can contain no further polynomials, *every relatively integral domain $\mathfrak{G}_{n\rho}$ is identical with the totality of the polynomials of the smallest containing field.*

Conversely, let $\mathfrak{K}_{n\sigma}$ be any field of rational functions (therefore not necessarily the smallest containing field of a given system), and let $\mathfrak{J}$ be the totality of the polynomials contained in $\mathfrak{K}_{n\sigma}$. $\mathfrak{J}$ satisfies the conditions 1, 2, 3 of an integrality domain and condition 4a), and so is a relatively integral domain; that is, *the totality of the polynomials contained in a field $\mathfrak{K}_{n\sigma}$ forms a relatively integral domain $\mathfrak{G}_{n\rho}$ $(\rho\leq\sigma)$.*[98]

We further show the identity of the relatively integral domains with Hilbert's "integrality domains of relatively integral functions."[99]

Let

$$G_1(x),\ldots,G_k(x) \tag{1}$$

be a rational basis of $\mathfrak{G}_{n\rho}$. By conditions 1, 2, 3, 4a, every rational combination of the polynomials (1) which is integral in the indeterminates, that is, becomes a polynomial, belongs to $\mathfrak{G}_{n\rho}$. Conversely, every polynomial from $\mathfrak{G}_{n\rho}$ can, by the notion of rational basis, be rationally expressed through the polynomials (1); hence $\mathfrak{G}_{n\rho}$ is *identical with the totality of those rational combinations of the polynomials (1) which are integral in the indeterminates, that is, are polynomials.* Thus we have Hilbert's definition starting from the special rational basis, whereas our definition uses only abstract properties of the domain.

For relatively integral domains, the definition of a common divisor given in § 8 for general integrality domains simplifies:

Let $F(x),G(x)$ be two polynomials from $\mathfrak{G}_{n\rho}$ and let $L(x)$ be a common divisor of these polynomials. Then $L(x)$ is called a *common divisor in* $\mathfrak{G}_{n\rho}$ if and only if $L(x)$ belongs to $\mathfrak{G}_{n\rho}$.

For then the membership of the quotients $F(x):L(x)$ and $G(x):L(x)$ follows from condition 4a).

[98] It may also be $\rho=0$; that is, the totality of the polynomials from $\mathfrak{K}_{n\sigma}$ consists of elements of the coefficient domain Ω.

[99] D. Hilbert: Mathematische Probleme. Lecture Paris 1900. Problem 14 (Göttinger Nachrichten 1900).

Another essential property of relatively integral domains is that the notion of being "algebraically integral with respect to $\mathfrak{G}_{n\rho}$" can be defined by means of any equation exactly as is known for algebraic number fields or for the field $\Omega(x_1, \dots, x_n)$.

Definition V: A quantity ξ is called algebraically integral with respect to $\mathfrak{G}_{n\rho}$ — or also relatively algebraically integral — if it satisfies some equation whose highest coefficient is the unit and whose remaining polynomials are from $\mathfrak{G}_{n\rho}$.

Suppose, namely, that the relatively algebraically integral quantity ξ satisfies, for example, the equation

$$L(z) = z^{\lambda} + G_1(x)z^{\lambda-1} + \cdots + G_{\lambda}(x) = 0.$$

$L(z)$ is divisible by the integral function irreducible with respect to the smallest containing field $\mathfrak{K}_{n\rho}$,

$$M(z) = z^{\mu} + H_1(x)z^{\mu-1} + \cdots + H_{\mu}(x).$$

But from this divisibility it follows by the known extension of Gauss's theorem[100] that the rational functions $H_1(x), \dots, H_{\mu}(x)$ are polynomials from $\mathfrak{K}_{n\rho}$, and therefore belong to $\mathfrak{G}_{n\rho}$. Thus the definition of the relatively algebraically integral quantity ξ is obtained from the irreducible equation. In particular, every polynomial $F(x)$ from $\mathfrak{G}_{n\rho}$ is also algebraically integral with respect to $\mathfrak{G}_{n\rho}$, since it satisfies the equation $z - F(x) = 0$.

It should further be mentioned that, for an integrality domain $\mathfrak{J}_{n\rho}$ of polynomials, analogously to the smallest containing field, the smallest containing relatively integral domain $\mathfrak{G}_{n\rho}$ is also defined by the requirement:

$\mathfrak{G}_{n\rho}$ contains, along with $\mathfrak{J}_{n\rho}$, all polynomials $H(x)$, and only those, which can be represented as quotients of two polynomials from $\mathfrak{J}_{n\rho}$.

From this definition it follows further: From every involution form and involution basis of $\mathfrak{J}_{n\rho}$ a corresponding one of $\mathfrak{G}_{n\rho}$ arises by splitting off, at most, a common divisor in $\mathfrak{G}_{n\rho}$ of all polynomials of the basis.

For $\mathfrak{J}_{n\rho}$ and $\mathfrak{G}_{n\rho}$ possess the same smallest containing field $\mathfrak{K}_{n\rho}$; and every involution form of $\mathfrak{J}_{n\rho}$ arises from an involution form of $\mathfrak{K}_{n\rho}$ by multiplication with a polynomial from $\mathfrak{J}_{n\rho}$, and so has coefficients from $\mathfrak{G}_{n\rho}$, which however may have a common divisor in $\mathfrak{G}_{n\rho}$.[101]

Finally it should be noted that the *mapping domain* of a relatively integral domain $\mathfrak{G}_{n\rho}$, which by § 7 is again an integrality domain $\mathfrak{J}^*_{\rho\rho}$ of polynomials, need not itself be a relatively integral domain. For if $F(x)$ and $G(x)$ are two polynomials from $\mathfrak{G}_{n\rho}$ whose quotient is not a polynomial and consequently does not belong to $\mathfrak{G}_{n\rho}$, then the quotient $F^*(x) : G^*(x)$ also does not belong to $\mathfrak{J}^*_{\rho\rho}$. But this quotient, arising by specialization of some indeterminates, may very well be a polynomial, and for $\mathfrak{J}^*_{\rho\rho}$ condition 4a) is then not fulfilled.[102]

§ 10. Relatively integral domains of the first kind and their integrality basis.

The notion introduced in § 9 (Definition V), "relatively algebraically integral," makes it possible to give a class of relatively integral domains which are finite integrality domains, and thereby to answer positively, at least for this class, a question raised by D. Hilbert (Math. Problems, Problem 14). We give the following

Definition VI: A finite number of polynomials from $\mathfrak{G}_{n\rho}$ is called an integrality basis if every polynomial from $\mathfrak{G}_{n\rho}$ can be represented as an integral rational combination of this finite number, with coefficients from Ω. An integrality domain of polynomials which possesses an integrality basis is called a finite integrality domain.

The class of relatively integral domains to be considered, for which the involution basis will prove to be an integrality basis, we designate as relatively integral domains of the first kind according to the following

Definition VII: A relatively integral domain $\mathfrak{G}_{n\rho}$ is called of the first kind if, as mapping domain, it possesses a relatively integral domain $\mathfrak{G}^*_{\rho\rho}$ such that every arbitrary polynomial in $x_1, \dots, x_\rho$ (with coefficients from Ω) is algebraically integral over $\mathfrak{G}^*_{\rho\rho}$.

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[100] Compare, for instance, J. König: Einleitung in die allgemeine Theorie der algebraischen Größen (Leipzig, B. G. Teubner, 1903), Kap. IX, § 2, or Weber (small edition), § 20.

[101] For example, for the integrality domain $\mathfrak{J}_{22}$ consisting of all integral rational combinations of x_1^2 and x_1x_2, the following is obtained as an involution form:

$$\Psi(z, u) = x_1^2z^2 - x_1^4u_1^2 - 2x_1^3x_2u_1u_2 - x_1^2x_2^2u_2^2.$$

Since x_1^2 belongs to $\mathfrak{J}_{22}$, and therefore all the more to the smallest containing relatively integral domain $\mathfrak{G}_{22}$, and consequently is a common divisor in $\mathfrak{G}_{22}$, the following results as an involution form of $\mathfrak{G}_{22}$:

$$\Psi(z, u) = z^2 - x_1^2u_1^2 - 2x_1x_2u_1u_2 - x_2^2u_2^2.$$

[102] Let $\mathfrak{G}_{22}$ consist, for example, of all polynomials which are rational combinations of $F = x_1^2x_2$ and $G = x_1^2(1 + x_3)$. The admissible specialization $x_3 = 0$ gives $F^* : G^* = x_2$, whereas $F : G$ is not a polynomial. $\mathfrak{J}^*_{22}$ is therefore not a relatively integral domain.

According to this definition, the indeterminates $x_1, \dots, x_\rho$, and consequently also the linear form $u(x) = u_1x_1 + \dots + u_\rho x_\rho$, are algebraically integral over $\mathfrak{G}^*_{\rho\rho}$. The irreducible integral function with zero $z = u(x)$ — the involution form of the smallest containing field $\mathfrak{K}^*_{\rho\rho}$ — therefore has the unit as highest coefficient and, as its further coefficients, polynomials from $\mathfrak{G}^*_{\rho\rho}$; it is therefore at the same time an involution form of $\mathfrak{G}^*_{\rho\rho}$, and in $\mathfrak{G}^*_{\rho\rho}$ no further involution form can exist. By the transfer principle, therefore, $\mathfrak{G}_{n\rho}$ also possesses an involution form whose highest coefficient is the unit; and since, by Theorem VI, in the rational representation of all polynomials from $\mathfrak{G}_{n\rho}$ by the corresponding involution basis only this highest coefficient enters into the denominator, the involution basis becomes an integrality basis. Thus:

Theorem VII. Every relatively integral domain of the first kind is a finite integrality domain; the integrality basis is given by the involution basis obtained from the characterizing mapping domain. In particular, the integrality basis of relatively integral domains of the first kind $\mathfrak{G}_{\rho\rho}$ is given by the uniquely defined involution basis of $\mathfrak{G}_{\rho\rho}$.

We add a criterion for relatively integral domains of the first kind. Let $\mathfrak{G}_{n\rho}$ be of the first kind and let an integrality basis of $\mathfrak{G}_{n\rho}$ be given by

$$F_1(x), F_2(x), \dots, F_k(x). \tag{1}$$

Then, by a Hilbert auxiliary theorem,[103] there exist ρ algebraically independent polynomials from $\mathfrak{G}_{n\rho}$,

$$G_1(x), G_2(x), \dots, G_\rho(x), \tag{2}$$

such that the polynomials (1), and hence every polynomial from $\mathfrak{G}_{n\rho}$, are algebraically integral over the polynomials (2). The same holds for every polynomial of the mapping domain $\mathfrak{G}^*_{\rho\rho}$ with respect to the corresponding polynomials

$$G^*_1(x), G^*_2(x), \dots, G^*_\rho(x). \tag{3}$$

By Definition VII, therefore, every arbitrary polynomial in $x_1, \dots, x_\rho$ is also algebraically integral over the polynomials (3).

Conversely, suppose a mapping domain $\mathfrak{J}^*_{\rho\rho}$ of an arbitrary relatively integral domain $\mathfrak{G}_{n\rho}$ contains ρ polynomials (3) over which every polynomial in $x_1, \dots, x_\rho$ is algebraically integral. Then we show that $\mathfrak{J}^*_{\rho\rho}$ thereby becomes a relatively integral domain $\mathfrak{G}^*_{\rho\rho}$, and further that $\mathfrak{G}^*_{\rho\rho}$, and consequently $\mathfrak{G}_{n\rho}$, are of the first kind.

Indeed, let $F^*(x)$ be a polynomial from the smallest field $\mathfrak{K}^*_{\rho\rho}$ containing $\mathfrak{J}^*_{\rho\rho}$. By assumption it is algebraically integral over the polynomials (3). Since the corresponding function from $\mathfrak{K}_{n\rho}$ is then algebraically integral over ρ polynomials from $\mathfrak{G}_{n\rho}$, it is a polynomial $F(x)$ and hence belongs to $\mathfrak{G}_{n\rho}$; therefore $F^*(x)$ belongs to $\mathfrak{J}^*_{\rho\rho}$. The mapping domain $\mathfrak{J}^*_{\rho\rho}$ therefore contains all polynomials of $\mathfrak{K}^*_{\rho\rho}$ and is a relatively integral domain $\mathfrak{G}^*_{\rho\rho}$. For $\mathfrak{G}^*_{\rho\rho}$, and hence also for $\mathfrak{G}_{n\rho}$, it follows immediately from Definitions V and VII and from the assumption that they are of the first kind; that is:

The relatively integral domains of the first kind $\mathfrak{G}_{n\rho}$ are characterized by the fact that in a mapping domain $\mathfrak{J}^*_{\rho\rho}$ there exist ρ polynomials over which every arbitrary polynomial in $x_1, \dots, x_\rho$ is algebraically integral. From this assumption the mapping domain itself results as a relatively integral domain.[104]

§ 11. Auxiliary theorem on algebraically integral dependence.

We now give a criterion for every arbitrary polynomial in $x_1, \dots, x_\rho$ to be algebraically integral over ρ polynomials in $x_1, \dots, x_\rho$,

$$G^*_1(x), \dots, G^*_\rho(x). \tag{1}$$

If the polynomials (1) are, in particular, homogeneous forms, then for every special system of values which makes (1) vanish, all algebraically integrally dependent forms vanish at the same time; hence in particular also $x_1, \dots, x_\rho$.

The forms (1), therefore, cannot vanish for any system of values other than

$$x_1 = 0, \dots, x_\rho = 0;$$

their resultant must be different from zero.

[103] D. Hilbert: Über die vollen Invariantensysteme, Math. Ann. 42 (1893), § 1. The auxiliary theorem stated there only for homogeneous polynomials holds likewise for inhomogeneous ones.

[104] For example, in the domain $\mathfrak{G}_{22}$ mentioned in the closing remark to § 9, the admissible specialization $x_1 = 1$ gives the two polynomials $F^* = x_2$, $G^* = 1 + x_3$, over which every arbitrary polynomial in x_2, x_3 is algebraically integral. $\mathfrak{G}_{22}$ is therefore of the first kind, indeed with $F(x)$ and $G(x)$ as integrality basis.

Conversely, this condition is also sufficient and admits an extension to inhomogeneous polynomials according to the following

Auxiliary theorem: A sufficient condition that every arbitrary polynomial in $x_1, \ldots, x_\rho$ *be algebraically integral over* ρ *polynomials in* $x_1, \ldots, x_\rho$

$$G_1^*(x), \ldots, G_\rho^*(x)$$

is the non-vanishing of the resultant, taken with respect to $x_1, \ldots, x_\rho$*, of the terms of highest dimension*

$$H_1^*(x), \ldots, H_\rho^*(x)$$

of the polynomials (1):

$$R_0 = \begin{bmatrix} H_1^* & \cdots & H_\rho^* \\ x_1 & \cdots & x_\rho \end{bmatrix} \neq 0. \tag{2}$$

For homogeneous forms (1) the condition (2) is necessary.[105]

For the proof we form the polynomials

$$\begin{gathered} \bar{G}_1^*(x) = G_1^*(x) - \lambda_1, \ldots, \bar{G}_\rho^*(x) = G_\rho^*(x) - \lambda_\rho, \\ \bar{\Phi}^*(x) = \Phi^*(x) - \mu, \end{gathered} \tag{3}$$

where $\lambda_1, \ldots, \lambda_\rho, \mu$ denote indeterminates and $\Phi^*(x)$ denotes an arbitrary polynomial in $x_1, \ldots, x_\rho$. Let the resultant of the polynomials (3) with respect to $x_1, \ldots, x_\rho, 1$[106] be given by

$$R(\lambda, \mu) = \begin{bmatrix} \bar{G}_1^* & \cdots & \bar{G}_\rho^* & \bar{\Phi}^* \\ x_1 & \cdots & x_\rho & 1 \end{bmatrix}; \tag{4}$$

it is an integral rational function of $\lambda_1, \ldots, \lambda_\rho, \mu$. According to F. Mertens, in the expansion of (4) by powers of μ, the highest coefficient can be explicitly specified;[107] one obtains

$$(-1)^\tau R(\lambda, \mu) = R_0^\sigma \mu^\tau + A_1(\lambda)\mu^{\tau-1} + \cdots + A_\tau(\lambda), \tag{5}$$

where $A_1(\lambda), \ldots, A_\tau(\lambda)$ denote integral integer-valued functions of $\lambda_1, \ldots, \lambda_\rho$ and the coefficients of the polynomials (3). This expansion first shows that under the assumption (2), $R_0 \neq 0$, also $R(\lambda, \mu)$ cannot vanish identically. The identity in $x_1, \ldots, x_\rho, \lambda_1, \ldots, \lambda_\rho, \mu$,

$$R(\lambda, \mu) \equiv 0 \pmod{\bar{G}_1^*(x), \ldots, \bar{G}_\rho^*(x), \bar{\Phi}^*(x)},$$

passes, by the substitution

$$\lambda_i = G_i^*(x), \qquad \mu = \Phi^*(x),$$

into

$$\begin{aligned} R(G_i^*(x), \Phi^*(x)) = R_0^\sigma \Phi^{*\tau}(x) + A_1(G_i^*(x))\Phi^{*\tau-1}(x) + \cdots \\ + A_\tau(G_i(x)) = 0, \end{aligned} \tag{6}$$

and this proves the auxiliary theorem, since R_0 is, by assumption, a number from Ω.

§ 12. The integrality basis of regular systems $\mathfrak{S}_{n\rho}$ consisting of polynomials.

The auxiliary theorem of § 11, together with the conclusion of § 10, immediately yields the fact:

If there exist, in a mapping domain $\mathfrak{J}_{\rho\rho}^*$ *of* $\mathfrak{S}_{n\rho}$*,* ρ *polynomials such that the homogeneous resultant of their terms of highest dimension does not vanish identically —* $R_0 \neq 0$ *—, then* $\mathfrak{S}_{n\rho}$ *is of the first kind and consequently is a finite integrality domain, with the involution basis as integrality basis.*

The condition $R_0 \neq 0$ now permits, by means of the auxiliary theorem, a second existence proof for the integrality basis, essentially due to D. Hilbert;[108] this proof, to be sure, does not make the involution basis recognizable as such, but it holds uniformly for all systems $\mathfrak{S}_{n\rho}$ with the indicated property.

[105] For resultant theory, compare F. Mertens, Zur Theorie der Elimination (I. and II. Teil), Sitzungsber. d. k. Ak. d. Wiss. Wien, Math.-naturw. Kl. Abt. IIa, Bd. 108 (1899). Partly reproduced in J. König, Theorie d. algebr. Größen, Kap. VI.

[106] That is, the homogeneous resultant with respect to $x_1, \ldots, x_\rho, x_0$, if the polynomials (3) are homogenized by means of an additional indeterminate x_0 so that the degree in the x is preserved.

[107] A. a. O. § 3 (proof of Theorem I), and explicitly § 6 (determination of the totality of all terms of the resultant R which contain a given power product $a_{\lambda\lambda}^{p_\lambda} a_{\mu\mu}^{p_\mu} \cdots a_{\sigma\sigma}^{p_\sigma}$ as factor). In our case $p_\lambda = 0$, $p_\mu = 0, \ldots, p_\rho = \tau$, $a_{\sigma\sigma} = (-\mu)$ is set. Reproduced in König, Kap. VI, § 9.

[108] Über die vollen Invariantensysteme, § 2. In Hilbert's case, under the assumption of the otherwise obtained existence of the integrality basis of the invariant field, the point is only an interpretation of this basis through the Kronecker fundamental system of the algebraically integral quantities of a field.

Indeed, let the ρ polynomials from $\mathfrak{J}^*_{\rho\rho}$ for which $R_0 \neq 0$, and which consequently are algebraically independent, be given by

$$G_1^*(x), \ldots, G_\rho^*(x). \tag{1}$$

By the auxiliary theorem, every arbitrary polynomial in $x_1, \ldots, x_\rho$, hence in particular every polynomial of the smallest field $\mathfrak{K}^*_{\rho\rho}$ containing $\mathfrak{J}^*_{\rho\rho}$, is algebraically integral over the polynomials (1). If the corresponding polynomials in $\mathfrak{S}_{n\rho}$ are therefore

$$G_1(x), \ldots, G_\rho(x), \tag{2}$$

then, by the transfer principle, every polynomial of the smallest field $\mathfrak{K}_{n\rho}$ containing $\mathfrak{S}_{n\rho}$, and hence in particular every polynomial from $\mathfrak{S}_{n\rho}$, is algebraically integral over the polynomials (2). Conversely, every function from $\mathfrak{K}_{n\rho}$ which is algebraically integral over the polynomials (2) is also a polynomial. Thus, if one regards $\mathfrak{K}_{n\rho}$ (according to § 4) as an algebraic field over $\Omega(G_1, \ldots, G_\rho)$, the totality of the polynomials from $\mathfrak{K}_{n\rho}$ — the relatively integral domain $\mathfrak{G}_{n\rho}$ — is identical with the algebraically integral quantities of this field. Let $G(x)$ be a polynomial from $\mathfrak{K}_{n\rho}$ which completes (2) to a rational basis, and let $D(x)$ be the discriminant of the irreducible equation of degree k which $G(x)$ satisfies with respect to $\Omega(G_1, \ldots, G_\rho)$. Then every polynomial from $\mathfrak{K}_{n\rho}$, and in particular every polynomial $F(x)$ from $\mathfrak{S}_{n\rho}$, as an algebraically integral quantity, admits the representation

$$F(x) = \frac{\Gamma_1(G_i)G(x)^{k-1} + \Gamma_2(G_i)G(x)^{k-2} + \cdots + \Gamma_k(G_i)}{D(x)}. \tag{3}$$

Let $F(x)$ now run through all polynomials from $\mathfrak{S}_{n\rho}$. Then the application of Hilbert's Theorem I on the module basis (Math. Ann. 36) to the system formed from the corresponding functions

$$v_1\Gamma_1(G_i) + v_2\Gamma_2(G_i) + \cdots + v_k\Gamma_k(G_i)$$

shows the existence of a finite number of polynomials from $\mathfrak{S}_{n\rho}$,

$$F_1(x), F_2(x), \ldots, F_\lambda(x), \tag{4}$$

such that every polynomial from $\mathfrak{S}_{n\rho}$ admits a representation

$$F(x) = A_1(G_i)F_1(x) + A_2(G_i)F_2(x) + \cdots + A_\lambda(G_i)F_\lambda(x), \tag{5}$$

where $A_1, \ldots, A_\lambda$ denote polynomials in the $G_i(x)$. The polynomials (2) and (4) therefore form an integrality basis of $\mathfrak{S}_{n\rho}$; that is:

Theorem VIII: If there exist, in a mapping system $\mathfrak{J}^*_{\rho\rho}$ of the system of polynomials $\mathfrak{S}_{n\rho}$, ρ polynomials such that the resultant of the terms of highest dimension does not vanish identically — $R_0 \neq 0$ —, then $\mathfrak{S}_{n\rho}$ possesses an integrality basis.

This theorem allows a slight generalization. It is enough that the ρ polynomials (1) for which $R_0 \neq 0$ belong to the integrality domain $\mathfrak{J}^*_{\rho\rho}$ which is the smallest containing domain of $\mathfrak{S}^*_{\rho\rho}$. In fact, the arguments leading to Theorem VIII show that the representation (5) still holds then. But by the definition, every polynomial $L(x)$ of the smallest containing integrality domain $\mathfrak{J}_{n\rho}$ can be represented integrally and rationally by a finite number — depending on $L(x)$ — of polynomials from $\mathfrak{S}_{n\rho}$; hence there exist σ polynomials from $\mathfrak{S}_{n\rho}$,

$$K_1(x), K_2(x), \ldots, K_\sigma(x), \tag{6}$$

such that the polynomials $G_i(x)$ become integral rational combinations of (6). Thus (6) and (4) give an integrality basis. Introducing the following terminology:

Definition VIII: A system $\mathfrak{S}_{n\rho}$ of polynomials is called regular if in a mapping domain $\mathfrak{J}^*_{\rho\rho}$ of the smallest containing integrality domain there exist ρ polynomials such that the resultant of their terms of highest dimension does not vanish identically — $R_0 \neq 0$ —

we therefore have:

Theorem IX: Every regular system $\mathfrak{S}_{n\rho}$ of polynomials possesses an integrality basis.[109]

We further give, analogously to § 5 for the involution, a geometric interpretation of regular systems. For this purpose we consider, as there in (7), the system of equations

$$L^*(x) - L^*(\xi) = 0, \tag{7}$$

[109]That there are non-regular systems without integrality basis was shown by Hilbert with an example (Gött. Nachr. 1891, p. 233). The following is even simpler:

$$\mathfrak{S}_{22} = x_1, x_1x_2, x_1x_2^2, \ldots, x_1x_2^\nu, \ldots \quad \text{in inf.}$$

where $L^*(x)$ runs through all polynomials from $\mathfrak{J}^*_{\rho\rho}$. Homogenizing by means of x_0, ξ_0, let (7) pass into the system of equations between homogeneous forms

$$\xi_0^m M^*(x) - x_0^m M^*(\xi) = 0, \tag{8}$$

where, if $H^*(x)$ denotes the terms of highest dimension of $L^*(x)$, the relation

$$H^*(x) = M^*(x) \pmod{x_0} \tag{9}$$

holds.

As *fundamental points* of $\mathfrak{J}^*_{\rho\rho}$ (compare the note to § 5), we designate the solutions of (8) that are independent of x — thus different from the rows conjugate to x. Fundamental points are therefore the systems of values different from $\xi_1 = 0, \ldots, \xi_\rho = 0$ for which simultaneously

$$\xi_0^m = 0, \qquad M^*(\xi) = 0$$

holds, or, by (9),

$$\xi_0 = 0, \qquad H^*(\xi) = 0.$$

If now there exist ρ forms $H^*(\xi)$ for which $R_0 \neq 0$, hence with no common system of solutions, then $\mathfrak{J}^*_{\rho\rho}$ can possess no fundamental point. If, in particular, $\mathfrak{S}^*_{\rho\rho}$ consists of homogeneous forms, then for $R_0 \neq 0$ also $\mathfrak{S}^*_{\rho\rho}$ can possess no fundamental point, since such a point would at the same time be a fundamental point of $\mathfrak{J}^*_{\rho\rho}$. But the converse also holds. For this we consider the system $\mathfrak{J}(H^*)$ formed from the totality of the forms $H^*(x)$; this is again an integrality domain.

From Hilbert's Theorem I on the module basis and Hilbert's auxiliary theorem (compare the citation in § 10), there follows the existence of ρ forms from $\mathfrak{J}(H^*)$ whose vanishing has the vanishing of all forms from $\mathfrak{J}(H^*)$ as consequence. If now $\mathfrak{J}^*_{\rho\rho}$ possesses no fundamental point, these ρ forms cannot vanish simultaneously; their resultant $R_0 \neq 0$.

Thus:

The regular systems $\mathfrak{S}_{n\rho}$ of polynomials are characterized by the fact that a mapping domain $\mathfrak{J}^*_{\rho\rho}$ of the smallest containing integrality domain possesses no fundamental point. For regular systems $\mathfrak{S}^*_{\rho\rho}$ of homogeneous forms, the system itself already possesses no fundamental point.[110]

§ 13. Examples of relatively integral domains of the first kind and of regular systems.

1. As a first example of a relatively integral domain of the first kind $\mathfrak{G}_{nn}$, we consider the *totality of the polynomials in $x_1, \ldots, x_n$ that admit a given permutation group G* — thus the totality of the polynomials of a Lagrange genus domain. Since $x_1, \ldots, x_n$, by virtue of the identity

$$x_k^n - \sigma_1(x)x_k^{n-1} + \sigma_2(x)x_k^{n-2} + \cdots \pm \sigma_n(x) = 0 \qquad (k = 1, 2, \ldots, n), \tag{1}$$

are algebraically integral over the elementary symmetric functions belonging to $\mathfrak{G}_{nn}$, and hence by Definition V are algebraically integral over $\mathfrak{G}_{nn}$, $\mathfrak{G}_{nn}$ is of the first kind. Moreover $\mathfrak{G}_{nn}$, as a domain of the first kind, since its elements are homogeneous forms and $n = \rho$, forms a regular system (by the conclusion of § 10). Indeed, because of (1), the resultant R_0 of the elementary symmetric functions cannot vanish identically; by the rules of calculation of resultant theory, R_0 is computed to be ± 1. The *involution form* of $\mathfrak{G}_{nn}$ is uniquely

The existence of the integrality basis here would be equivalent to the existence of a positive integer ν such that the identities

$$x_1 x_2^{\nu+\sigma} = \sum C \cdot x_1^{i_0}(x_1x_2)^{i_1}(x_1x_2^2)^{i_2}\cdots(x_1x_2^\nu)^{i_\nu} \quad (\sigma = 1, 2, \ldots)$$

hold. Comparing the dimensions in x_1, x_2 already for $\sigma = 1$ leads to the two conditional equations for the exponents

$$1 = i_0 + i_1 + \cdots + i_\nu, \qquad \nu + 1 = i_1 + 2i_2 + \cdots + \nu i_\nu,$$

which plainly have no solution in non-negative integers. Thus $\mathfrak{S}_{22}$ possesses no integrality basis.

[110] The non-regular system $\mathfrak{S}_{22}$ given in the preceding note possesses the fundamental point

$$\xi_1 : \xi_2 : \xi_0 = 0 : 1 : 0.$$

determined — since $\mathfrak{G}_{nn}$ is of the first kind and $n = \rho$ — as the involution form of the corresponding Lagrange genus domain. Since the quantities conjugate to $x_1, \ldots, x_n$ with respect to this domain arise through the permutations of the group G, this involution form is therefore given by

$$\begin{aligned} \Psi(z,u) &= \prod_i (z - u_1 x_{i_1} - u_2 x_{i_2} - \cdots - u_n x_{i_n}) \\ &= z^i + \sum{}' G_{\alpha\alpha_1\ldots\alpha_n}(x) z^\alpha u_1^{\alpha_1} \cdots u_n^{\alpha_n}, \end{aligned} \tag{2}$$

where the product is to be extended over all permutations of G. Formula (2) represents the "Galois form of the group G," and hence the following fact holds:

All polynomials in $x_1, \ldots, x_n$ which admit a permutation group G are integral rational combinations of the coefficients $G_{\alpha\alpha_1\ldots\alpha_n}(x)$ of the Galois form of the group.

2. As an example of *regular systems*, consider the systems $\mathfrak{S}_{n1}$ of polynomials, that is, systems which contain only one algebraically independent polynomial.

Indeed, let $\mathfrak{S}_{11}^*$ be a mapping system of $\mathfrak{S}_{n1}$ and let

$$F^*(x) = a_0 x^k + a_1 x^{k-1} + \cdots + a_k \qquad (a_0 \neq 0)$$

be a polynomial from $\mathfrak{S}_{11}^*$. Then

$$R_0 = a_0 \neq 0. \tag{3}$$

Thus $\mathfrak{S}_{n1}$ is regular by Definition VIII, and consequently:

Every system $\mathfrak{S}_{n1}$ of polynomials, in particular every system of infinitely many polynomials in one indeterminate, possesses an integrality basis.[111]

3. We now specialize the systems $\mathfrak{S}_{n1}$ to relatively integral domains $\mathfrak{G}_{n1}$, which, as regular systems by § 12, are relatively integral domains of the first kind. Their integrality basis is therefore given by an involution basis, which is at the same time the involution basis of the smallest containing field $\mathfrak{K}_{n1}$. But by § 6, any function of the involution basis of $\mathfrak{K}_{n1}$ is at the same time a minimal basis — we called it the Lüroth function of the field — and every function of the involution basis depends on this Lüroth function by a fractional-linear transformation. In the present case the Lüroth function, since it belongs to $\mathfrak{G}_{n1}$, is a polynomial; every further polynomial of the involution basis therefore depends fractional-linearly, and by (3) algebraically integrally, hence integrally and linearly, on this Lüroth function $L(x)$, which thus becomes the integrality basis. The involution form of $\mathfrak{G}_{11}^*$, setting also $(u_1 = 1)$, is

$$\Psi^*(z) = L^*(z) - L^*(x),$$

from which $L(x)$ again appears as integrality basis. Thus:

The relatively integral domains $\mathfrak{G}_{n1}$ are of the first kind; their integrality basis consists of one polynomial, the Lüroth function of the smallest containing field.

4. As an example of a relatively integral domain $\mathfrak{G}_{n1}$, let us mention specifically the integral rational projective invariants of a quadratic form in s variables. In agreement with the preceding, it is well known that they can be represented as integral rational combinations of the determinant $|a_{ik}|$ of the form; and this determinant is at the same time the Lüroth function of the corresponding invariant field.

§ 14. Systems consisting of integer polynomials.

The basis theorems obtained are now to be sharpened with respect to integrality over the integers.

The coefficient domain Ω will therefore in what follows be assumed to be a (finite) algebraic number field; and let $[\Omega]$ denote the totality of the algebraic integers from Ω, which we shall briefly call whole numbers. By an integer polynomial is meant a polynomial with coefficients from $[\Omega]$; from now on $F(x), G(x), \ldots$ shall denote only integer polynomials.

In analogy with § 7, we consider the different systems made up of integer polynomials:

An arbitrary system of integer polynomials shall be denoted as an integer system $\mathfrak{S}_{n\rho}$.

The integer linear family $\mathfrak{L}_{n\rho}$ is an integer system which, along with $F_1(x)$ and $F_2(x)$, always contains $c_1 F_1(x) + c_2 F_2(x)$, where c_1, c_2 are arbitrary whole numbers from $[\Omega]$.

[111] The simplest possible non-regular systems without integrality basis are therefore systems $\mathfrak{S}_{22}$. That such systems actually exist is shown by Hilbert's example and by the example given in the note to § 12.

The integer integrality domain $\mathfrak{J}_{n\rho}$ is an integer linear family which, along with $F(x)$ and $G(x)$, always contains $F(x) \cdot G(x)$.

The integer relatively integral domain $\mathfrak{G}_{n\rho}$ is an integrality domain which, along with $F(x)$ and $G(x)$ — where $G(x) \neq 0$ — always and only contains the quotient $F(x) : G(x)$ when this quotient is an integer polynomial.

Correspondingly as in § 7, the smallest containing integer integrality domain $\mathfrak{J}_{n\rho}$ of an integer system $\mathfrak{S}_{n\rho}$ consists of all polynomials $H(x)$ which are integral rational, integer combinations of a finite number of polynomials from $\mathfrak{S}_{n\rho}$;

the smallest containing integer relatively integral domain $\mathfrak{G}_{n\rho}$ consists of all integer polynomials $H(x)$ which are rational combinations of a finite number of polynomials from $\mathfrak{S}_{n\rho}$.

We now pass to the transfer of the basis theorems. With respect to the rational basis, it is to be noted that the notions of field, smallest containing field, and rational basis, which rest only on "rationality," undergo no change. Furthermore, the mapping system $\mathfrak{S}^*_{\rho\rho}$ of an integer system $\mathfrak{S}_{n\rho}$ can in arbitrarily many ways again be chosen as an integer system. Since, also for the integer linear family, the arguments leading to the bound on the number of generators remain valid, one obtains:

Theorem V′: Every integer system $\mathfrak{S}_{n\rho}$ possesses a rational basis; the rational basis of an integer linear family $\mathfrak{L}_{n\rho}$ can in particular be chosen so that it contains at most $\rho + 1$ polynomials.

We now investigate the analogue of the involution basis of integrality domains of polynomials (§ 8). For this purpose the theorem on *symmetric functions of rows of quantities* must be extended with respect to *integrality over the integers* by the following

Auxiliary theorem. The integral integer symmetric functions of n rows of quantities $(yz \cdots t)$:

$$\begin{matrix} y_1 & z_1 & \cdots & t_1 \\ y_2 & z_2 & \cdots & t_2 \\ \vdots & \vdots & & \vdots \\ y_n & z_n & \cdots & t_n \end{matrix} \tag{1}$$

are integral integer functions of a finite number among them, namely of the elementary symmetric functions

$$\begin{gathered} \sigma_1(y), \sigma_2(y), \ldots, \sigma_n(y); \quad \sigma_1(z), \sigma_2(z), \ldots, \sigma_n(z); \quad \ldots; \\ \sigma_1(t), \ldots, \sigma_n(t); \end{gathered} \tag{2}$$

and of the functions

$$\chi_1(y, z, \ldots, t), \ldots, \chi_k(y, z, \ldots, t), \tag{3}$$

which are algebraically integral and integer over the functions (2).

Indeed, the n quantities $y_1, \ldots, y_n$ depend algebraically integrally and integrally over the integers on $\sigma_1(y), \ldots, \sigma_n(y)$; the corresponding assertion holds for $z_1, \ldots, z_n$ with respect to $\sigma_1(z), \ldots, \sigma_n(z)$, and so on. The field of symmetric functions of the rows of quantities (1) therefore forms an algebraic field over the field determined by the functions (2), in such a way that the algebraically integral and integer quantities of this field are identical with the integral integer symmetric functions of the rows of quantities (1). If one therefore takes (3) as Kronecker's fundamental system, the auxiliary theorem is proved.

Let now $\mathfrak{J}_{n\rho}$ be an integer integrality domain, $\mathfrak{J}^*_{\rho\rho}$ an integer mapping domain, and

$$\Psi^*(z, u) = G^*(x) z^i + \sum{}' G^*_{\alpha\alpha_1 \ldots \alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}$$

an involution form of $\mathfrak{J}^*_{\rho\rho}$; here $G^*(x)$ may also have dimension zero, that is, be a number from $[\Omega]$. The elementary symmetric functions corresponding to the functions (2) are then given by certain ones among the quotients

$$G^*_{\alpha\alpha_1 \ldots \alpha_\rho}(x) : G^*(x). \tag{4}$$

Since by the auxiliary theorem the functions (3) depend algebraically integrally and integrally over the integers on (2), the functions from $\mathfrak{K}^*_{\rho\rho}$ corresponding to (3) are given — by the extension of Gauss's theorem to polynomials with algebraically integral coefficients — by

$$K^*_\nu(x) : (G^*(x))^\sigma, \tag{5}$$

where $K^*_\nu(x)$ are integer polynomials from $\mathfrak{J}^*_{\rho\rho}$. Thus $K^*_\nu(x)$ belong to the domain if the case is that of relatively integral domains; in general they are quotients of polynomials from $\mathfrak{J}^*_{\rho\rho}$.

If now $F^*(x)$ is an arbitrary integer polynomial from $\mathfrak{J}^*_{\rho\rho}$, then one has

$$F^*(x) = \frac{1}{i}\Big(F^*(x) + F^*(x^{(1)}) + \cdots + F^*(x^{(i-1)})\Big),$$

and therefore $i \cdot F^*(x)$ becomes an integral integer symmetric function of the rows of quantities $x, x^{(1)}, \ldots, x^{(i-1)}$.

By the auxiliary theorem and the transfer principle, therefore:

Theorem VI′: For integer integrality domains $\mathfrak{J}_{n\rho}$, each involution form determines a distinguished rational basis in such a way that every polynomial $F(x)$ from $\mathfrak{J}_{n\rho}$ admits a representation

$$F(x) = \frac{\Gamma(H(x), H_1(x), \ldots, H_s(x))}{i \cdot H(x)^i},$$

where Γ denotes an integral integer function. For integer relatively integral domains $\mathfrak{G}_{n\rho}$, with integer relatively integral domain $\mathfrak{G}^*_{\rho\rho}$ as mapping domain, the denominator $H(x)$ is given by the highest coefficient of the corresponding involution form.

§ 15. Integer relatively integral domains of the first kind and regular systems.

For integer systems, the integer integrality basis takes the place of the integrality basis; that is, a finite number of integer polynomials of the system such that every integer polynomial of the system can be represented as an integral integer combination of this finite number.

The theorems on the integer integrality basis run essentially parallel to the earlier ones, and only the proofs require slight modifications.

First, Definition V (§ 9) transfers directly.

Definition V′: A quantity ξ is called algebraically integral and integer with respect to the integer relatively integral domain $\mathfrak{G}_{n\rho}$ if it satisfies some equation whose highest coefficient is a unit from $[\Omega]$ and whose remaining polynomials are from $\mathfrak{G}_{n\rho}$.

The extension of Gauss's theorem to polynomials with algebraically integral coefficients shows, as in § 9, that the definition at the irreducible equation can be obtained from this.

From Definition V′ one further obtains the transfer of Definition VII.

Definition VII′: An integer relatively integral domain $\mathfrak{G}_{n\rho}$ is called of the first kind if, as mapping domain, it possesses a relatively integral domain $\mathfrak{G}^*_{\rho\rho}$ such that every arbitrary integer polynomial in $x_1, \ldots, x_\rho$ is algebraically integral and integer over $\mathfrak{G}^*_{\rho\rho}$.

To show the finiteness of these domains of the first kind, we first note that, as in § 10, it follows from the definition that the highest coefficient of the involution form is a unit from $[\Omega]$.

By Theorem VI′ (§ 14), every polynomial $F(x)$ from $\mathfrak{G}_{n\rho}$ therefore admits a representation

$$F(x) = \frac{\Gamma(H_1(x), \ldots, H_\sigma(x))}{i}.$$

Applying here to the system formed from the integer polynomials $\Gamma(H_1, \ldots, H_\sigma)$ Hilbert's Theorem II on the integer module basis (Math. Ann. 36), originally stated for integral rational numerical coefficients, in its extension to algebraically integral polynomials,[112] the existence of the integer integrality basis follows; that is:

Theorem VII′: Every integer relatively integral domain of the first kind possesses an integer integrality basis.

We further transfer the definition of regular systems:

Definition VIII′: An integer system $\mathfrak{G}_{n\rho}$ is called integer-regular if, in a mapping domain $\mathfrak{J}^*_{\rho\rho}$ of the smallest containing integer integrality domain, there exist ρ polynomials such that the resultant of the terms of highest dimension is a unit from $[\Omega]$ — $R_0 = \varepsilon$.

In order to transfer the existence proof of the integrality basis, we note that the auxiliary theorem of § 11 admits the following sharper formulation, which follows from equations (5) and (6) in § 11 and from the fact that $A_1(\lambda), \ldots, A_\tau(\lambda)$ are integral integer functions of $\lambda_1, \ldots, \lambda_\rho$:

[112] If $w_1, \ldots, w_k$ denotes a fundamental system of the algebraic integers from Ω, and one sets

$$\Gamma(H_1, \ldots, H_\sigma) = \Gamma_1(H)w_1 + \cdots + \Gamma_k(H)w_k,$$

then $\Gamma_1(H), \ldots, \Gamma_k(H)$ have integral rational numerical coefficients. Applying Theorem II to the system formed from the forms $v_1\Gamma_1(H) + \cdots + v_k\Gamma_k(H)$ directly shows the validity of the extension.

A sufficient condition that every arbitrary integer polynomial in $x_1, \ldots, x_\rho$ depend algebraically integrally and integrally over ρ integer polynomials is that the resultant of the terms of highest dimension of these polynomials be a unit from $[\Omega]$ — $R_0 = \varepsilon$.

From this auxiliary theorem, exactly as in § 12, for every polynomial $F(x)$ of an integer-regular system, representation (3) of § 12 follows, where now $\Gamma_1(G_i), \ldots, \Gamma_k(G_i)$ are integer polynomials in the G_i. Furthermore, as in § 12, the application of Hilbert's Theorem II (Math. Ann. 36) in its extension to algebraically integral polynomials, together with the definition of the smallest containing integer integrality domain, gives the existence of the integer integrality basis; that is:

Theorem IX′: Every integer-regular system possesses an integer integrality basis.

As an example of an integer relatively integral domain of the first kind, in analogy with example 1 in § 13, we refer to the totality $\mathfrak{G}_{nn}$ of integer polynomials of a Lagrange genus domain. That $\mathfrak{G}_{nn}$ is integer of the first kind follows from the identity

$$x_k^n - \sigma_1(x)x_k^{n-1} + \cdots \pm \sigma_n(x) = 0 \qquad (k = 1, 2, \ldots, n);$$

since the resultant of the elementary symmetric functions has the value ± 1, $\mathfrak{G}_{nn}$ is also an integer regular system.

A further example is given, by the auxiliary theorem of § 14, by the totality of the integral integer symmetric functions of n rows of quantities.

Erlangen, May 1914.

7. The finiteness theorem for the invariants of finite groups

Math. Ann. 77 (1916), pp. 89–92

In what follows an entirely elementary proof of finiteness for the invariants of finite groups is to be given — one based only on the theory of symmetric functions — and at the same time it gives an actual specification of the full system; whereas the usual proof, based on Hilbert's theorem on module bases (Ann. 36), is only an existence proof.[113]

Let the finite group $\mathfrak{H}$ consist of the h linear transformations (with non-vanishing determinant) $A_1, \ldots, A_h$, where A_k denotes the linear transformation

$$x_1^{(k)} = \sum_{\nu=1}^{n} a_{1\nu}^{(k)} x_\nu, \ldots, x_n^{(k)} = \sum_{\nu=1}^{n} a_{n\nu}^{(k)} x_\nu,$$

or, in abbreviated form, $(x^{(k)}) = A_k(x)$. The group $\mathfrak{H}$ therefore carries the row (x) with elements $x_1, \ldots, x_n$ into the rows $(x^{(k)})$ with elements $x_1^{(k)}, \ldots, x_n^{(k)}$. Since the identity must be contained among $A_1, \ldots, A_h$, the row (x) is also contained among the rows $(x^{(k)})$. By an integral rational (absolute) invariant of the group is meant an integral rational function of $x_1, \ldots, x_n$ which remains identically unchanged under the application of $A_1, \ldots, A_h$; for such an invariant $f(x)$ one therefore has

$$f(x) = f(x^{(1)}) = \cdots = f(x^{(h)}) = \frac{1}{h} \sum_{k=1}^{h} f(x^{(k)}). \tag{1}$$

1.

Formula (1) says that $f(x)$ is an integral rational symmetric function of the rows of quantities $(x^{(1)}), \ldots, (x^{(h)})$ — and indeed, since each summand $f(x^{(k)})$ contains only the one row $(x^{(k)})$, this is the simplest case, usually called the *one-form* case. By the known theorem on symmetric functions of rows of quantities,[114] $f(x)$ is therefore integrally and rationally representable by the elementary symmetric functions of these rows, that is, by the coefficients $G_{\alpha\alpha_1\ldots\alpha_n}(x)$ of the "Galois resolvent"

$$\begin{aligned} \Phi(z,u) &= \prod_{k=1}^{h} \left(z + u_1 x_1^{(k)} + \cdots + u_n x_n^{(k)}\right) \\ &= z^h + \sum G_{\alpha\alpha_1\ldots\alpha_n}(x) z^\alpha u_1^{\alpha_1} \cdots u_n^{\alpha_n}, \end{aligned} \qquad \left(\begin{matrix} \alpha + \alpha_1 + \cdots + \alpha_n = h \\ \alpha \neq h \end{matrix} \right),$$

where the $G_{\alpha\alpha_1\ldots\alpha_n}(x)$ are invariants of degree $\alpha_1 + \cdots + \alpha_n$ in the x's. Thus it is proved:

The coefficients $G_{\alpha\alpha_1\ldots\alpha_n}(x)$ of the Galois resolvent form a full system of invariants of the group, in such a way that every invariant can be represented integrally and rationally by these finitely many invariants.

2.

Starting from (1), one may also make the following still more elementary consideration,[115] which at the same time leads to a second full system. Put

$$f(x) = a + b x_1^{\mu_1} \cdots x_n^{\mu_n} + \cdots + c x_1^{\nu_1} \cdots x_n^{\nu_n},$$

where $a, b, \ldots, c$ denote constants. Then by (1) one has

$$\begin{aligned} h \cdot f(x) &= h \cdot a + b \sum_{k=1}^{h} (x_1^{(k)})^{\mu_1} \cdots (x_n^{(k)})^{\mu_n} + \cdots + c \sum_{k=1}^{h} (x_1^{(k)})^{\nu_1} \cdots (x_n^{(k)})^{\nu_n} \\ &= h \cdot a + b \cdot J_{\mu_1\ldots\mu_n} + \cdots + c \cdot J_{\nu_1\ldots\nu_n}. \end{aligned}$$

Every invariant can therefore be composed integrally and linearly from the special invariants

$$J_{\mu_1\ldots\mu_n} = \sum_{k=1}^{h} (x_1^{(k)})^{\mu_1} \cdots (x_n^{(k)})^{\mu_n},$$

[113] Cf., for example, Weber, *Lehrbuch der Algebra* (2nd ed.), vol. 2, § 57.

[114] Cf. the note to 2.

[115] This amounts to a proof of the theorem on symmetric functions of rows of quantities in the above-mentioned one-form case.

and it is enough to prove the assertion for these special ones. But, apart from a numerical factor, $J_{\mu_1\ldots\mu_n}$ is the coefficient of $u_1^{\mu_1}\cdots u_n^{\mu_n}$ in the expression

$$S_\mu = \sum_{k=1}^{h}\bigl(u_1x_1^{(k)} + \cdots + u_nx_n^{(k)}\bigr)^\mu,$$

where $\mu = \mu_1 + \cdots + \mu_n$, and this expression is the μ-th power sum of the h linear forms

$$\xi_1 = u_1x_1^{(1)} + \cdots + u_nx_n^{(1)}, \ldots, \xi_h = u_1x_1^{(h)} + \cdots + u_nx_n^{(h)}.$$

It is known that the infinitely many power sums S_μ are integral rational combinations of

$$S_1, S_2, \ldots, S_h,$$

whose coefficients are given by the invariants

$$J_{\mu_1\ldots\mu_n} \qquad (\mu_1 + \cdots + \mu_n \leq h).$$

Thus a second full system has been obtained:

A full system of invariants of the group is given by all invariants $J_{\mu_1\ldots\mu_n}$ for which $\mu_1 + \cdots + \mu_n \leq h$, where h denotes the order of the group.

The connection with 1. is established by observing that the power sums $S_1, \ldots, S_h$ can be replaced by the elementary symmetric functions of $\xi_1, \ldots, \xi_h$, which leads to the full system of the $G_{\alpha\alpha_1\ldots\alpha_n}(x)$ given there. Both results show that all invariants can be expressed integrally and rationally by those whose degree in the x's does not exceed the order h of the group.

3.

Consequences for rational representation can be drawn from what has just been proved. Every rational absolute invariant can be represented — as is seen immediately by multiplying numerator and denominator by the conjugates of the denominator with respect to the group — as the quotient of two integral rational absolute invariants, not necessarily relatively prime. It follows that every rational absolute invariant can be expressed rationally by the coefficients $G_{\alpha\alpha_1\ldots\alpha_n}(x)$ of the Galois resolvent $\Phi(z, u)$. This theorem can also be proved easily in various ways without passing through the finiteness theorem; it is already formulated in Weber II, § 58.[116]

4.

Finally, it should be mentioned that the results derived here are implicitly contained in my paper "Fields and systems of rational functions";[117] more precisely, the full system given in 1. is contained in Theorems VI and VII, and a proof of rational representation independent of this finiteness theorem is contained in Theorem III.[118] In the present special case, however, the course of the proof and the result simplify considerably in comparison with the general theory. I was led to apply the general investigations to the invariants of finite groups by conversations with E. Fischer, from whom, in substance, the proof given under 2 — in the general case the full system that appears there is not rationally definable — and the note to 3 also originate.

Erlangen, May 1915.

[116] The proof given there, however, is not sound. It is only shown that the function $\Psi(t)$ in formula (7) has invariants as coefficients, but not that these invariants are rationally expressible by the coefficients of $\Phi(t)$. This gap is avoided if, in representing the $x_i^{(k)}$ by ξ_k, one uses a known differentiation procedure instead of Lagrange's interpolation formula. This is the relation obtained by differentiating the identity $\Phi(-\xi_k, u) = 0$ with respect to all u:

$$\left[\frac{\partial\Phi}{\partial u_i} - x_i^{(k)}\frac{\partial\Phi}{\partial z}\right]_{z=-\xi_k} = 0.$$

Accordingly, for each rational function $\omega(x)$ the following formula must replace formulas (7) and (8) in Weber:

$$\omega(x_1^{(k)}, \ldots, x_n^{(k)}) = \omega\left(\frac{\partial\Phi/\partial u_1}{\partial\Phi/\partial z}, \ldots, \frac{\partial\Phi/\partial u_n}{\partial\Phi/\partial z}\right)\bigg|_{z=-\xi_k},$$

which contains only coefficients of $\Phi(z, u)$, and whose summation over all k, for invariants $\omega(x)$, gives the desired representation.

[117] Math. Ann. 76, p. 161 (1915).

[118] For relative invariants, Theorems VIII and IX contain a proof of finiteness which is likewise based on a different foundation from the usual one, but is also only an existence proof; the reason is that the relative invariants do not form a field.

8. On integral rational representation of the invariants of a system of arbitrarily many ground forms

Math. Ann. 77 (1916), pp. 93–102

At the end of his contribution to the Schwarz Festschrift, "On the invariants of a system of arbitrarily many ground forms", D. Hilbert states the conjecture that these invariants can be represented integrally and rationally by the finitely many invariants of the system (J, PJ), where J denotes the full system of invariants of the linearly independent ground forms, and the invariants PJ arise from J by polar processes.

I show below, in I, that this conjecture is in fact correct, and indeed as a simple consequence of the reduction theorem that every form in arbitrarily many rows of n variables each can be represented as a sum of polars of forms which contain only n fixed rows and which themselves are derived from the original form by polar processes. This reduction theorem is a special case, first formulated by F. Mertens, of the generalization, given by A. Capelli, F. Mertens, and J. Deruyts, of the Clebsch–Gordan series expansion to forms in arbitrarily many rows of n variables each; this generalization is proved by formal transformation of the differentiation processes.[119]

In II I also give a more conceptual proof of the reduction theorem, by treating the question as a problem concerning the equivalence of linear families of forms. This point of view, and also an essential part of the considerations used, I take from a paper "On algebraic module systems"[120] and from unpublished theorems of E. Fischer adjoining it, whose use he kindly permitted me.

Finally, in III, I show how the corresponding considerations also lead to the most general series expansions.

I.

Let θ be a form homogeneous in each of the $N > n$ rows

$$A_1, A_2, \ldots, A_N,$$

where in general the row A_k consists of the n elements

$$A_k^{(1)}, A_k^{(2)}, \ldots, A_k^{(n)}.$$

The coefficients of θ may be indeterminates or quantities of a given domain of rationality. Let P further denote an integral rational function — with rational integer coefficients — of the polar processes

$$P_{hk} = \left(A_h \frac{\partial}{\partial A_k}\right) = A_h^{(1)} \frac{\partial}{\partial A_k^{(1)}} + A_h^{(2)} \frac{\partial}{\partial A_k^{(2)}} + \cdots + A_h^{(n)} \frac{\partial}{\partial A_k^{(n)}},$$

where the product $P_{hk} P_{ij} \theta$ is to be understood as the application of P_{hk} to $P_{ij}\theta$.

Then the reduction theorem just mentioned is given by the identity

$$\theta = \sum PZ, \tag{1}$$

where the forms Z contain only the rows

$$A_1, A_2, \ldots, A_n$$

and are derived from θ by polar processes P.

We now consider the system of ground forms (F'), consisting of the N forms of the same order and the same number of variables

$$F_1, F_2, \ldots, F_N,$$

whose coefficients are taken respectively as the N rows

$$A_1, A_2, \ldots, A_N.$$

[119] F. Mertens: "Über eine Formel der Determinantentheorie." (Formula 5), Sitzb. d. Ak. d. Wiss. Wien, vol. 91, II. Abt. (1885), and more fully (with applications) in: "Über ganze Funktionen von m Systemen von je n Unbestimmten". Monatsh. f. Math. u. Phys. 4th year (1893). For some of Capelli's proofs (since 1882) cf., for example, Math. Ann. 37, or the autographed "Lezioni sulla teoria delle forme algebriche" (1902). J. Deruyts: *Essai d'une théorie générale des formes algébriques*, Mém. d. 1. Soc. Roy. des Sciences de Liège (2) 17 (1892).

[120] E. Fischer: Über algebraische Modulsysteme und lineare homogene partielle Differentialgleichungen mit konstanten Koeffizienten, § 1–4, J. f. M. 140 (1911).

Let (J) denote the full system — consisting of finitely many invariants — of $F_1, \ldots, F_n$, in such a way that every invariant of $F_1, \ldots, F_n$ can be expressed integrally and rationally by the invariants (J). Hilbert's conjecture to be proved is then that, for the simultaneous invariants S of $F_1, \ldots, F_N$, a full system is given by the finitely many invariants (J, PJ), where PJ denotes all invariants derivable from J by polar processes P.

Indeed, every simultaneous invariant S with rational integer coefficients of (F') is a form homogeneous in each of the N rows $A_1, \ldots, A_N$ — with rational integer coefficients — and identity (1) can therefore be applied to it. Since application of the polar processes P to S is known to produce invariants again, the forms Z are also invariants, namely simultaneous invariants of $F_1, \ldots, F_n$, since they contain only the rows $A_1, \ldots, A_n$; hence, by assumption, they are integral rational functions of the invariants (J). Thus identity (1) becomes

$$S = \sum PY,$$

where the Y denote power products of the invariants (J). But, by the definition of P, the actual execution of the processes P on Y consists in the successive, finitely repeated execution of the simple polar operations P_{hk}, which, by the differentiation rule for products, leads to sums of products of invariants (J) and (PJ). The same holds for the application of P_{hk} to a product from (J) and (PJ); hence PY too yields only integral rational combinations of (J, PJ). *The integral rational representability of all invariants S by (J, PJ) is thereby proved.*

II.

Before proving the reduction theorem, we recall some facts about linear families of forms. By a *linear family of forms over* K we mean a system of forms of the same dimension, complete in the sense that together with θ_1 and θ_2 it always contains $c_1\theta_1 + c_2\theta_2$, where c_1, c_2 are quantities of a prescribed domain of rationality K, and where K contains the coefficient domain of the θ's. Among these forms there is always a finite number — say ρ — linearly independent over K, while any $\rho + 1$ forms satisfy a linear relation with coefficients from K; the family has *rank* ρ with respect to K.[121] We shall use the easily seen fact that if $\mathcal{L}$ and $\mathcal{T}$ are two linear families over K of the same rank ρ, and if $\mathcal{T}$ is a subfamily of $\mathcal{L}$, then $\mathcal{L}$ and $\mathcal{T}$ are identical (equivalent).

With this in hand, we pass to the reduction theorem. The identity from I,

$$\theta = \sum PZ,$$

passes into an analogous identity for $P\theta$ when one applies the same polar process P to both sides; thus the reduction theorem at the same time gives a representation of all polars of θ by sums of the special polars PZ. Since, conversely, these special polars are surely contained among all polars, the reduction theorem asserts the equivalence of these two linear families of forms; in particular, it also asserts the equivalence of those subfamilies whose forms have, in each individual row, the same dimension as θ. To prove this equivalence we insert an intermediate family determined by the forms Z. For simplicity we first give the proof for the case of three binary rows ($N = 3, n = 2$), and then briefly indicate the completely parallel general proof.

Thus let $\theta(x, y, z)$ have dimensions α, β, p respectively in the rows

$$x_1x_2; \quad y_1y_2; \quad z_1z_2;$$

and let the coefficients of θ first be indeterminates a (or rational numbers). We consider the following three linear families of forms.

1. The family $\mathcal{L}$, consisting of all forms $f(x, y, z)$ which arise from θ by polar processes P and which, in the individual rows x, y, z, have the same dimension as θ; $\mathcal{L}$ contains θ in particular. If $f(x, y, z)$ is regarded as a form in x, y, z and the indeterminates a, then the coefficient field is the field R of rational numbers; for K as well one must take R, since only for rational integral θ_1, θ_2 does $c_1P_1 + c_2P_2$ again become a process P; let $\mathcal{L}$ have rank ρ with respect to R.

2. The family $\mathcal{S}$, consisting of all forms $g(\lambda; x, y)$ defined by

$$g(\lambda; x, y) = f(x, y, \lambda_1x + \lambda_2y),$$

or, expressed by polar processes,

$$\begin{aligned} g(\lambda; x, y) &= \frac{1}{p!}\left[(\lambda_1x + \lambda_2y)\frac{\partial}{\partial z}\right]^p f(x, y, z) \\ &= g_0(xy)\lambda_1^p + g_1(x, y)\lambda_1^{p-1}\lambda_2 + \cdots + g_p(xy)\lambda_2^p, \end{aligned}$$

[121] More general linear families of forms — whose rank need not be finite — are obtained by dropping the restriction of equal dimension, or also by dropping the restriction that K contain the coefficient domain of the θ's.

where

$$i!(p-i)!\, g_i(xy) = \left(y\frac{\partial}{\partial z}\right)^i \left(x\frac{\partial}{\partial z}\right)^{p-i} f(xyz).$$

[122] Thus the $g_i(xy)$ give forms Z of identity (1). The g are rational-integer forms in x, y, λ and the indeterminates a; let $\mathcal{S}$ have rank σ with respect to R.

3. The family $\mathcal{T}$, obtained from $\mathcal{S}$ by the inverse polar process just as $\mathcal{S}$ is obtained from $\mathcal{L}$; the forms h of $\mathcal{T}$ are defined by

$$\begin{aligned} h(x,y,z) &= \frac{1}{p!}\left[z\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right)\right]^p g(\lambda; xy) \\ &= h_0(xyz) + h_1(xyz) + \cdots + h_p(xyz), \end{aligned}$$

where, by 2.,

$$h_i(xyz) = \left(z\frac{\partial}{\partial x}\right)^{p-i} \left(z\frac{\partial}{\partial y}\right)^i g_i(xy).$$

The individual h_i, and consequently also h, are therefore forms PZ and sums of them; let $\mathcal{T}$ have rank τ with respect to R.

As the construction of the forms $h(x,y,z)$ of $\mathcal{T}$ shows, these forms arise from θ by polar processes P, and in the rows x, y, z have individually the same dimension as θ; hence the family $\mathcal{T}$ is a subfamily of $\mathcal{L}$. By the fact mentioned above, it remains only to prove equality of the ranks. In this proof no further use is made of the special structure of the $f(xyz)$ — namely, that they are polars of one and the same form θ.

a) For comparing the ranks of $\mathcal{L}$ and $\mathcal{S}$ we use Lemma a): from $f(x,y,\lambda_1 x + \lambda_2 y) = 0$ it follows necessarily that $f(x,y,z) = 0$. This follows immediately from the identical relation between three binary rows

$$(xy)z + (yz)x + (zx)y = 0, \qquad (xy) = (x_1y_2 - x_2y_1), \ldots,$$

which implies

$$f[x, y, (xy)x + (xz)y] = (xy)^p f(x,y,z).$$

Thus from $f(x,y,\lambda_1 x + \lambda_2 y) = 0$ (identically in λ) it follows, as is shown by the substitution $\lambda_1 = (zy)$, $\lambda_2 = (xz)$, and since $(xy) \neq 0$, that necessarily $f(x,y,z) = 0$.

By this lemma there is a one-to-one correspondence between the forms of $\mathcal{S}$ and those of $\mathcal{L}$. For from

$$f_1(x,y,\lambda_1 x + \lambda_2 y) = g(\lambda; xy), \qquad f_2(x,y,\lambda_1 x + \lambda_2 y) = g(\lambda; xy)$$

it follows necessarily that

$$f_1(xyz) = f_2(xyz).$$

Furthermore, every relation in $\mathcal{S}$ is preserved between the uniquely corresponding forms in $\mathcal{L}$ (and conversely). For from

$$c_1 g^{(1)}(\lambda; xy) + c_2 g^{(2)}(\lambda; xy) + \cdots + c_r g^{(r)}(\lambda; xy) = 0$$

it follows by the lemma that necessarily

$$c_1 f^{(1)}(x,y,z) + c_2 f^{(2)}(x,y,z) + \cdots + c_r f^{(r)}(x,y,z) = 0.$$

This proves equality of the ranks of $\mathcal{L}$ and $\mathcal{S}$; $\rho = \sigma$.

b) For comparing the ranks of $\mathcal{S}$ and $\mathcal{T}$ we use the corresponding Lemma b): from $h(x,y,z) = 0$ it follows necessarily that $g(\lambda; xy) = 0$. For the proof, observe that by 3. $h(x,y,z) = 0$ is equivalent to the vanishing of the individual power products

$$\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_1} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_1}\right)^{p_1} \left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_2} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_2}\right)^{p_2} g(\lambda; x, y), \qquad p_1 + p_2 = p.$$

[122] As in I,

$$\left(t\frac{\partial}{\partial x}\right) = t_1\frac{\partial}{\partial x_1} + t_2\frac{\partial}{\partial x_2},$$

and hence in particular

$$\left[(\lambda_1 x + \lambda_2 y)\frac{\partial}{\partial z}\right] = (\lambda_1 x_1 + \lambda_2 y_1)\frac{\partial}{\partial z_1} + (\lambda_1 x_2 + \lambda_2 y_2)\frac{\partial}{\partial z_2},$$

$$\left[z\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right)\right] = z_1\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_1} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_1}\right) + z_2\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_2} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_2}\right).$$

Further differentiation then also yields the vanishing of the power products

$$\left(\frac{\partial}{\partial x_1}\right)^{\alpha_1}\left(\frac{\partial}{\partial x_2}\right)^{\alpha_2}\left(\frac{\partial}{\partial y_1}\right)^{\beta_1}\left(\frac{\partial}{\partial y_2}\right)^{\beta_2}$$
$$\cdot\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_1}+\frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_1}\right)^{p_1}\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_2}+\frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_2}\right)^{p_2} g(\lambda;xy).$$

Consequently every linear combination of these power products vanishes, and hence every form

$$f\left(\frac{\partial}{\partial x},\frac{\partial}{\partial y},\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x}+\frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right) g(\lambda,xy).$$

Choosing f in particular so that

$$f(x,y,\lambda_1 x+\lambda_2 y)=g(\lambda;xy),$$

one also obtains

$$g\left(\frac{\partial}{\partial x};\frac{\partial}{\partial x},\frac{\partial}{\partial y}\right) g(\lambda,x,y)=0;$$

or, if

$$g(\lambda,xy)=\sum G_i\lambda_1^{\nu_1}\lambda_2^{\nu_2}x_1^{\alpha_1}x_2^{\alpha_2}y_1^{\beta_1}y_2^{\beta_2},$$

where the G_i are rational-integer linear forms in the indeterminates a, then

$$\sum \nu_1!\nu_2!\alpha_1!\alpha_2!\beta_1!\beta_2!\,G_i^2=0,$$

and hence $g(\lambda;xy)=0$.

It follows from Lemma b), exactly as in a), that *the ranks of $\mathcal{S}$ and $\mathcal{T}$ are equal; $\sigma=\tau$.*

Thus by a) and b) one has $\rho=\tau$, and consequently — since $\mathcal{T}$ is a subfamily of $\mathcal{L}$ — the equivalence of these two linear families is proved. Hence every form of $\mathcal{L}$, in particular also θ, can be represented as a linear rational-integer combination of ρ linearly independent forms $h(x,y,z)$:

$$\theta=\sum c_i h^{(i)}(x,y,z)=\sum PZ.$$

This identity, derived for indeterminate coefficients a of θ, remains valid when the indeterminates a are replaced by quantities of any domain of rationality; hence *the reduction theorem is generally proved for the case of three binary rows.*

The general proof, for N rows of n variables each, is completely parallel. Let $\theta(A)$ have dimension α_i in the row A_i. We again consider the three linear families of forms:

1. the family $\mathcal{L}$, consisting of all forms $f(A)$ which arise from $\theta(A)$ by polar processes P and, in each individual row A_i, have the same dimension as $\theta(A)$;
2. the family $\mathcal{S}$, consisting of all forms $g(\lambda;A_1\cdots A_n)$ which arise from $f(A)$ by the substitutions

$$A_{n+1}=\lambda_{n+1}^{(1)}A_1+\lambda_{n+1}^{(2)}A_2+\cdots+\lambda_{n+1}^{(n)}A_n,$$
$$\vdots$$
$$A_N=\lambda_N^{(1)}A_1+\lambda_N^{(2)}A_2+\cdots+\lambda_N^{(n)}A_n;$$

in this process, the coefficients of the individual power products of the λ's in $g(\lambda;A_1\cdots A_n)$ give forms Z of identity (1);

3. the family $\mathcal{T}$, consisting of all forms $h(A)$ defined by

$$h(A)=\left[A_{n+1}\left(\frac{\partial}{\partial\lambda_{n+1}^{(1)}}\frac{\partial}{\partial A_1}+\cdots+\frac{\partial}{\partial\lambda_{n+1}^{(n)}}\frac{\partial}{\partial A_n}\right)\right]^{\alpha_{n+1}}$$
$$\cdots\left[A_N\left(\frac{\partial}{\partial\lambda_N^{(1)}}\frac{\partial}{\partial A_1}+\cdots+\frac{\partial}{\partial\lambda_N^{(n)}}\frac{\partial}{\partial A_n}\right)\right]^{\alpha_N} g(\lambda;A_1\cdots A_n).$$

Here $h(A)$ is a sum of forms PZ.

Because of the identical relation between any $n+1$ rows, Lemma a) remains valid; Lemma b) likewise remains valid, since the number of variables did not enter into its proof. Thus $\mathcal{L}$ and $\mathcal{T}$ have the same rank and — since $\mathcal{T}$ is a subfamily of $\mathcal{L}$ — are identical. Consequently the identity

$$\theta = \sum PZ$$

holds for indeterminate coefficients, and hence for coefficients that are quantities of any domain of rationality; *the reduction theorem is thereby proved in general.*

III.

We still indicate briefly how analogous considerations also yield the *general series expansion* (for $N \le n$). Let Z be a separately homogeneous form in the n rows $A_1, \dots, A_n$ of n quantities each; and let Δ be the determinant of these rows. We first prove the identity corresponding to (1):

$$Z = \sum PH \pmod{\Delta}, \tag{2}$$

where the forms H contain only the rows $A_1, \dots, A_{n-1}$ and are derived from Z by processes P.

The proof consists in converting the relations derived in II into congruences modulo Δ. For simplicity take three ternary rows x, y, z; the coefficients are assumed to be indeterminates.

The three linear families $\mathcal{L}, \mathcal{S}, \mathcal{T}$ are defined as in II for the case of binary rows; let their ranks be ρ, σ, τ, while ρ^* and τ^* denote the ranks of $\mathcal{L}$ and $\mathcal{T}$ modulo Δ (that is, there are ρ^*, but not ρ^*+1, forms of $\mathcal{L}$ that are linearly independent mod Δ). Then, as in II, one has

Lemma a): from $f(x, y, \lambda_1 x + \lambda_2 y) = 0$ *it follows necessarily that* $f(xyz) \equiv 0 \pmod{\Delta}$ *(and conversely).* For, by the relation between four ternary rows,

$$\Delta_1 x - \Delta_2 y + \Delta_3 z = \Delta t, \qquad \Delta_1 = (yzt), \dots,$$

there is always a linear relation modulo Δ between three ternary rows:

$$\Delta_1 x - \Delta_2 y + \Delta_3 z \equiv 0 \pmod{\Delta},$$

and hence, as in II,

$$f(x, y, -\Delta_1 x + \Delta_2 y) = \Delta_3^p f(x, y, z) \pmod{\Delta}.$$

It follows from $f(x, y, \lambda_1 x + \lambda_2 y) = 0$ (identically in λ), since Δ is irreducible and Δ_3 is not divisible by Δ, that necessarily $f(x, y, z) \equiv 0 \pmod{\Delta}$. The converse is clear from $\Delta(x, y, \lambda_1 x + \lambda_2 y) = 0$. Hence, as in II, $\rho^* = \sigma$.

Lemma b) remains valid with its proof, and one has therefore $\sigma = \tau$.

For proving $\tau = \tau^*$ we use

Lemma c): from $h(x, y, z) \equiv 0 \pmod{\Delta}$ *it follows necessarily that* $h(x, y, z) = 0$. Namely, as a polar, $h(x, y, z)$ vanishes under the application of the known Ω-process; this is expressed by the differential equation

$$\Delta\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) h(x, y, z) = 0.$$

If now $h(x, y, z) = \Delta(x, y, z) \cdot k(x, y, z)$, then by further differentiation one also has the vanishing of

$$k\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \Delta\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) = h\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) h(x, y, z),$$

and hence of $h(x, y, z)$. Thus $\rho^* = \sigma = \tau = \tau^*$; since $\mathcal{T}$ is a subfamily of $\mathcal{L}$, identity (2) is proved. The same argument gives the proof for n variables.[123]

Applying identity (2) to the multiplier of Δ, one obtains an expansion

$$Z = \varphi_0 + \Delta\varphi_1 + \Delta^2\varphi_2 + \dots + \Delta^\mu \varphi_\mu,$$

[123] After 3. one has the corresponding relation, vanishing because of the determinant factor, for $\Omega(h)$; in operator notation it is the equation arising from products of

$$\left(\lambda_1 \frac{\partial}{\partial \xi} + \lambda_2 \frac{\partial}{\partial \eta}\right), \qquad \left[z\left(\frac{\partial}{\partial \lambda_1}\frac{\partial}{\partial \xi} + \frac{\partial}{\partial \lambda_2}\frac{\partial}{\partial \eta}\right)\right],$$

and it vanishes because of the vanishing of the determinant factor.

where the φ_i each vanish under application of the Ω-process. The i-fold repetition of the process gives, since in general $\Omega(\Delta \cdot \psi) = c_1\psi + c_2\Delta \cdot \Omega(\psi)$,

$$c \cdot \Omega^i(Z) \equiv \varphi_i \pmod{\Delta};$$

on the other hand, by (2) one has

$$c \cdot \Omega^i(Z) \equiv \sum PH_i \pmod{\Delta},$$

where the H_i are derived from the form $\Omega^i(Z)$ in the same way as H is derived from Z. By Lemma c) (extended to n variables) one therefore gets $\varphi_i = \sum PH_i$, and consequently

$$Z = \sum PH + \Delta \cdot \sum PH_1 + \Delta^2 \cdot \sum PH_2 + \cdots + \Delta^\mu \cdot \sum PH_\mu \tag{3}$$

as the most general series expansion of a form in n rows of n variables each, according to polars of forms in only $n-1$ rows.

The series expansion for forms in N rows of n variables ($N < n$) is then obtained, as is known, by a simple substitution. We put (for $N = 3$, $n > 3$)

$$u = \xi_1 x + \xi_2 y + \xi_3 z, \quad v = \eta_1 x + \eta_2 y + \eta_3 z, \quad w = \zeta_1 x + \zeta_2 y + \zeta_3 z,$$

and expand $H(u, v, w) = Z(\xi, \eta, \zeta)$ according to (3). Then, since ξ, η, ζ occur only in the combinations u, v, w, one has

$$\left(\eta\frac{\partial}{\partial\xi}\right) = \left(v\frac{\partial}{\partial u}\right)\cdots, \qquad \Omega = \sum_{ikl}\left(\frac{\partial}{\partial u}\frac{\partial}{\partial v}\frac{\partial}{\partial w}\right)_{ikl}(xyz)_{ikl} = \nabla_{uvw}(xyz).$$

Under the specialization $\xi_1 = \eta_2 = \zeta_3 = 1$; $\xi_2 = \xi_3 = \cdots = \zeta_2 = 0$, $H(u, v, w)$ goes over into $H(x, y, z)$, and $\nabla_{uvw}(xyz)$ into $\nabla_{xyz}(xyz) = \nabla_{xyz}$ up to a numerical factor, since the application of $(y\partial/\partial x)$ etc. to the determinants $(xyz)_{ikl}$ makes them vanish. Since the corresponding assertion holds for N rows, (3) gives the expansion

$$H = \sum P\Phi + \sum P\Phi_1 + \cdots + \sum P\Phi_\mu, \tag{4}$$

where the Φ_i arise from $\nabla^i_{A_1 \ldots A_N} H$ by processes P and contain only the rows $A_1, \ldots, A_{N-1}$, apart from the determinant combinations occurring in ∇^i, which, as mentioned, do not come into consideration under polarization.

If these determinants in ∇^i are then replaced by indeterminates p, the resulting forms can again be expanded according to (4); in this way one finally obtains an expansion

$$H = \left[\sum P\psi(p)\right]_{p=\text{determinants of } A_1 \cdots A_N}, \tag{5}$$

where the ψ arise from the original form by the processes P and $\nabla_{A_1 \ldots A_N}(p)$, $\nabla_{A_1 \ldots A_{N-1}}(p)$, and so on. With these expansions and their combinations — for example by applying (3), and then (4) and (5), to the forms Z occurring in (1) — all known expansions for forms in arbitrarily many rows of n cogredient variables each are exhausted.

Erlangen, 5 January 1915.

9. The Most General Domains from Integral Transcendental Numbers

Math. Ann. 77 (1916), pp. 103–128

By means of the well-ordering theorem, E. Zermelo constructed a domain of "integral transcendental numbers," that is, a numerical domain $\mathfrak{G}$ having the following abstract properties:

I. The sum, difference, and product of two numbers from $\mathfrak{G}$ is again a number from $\mathfrak{G}$.

II. Every real or complex number is the quotient of two numbers from $\mathfrak{G}$.

III. Every integral rational (respectively algebraic) number is an element of $\mathfrak{G}$.

IV. No non-integral rational (respectively algebraic) number belongs to $\mathfrak{G}$.[124]

The construction rests on the fact that the well-ordering theorem implies the existence of an *algebraic basis* of all numbers: namely, a *system H of numbers* η among which no algebraic relations exist, but by means of which all remaining numbers can be expressed algebraically. Because of their algebraic independence, these basis numbers η can be regarded as indeterminates, and the desired domain therefore becomes an integral domain of rational and algebraic functions of indeterminates η,[125] whose coefficients belong to the field K of all algebraic numbers. The special domain $\mathfrak{G}_\eta$ constructed by Zermelo is then characterized by the following basis properties:

$\mathfrak{G}_\eta$ contains:

1) all basis numbers η,

2) all polynomials in the η with integral[126] coefficients,

3) all integral algebraic functions of the η with integral coefficients.

$\mathfrak{G}_\eta$ excludes:

4) all rationally fractional functions of the η with integral coefficients,

5) all algebraically fractional functions of the η with integral coefficients.[127]

It is now easy to see (compare the examples in §§ 2 and 3) that there are domains $\mathfrak{G}$ essentially different from the Zermelo domain $\mathfrak{G}_\eta$: that is, domains $\mathfrak{G}$ for which, under no well-ordering, do all basis properties 1)–5) hold, and which consequently cannot be generated by Zermelo's construction under any well-ordering. In other words, there are domains $\mathfrak{G}$ that cannot be mapped one-to-one and isomorphically onto $\mathfrak{G}_\eta$. Thus the question arises of the most general domains from integral transcendental numbers; it will be answered completely in what follows.

For this purpose one must first determine which of the basis properties are consequences of the abstract defining properties and which are caused by the special construction. It is shown (§ 1) that for every prescribed domain $\mathfrak{G}$ there is a well-ordering such that basis properties 1) and 2) are satisfied; hence every domain $\mathfrak{G}$ can be constructed by means of an algebraic basis of all numbers. None of the further basis properties need be satisfied (§§ 2 and 3). In particular it follows that a further abstract property of the Zermelo domain does not belong to all domains $\mathfrak{G}$: algebraic integral closedness. It can be formulated as follows:

V. Every number that depends algebraically-integrally and integrally on finitely many numbers from $\mathfrak{G}$ belongs to $\mathfrak{G}$.

The special domains for which V holds in addition to I–IV will be called domains $\mathfrak{H}$ from algebraically-integral transcendental numbers. For the most general domains $\mathfrak{H}$ one obtains the simple results (§ 4):

[124] E. Zermelo, Über ganze transzendente Zahlen, Math. Ann. 75, p. 434 (1914).

[125] For this reason the considerations of the present paper partly run parallel to those of the author's earlier paper: Körper und Systeme rationaler Funktionen, Math. Ann. 76, p. 161 (1915).

[126] Throughout, "integral" is to mean that the coefficients belong to the integral domain $[K]$ of all algebraic integers. For the definition of $\mathfrak{G}_\eta$, it would also suffice in 1)–5) to consider only rational integer coefficients; for more general domains, however, the basis properties would then become less simple.

[127] Zermelo, loc. cit., § 3.

a) The intersection of all domains $\mathfrak{H}$ constructible by means of the same basis H is given by all integral algebraic functions of the basis, that is, by the Zermelo domain $\mathfrak{G}_\eta$, which is itself a domain $\mathfrak{H}$. (This also gives an abstract characterization of the Zermelo domain.)

b) Every domain $\mathfrak{H}$ arises by algebraically-integral adjunction[128] of an arbitrary system $\mathfrak{S}(\eta)$ to $\mathfrak{G}_\eta$, where $\mathfrak{S}(\eta)$ has to satisfy only the single condition that no algebraically fractional number enters the domain through the adjunction.

The results for the most general domains $\mathfrak{G}$, when an algebraic basis is taken as foundation, are less simple. Here the intersection of all domains $\mathfrak{G}$ is itself not a domain $\mathfrak{G}$, and the construction of the most general domains is consequently harder to survey (§ 5).

In order to obtain results analogous to those for the domains $\mathfrak{H}$, the algebraic basis is replaced by a *rational basis* Θ, that is, by a *system* Θ *of numbers* ϑ which permits all numbers to be expressed rationally, while under at least one well-ordering no basis number can be expressed rationally by finitely many preceding ones. This rational basis Θ must still be subject to the side condition that the integral domain $[\Theta]$ consisting of all integral polynomials in the ϑ contains no algebraically fractional number. (The construction of such a basis with the side condition is shown in § 7.) Then for the most general domains $\mathfrak{G}$ — which one could also call domains from rationally integral transcendental numbers — one again obtains the simple results (§ 6):

a) The intersection of all domains $\mathfrak{G}$ constructible by means of the same rational basis Θ is given by all integral rational functions of the basis, that is, by the domain $[\Theta]$, which is itself a domain $\mathfrak{G}$.

b) Every domain $\mathfrak{G}$ arises by rationally integral adjunction of an arbitrary system $\mathfrak{S}(\vartheta)$ to $[\Theta]$, where $\mathfrak{S}(\vartheta)$ has to satisfy only the single condition that no algebraically fractional number enters the domain through the adjunction.

To have complete analogy with the domains $\mathfrak{H}$, one would have to omit the algebraic numbers from the defining properties III and IV for the domains $\mathfrak{G}$. With this modification of III and IV, the construction of the integral elements by means of a rational basis can be transferred to an arbitrary field (§ 10), whereas Zermelo's construction presupposes the algebraic closedness of the field.

As H runs through every algebraic basis, respectively as Θ runs through every rational basis satisfying the side condition, the results a) and b) give the totality of all domains $\mathfrak{H}$ and $\mathfrak{G}$. If all isomorphic domains are collected into a class, then from this totality one can select a subset which represents the totality of all classes — at least countably infinitely many by § 9 (§ 8).

§ 1. Proof of basis properties 1) and 2) for all domains $\mathfrak{G}$.

Let $\mathfrak{G}$ be any prescribed domain from integral transcendental numbers, so that it has the abstract properties I–IV. Following the procedure applied by E. Zermelo to the field of all complex numbers, we select from $\mathfrak{G}$ an *algebraic basis* H *of all numbers from* $\mathfrak{G}$. Since by II every real or complex number can be represented as the quotient of two numbers from $\mathfrak{G}$, H is at the same time an algebraic basis of *all* numbers. Thus, for every prescribed domain $\mathfrak{G}$, the algebraic basis of all numbers can be chosen so that it belongs to $\mathfrak{G}$; hence basis property 1) is satisfied. Moreover, by III, $\mathfrak{G}$ contains the integral domain $[K]$ of all algebraic integers, and therefore, by I, also all polynomials in the basis elements η with coefficients from $[K]$, that is, all integral polynomials in the η, which form the integral domain $[H]$. Hence basis property 2) is also always satisfied; in other words, every domain $\mathfrak{G}$ can be constructed by means of an algebraic basis H of all numbers in such a way that $\mathfrak{G}$ contains the integral domain $[H]$ as a divisor.

Remark. Nevertheless, starting from a basis H, one can of course construct a domain $\mathfrak{G}$ for which basis properties 1) and 2) do not hold with respect to H itself, but do hold with respect to some other basis H'. For example, let $\mathfrak{G}'$ arise from the Zermelo domain $\mathfrak{G}_\eta$ by replacing, throughout the whole construction, some basis element η by a polynomial $f(\eta)$. Then $\mathfrak{G}'$ contains neither η nor $[H]$; but if in the basis H the basis element η is replaced by $\xi = f(\eta)$ — whereby H again goes over into an algebraic basis H' —, then basis properties 1) and 2) are satisfied for $\mathfrak{G}'$ with respect to H'.

§ 2. Exclusion of basis properties 4) and 5).

We now show that the domains $\mathfrak{G}$ need not satisfy any of the further basis properties. First 4) and 5) are excluded by replacing, in Zermelo's construction, the integral domain $[H]$ by a domain of integral rational

[128] Explanation of the expression in § 4.

"functionals."

According to Weber,[129] a *rational functional* means every rational function with rational number coefficients in the following uniquely determined representation:

$$A(\eta_1 \cdots \eta_t) = a \cdot \frac{E_1(\eta_1 \cdots \eta_t)}{E_2(\eta_1 \cdots \eta_t)},$$

where E_1, E_2 are relatively prime primitive polynomials with rational integer coefficients, and a is a positive rational number (the absolute value of A). If, in particular, a is a *rational integer*, then A is called an *integral rational functional.* As special cases, the integral rational functionals include the rational integers and the polynomials with rational integer coefficients, while rationally fractional numbers and polynomials with rationally fractional coefficients are to be counted among the fractional rational functionals.

Let the domain $\mathfrak{G}$ now consist of the totality $\mathfrak{F}$ of all integral rational functionals of finitely many basis elements η at a time, and of all functions which depend algebraically-integrally on $\mathfrak{F}$ — that is, which satisfy some equation whose leading coefficient is unity and whose remaining coefficients are integral rational functionals.

The proof of properties I–IV for $\mathfrak{G}$ runs exactly parallel to the corresponding proof for Zermelo and will only be indicated briefly. First, II and III are clear, since $\mathfrak{G}$ contains the Zermelo domain $\mathfrak{G}_\eta$ as a divisor. By Gauss's theorem on polynomials with rational integer coefficients, sums, differences, and products of integral rational functionals are again integral rational functionals; hence $\mathfrak{F}$ forms an integral domain, and by familiar arguments[130] $\mathfrak{G}$ also becomes an integral domain; thus I is fulfilled. Further, from the extended Gauss theorem[131] follow the two auxiliary propositions: "If z depends algebraically-integrally on $\mathfrak{F}$ by means of *some* equation, then it also does so by means of the *irreducible* equation which z satisfies with respect to the field of all rational functionals,"[132] and "the equation irreducible over the field of all rational numbers and satisfied by an algebraic number is also irreducible over the field of all rational functionals." Thus $\mathfrak{G}$ can contain no fractional algebraic number; hence IV, and therefore all abstract properties I–IV, have been verified.

On the other hand, under no well-ordering can a basis be selected from $\mathfrak{G}$ in such a way that $\mathfrak{G}$ excludes all rationally and algebraically fractional functions of the basis elements. For let $z(\eta)$ be any function of the domain to be chosen as a basis element, hence defined by an equation

$$z^k + A_1(\eta) z^{k-1} + \cdots + A_k(\eta) = 0,$$

where

$$A_i(\eta) = a_i \cdot \frac{E_1(\eta)}{E_2(\eta)}$$

are integral rational functionals. Then the function $\frac{a_k}{z}$ belongs to $\mathfrak{G}$, and likewise $\frac{a_k}{\sqrt{z}}$, $\frac{a_k}{\sqrt[3]{z}}$, etc.[133] Hence basis properties 4) and 5) are excluded for the totality of the domains $\mathfrak{G}$.

Remark. An example in § 3 will show that $\mathfrak{G}$ may, under every well-ordering, contain rationally fractional functionals without containing a rationally or algebraically fractional number.

§ 3. Exclusion of basis property 3).

For excluding basis property 3), we use a generalized *module domain* in which the arguments of the polynomials of the domain are no longer the indeterminates, but all integral algebraic functions of the indeterminates, [134] while the indeterminates themselves take over the role of a module basis.

Let $\mathfrak{G}$ consist of *all elements*

$$z = \alpha + g_1(\eta)\eta_1 + g_2(\eta)\eta_2 + \cdots + g_t(\eta)\eta_t, \tag{1}$$

where α runs through all algebraic integers, $\eta_1 \cdots \eta_t$ through all basis elements, and for each fixed combination $\eta_1 \cdots \eta_t$, the functions $g_1(\eta) \cdots g_t(\eta)$ individually run through all integral algebraic functions of the η.[135]

[129] H. Weber, Lehrbuch der Algebra. Kleine Ausgabe §§ 96 and 97.

[130] Compare, for example, Weber, § 97, 8.

[131] Weber, § 20, 5.

[132] Compare Weber, § 97, 4.

[133] Quite analogously, every algebraic function of the η can be transformed into an element of the functional domain $\mathfrak{G}$ by multiplication by a rational integer. Thus this domain has the property that every complex number can be represented as the quotient of an element from $\mathfrak{G}$ and a rational integer, in analogy with the algebraic numbers.

[134] By "integral algebraic functions" are always meant functions with *integral* coefficients; that is, functions satisfying some equation whose leading coefficient is unity and whose remaining coefficients are polynomials in the η with algebraic-integer coefficients — polynomials from $[H]$. In particular, all polynomials from $[H]$ belong to the integral algebraic functions.

[135] The module property belongs to the elements $z - \alpha$.

It is easy to verify that properties I–IV hold for this module domain. First I holds, since the sum, difference, and product of two elements z again takes the form (1). The domain $\mathfrak{G}$ also contains all basis elements η, and besides all elements $\eta \cdot g(\eta)$, where $g(\eta)$ runs through all integral algebraic functions of the η. Hence all integral algebraic functions of the η, and therefore all algebraic functions of the η altogether, can be represented as quotients of two elements of $\mathfrak{G}$, proving II. In the representation (1), all algebraic integers are also included in particular — for $g_1 = 0 \cdots g_t = 0$ —; but z cannot be equal to a fractional algebraic number. For if z represents an algebraic number β, then from

$$\beta = \alpha + g_1(\eta)\eta_1 + \cdots + g_t(\eta)\eta_t$$

identically in the η, necessarily $\beta = \alpha$. This is shown by the specialization

$$\eta_1 = 0 \cdots \eta_t = 0,$$

under which the multipliers $g_1(\eta) \cdots g_t(\eta)$ remain finite as integral algebraic functions, since they go over into integral algebraic functions or numbers.[136] Thus conditions I–IV are satisfied for $\mathfrak{G}$.

On the other hand, under no well-ordering can a basis be selected from $\mathfrak{G}$ in such a way that all integral algebraic functions of the basis elements belong to $\mathfrak{G}$. For let $z(\eta)$ be any function of the domain to be chosen as a basis element — which therefore must actually contain the indeterminates η —. We shall show that, from a certain finite value of ν onward, depending on z, the functions

$$\xi = \sqrt[\nu]{z - \alpha}$$

can no longer belong to $\mathfrak{G}$.

Indeed, suppose that ξ belongs to $\mathfrak{G}$. Then, by (1), it is of the form

$$\xi = \gamma + h_1(\eta)\eta_{i_1} + h_2(\eta)\eta_{i_2} + \cdots + h_\tau(\eta)\eta_{i_\tau},$$

where again $h_1(\eta) \cdots h_\tau(\eta)$ are integral algebraic functions of the η, and $\eta_{i_1} \cdots \eta_{i_\tau}$ are arbitrary basis elements, which in particular may also coincide with $\eta_1 \cdots \eta_t$. First it must be shown that the algebraic integer γ is zero. Since $h_1(\eta) \cdots h_\tau(\eta)$ are integral algebraic functions, ξ becomes equal to γ for $\eta_{i_1} = 0 \cdots \eta_{i_\tau} = 0$ with arbitrary values of the remaining η; in particular, then, ξ is equal to γ for

$$\eta_{i_1} = 0 \cdots \eta_{i_\tau} = 0; \qquad \eta_1 = 0 \cdots \eta_t = 0.$$

But under this specialization the function $(z - \alpha)$ vanishes by (1), and therefore so does ξ; hence $\gamma = 0$, and one has

$$\xi = h_1(\eta)\eta_{i_1} + h_2(\eta)\eta_{i_2} + \cdots + h_\tau(\eta)\eta_{i_\tau}.$$

Raising this to the ν-th power gives

$$\xi^\nu = z - \alpha = F_\nu(\eta),$$

where $F_\nu(\eta)$ denotes a homogeneous, not identically vanishing, form of ν-th dimension in $\eta_{i_1} \cdots \eta_{i_\tau}$, whose coefficients are integral algebraic functions of the η. Now $z - \alpha$, which by (1) is an integral algebraic function of the η, satisfies an equation irreducible with respect to $[H]$:

$$(z - \alpha)^\chi + B(\eta)(z - \alpha)^{\chi - 1} + \cdots + C(\eta) = 0,$$

where the last coefficient $C(\eta)$ — the product of $(z - \alpha)$ with its conjugates — is a polynomial; let its degree be λ. On the other hand, from $z - \alpha = F_\nu(\eta)$, by multiplying $F_\nu(\eta)$ by the corresponding conjugates, one obtains

$$C(\eta) = G_{\nu\chi}(\eta),$$

where $G_{\nu\chi}(\eta)$ denotes a homogeneous form of $\nu\chi$-th dimension in $\eta_{i_1} \cdots \eta_{i_\tau}$, whose coefficients are polynomials in the η. Thus $G_{\nu\chi}$ is at least of degree $\nu\chi$ in η, or $\lambda \geq \nu\chi$; that is, the functions $\sqrt[\nu]{z - \alpha}$ for which $\nu > \lambda/\chi$ do not belong to $\mathfrak{G}$. Basis property 3) is therefore excluded for the totality of the domains $\mathfrak{G}$.

Remark. A slight generalization of the domain just constructed leads to a domain $\mathfrak{G}$ which, under every well-ordering, also contains fractional functionals. Let $\mathfrak{G}$ consist of all elements

$$z = \alpha + \gamma_1(\eta)\eta_1 + \gamma_2(\eta)\eta_2 + \cdots + \gamma_t(\eta)\eta_t,$$

where now $\gamma(\eta)$ runs through all algebraically integral, but not necessarily integral, functions of the η. The proof of properties I–IV remains the same as above. But since $\mathfrak{G}$, besides every function z, also contains $a \cdot (z - \alpha)$, where a denotes any rational or algebraically fractional number, fractional functionals also occur under every well-ordering.

[136] This more detailed proof of IV is given because of the expanded example at the end of the paragraph. Here the remark would suffice that, by construction, z is an integral algebraic function.

§ 4. The most general domains from algebraically-integral transcendental numbers.

For a more convenient formulation of what follows, introduce the expression: "an element z depends algebraically-integrally on $\mathfrak{T}$," meaning that z satisfies some equation whose leading coefficient is unity and whose remaining coefficients are integral polynomials in the elements of $\mathfrak{T}$, where $\mathfrak{T}$ denotes any prescribed domain.[137]

A domain $\mathfrak{T}$ is called *algebraically-integrally closed* if it satisfies the condition (compare the introduction):

V. Every element that depends algebraically-integrally on $\mathfrak{T}$ belongs to $\mathfrak{T}$.

We first show that the abstract property V belongs to the Zermelo domain $\mathfrak{G}_\eta$. Let an element z depend algebraically-integrally on $\mathfrak{G}_\eta$ by means of an equation $f(z) = 0$. Then multiplication of $f(z)$ by its conjugates with respect to $[H]$ shows that z also depends algebraically-integrally on $[H]$, and hence belongs to $\mathfrak{G}_\eta$. The algebraically-integral closedness of the functional domain considered in § 2 is shown in the same way.

That property V does not belong to every domain $\mathfrak{G}$ is shown by the module domain considered in § 3, which indeed does not possess basis property 3). More precisely, § 3 showed that the module domain is algebraically-integrally closed with respect to no transcendental element of the domain, whereas, by III and IV, this is of course the case with respect to every algebraic element of the domain.

This leads us first to select, from the totality of domains $\mathfrak{G}$, those which also possess the abstract property V; these are to be called "domains $\mathfrak{H}$ from algebraically-integral transcendental numbers."

Let $\mathfrak{H}$ be such a domain, and let H be an algebraic basis of all numbers from $\mathfrak{H}$; by II, H is at the same time an algebraic basis of all numbers. By III, I, and V, $\mathfrak{H}$ contains, besides H, all integral algebraic functions of the η, that is, the Zermelo domain $\mathfrak{G}_\eta$. Thus, in analogy with § 1: every domain $\mathfrak{H}$ from algebraically-integral transcendental numbers can be constructed by means of an algebraic basis of all numbers in such a way that $\mathfrak{H}$ contains the Zermelo domain $\mathfrak{G}_\eta$ as a divisor.

Now let H be any algebraic basis of all numbers, and let $\mathfrak{M}(H)$ denote the totality of all domains $\mathfrak{H}$ containing H. Every domain $\mathfrak{H}$ in $\mathfrak{M}(H)$ contains, as just proved, the Zermelo domain $\mathfrak{G}_\eta$ as a divisor; and since $\mathfrak{G}_\eta$ itself is a domain $\mathfrak{H}$, $\mathfrak{G}_\eta$ is the greatest common divisor, or intersection, of all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$. Thus $\mathfrak{G}_\eta$ has been defined abstractly. It also follows directly that $\mathfrak{M}(H)$ is closed under intersection: that is, the intersection of all domains $\mathfrak{H}$ of any subsystem of $\mathfrak{M}(H)$ again belongs to $\mathfrak{M}(H)$. For, as an intersection, I and V are satisfied; II and III follow because $\mathfrak{G}_\eta$ is a divisor; IV follows because the intersection is a divisor of $\mathfrak{H}$.

As the example of the functional domain in § 2 shows, the domains $\mathfrak{H}$ in general contain further functions besides $\mathfrak{G}_\eta$. Let $\psi(\eta)$ be such a rational or algebraic function of the η. Then, by I, $\mathfrak{H}$ also contains all integral rational combinations of ψ and finitely many functions from $\mathfrak{G}_\eta$ at a time. The integral domain so arising — consisting of all polynomials in ψ with coefficients from $\mathfrak{G}_\eta$ — will be denoted by $[\mathfrak{G}_\eta, \psi]$, and the process leading to it will be called "rationally integral adjunction of ψ to $\mathfrak{G}_\eta$." By V, $\mathfrak{H}$ further contains all functions which depend algebraically-integrally on $[\mathfrak{G}_\eta, \psi]$; the domain so arising will be denoted by $\{\mathfrak{G}_\eta, \psi\}$, and the process leading to it by "algebraically integral adjunction of ψ to $\mathfrak{G}_\eta$." The domain $\{\mathfrak{G}_\eta, \psi\}$ consists of all integral algebraic functions of ψ with coefficients from $\mathfrak{G}_\eta$, and is therefore an integral domain by familiar arguments.[138]

We show that $\{\mathfrak{G}_\eta, \psi\}$ is also algebraically-integrally closed, that is, satisfies condition V. Indeed, let z depend algebraically-integrally on $\{\mathfrak{G}_\eta, \psi\}$ by means of an equation

$$f(z; a \cdots c) = z^\nu + az^{\nu-1} + \cdots + c = 0,$$

where $a \cdots c$, as elements of $\{\mathfrak{G}_\eta, \psi\}$, satisfy equations with coefficients from $[\mathfrak{G}_\eta, \psi]$:

$$a^\lambda + A_1 a^{\lambda-1} + \cdots + A_\lambda = 0,$$
$$\vdots$$
$$c^\mu + C_1 c^{\mu-1} + \cdots + C_\mu = 0.$$

Let their roots be denoted by $a, a' \cdots a^{(\lambda-1)} \cdots c, c' \cdots c^{(\mu-1)}$. If one now forms the product over all combinations of the roots,

$$g(z) = f(z; a \cdots c) f(z; a' \cdots c') \cdots f(z; a^{(\lambda-1)} \cdots c^{(\mu-1)}),$$

[137] For an integral domain $\mathfrak{T}$, the leading coefficient is unity and the remaining coefficients are elements of $\mathfrak{T}$.

[138] Compare, for example, Weber, § 97, 6 and 7.

then $g(z)$ has unity as leading coefficient and elements of $[\mathfrak{G}_\eta, \psi]$ as its remaining coefficients. Consequently z belongs to $\{\mathfrak{G}_\eta, \psi\}$.[139]

Thus the domain $\{\mathfrak{G}_\eta, \psi\}$ has properties I and V; and, since $\mathfrak{G}_\eta$ is a divisor of the domain, it also has II and III. Whether IV is satisfied depends on the choice of ψ; for example, IV is not satisfied for $\psi = \frac{1}{n \cdot \eta}$, since then $\frac{1}{n}$ belongs to the domain. It is satisfied, however, by § 2 for $\psi = \frac{1}{\eta}$, or also for $\psi = \frac{\eta}{n}$. For in the latter case, if a function from $\{\mathfrak{G}_\eta, \psi\}$ represents an algebraic number, its value remains unchanged for $\eta = 0$; thereby it passes into a function from $\mathfrak{G}_\eta$, and consequently into an algebraic integer.

Thus $\{\mathfrak{G}_\eta, \psi\}$ becomes a domain $\mathfrak{H}$ as soon as one imposes on ψ the condition that no algebraically fractional number enters the domain through the algebraically integral adjunction.

Quite analogously one defines the rationally integral, respectively algebraically integral, adjunction of an arbitrary system $\mathfrak{S}$ of rational or algebraic functions of the η to $\mathfrak{G}_\eta$. The rationally integral adjunction leads to the integral domain $[\mathfrak{G}_\eta, \mathfrak{S}]$, consisting of all integral polynomials in finitely many elements from $\mathfrak{G}_\eta$ and $\mathfrak{S}$ at a time. The algebraically integral adjunction gives the domain $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ consisting of all elements that depend algebraically-integrally on $[\mathfrak{G}_\eta, \mathfrak{S}]$. Exactly as above, one shows that the integral domain $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ is algebraically-integrally closed and also satisfies II and III. Thus, as above:

$\{\mathfrak{G}_\eta, \mathfrak{S}\}$ becomes a domain $\mathfrak{H}$ as soon as one imposes on $\mathfrak{S}$ the condition that no algebraically fractional number enters the domain through the algebraically integral adjunction — that is, as soon as $\mathfrak{S}$ becomes an admissible system.[140]

It is now important that the converse also holds, so that the domains $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ actually exhaust all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$ as $\mathfrak{S}$ runs through all admissible systems. Indeed, let $\mathfrak{H}$ be any domain from $\mathfrak{M}(H)$ and denote by $\mathfrak{L} = \mathfrak{H} - \mathfrak{G}_\eta$ the system consisting of all functions from $\mathfrak{H}$ not belonging to $\mathfrak{G}_\eta$. By V, $\mathfrak{H}$ contains the domain $\{\mathfrak{G}_\eta, \mathfrak{L}\}$ as a divisor; but $\{\mathfrak{G}_\eta, \mathfrak{L}\}$ is also divisible by $\mathfrak{H}$, because it contains $\mathfrak{G}_\eta$ and $\mathfrak{L}$, and since these have no common element, it contains $\mathfrak{H}$ as well. Hence $\mathfrak{H} = \{\mathfrak{G}_\eta, \mathfrak{L}\}$; and since IV is satisfied for $\mathfrak{H}$, $\mathfrak{L}$ is of course an admissible system.[141]

As H runs through every possible algebraic basis, $\mathfrak{M}(H)$ runs, as proved at the beginning, through the totality of all domains $\mathfrak{H}$. Consequently the question of the most general domains $\mathfrak{H}$ from algebraically-integral transcendental numbers is completely answered by the results of this paragraph:

a) The intersection of all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$ is given by the Zermelo domain $\mathfrak{G}_\eta$, which itself is a domain $\mathfrak{H}$; $\mathfrak{M}(H)$ is closed with respect to forming intersections.

b) Every domain $\mathfrak{H}$ from $\mathfrak{M}(H)$ arises by algebraically integral adjunction of an admissible system $\mathfrak{S}$ to $\mathfrak{G}_\eta$. As H runs through every possible algebraic basis, $\mathfrak{M}(H)$ runs through the totality of all domains $\mathfrak{H}$.[142]

[139] Thus, by defining algebraic integrality by means of *some* equation, the concept of irreducibility with respect to a ground field becomes unnecessary for the proof of I and V. Compare also § 2, where, only for the proof of IV, the auxiliary proposition is needed which passes from the definition of algebraic integrality by an arbitrary equation to the irreducible equation. Since this auxiliary proposition no longer holds in general in the present case, IV has to be imposed on the function ψ as a special condition.

[140] More generally, one shows analogously: if $\mathfrak{T}$ is any domain and $\{\mathfrak{T}\}$ denotes the totality of elements depending algebraically-integrally on $\mathfrak{T}$, then $\{\mathfrak{T}\}$ is algebraically-integrally closed.

[141] The adjunction of a subsystem of $\mathfrak{L}$ can already suffice; for example, the functional domain of § 2 arises by algebraically integral adjunction of all integral rational functionals — with the exception of the polynomials — to $\mathfrak{G}_\eta$.

[142] The admissible systems $\mathfrak{S}$ can be found as follows. Put all algebraically fractional functions of the η under some well-ordering Ω, and take into $\mathfrak{S}$ all functions except those which, when algebraically-integrally adjoined to $\mathfrak{G}_\eta$ and the previously admitted functions, generate an algebraically fractional number. This procedure can be broken off at any arbitrary place. As Ω runs through every well-ordering, and as the process is broken off at every arbitrary place for each fixed Ω, all admissible systems $\mathfrak{S}$ are thereby exhausted.